AK High-Dimensional Projection Structural Theory

Appendix NC — Complete Re-audited Edition with Integrated SC.1 Successor Research

NC.1–NC.23 Frozen Theorem DAG + Part G Silent-Collapse / TFEP Closing-Sprint Layer

Atsushi Kobayashi
AK-HPDST Project

20 October 2026

Abstract

This integrated artifact preserves the complete 18 October 2026 NC.1–NC.23 re-audited source as the frozen theorem-bearing baseline and physically integrates the 20 October 2026 SC.1 Silent-Collapse / Torsion-Free Enveloping Projectivity research continuation as Part G. The integration is one-way in logical authority: NC.1–NC.23 may be consumed by SC.1, but no SC.1 theorem, conditional criterion, open bridge, or research-management status is permitted to flow backward into the closed NC theorem DAG. The unrestricted Nakai resolution ownership remains Class III external theory at the NC evidence ceiling. SC.1 remains unresolved after the Closing Sprint, but its residual structure is sharpened by projectivity-defect reflexive closure, associated-graded compression torsion, a stable Lie–Rinehart action, the canonical projectivity ideal and differential-core obstruction, a two-dimensional Gorenstein scalar/symmetric split, and an order-two lci Hessian quantization filtration. The exact remaining HOLD obligations are determinant-to-symmetric coupling, compression elimination, Hessian/higher-order transgression, stable-to-differential quantization, or construction of an actual Silent-Collapse counterexample.

Integration constitution and precedence

(I1) **Frozen source sovereignty.** The complete NC.1–NC.23 PDF dated 18 October 2026 is reproduced without textual rewriting. Its theorem statements, proofs, evidence levels, external-source bindings, non-claims, and terminal closure retain their exact source strength.

(I2) **One-way dependency.** Part G starts from the NC.21–NC.23 chain

$$W_q \subseteq M_q = \mathrm{Sat}_A(W_q; \Delta) \subseteq D_q = M_q^{**},$$

together with the torsion-free enveloping model. Part G is not a premise of NC.1–NC.23.

(I3) **Successor means logical, not physical.** The NC.23 language exporting SC.1 as a successor problem remains correct. In this integrated edition the successor layer is co-located in the same artifact while remaining outside the closed NC.1–NC.23 theorem DAG.

(I4) **Status nonpromotion.** "Proved in Part G", "conditional", "open", "rejected route", and "HOLD" are distinct statuses. HOLD is research management only and has no truth-value force.

(I5) **No silent converse.** No statement in this artifact asserts $D = \Delta$ if and only if $\mathrm{Der}_k(A)$ is projective. The reverse implication is SC.1 and remains open.

(I6) **Chronological supersession.** The 18 October NC.23 statement that SC.1 was unresolved remains mathematically correct; Part G supersedes it only by adding the 20 October reduction structure and the explicit HOLD/re-entry ledger.

NC-to-SC.1 complete dependency map

Frozen NC owner	Consumed boundary	Part G continuation
NC.21	$D_q = M_q^{**}$ and $\mathfrak{B}_q = D_q/M_q$	Projectivity defect is placed in the same reflexive-hull defect geometry; Leakage/RDS equivalences are audited.
NC.22	$W_q \subseteq M_q \subseteq D_q$, $\mathfrak{C}_q = M_q/W_q$, projectivity defect/locus separation	Compression-hidden versus ultra-silent branching; two-dimensional scalar/symmetric projectivity splitting.
NC.23	$M_q = \mathrm{Sat}_A(W_q;\Delta)$, $\ker(U \to D) = \tau_A(U)$, $\Delta \cong U/\tau_A(U)$	Associated-graded torsion identity, PBW torsion survivors, stable Lie–Rinehart defect action, projectivity ideal, differential-core obstruction.
NC.18 + NC.21–23 interface	Rees/quantization does not automatically lift	Negative-degree Rees retained only as a conditional transverse criterion; universal negative-degree supply is rejected.
NC.22–23 residual fourth quadrant	Silent-Collapse / TFEP	2D Gorenstein determinant ideal, lci order-two principal-parts boundary, Hessian quantization defect, exact quadratic filtration, corrected counterexample package.

Integrated SC.1 status register

Item	Status	Exact ceiling
$E(L)^{**} = \mathrm{End}_A(L)$ and $\mathfrak{P} = E(L)^{**}/E(L)$	proved	Standing normal finite-type domain hypotheses.
Compression $\leftrightarrow$ torsion in $\mathrm{gr}^W \Delta$	proved	Uses NC.23 saturation identity and domain order strictness.
PBW torsion survivor formula	proved	Detects compression only; not projectivity defect universally.
Stable Lie–Rinehart action on $\mathfrak{P}$	proved	Does not automatically descend to a full D-module on all of $\mathfrak{P}$.
Canonical J_L, $V(J_L) = \mathrm{NF}(L)$, $J_L\mathfrak{P} = 0$	proved	Stable-annihilator theorem; no Collapse required for annihilation.

A/J_L descends to Δ; J_L is D-stable under Collapse	proved	Uses horizontal identity class and NC.23 action kernel.
Minimal local Silent-Collapse $\Rightarrow \mathrm{core}_D(\mathfrak{m}) \neq 0$	proved	Punctured-projective local minimal regime.
2D normal Gorenstein $J_L = I_L$	proved	Relative to determinant trivialization; ideal is unit-independent.
$\mathfrak{P} \cong A/I_L \oplus \mathfrak{P}_{\mathrm{sym}}$	proved	Characteristic zero / rank-two Gorenstein surface regime.
$I_L \neq A \Rightarrow \mathfrak{P}_{\mathrm{sym}} \neq 0$	open	Rejected determinant-minor shortcut is insufficient.
Artin quotient lifting obstruction	conditional proved	Requires local residue field $A/\mathfrak{m} = k$ in the stated form.
Negative-degree Rees transverse criterion	conditional proved	Sufficient if an actual negative-degree Rees quantization exists; not a universal supply theorem.
2D Gorenstein lci Hessian boundary and quadratic filtration	proved in declared source class	$\mathcal{H}_2$ is failed order-two quantization; nonzero $\mathcal{H}_2$ is not Escape by itself.
Self-Ext / stable-to-differential transversality	open	No proved map to any $\mathfrak{B}_q$.
Leakage and RDS universal implications	equivalent target	They are SC.1 itself in defect notation, not weaker reductions.
SC.1 / TFEP	unresolved / HOLD	No universal exclusion of all Silent-Collapse branches and no counterexample constructed.

Part I — Frozen Appendix NC.1–NC.23 source

The following 183 pages are the complete re-audited 18 October 2026 NC source. They are embedded as the immutable theorem-bearing baseline; no formula is retyped or reconstructed in this wrapper.

AK High-Dimensional Projection Structural Theory

Appendix NC — Complete Re-audited Edition

NC.1–NC.23: Nakai Resolution Ownership, Escape Theory, Rees Transfer, Collapse Mechanisms, and the Escape–Collapse Boundary

Atsushi Kobayashi
AK-HPDST Project

18 October 2026
Complete-source reassembly and final ownership correction

Abstract

This edition is the complete-source reassembly of Appendix NC.1–NC.23. It preserves the detailed theorem, lemma, proposition, definition, proof, policy, declaration, computation-control, failure-route, and claim-register content of the audited NC source files while imposing one final authority correction on the ownership of the unrestricted Nakai resolution.

Write

$$D_{A/k} = \bigcup_{q\geq 0} D^q_{A/k}, \qquad \Delta_A = A\langle \mathrm{Der}_k(A)\rangle.$$

The unrestricted Nakai conjecture asserted that $D_{A/k} = \Delta_A$ forces regularity. The final resolution classification of this edition is **Class III: external theory**. Assuming the correctness of OpenAI Family 048, its singular normal affine complex surface with free tangent module, combined with the classical Becker–Rego implication connecting Nakai to Zariski–Lipman, already yields the negative unrestricted Nakai conclusion without requiring any AK-specific theorem. NC.19–NC.20 therefore constitute an AK reconstruction, transfer, and audit of that route rather than the ownership basis of the solution.

The independent AK/NC mathematical content is retained separately. NC.1–NC.17 build the positive Escape program, culminating in the C2-high homogeneous complete-intersection master theorem candidate of NC.17. NC.18 gives the exact Rees quantization bridge and explains why graded or tangent-cone Escape does not automatically lift. NC.21–NC.23 then recast the post-resolution problem as the structural classification of Escape versus Collapse, isolating reflexive, saturation, order-compression, and tangent-projectivity defects and exporting the strictly stronger Silent-Collapse/TFEP converse as Successor Problem SC.1.

No theorem is strengthened merely by this assembly. Historical statements that the unrestricted Nakai problem was still open are retained as research-history text but are superseded by the authority declaration below. Likewise, no statement in this edition asserts "Collapse iff projective tangent."

Contents

Final Authority and Nakai Resolution Classification . . . 8
Archival source abstract for NC.1–NC.14 . . . 9
NC.1 Scope, Authority, and Constitutional Boundary . . . 10
NC.1.1 Role and release relation . . . 10
NC.1.2 Target sovereignty . . . 10
NC.1.3 Strong and weak non-generation . . . 10
NC.1.4 NC claim levels . . . 11
NC.1.5 Explicit non-claims . . . 11
NC.1.6 Construction order . . . 12
NC.2 Exact Target, Research Status, and Route Separation . . . 12
NC.2.1 Exact target statement . . . 12
NC.2.2 Current NC status freeze . . . 12

NC.2.3 Three noninterchangeable proof routes . . . 13
NC.2.4 Failure routes retained as mathematical data . . . 13
NC.2.5 Provisional literature binding for the derivation theorem . . . 13
NC.2.6 Publication and promotion boundary . . . 14
NC.3 Graded Complete Intersections and Derivation Degree Structure . . . 14
NC.3.1 Standing graded notation . . . 14
NC.3.2 Internal degree and differential order . . . 15
NC.3.3 Euler derivation . . . 15
NC.3.4 Imported local derivation-generation theorem . . . 16
NC.3.5 Degree of the standard-graded trivial generators . . . 16
NC.3.6 Local-to-graded transfer . . . 16
NC.3.7 Homogeneous generator reduction and degree budget . . . 17
NC.3.8 NC.3 closure and next obligation . . . 18
NC.4 Projectivized Operators and the Critical-Window Theorem . . . 18
NC.4.1 Purpose of the section . . . 18
NC.4.2 Localization of finite-order differential operators . . . 18
NC.4.3 Homogeneous sheafification on Proj A . . . 19
NC.4.4 Projective principal symbols . . . 19
NC.4.5 Euler scalarization . . . 20
NC.4.6 Projective order of an arbitrary generated word . . . 20
NC.4.7 The critical-window theorem . . . 21
NC.4.8 Strong escape interface . . . 22
NC.4.9 Adversarial audit of the long-word issue . . . 22
NC.4.10 Scope and evidence status . . . 23
NC.5 The Type $(2, d)$ Explicit Route: Audited Corank-One Branch . . . 23
NC.5.1 Purpose and exact scope . . . 23
NC.5.2 The geometric corank dichotomy . . . 24
NC.5.3 Homogeneous differential identities . . . 24
NC.5.4 Principal tangency identities . . . 25
NC.5.5 Lower-order correction and descent . . . 25
NC.5.6 Critical internal degree . . . 26
NC.5.7 Generic-chart certificate for a nonzero projective symbol . . . 27
NC.5.8 Strong escape for the corank-one branch . . . 28
NC.5.9 Adversarial audit and remaining all-$(2, d)$ obligation . . . 28
NC.6 The Nondegenerate Type $(2, d)$ Branch and Full Smooth Closure . . . 29
NC.6.1 Purpose . . . 29
NC.6.2 Coordinate-free setup for a nondegenerate quadric . . . 29
NC.6.3 The one-degree-lower cofactor tensor . . . 29
NC.6.4 Trace and Hessian contractions . . . 30
NC.6.5 Lower-order correction . . . 31
NC.6.6 The actual ambient differential operator . . . 31
NC.6.7 Critical degree . . . 32
NC.6.8 Nonzero projective symbol . . . 32
NC.6.9 Strong escape in the nondegenerate branch . . . 33
NC.6.10 Faithfully flat field-extension detection . . . 34
NC.6.11 Closure of the full smooth type-$(2, d)$ branch . . . 34
NC.6.12 Adversarial status and nonclaim ceiling . . . 35
NC.7 Jacobian Stress, Hessian Obstruction, and the Second-Order Quantization Gate . . . 36
NC.7.1 Purpose and presentation level . . . 36
NC.7.2 The degreewise stress and correction spaces . . . 36
NC.7.3 The exact ambient Leibniz identity . . . 37
NC.7.4 Exact descent criterion . . . 38
NC.7.5 Cotangent-cohomology target . . . 38

NC.7.6 The Hessian obstruction map . . . 39
NC.7.7 Parameter-space exact sequence . . . 40
NC.7.8 Presentation dependence and invariance ceiling . . . 41
NC.7.9 Diagonal quadratic collapse of the obstruction . . . 41
NC.7.10 Finite linear-algebra character . . . 42
NC.7.11 NC.7 closure . . . 42
NC.8 Critical Quantized Stress Defect and the NC Master Escape Bridge . . . 43
NC.8.1 Purpose . . . 43
NC.8.2 The projective symbol map for quantized stresses . . . 43
NC.8.3 Projectively invisible and visible quantized stresses . . . 44
NC.8.4 The generated projective-symbol sector . . . 45
NC.8.5 The Quantized Stress escape defect . . . 45
NC.8.6 Critical Quantized Stress defect . . . 46
NC.8.7 The new NC Master Bridge . . . 47
NC.8.8 Diagonal quadrics as the first natural source class . . . 47
NC.8.9 A concrete horizontal witness for the canonical $(2,2)$ control . . . 48
NC.8.10 Finite computation and evidence discipline . . . 48
NC.8.11 NC.8 closure . . . 49
NC.9 Higher-Codimension Diagonal Quadratic Stress Escape . . . 49
NC.9.1 Scope and objective . . . 49
NC.9.2 Normal-domain closure from projective smoothness . . . 50
NC.9.3 Coefficient rank . . . 50
NC.9.4 The finite stress kernel . . . 50
NC.9.5 The dimension criterion . . . 52
NC.9.6 Critical polynomial stress . . . 52
NC.9.7 Jacobian stress and automatic quantization . . . 53
NC.9.8 Projective visibility . . . 54
NC.9.9 Higher-codimension diagonal quadratic theorem . . . 55
NC.9.10 Explicit Vandermonde family . . . 55
NC.9.11 Evidence status and novelty ceiling . . . 58
NC.10 General Quadratic ICIS: Exact-Stress Divergence Quantization and the One-Degree-Lower Barrier 59
NC.10.1 Purpose and current boundary . . . 59
NC.10.2 Exact quadratic stresses . . . 60
NC.10.3 Divergence identity and automatic quantization . . . 60
NC.10.4 Divergence-form interpretation . . . 61
NC.10.5 Critical exact-stress escape criterion . . . 62
NC.10.6 A universal exact stress at the threshold . . . 63
NC.10.7 Two-lane architecture for arbitrary quadrics . . . 64
NC.10.8 What is already solved inside the quadratic class . . . 64
NC.10.9 The exact remaining obstruction for all quadratic ICIS . . . 65
NC.10.10 Research consequences of the Gram benchmark . . . 65
NC.10.11 NC.10 closure . . . 66
NC.11 Cofactor Syzygies, Buchsbaum–Rim Symmetrization, and the Radial Trap . . . 66
NC.11.1 Purpose . . . 66
NC.11.2 The linear-gradient matrix . . . 67
NC.11.3 Cofactor syzygies . . . 67
NC.11.4 Raw dimension count cannot force a critical exact stress . . . 68
NC.11.5 The radial cofactor trap . . . 69
NC.11.6 Cofactor matrix and nonradial symmetrization . . . 70
NC.11.7 Buchsbaum–Rim completeness gate . . . 72
NC.11.8 Why Buchsbaum–Rim alone is not yet the all-quadratic proof . . . 73
NC.11.9 Finite linear-algebra realization . . . 73
NC.11.10 NC.11 closure . . . 74

NC.12 Exact Non-Diagonal Quadratic Controls and Branch Separation . . . 75
NC.12.1 Purpose . . . 75
NC.12.2 Control C_5: a smooth non-simultaneously-diagonal curve . . . 75
NC.12.3 Control C_6: a smooth non-simultaneously-diagonal surface . . . 76
NC.12.4 Exact graded quotient bases . . . 76
NC.12.5 The critical quantization matrix . . . 77
NC.12.6 Radial sector and a horizontal projective detector . . . 77
NC.12.7 Exact cofactor lane . . . 78
NC.12.8 The branch-separation theorem . . . 79
NC.12.9 Interpretation and evidence ceiling . . . 79
NC.13 Determinantal Rank Geometry on a Fixed Normal-Form Slice . . . 80
NC.13.1 Purpose . . . 80
NC.13.2 A fixed quadratic Gröbner normal-form slice . . . 81
NC.13.3 Universal critical matrices . . . 82
NC.13.4 Rank loci . . . 82
NC.13.5 The visibility rank increment . . . 83
NC.13.6 Rank-stratified openness . . . 84
NC.13.7 Smoothness and non-diagonal loci . . . 84
NC.13.8 The NC.12 surface control as a rank-stratum seed . . . 85
NC.13.9 The generic promotion gate . . . 86
NC.13.10 Why generic is still not all . . . 87
NC.13.11 NC.13 closure . . . 87
NC.14 K3 Critical Cohomology and the All-Smooth $(2,2,2)$ Theorem . . . 88
NC.14.1 Global re-audit register before NC.14 . . . 88
NC.14.2 The K3 geometry of a smooth triple quadric . . . 88
NC.14.3 Universal vanishing of the critical T^1 slice . . . 89
NC.14.4 The critical stress/projective-tensor exact sequence . . . 91
NC.14.5 Riemann–Roch for the critical K3 tensor bundle . . . 93
NC.14.6 Vanishing of the top cohomology . . . 94
NC.14.7 The all-smooth triple-quadric theorem . . . 95
NC.14.8 Reinterpretation of the NC.12 number 22 . . . 95
NC.14.9 Relation to explicit symmetric-cotangent cohomology . . . 96
NC.14.10 Architectural consequences . . . 97
NC.15 Quadratic Completion: Universal Critical Quantization and Intrinsic Cofactor Escape . . . 98
NC.15.1 Purpose, hypotheses, and claim ceiling . . . 98
NC.15.2 Critical numerical data . . . 99
NC.15.3 Gate Q: the critical T^1 slice . . . 99
NC.15.4 Intrinsic interpretation and numerical transport . . . 101
NC.15.5 Gate N: intrinsic adjugate tensors . . . 102
NC.15.6 Universal quadratic stress/projective-tensor sequence . . . 103
NC.15.7 All-quadratic strong escape . . . 105
NC.15.8 Low-dimensional compatibility checks . . . 105
NC.15.9 NC.15 status freeze . . . 106
NC.16 General-Degree Complete Intersections: Intrinsic Critical Bundles, Curve Closure, and the Order-Selection Frontier . . . 106
NC.16.1 General homogeneous setup . . . 106
NC.16.2 Universal critical-bundle identity . . . 107
NC.16.3 General critical reduction . . . 107
NC.16.4 Complete closure in projective dimension one . . . 108
NC.16.5 Equidegree adjugates and the matching index . . . 109
NC.16.6 The same index on the diagonal Quantization Gate . . . 110
NC.16.7 First post-quadratic boundary: cubic surfaces . . . 111
NC.16.8 Beyond the order-two wall . . . 112

NC.16.9 Mixed multidegrees . . . 112
NC.16.10 NC.16 architectural closure . . . 112
NC.17 General Homogeneous-ICIS Completion: Strong Escape Master Theorem . . . 113
NC.17.1 Scope, authority, and integration policy . . . 115
NC.17.1.1 Target and source class . . . 115
NC.17.1.2 Strong escape versus order-wise escape . . . 116
NC.17.1.3 Legacy-to-integrated map . . . 116
NC.17.2 Graded ICIS derivations and the robust Critical Window . . . 116
NC.17.2.1 The derivation threshold . . . 116
NC.17.2.2 Imported local theorem and exact graded transfer . . . 117
NC.17.2.3 Internal-degree factor budget . . . 118
NC.17.2.4 Projectivization and Euler scalarization . . . 118
NC.17.3 The universal critical bundle and projective convention . . . 120
NC.17.3.1 Projectivization convention . . . 120
NC.17.4 Universal positivity of the top complete-symmetric class . . . 121
NC.17.4.1 K-theory and generating function . . . 121
NC.17.4.2 Exact two-layer decomposition . . . 121
NC.17.4.3 The coefficient inequality . . . 122
NC.17.5 Asymptotic abundance of critical projective symbols . . . 123
NC.17.6 Lower symbol layers: deep vanishing and boundary growth . . . 124
NC.17.6.1 Canonical bundle of the projective bundle . . . 124
NC.17.7 The cone-tangent filtration and projective lifting . . . 125
NC.17.8 Quantization on the punctured cone . . . 126
NC.17.8.1 Equivariant tangent identification . . . 126
NC.17.8.2 Principal-symbol filtration . . . 127
NC.17.9 Extension across the cone vertex . . . 128
NC.17.10 The curve case: exact order-two closure . . . 128
NC.17.11 Master theorem: homogeneous complete intersections . . . 129
NC.17.12 Legacy refinements, negative controls, and dependency reduction . . . 130
NC.17.12.1 NC.14: the K3 quantitative refinement . . . 130
NC.17.12.2 NC.15: quadrics . . . 130
NC.17.12.3 NC.16: surfaces and the cubic order-two failure . . . 130
NC.17.12.4 Threefold and fourfold asymptotics . . . 130
NC.17.13 Audit register, imported edges, and publication posture . . . 131
NC.17.13.1 Imported edges actually consumed . . . 131
NC.17.13.2 Internally proved edges . . . 131
NC.17.13.3 Adversarial checks retained . . . 132
NC.17.13.4 Prior-art ceiling . . . 132
NC.17.14 Research consequences and next frontier . . . 132
.1 Specialization checks in dimensions three and four . . . 133
.2 Claim register for the integrated edition . . . 133
.3 References and exact source notes . . . 134
A General Nakai Reduction and Exact Rees Quantization Gate . . . 134
A.1 General Nakai reduction and Rees quantization bridge . . . 135
A.1.1 Scope, target, and relation to NC.17 . . . 135
A.1.2 Principal parts and localization . . . 135
A.1.3 Reduction to an isolated local singularity . . . 136
A.1.4 Derivation invariance of the isolated maximal ideal . . . 137
A.1.5 Extended Rees algebra and tangent cone . . . 137
A.1.6 Principal parts under the Rees degeneration . . . 138
A.1.7 The exact Rees Quantization Gate . . . 138
A.1.8 Lifting first-order derivations . . . 139
A.1.9 Saturation and the derivation shadow . . . 140

A.1.10 Exact escape transfer . . . 140
A.1.11 The Rees defect quotient . . . 141
A.1.12 Graded sectors and a dimension-surplus theorem . . . 141
A.1.13 Why the old Critical Window does not automatically globalize . . . 143
A.1.14 The universal Shadow-Defect Gate . . . 143
A.1.15 Three attack routes for the remaining gate . . . 143
A.1.16 A sharper contradiction template . . . 144
A.1.17 Audit: what NC.18 proves and what it does not . . . 144
A.1.18 Immediate NC continuation . . . 145
References and source bindings . . . 145
B Free-Tangent Collapse and Final-Gate Bifurcation . . . 145
B.1 NC.19 scope and the final-gate question . . . 146
B.2 Differential operators, the generated algebra, and localization . . . 146
B.3 Reflexive extension package . . . 146
B.4 The free-tangent collapse theorem . . . 147
B.5 Immediate negative criterion for Nakai . . . 148
B.6 Impact on the NC.18 Universal Shadow-Defect Gate . . . 148
B.7 The September 2026 free-tangent surface claim . . . 149
B.8 Final-gate bifurcation . . . 149
B.9 What survives regardless of the external fork . . . 149
B.10 Audit register . . . 150
B.11 Required next audit and claim ceiling . . . 150
C Family 048 Audit and Negative Nakai Transfer . . . 150
C.1 Exact logical target . . . 151
C.2 The external witness theorem . . . 151
C.3 Audit layer I: curve and residue engine . . . 151
C.4 Audit layer II: low-order jet obstruction . . . 152
C.5 Audit layer III: analytic realization . . . 153
C.6 Audit layer IV: algebraization and descent . . . 153
C.7 Audit layer V: Nakai transfer . . . 154
C.8 Claim register and status . . . 154
C.9 Consequences for Appendix NC . . . 155
D Reflexive Escape–Collapse Boundary Theory . . . 155
D.1 NC.21 scope, standing hypotheses, and audit ceiling . . . 156
D.1.1 Standing hypotheses . . . 156
D.1.2 Why the order filtration must be used . . . 156
D.2 Reflexivity of finite-order differential operators . . . 156
D.2.1 Principal parts . . . 156
D.3 Codimension-one coincidence . . . 157
D.3.1 Localization interfaces . . . 157
D.3.2 Height-one regularity . . . 157
D.4 The exact Escape–Collapse Boundary Theorem . . . 157
D.5 Local cohomology and the isolated-singularity form . . . 158
D.6 Collapse region: from free to projective tangent modules . . . 159
D.6.1 The Becker–Rego mechanism . . . 159
D.6.2 Two tempting weakenings that do not distinguish the boundary . . . 160
D.7 Relation to the NC.18 Rees–Nakai defect . . . 160
D.8 Known and AK-controlled regions . . . 161
D.9 Boundary invariants . . . 161
D.10 The actual unresolved boundary . . . 162
D.11 Adversarial checks and failure modes . . . 163
D.11.1 Why normality is essential to the exact theorem . . . 163
D.11.2 Why the word filtration is not substituted . . . 163

D.11.3 Why Der(A) reflexive does not imply Collapse . . . 163
D.11.4 Why codimension-one agreement alone is not Collapse . . . 163
D.12 Prior-art position . . . 163
D.13 Claim register . . . 163
D.14 Final synthesis . . . 164
E Projective Exhaustion and Silent-Collapse Boundary . . . 164
E.1 Scope, standing hypotheses, and inherited boundary . . . 165
E.1.1 Standing hypotheses . . . 165
E.2 Word filtration and the order-compression defect . . . 166
E.3 The tangent projectivity defect . . . 166
E.4 Projective tangent annihilates both differential defects . . . 167
E.5 Escape, compression, and nonprojectivity loci . . . 167
E.6 Projective Exhaustion and exact class-level equivalences . . . 168
E.7 The isolated-singularity four-quadrant table . . . 168
E.8 Compatibility with the negative resolution of unrestricted Nakai . . . 169
E.9 Why PBW cannot be reversed at NC.22 . . . 169
E.10 NC.22 claim register and terminal handoff . . . 169
F Torsion-Free Enveloping Reduction and Final NC Closure . . . 170
F.1 Scope, notation, and inherited NC results . . . 171
F.2 PBW without projective tangent . . . 171
F.3 The generic Lie–Rinehart action . . . 171
F.4 The action-kernel theorem . . . 172
F.5 Generic detection of differential order . . . 173
F.6 Filtration saturation . . . 173
F.7 The two-stage singular correction . . . 174
F.8 Projective tangent and the residual implication . . . 175
F.9 A minimal Silent-Collapse core . . . 175
F.10 Exclusion mechanisms and known regions . . . 175
F.10.1 D-simple singularities . . . 176
F.10.2 Normal toric singularities . . . 176
F.10.3 Isolated homogeneous hypersurfaces . . . 176
F.10.4 Homogeneous complete intersections of codimension at least two . . . 176
F.10.5 Projective tangent and Family 048 . . . 176
F.11 Finite-generation obstruction . . . 176
F.12 Why three tempting completion routes fail . . . 177
F.12.1 PBW failure is not the final obstruction . . . 177
F.12.2 A D-module structure does not force local freeness . . . 177
F.12.3 Reflexivity is too weak . . . 177
F.13 The exact residual problem and its successor status . . . 177
F.14 NC.1–NC.23 terminal closure theorem . . . 178
F.15 Final quadrant and locus register . . . 179
F.16 Final NC.23 claim register . . . 179
F.17 Research freeze and successor protocol . . . 179
Unified References and Source Bindings . . . 181

Final Authority and Nakai Resolution Classification

Declaration NC.0.1 (Three-way resolution classification)**.** For the purpose of attribution and theorem ownership, the target "resolve the unrestricted Nakai conjecture" is classified into exactly one of the following lanes:

(I) **AK-only resolution**: the decisive theorem chain requires an AK-owned mathematical theorem;

(II) **AK-core plus external theory**: an AK-owned core theorem is indispensable, with external theorems supplying additional edges;

(III) **External-theory resolution**: external results already close the target without any indispensable AK-owned theorem.

The unrestricted Nakai conjecture is assigned to **Class III** in this final edition.

Theorem NC.0.2 (Final ownership statement for the unrestricted Nakai resolution)**.** *Assume the Family 048 theorem is correct at its declared external evidence level. Then the unrestricted Nakai conjecture is false by an external theorem chain: the Family 048 counterexample to the characteristic-zero Zariski–Lipman expectation, together with the classical Becker–Rego implication relating Nakai generation to Zariski–Lipman regularity, is sufficient. Consequently no AK theorem is logically necessary for the negative resolution itself.*

Proof. The external Family 048 theorem supplies a singular normal affine complex algebra whose tangent module is free. The Becker–Rego mechanism is precisely the classical bridge under which the Nakai generation assertion would force the corresponding Zariski–Lipman regularity statement. Hence a valid characteristic-zero Zariski–Lipman counterexample rules out unrestricted Nakai. NC.19 reconstructs the collapse mechanism and NC.20 audits the external witness, but those AK layers are not logically indispensable to the external implication. □

Declaration NC.0.3 (What remains AK/NC-owned)**.** The ownership correction does not erase the NC research program. The surviving AK/NC objects are separated as follows: NC.1–NC.16 build the Critical-Window/Quantized-Stress positive program; NC.17 is the terminal homogeneous-ICIS Escape master theorem candidate; NC.18 is the Rees quantization and transfer bridge; NC.19 is an internal reconstruction of the classical collapse mechanism; NC.20 is an external-source audit and transfer; and NC.21–NC.23 form the post-resolution Escape–Collapse boundary theory. Novelty and publication priority for the AK-owned candidate theorems remain subject to dedicated prior-art review and independent refereeing.

Archival source abstract for NC.1–NC.14

This appendix develops a post-v21 research program for the Nakai conjecture through graded differential operators on singular homogeneous complete intersections. Its immediate objective is not to claim the general Nakai conjecture, but to isolate, prove, and audit reusable mechanisms that can force

$$D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle$$

for singular graded complete intersections. The appendix separates five logically distinct layers: graded derivation structure, a critical internal-degree window, existence of non-radial principal symbols, quantization of those symbols to actual quotient differential operators, and strong escape from the algebra generated by first-order derivations. The new Quantized Stress calculus rewrites the second-order quantization problem as a Jacobian tangency equation together with a Hessian obstruction class. For diagonal quadratic complete intersections the Hessian obstruction vanishes on a natural zero-diagonal stress family, giving a concrete route toward a scoped strong non-generation theorem. All such results remain Appendix-NC statements until their hypotheses, imported source bindings, exact algebraic proofs, and independent replay obligations have been discharged. No statement in this construction draft asserts a solution of the general Nakai conjecture or of all homogeneous complete intersections.

NC.1 Scope, Authority, and Constitutional Boundary

NC.1.1 Role and release relation

Definition NC.1.1 (NC research-supplement role)**.** Appendix NC is a post-v21 research supplement devoted to the Nakai conjecture and to theorem-producing mechanisms for differential-operator non-generation on singular graded complete intersections. It does not modify, overwrite, or silently strengthen any statement of AK-HPDST v21.0.0. Every theorem, lemma, construction, conjecture, computation, or failure record in this appendix receives an NC-local identity and is interpreted at exactly the strength printed here.

Policy 1 (Frozen-v21 boundary)**.** *The complete v21 source set, its Claim Register, and all v21 theorem identities are treated as immutable lower-baseline material. An NC argument may invoke a v21 theorem only at its registered hypotheses and conclusion strength. An NC theorem is not a successor-Core theorem merely because it reuses a v21 mechanism such as a proof DAG, a finite certificate, a projection/Return pattern, or a structural-discovery rule. Any future promotion requires an explicit successor-release admission theorem and a distinct claim identity.*

Remark NC.1.2. The central NC argument is primarily commutative-algebraic and differential-operator-theoretic. Core-P, Core-DG, persistence, descent, and Bridge machinery are therefore *not* invoked by default. AK-SDR may be used as a research-design discipline—for example to identify a missing structure or a separating witness—but SDR methodology never substitutes for a missing algebraic theorem.

NC.1.2 Target sovereignty

Fix throughout a field k of characteristic 0, unless a theorem states a different coefficient regime.

Definition NC.1.3 (Nakai target)**.** For a finitely generated k-algebra A, write

$$D_{A/k} := D_{A/k}, \qquad \Delta_A := A\langle \mathrm{Der}_k(A)\rangle \subseteq D_{A/k}.$$

The Nakai conjecture asks whether

$$D_{A/k} = \Delta_A \quad \Longrightarrow \quad A \text{ is regular.}$$

Equivalently, its contrapositive asks whether every singular A in the declared class satisfies

$$D_{A/k} \neq \Delta_A.$$

Definition NC.1.4 (NC primary target class)**.** The primary construction class of NC 1.0 consists of standard-graded homogeneous isolated complete-intersection singularities

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_c), \qquad \deg f_\alpha = d_\alpha \geq 2,$$

subject to the normality, domain, quasihomogeneity, and projective smoothness hypotheses explicitly required by each theorem. No theorem may inherit any of these hypotheses merely from the phrase "homogeneous ICIS".

NC.1.3 Strong and weak non-generation

Definition NC.1.5 (Length-truncated generated sector)**.** For $q \geq 1$, define

$$\Delta_A^{[\leq q]} := A + \sum_{j=1}^{q} A\text{-span}\left\{\delta_1 \cdots \delta_j : \delta_i \in \mathrm{Der}_k(A)\right\} \subseteq D^q_{A/k}.$$

This sector limits the number of first-order derivation factors to at most q. It must not be confused with

$$\Delta_A \cap D^q_{A/k},$$

because an arbitrarily long word in first-order derivations may have differential order at most q after commutation or cancellation.

Definition NC.1.6 (Weak order-wise escape)**.** A weak order-q escape is an operator

$$D \in D^{q}_{A/k} \setminus \Delta_A^{[\leq q]}.$$

This proves failure of the length-truncated generation statement only. It does *not* imply $D \notin \Delta_A$.

Definition NC.1.7 (Strong escape)**.** A strong escape is an actual operator

$$D \in D_{A/k} \setminus \Delta_A.$$

Equivalently, for an operator already known to lie in $D^{q}_{A/k}$, it is enough to prove

$$D \notin \Delta_A \cap D^{q}_{A/k}.$$

Existence of a strong escape immediately gives

$$D_{A/k} \neq \Delta_A.$$

Policy 2 (No weak-to-strong promotion)**.** *A weak order-wise escape is never reported as strong ring non-generation unless a separate argument excludes* all *words in $A\langle \mathrm{Der}_k(A)\rangle$, including arbitrarily long words whose differential order may drop after commutation or cancellation.*

NC.1.4 NC claim levels

Definition NC.1.8 (NC evidence levels)**.** Every substantive NC claim is assigned one of the following evidence levels.

- C0 **Algebraic identity.** A finite exact identity established directly from definitions, linear algebra, Leibniz expansion, adjunction, or another explicitly displayed argument.
- C1 **Imported literature theorem.** A theorem whose original or authoritative source has been identified and whose hypotheses have been checked against the NC invocation.
- C2 **Internal proof candidate.** A complete or nearly complete NC proof whose independent publication-grade audit has not yet closed.
- C3 **Exact computation.** An exact rational/integer, Gröbner, symbolic, or finite-algebra computation with a replayable artifact.
- C4 **Modular or finite experimental evidence.** Strong computational evidence that is not promoted to theorem status.
- C5 **Conjecture or discovery target.** A mathematically explicit but unproved statement.

No favorable level on one claim promotes another claim.

Policy 3 (Weakest supported reading)**.** *If a prose summary, computation log, theorem draft, source note, or later consumer disagrees about the status, hypotheses, or strength of an NC statement, the weakest source-supported type-correct reading governs until a new audited revision repairs the discrepancy.*

NC.1.5 Explicit non-claims

Declaration NC.1.9 (NC.1 non-claim register)**.** NC 1.0 does not presently claim any of the following.

(i) a proof of the general Nakai conjecture;

(ii) a proof for all homogeneous isolated complete intersections;

(iii) a proof for all codimension-two complete intersections of arbitrary degree;

(iv) automatic quantization of every projective critical tensor;

(v) universal vanishing of a critical graded piece of T^1_A;

(vi) equivalence between existence of a projective symbol and existence of an ambient polynomial differential operator;

(vii) novelty or literature priority for any NC theorem before a dedicated prior-art audit;

(viii) promotion of any NC-local theorem into the stable v21 Core.

NC.1.6 Construction order

Policy 4 (Proof-first construction order)**.** *The NC construction is developed in the following dependency order:*

graded ICIS and derivation structure
$\longrightarrow$ *critical internal-degree window*
$\longrightarrow$ *actual quotient differential operator*
$\longrightarrow$ *non-radial principal-symbol certificate*
$\longrightarrow$ *strong escape from* Δ_A.

Projective tensor existence, Schur-functor nonvanishing, and higher-order enrichment are introduced only when the lower-order direct route does not already supply the required strong escape.

NC.2 Exact Target, Research Status, and Route Separation

NC.2.1 Exact target statement

Definition NC.2.1 (NC master implication)**.** For a declared source class $\mathscr{C}$ of singular graded k-algebras, the NC master target is the quantified implication

$$A \in \mathscr{C},\ A \text{ singular} \quad \Longrightarrow \quad D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.$$

A theorem for a proper subclass $\mathscr{C}_0 \subsetneq \mathscr{C}$ proves only the corresponding scoped Nakai statement on $\mathscr{C}_0$.

Remark NC.2.2. The distinction between a scoped theorem and the general conjecture is structural, not editorial. For example, a theorem for diagonal quadratic complete intersections does not become a theorem for arbitrary quadrics, and a theorem for complete intersections does not become a theorem for arbitrary finitely generated singular algebras.

NC.2.2 Current NC status freeze

Declaration NC.2.3 (NC 1.0 initial status freeze)**.** At the start of NC 1.0 construction, the research state is recorded as follows.

(a) The general Nakai conjecture remains an external open target.

(b) The general homogeneous-ICIS target remains open inside NC.

(c) A type-$(2, d)$ explicit strong-escape route exists as an internal theorem candidate and must be independently frozen before publication use.

(d) A general critical-window principle has a high-confidence internal proof candidate, but its long-word and symbol-factor statements require a dedicated proof in the NC namespace.

(e) Projective critical-symbol existence and quantization must not be conflated.

(f) The former regularity/H^1-shortcut quantization argument is rejected.

(g) Universal critical T^1-vanishing is rejected as a general route.

(h) The new second-order Quantized Stress route is provisionally organized around an exact Jacobian tangency equation plus a Hessian obstruction equation; the identities themselves are not consumed as theorem evidence until they are proved in the later NC sections.

(i) For diagonal quadratic complete intersections, a broad stress family is a candidate for vanishing Hessian obstruction and strong escape; this remains a downstream C2 target until the quantization identities and the critical-window theorem are independently closed.

NC.2.3 Three noninterchangeable proof routes

Definition NC.2.4 (Explicit-operator route)**.** The explicit-operator route constructs an ambient differential operator

$$\widetilde{D} \in D_{k[x_1,\dots,x_n]/k}$$

such that $\widetilde{D}(I) \subseteq I$, descends it to $D \in D_{A/k}$, and proves directly that $D \notin \Delta_A$. The type-$(2, d)$ construction and the quadratic-stress construction belong to this route.

Definition NC.2.5 (Projective-symbol route)**.** The projective-symbol route first produces a nonzero class in a sheaf-theoretic symbol space such as

$$H^0(X, \mathrm{Sym}^q T_X(r))$$

or an equivalent Schur-functor model and then requires a separate quantization theorem showing that at least one such class comes from an actual operator on the affine cone. Existence of the projective class alone is not an NC escape theorem.

Definition NC.2.6 (Quantized-stress route)**.** The Quantized Stress route searches directly for a principal-symbol matrix or higher tensor that already satisfies the descent equations needed for an actual quotient differential operator. For order two this route is governed by the Jacobian tangency and Hessian obstruction equations developed later in Appendix NC. It is therefore a direct quantization-first route rather than a projective-existence-then-lift route.

Policy 5 (Route noncompensation)**.** *Success of one route does not fill an unproved edge in another route. In particular, a successful explicit operator does not prove surjectivity of a global symbol map, and a nonzero projective tensor does not prove the existence of an ambient polynomial differential operator without a quantization theorem.*

NC.2.4 Failure routes retained as mathematical data

Definition NC.2.7 (NC failure classes)**.** The construction records at least the following failure classes:

$\mathsf{F}_{\mathrm{deg}}$	critical internal degree incompatible with the proposed construction,
$\mathsf{F}_{\mathrm{tan}}$	Jacobian tangency equation has no required nonzero solution,
$\mathsf{F}_{\mathrm{Hess}}$	Hessian obstruction does not lie in the correction image,
$\mathsf{F}_{\mathrm{rad}}$	every quantizable symbol in the tested family is Euler/radial,
F_{CW}	critical-window exclusion from Δ_A not proved,
$\mathsf{F}_{\mathrm{scope}}$	normality, domain, ICIS, grading, or source hypotheses fail,
$\mathsf{F}_{\mathrm{source}}$	an imported theorem lacks an exact source/hypothesis binding.

A failure of one route is not automatically a counterexample to the Nakai conjecture and is not automatically a proof that no other NC route can succeed.

NC.2.5 Provisional literature binding for the derivation theorem

Declaration NC.2.8 (Kersken source-binding record)**.** The derivation-generation input used later in NC.3 has the following current source hierarchy.

(i) The primary attribution is M. Kersken, *Reguläre Differentialformen*, Manuscripta Math. **46** (1984), 1–25, cited in later literature as the source of the Euler-plus-trivial generation theorem for quasihomogeneous normal ICIS.

(ii) An explicit modern statement is M. Granger and M. Schulze, *Derivations of negative degree on quasihomogeneous isolated complete intersection singularities*, arXiv:1403.3844, Theorem 10, which states that the module of derivations of a quasihomogeneous normal ICIS is generated by the Euler derivation and the trivial derivations; its equation (2.8) gives the weighted degree of the trivial generators.

(iii) J. Herzog and A. Martsinkovsky, *Gluing Cohen–Macaulay modules with applications to quasihomogeneous complete intersections with isolated singularities*, Comment. Math. Helv. **68** (1993), 365–384, Corollary 4.3 and Remark 4.4, gives a local algebraic formulation over a field of characteristic zero in which the trivial derivations together with the Euler derivation minimally generate the derivation module.

These sources are local/analytic or local-algebraic statements. Appendix NC does not silently identify them with the global standard-graded module $\mathrm{Der}_k(A)$. The local-to-graded transfer required for the present affine cone is proved separately in NC.3.

NC.2.6 Publication and promotion boundary

Policy 6 (Publication gate)**.** *An NC theorem may be designated publication-ready only after:*

(i) every imported theorem has an exact source binding;

(ii) every grading convention and degree shift is explicit;

(iii) every quotient-operator descent step is proved;

(iv) strong escape is distinguished from order-wise escape;

(v) computational evidence used in the proof has an exact replay artifact;

(vi) all stated base-field, normality, domain, and smoothness hypotheses are checked for necessity or explicitly retained; and

(vii) a dedicated prior-art search has been completed.

Compilation, replay, or successful symbolic computation is not by itself mathematical proof of an unproved theorem edge.

NC.3 Graded Complete Intersections and Derivation Degree Structure

NC.3.1 Standing graded notation

Let

$$P := k[x_1, \dots, x_n]$$

with the standard grading $\deg x_i = 1$, and let

$$I := (f_1, \dots, f_c) \subset P$$

be generated by a homogeneous regular sequence with

$$\deg f_\alpha = d_\alpha \geq 2.$$

Set

$$A := P/I, \qquad X := \mathrm{Proj}\, A.$$

Whenever X is nonempty,

$$p := \dim X = n - c - 1, \qquad \dim A = p + 1.$$

Definition NC.3.1 (NC derivation threshold)**.** Define

$$\boxed{m := \sum_{\alpha=1}^{c} d_\alpha - (c+1).}$$

The role of m is not assumed from notation alone. It becomes a derivation-degree threshold only under the imported Kersken-type generation theorem stated below.

NC.3.2 Internal degree and differential order

Definition NC.3.2 (Internal degree)**.** A k-linear homogeneous operator

$$D : A \to A$$

has internal degree r if

$$D(A_t) \subseteq A_{t+r}$$

for every t. Write $D^q_{A/k,r}$ for differential operators of order at most q and internal degree r.

Definition NC.3.3 (Generated algebra)**.** Set

$$\Delta_A := A\langle \mathrm{Der}_k(A) \rangle \subseteq D_{A/k}.$$

This is the full subalgebra generated by multiplication operators and all first-order k-derivations. It permits arbitrary word length. For each internal degree r,

$$(\Delta_A)_r$$

denotes its homogeneous degree-r part.

Remark NC.3.4. Word length, differential order, and internal degree are three different filtrations or gradings. The NC critical-window argument uses internal degree to bound how many positive-degree non-Euler derivations can occur, and differential order to inspect the surviving principal symbol. No argument may replace one of these quantities by another without proof.

NC.3.3 Euler derivation

Definition NC.3.5 (Euler derivation)**.** The Euler derivation is

$$E := \sum_{i=1}^{n} x_i \partial_i.$$

It has internal degree 0, and for every homogeneous $a \in A_t$,

$$E(a) = t\, a.$$

Lemma NC.3.6 (Euler commutator identities)**.** *Let $a \in A$ be homogeneous of degree t, and let $\delta \in \mathrm{Der}_k(A)$ be homogeneous of internal degree s. Then*

$$[E, a] = t\, a, \qquad [E, \delta] = s\, \delta$$

as operators on A.

Proof. For homogeneous $b \in A_u$,

$$[E, a](b) = E(ab) - aE(b) = (t + u)ab - uab = tab.$$

This proves the first identity. For the second,

$$[E, \delta](b) = E(\delta b) - \delta(Eb).$$

Since $\delta b \in A_{u+s}$,

$$E(\delta b) = (u + s)\delta b, \qquad \delta(Eb) = u\delta b,$$

hence

$$[E, \delta](b) = s\delta b.$$

□

Remark NC.3.7. Lemma NC.3.6 is an NC-local algebraic identity. It will later permit Euler factors to be reordered inside arbitrary words without changing the number of positive-degree non-Euler derivation factors. This is one of the ingredients needed to close the long-word issue in the critical-window theorem.

NC.3.4 Imported local derivation-generation theorem

Theorem NC.3.8 (Local Euler–trivial generation; imported)**.** *Let $(R, \mathfrak{m}, k)$ be a quasihomogeneous normal isolated complete intersection singularity over a characteristic-zero field in the regime of the sources bound in NC.2. Then*

$$\mathrm{Der}_k(R)$$

is generated as an R-module by an Euler derivation together with the trivial derivations.

Remark NC.3.9 (Evidence status and scope)**.** Theorem NC.3.8 is a C1 imported theorem at the local ICIS level. Its use for the global standard-graded affine cone requires the transfer lemmas below. No global graded conclusion is imported merely by notation.

NC.3.5 Degree of the standard-graded trivial generators

Lemma NC.3.10 (Degree of a trivial derivation)**.** *Let*

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

be standard graded with $\deg f_\alpha = d_\alpha$*. For a choice of distinct indices $\nu_0, \dots, \nu_c$, let δ_ν be the standard trivial derivation obtained by expanding the $(c+1) \times (c+1)$ determinant whose first row is*

$$(\partial_{\nu_0}, \dots, \partial_{\nu_c})$$

and whose remaining rows are the corresponding partial derivatives of $f_1, \dots, f_c$. Then δ_ν is homogeneous of internal degree

$$\boxed{\sum_{\alpha=1}^{c} d_\alpha - (c+1) = m.}$$

Proof. Every coefficient of ∂_{ν_j} is a $c \times c$ Jacobian minor. Its degree is

$$\sum_{\alpha=1}^{c} (d_\alpha - 1) = \sum_{\alpha=1}^{c} d_\alpha - c.$$

Since ∂_{ν_j} has internal degree -1, every summand of δ_ν has internal degree

$$\sum_{\alpha=1}^{c} d_\alpha - c - 1 = m.$$

□

NC.3.6 Local-to-graded transfer

Let

$$\mathfrak{m} := A_+ = \bigoplus_{t>0} A_t.$$

Because $A_0 = k$, this is a maximal ideal.

Lemma NC.3.11 (Localization of derivations)**.** *For a finitely generated k-algebra A and any multiplicative set $S \subseteq A$, the natural map gives*

$$S^{-1} \mathrm{Der}_k(A) \cong \mathrm{Der}_k(S^{-1}A).$$

In particular,

$$\mathrm{Der}_k(A)_{\mathfrak{m}} \cong \mathrm{Der}_k(A_{\mathfrak{m}}).$$

Proof. For finite type A/k, the Kähler module $\Omega^1_{A/k}$ is finitely presented and

$$\mathrm{Der}_k(A) = \mathrm{Hom}_A(\Omega^1_{A/k}, A).$$

Kähler differentials commute with localization, and localization of Hom out of a finitely presented module gives

$$S^{-1}\,\mathrm{Hom}_A(\Omega^1_{A/k}, A) \cong \mathrm{Hom}_{S^{-1}A}(S^{-1}\Omega^1_{A/k}, S^{-1}A).$$

This is the asserted isomorphism. □

Lemma NC.3.12 (The irrelevant maximal ideal detects graded modules)**.** *Let M be a graded A-module. If*

$$M_{\mathfrak{m}} = 0,$$

then $M = 0$.

Proof. Assume that a nonzero homogeneous element $u \in M_d$ exists. Since $u/1 = 0$ in $M_{\mathfrak{m}}$, there is an element $s \in A \setminus \mathfrak{m}$ with $su = 0$. Write

$$s = s_0 + s_+, \qquad s_0 \in k^\times, \qquad s_+ \in \mathfrak{m}.$$

The degree-d homogeneous component of su is $s_0 u$, because every homogeneous component of $s_+ u$ has degree strictly larger than d. Hence $s_0 u = 0$, and therefore $u = 0$, a contradiction. □

Proposition NC.3.13 (Graded Euler–trivial generation)**.** *Let*

$$A = k[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$$

be a standard-graded homogeneous ICIS such that the local ring $A_{\mathfrak{m}}$ satisfies the hypotheses of Theorem NC.3.8. Let $T \subseteq \mathrm{Der}_k(A)$ be the graded A-submodule generated by the global trivial derivations. Then

$$\boxed{\mathrm{Der}_k(A) = AE + T.}$$

Moreover every basic trivial generator occurring here has internal degree m.

Proof. By Lemma NC.3.11, localization identifies $\mathrm{Der}_k(A)_{\mathfrak{m}}$ with $\mathrm{Der}_k(A_{\mathfrak{m}})$, and the Euler and trivial derivations localize to their corresponding local derivations. Hence Theorem NC.3.8 gives

$$(\mathrm{Der}_k(A)/(AE + T))_{\mathfrak{m}} = 0.$$

The quotient is graded. Lemma NC.3.12 therefore gives

$$\mathrm{Der}_k(A)/(AE + T) = 0.$$

The degree assertion is Lemma NC.3.10. □

NC.3.7 Homogeneous generator reduction and degree budget

Lemma NC.3.14 (Homogeneous generator reduction)**.** *Assume Proposition NC.3.13. Every homogeneous element of Δ_A is a finite k-linear combination of homogeneous words in homogeneous multiplication operators, the Euler derivation E, and the basic homogeneous trivial derivations. Each word can be chosen with the same total internal degree as the homogeneous element being represented.*

Proof. Expand every first-order derivation using $\mathrm{Der}_k(A) = AE + T$, expand the resulting products distributively, and decompose all coefficients into homogeneous components. Since composition of homogeneous operators adds internal degree, the homogeneous component of the required total degree is the sum of precisely the words of that degree. □

Lemma NC.3.15 (Non-Euler factor budget)**.** *Assume Proposition NC.3.13. Let a homogeneous word in multiplication operators, E, and basic homogeneous trivial derivations have internal degree r. If the word contains s trivial-derivation factors, then*

$$r \geq sm.$$

Consequently,

$$r < qm \quad \Longrightarrow \quad s \leq q - 1.$$

Proof. Every basic trivial-derivation factor contributes internal degree m, every Euler factor contributes degree 0, and every homogeneous coefficient multiplication has nonnegative degree because A is standard graded. Internal degrees add under composition of homogeneous operators. Therefore

$$r \geq sm.$$

If $r < qm$, then $sm < qm$, hence $s \leq q - 1$. □

Corollary NC.3.16 (Second-order pre-threshold budget)**.** *Under the same hypotheses, if $r < 2m$, every homogeneous word in the generator expansion of Lemma NC.3.14 contains at most one trivial-derivation factor.*

NC.3.8 NC.3 closure and next obligation

Declaration NC.3.17 (NC.3 closure)**.** Sections NC.1–NC.3 establish the constitutional boundary, exact target, route separation, graded ICIS notation, Euler commutator calculus, an explicit local-to-graded transfer for the Euler–trivial derivation theorem, and the internal-degree factor budget needed by the later critical-window argument. No strong Nakai theorem has yet been claimed. The next theorem-bearing obligation is to prove, without relying on projective handwaving, that in the degree range

$$r < 2m$$

every order-two principal symbol arising from Δ_A carries an Euler/radial factor in a canonical ring-theoretic sense. That statement will be the core of Section NC.4.

NC.4 Projectivized Operators and the Critical-Window Theorem

NC.4.1 Purpose of the section

Section NC.4 closes the principal logical gap left by NC.3. The factor-budget lemma controls the number of positive-degree non-Euler derivations in a generated word, but by itself it does not control arbitrary word length, differential-order drop, or cancellation among long words. The purpose of this section is to pass homogeneous operators to the projective quotient, where every Euler factor becomes order zero. This turns the internal-degree budget into an actual projective-order bound and yields the critical-window escape theorem without any bound on affine word length.

NC.4.2 Localization of finite-order differential operators

For a commutative k-algebra B, let

$$\mathcal{P}^q_{B/k} := (B \otimes_k B)/I_\Delta^{q+1}, \qquad I_\Delta := \ker(B \otimes_k B \to B),$$

be the module of principal parts of order q, viewed as a left B-module through the first tensor factor.

Lemma NC.4.1 (Localization of differential operators)**.** *Let B be a finitely generated commutative k-algebra, let $S \subseteq B$ be multiplicative, and let $D \in D^q_{B/k}$. Then there is a unique operator*

$$D_S \in D^q_{S^{-1}B/k}$$

whose restriction to B agrees with the localization of D. The construction is compatible with further localization and with composition. If B is graded, S is generated by homogeneous elements, and D is homogeneous of internal degree r, then D_S is homogeneous of degree r for the induced $\mathbb{Z}$-grading.

Proof. A differential operator $D \in D^q_{B/k}$ is equivalently a left B-linear map

$$\varphi_D : \mathcal{P}^q_{B/k} \longrightarrow B$$

satisfying $D(b) = \varphi_D(1 \otimes b)$. Localize the left B-module $\mathcal{P}^q_{B/k}$ at S. In $S^{-1}\mathcal{P}^q_{B/k}$, every element $1 \otimes s$, $s \in S$, is invertible: it differs from $s \otimes 1$ by an element of the nilpotent ideal $I_\Delta / I_\Delta^{q+1}$, so its inverse is the finite geometric-series inverse of a nilpotent perturbation of the invertible element $s \otimes 1$. Consequently

$$S^{-1}\mathcal{P}^q_{B/k} \cong \mathcal{P}^q_{S^{-1}B/k}.$$

Tensoring φ_D with $S^{-1}B$ therefore gives a unique $S^{-1}B$-linear map

$$\mathcal{P}^q_{S^{-1}B/k} \longrightarrow S^{-1}B,$$

which corresponds to the required operator D_S. Functoriality of localization gives compatibility with further localization and composition. In the graded case all maps above are homogeneous, so internal degree is preserved. □

NC.4.3 Homogeneous sheafification on Proj A

Let A be the standard-graded algebra of NC.3 and set

$$X := \operatorname{Proj} A.$$

For a homogeneous $f \in A_{>0}$, write $D_+(f) \subseteq X$ for the standard affine open.

Definition NC.4.2 (Projectivized homogeneous operator)**.** Let $D \in D^q_{A/k,r}$ be homogeneous of internal degree r. For each integer t, Lemma NC.4.1 restricts on every $D_+(f)$ to a differential operator

$$(D_f)_t : (A_f)_t \longrightarrow (A_f)_{t+r}.$$

These maps glue to a sheaf differential operator

$$\widetilde{D}_t : \mathcal{O}_X(t) \longrightarrow \mathcal{O}_X(t + r)$$

of order at most q. We call $\widetilde{D}_t$ the projectivization of D at twist t.

Lemma NC.4.3 (Compatibility on overlaps)**.** *Definition NC.4.2 is independent of the chosen standard affine cover and is compatible with restriction to intersections.*

Proof. On an overlap $D_+(f) \cap D_+(g)$, both local operators are obtained from D by localization to A_{fg}. Compatibility with further localization in Lemma NC.4.1 therefore identifies the two restrictions. The degree-t submodules glue to $\mathcal{O}_X(t)$, proving the claim. □

NC.4.4 Projective principal symbols

Definition NC.4.4 (Projective q-symbol)**.** Let $D \in D^q_{A/k,r}$. The order-q symbol of $\widetilde{D}_t$ is denoted

$$\sigma^X_{q,t}(D) \in \mathcal{H}om_{\mathcal{O}_X}\left(\operatorname{Sym}^q \Omega^1_{X/k} \otimes \mathcal{O}_X(t), \mathcal{O}_X(t + r)\right).$$

If X is smooth, the usual identification gives

$$\sigma^X_{q,t}(D) \in H^0(X, \operatorname{Sym}^q T_X(r)).$$

Only vanishing or nonvanishing of this symbol is used in the critical-window argument; no identification across different twists is required.

Lemma NC.4.5 (Order detection by the principal symbol)**.** *For a sheaf differential operator of order at most q, its order-q principal symbol vanishes if the operator has order at most $q - 1$.*

Proof. This is the defining property of the order filtration on sheaf differential operators, equivalently of the associated graded of the sheaf of principal parts. □

NC.4.5 Euler scalarization

Lemma NC.4.6 (Euler becomes scalar on every twist)**.** *Let E be the Euler derivation. On every standard affine open $D_+(f)$ and every homogeneous element $u \in (A_f)_t$,*

$$E_f(u) = t\,u.$$

Consequently the projectivization

$$\widetilde{E}_t : \mathcal{O}_X(t) \longrightarrow \mathcal{O}_X(t)$$

is multiplication by the scalar t and has differential order 0*.*

Proof. Every homogeneous fraction can be written

$$u = a/f^N$$

with a homogeneous and $\deg a - N \deg f = t$. The localized Euler derivation satisfies

$$E_f(a/f^N) = \frac{E(a)}{f^N} - N\frac{aE(f)}{f^{N+1}} = \big(\deg a - N \deg f\big)\frac{a}{f^N} = t\,u.$$

Thus the induced sheaf operator is scalar multiplication. □

Lemma NC.4.7 (Projective order of the basic generators)**.** *Let $a \in A_d$ be homogeneous, let E be Euler, and let $\tau \in \mathrm{Der}_k(A)$ be a homogeneous trivial derivation of internal degree m. On every twist:*

(i) multiplication by a induces an order-zero map $\mathcal{O}_X(t) \to \mathcal{O}_X(t+d)$;

(ii) E induces an order-zero map by Lemma NC.4.6;

(iii) τ induces a differential operator of order at most one

$$\mathcal{O}_X(t) \longrightarrow \mathcal{O}_X(t+m).$$

Proof. Items (i) and (ii) are immediate. A derivation is a differential operator of order at most one, and localization preserves differential order by Lemma NC.4.1, giving (iii). □

NC.4.6 Projective order of an arbitrary generated word

Lemma NC.4.8 (Word projective-order bound)**.** *Assume Proposition NC.3.13. Let W be a homogeneous word in homogeneous multiplication operators, Euler factors, and basic trivial derivations. If W contains exactly s trivial-derivation factors, then for every twist t the induced sheaf operator*

$$\widetilde{W}_t$$

has differential order at most s.

Proof. Read the word from right to left. After each homogeneous prefix, the current source is a fixed twist $\mathcal{O}_X(t')$. At that twist every Euler factor is scalar by Lemma NC.4.6; multiplication factors are also order zero. The only factors capable of increasing sheaf differential order are the s trivial derivations, each of order at most one. Differential order is subadditive under composition, hence $\mathrm{ord}(\widetilde{W}_t) \le s$. No commutation or normal-form transformation of the affine word is used. □

Remark NC.4.9 (Long words are harmless after projectivization)**.** Lemma NC.4.8 permits arbitrarily many Euler factors and arbitrarily many coefficient multiplications. Thus affine word length may be unbounded even when projective differential order is small. This is precisely the mechanism needed to prevent the long-word/order-drop problem from reappearing in the critical-window argument.

NC.4.7 The critical-window theorem

Theorem NC.4.10 (Projective order budget)**.** *Assume Proposition NC.3.13 and suppose $m > 0$. Let $P \in (\Delta_A)_r$ be homogeneous of internal degree $r \geq 0$. Then for every twist t,*

$$\boxed{\operatorname{ord}(\widetilde{P}_t) \leq \left\lfloor \frac{r}{m} \right\rfloor.}$$

Proof. By Lemma NC.3.14, write P as a finite sum of homogeneous generator words, each of total internal degree r. If one such word contains s trivial derivations, Lemma NC.3.15 gives

$$sm \leq r, \qquad s \leq \left\lfloor \frac{r}{m} \right\rfloor.$$

Lemma NC.4.8 shows that the projectivization of each word has order at most s, hence at most $\lfloor r/m \rfloor$. A finite sum of operators of order at most that number again has order at most that number. □

Theorem NC.4.11 (NC Critical-Window Theorem)**.** *Under the hypotheses of Theorem NC.4.10, let $q \geq 2$. If*

$$r < qm,$$

then every homogeneous element $P \in (\Delta_A)_r$ satisfies

$$\boxed{\operatorname{ord}(\widetilde{P}_t) \leq q - 1 \quad \textit{for every } t.}$$

Consequently, whenever $P \in D^q_{A/k,r}$,

$$\boxed{\sigma^X_{q,t}(P) = 0 \quad \textit{for every } t.} \tag{CW}$$

The largest integral degree strictly below the threshold is

$$\boxed{r^*_q = qm - 1.}$$

Proof. The inequality $r < qm$ implies

$$\left\lfloor \frac{r}{m} \right\rfloor \leq q - 1.$$

Apply Theorem NC.4.10. The symbol statement follows from Lemma NC.4.5. The formula for r^*_q is immediate because internal degrees are integral. □

Corollary NC.4.12 (Second-order critical window)**.** *Assume the same hypotheses. If*

$$r < 2m,$$

then every $P \in (\Delta_A)_r$ has projective order at most one, and every $P \in (\Delta_A)_r \cap D^2_{A/k}$ satisfies

$$\boxed{\sigma^X_{2,t}(P) = 0 \quad \textit{for every } t.}$$

The maximal second-order pre-threshold degree is

$$\boxed{r^*_2 = 2m - 1.}$$

NC.4.8 Strong escape interface

Theorem NC.4.13 (Critical-symbol strong escape)**.** *Assume the hypotheses of Theorem NC.4.11. Let*

$$D \in D^q_{A/k,r}$$

be homogeneous with $r < qm$. If for some twist t

$$\boxed{\sigma^X_{q,t}(D) \neq 0,}$$

then

$$\boxed{D \notin \Delta_A.}$$

In particular,

$$D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.$$

Proof. Suppose toward a contradiction that $D \in \Delta_A$. Since D is homogeneous of degree r, it lies in $(\Delta_A)_r$. The Critical-Window Theorem then gives

$$\sigma^X_{q,t}(D) = 0$$

for every t, contradicting the assumed nonvanishing. □

Corollary NC.4.14 (Second-order strong escape interface)**.** *If*

$$D \in D^2_{A/k,r}, \qquad r < 2m,$$

and $\sigma^X_{2,t}(D) \neq 0$ for at least one twist t, then

$$D \in D_{A/k} \setminus \Delta_A.$$

NC.4.9 Adversarial audit of the long-word issue

Proposition NC.4.15 (Long-word/order-drop immunity)**.** *The proof of Theorem NC.4.11 does not assume any of the following:*

(i) a bound on the affine word length of an element of Δ_A;

(ii) equality between affine word length and differential order;

(iii) absence of cancellation among high affine-order terms;

(iv) a PBW theorem for Δ_A;

(v) commutation of Euler factors through the word.

Hence cancellation or commutator-induced order drop in the affine cone cannot create a projective order-q symbol inside the window $r < qm$.

Proof. After homogeneous generator reduction, the proof operates word by word only on the projectivized action. Multiplications and every Euler occurrence are order zero there, regardless of their number or position, while each trivial derivation contributes at most one to projective differential order. The number of trivial factors is bounded solely by internal degree. Summation and cancellation can only preserve or lower the resulting projective-order bound. None of the five listed assumptions enters the argument. □

NC.4.10 Scope and evidence status

Remark NC.4.16 (What NC.4 has and has not proved)**.** NC.4 proves a conditional critical-window theorem from the graded Euler-plus-trivial generation result of NC.3. The localization, Euler scalarization, word-order bound, and critical-window deduction are internal algebraic arguments. The sole imported mathematical input is the derivation structure ultimately traced to the Kersken-type theorem. Accordingly, until the literature binding and all source hypotheses are publication-audited, the complete NC.4 package remains a C2-high internal theorem candidate rather than a new C1 literature theorem.

NC.4 does *not* prove that a nonzero critical projective symbol exists for a given singularity, nor that an arbitrary projective symbol quantizes to a cone operator. Those are separate obligations. What NC.4 proves is the exclusion half: once an actual homogeneous cone operator of degree $r < qm$ has nonzero projective q-symbol, that operator cannot belong to Δ_A.

Declaration NC.4.17 (NC.4 closure)**.** Subject to the imported derivation-generation theorem and its graded transfer, the long-word/order-drop gap in the Critical Window is closed by projective-order budgeting:

$$\boxed{r < qm \Longrightarrow \operatorname{ord}(\widetilde{P}_t) \le q - 1 \quad (P \in (\Delta_A)_r).}$$

Thus NC.4 supplies a strong-escape interface for every later explicit or quantized-stress construction. The next construction stage may therefore focus on producing actual quotient differential operators with a nonzero critical projective symbol, rather than re-solving the generated-word problem.

NC.5 The Type $(2, d)$ Explicit Route: Audited Corank-One Branch

NC.5.1 Purpose and exact scope

The purpose of NC.5 is to connect the Critical-Window theorem of NC.4 to an actual explicit differential operator for a codimension-two complete intersection of type $(2, d)$. The present section fully audits the *corank-one quadric* branch. It does not yet freeze the nondegenerate-quadric branch, because an independently replayable formula for that branch is not present in the current NC source corpus.

Policy 7 (No all-$(2, d)$ promotion from one branch)**.** *A theorem proved in NC.5 for the corank-one quadric branch is not reported as a theorem for every smooth type-$(2, d)$ complete intersection. The full all-$(2, d)$ statement remains an NC theorem candidate until the nondegenerate quadric branch is independently reconstructed and audited.*

Let

$$P = k[y_1, \dots, y_{n-1}, z], \qquad q = \sum_{i=1}^{n-1} y_i^2, \qquad A = P/(q, g),$$

where g is homogeneous of degree $d \ge 2$. Put

$$X := \operatorname{Proj} A \subset \mathbf{P}^{n-1}.$$

Throughout the theorem-bearing part of this section we assume:

(i) $\operatorname{char} k = 0$;

(ii) $n \ge 4$;

(iii) A is a normal domain and X is smooth;

(iv) the graded Kersken-type conclusion of Proposition NC.3.13 applies to A;

(v) the displayed split corank-one normal form for q is valid over k.

The last hypothesis holds, in particular, after the usual diagonalization over an algebraically closed characteristic-zero field.

NC.5.2 The geometric corank dichotomy

Lemma NC.5.1 (Smooth type-$(2,d)$ complete intersections have quadric corank at most one)**.** *Assume k is algebraically closed and*

$$X = V(q,g) \subset \mathbf{P}^{n-1}$$

is a smooth complete intersection with $\deg q = 2$ *and* $\deg g = d \geq 2$*. Then*

$$\operatorname{corank}(q) \leq 1.$$

Proof. If $\operatorname{corank}(q) \geq 2$, then the projective singular locus

$$\mathbf{P}(\ker q)$$

has positive dimension. The restriction of the positive-degree homogeneous polynomial g to this projective linear space has a zero. At such a point x, one has

$$q(x) = g(x) = 0, \qquad \nabla q(x) = 0.$$

Hence the Jacobian matrix of (q,g) has rank at most one at x, contradicting smoothness of the codimension-two complete intersection. □

Remark NC.5.2. Lemma NC.5.1 reduces a smooth type-$(2,d)$ proof to two branches: a nondegenerate quadric and a corank-one quadric. NC.5 closes only the second branch.

NC.5.3 Homogeneous differential identities

Define

$$a := \frac{1}{d}\nabla_y g, \qquad c := \frac{1}{d} g_z,$$

$$J := \frac{1}{d}\operatorname{Hess}_{yy}(g), \qquad r := \frac{1}{d}\nabla_y(g_z), \qquad h := \frac{1}{d} g_{zz}, \qquad \tau := \operatorname{tr} J.$$

Thus a and c have degree $d-1$, while J, r, h have degree $d-2$.

Lemma NC.5.3 (Euler–Hessian identities)**.** *The following polynomial identities hold in P:*

$$\boxed{y \cdot a + zc = g,}$$

$$\boxed{Jy + zr = (d-1)a,} \qquad \boxed{y \cdot r + zh = (d-1)c.}$$

Proof. The first identity is Euler's homogeneous identity divided by d. Differentiate Euler's identity with respect to the y-variables and to z, respectively, and divide by d. □

Fix a nonzero constant vector

$$w \in k^{n-1}$$

and put

$$S := w \cdot y, \qquad T := w \cdot a, \qquad R := a \cdot a.$$

Define a symmetric block matrix

$$M = \begin{pmatrix} B & b \\ b^T & \delta \end{pmatrix}$$

by

$$\boxed{B = c^2\left(SI - yw^T - wy^T\right),} \tag{NC.5.B}$$

$$\boxed{b = c\left(Ty - Sa - czw\right),} \tag{NC.5.b}$$

$$\boxed{\delta = SR + 2Tzc.} \tag{NC.5.d}$$

Define the first-order correction

$$\boxed{\gamma_y = S(cr - ha) - (n + d - 4)c^2 w,} \tag{NC.5.gy}$$

$$\boxed{\gamma_z = S(r \cdot a - \tau c) + (n + d - 4)Tc.} \tag{NC.5.gz}$$

Finally set

$$\boxed{D_w := \gamma \cdot \nabla + \sum_{i,j=1}^{n} M_{ij}\partial_i\partial_j,} \tag{NC.5.D}$$

where the final coordinate is z, and the second-order sum uses the symmetric matrix M with both off-diagonal entries present.

NC.5.4 Principal tangency identities

Lemma NC.5.4 (Two exact matrix identities)**.** *In P^n,*

$$\boxed{M(y, 0)^T = (-c^2 qw,\ c(Tq - Sg)),} \tag{NC.5.1}$$

and

$$\boxed{M(a, c)^T = (-c^2 gw,\ cTg).} \tag{NC.5.2}$$

Consequently,

$$M\nabla q \in (q, g)P^n, \qquad M\nabla g \in (q, g)P^n.$$

Proof. Since $q = y \cdot y$, $S = w \cdot y$, and $g = y \cdot a + zc$,

$$By = c^2(Sy - yS - wq) = -c^2 qw,$$

while

$$b \cdot y = c(Tq - Sa \cdot y - czS) = c(Tq - Sg).$$

This gives (NC.5.1). Similarly,

$$Ba + bc = c^2(Sa - yT - w(a \cdot y)) + c^2(Ty - Sa - czw) = -c^2 gw,$$

and

$$b \cdot a + \delta c = cT(a \cdot y + zc) = cTg,$$

which gives (NC.5.2). Since

$$\nabla q = 2(y, 0), \qquad \nabla g = d(a, c),$$

the final containment follows. □

NC.5.5 Lower-order correction and descent

Lemma NC.5.5 (Exact action on the defining equations)**.** *The operator D_w satisfies*

$$\boxed{D_w(q) = -2Shg,} \tag{NC.5.3}$$

and

$$\boxed{D_w(g) = 0.} \tag{NC.5.4}$$

Proof. Because $\mathrm{Hess}(q)$ is twice the identity on the y-block and zero in the z-direction,

$$D_w(q) = 2\gamma_y \cdot y + 2\,\mathrm{tr}\, B.$$

Lemma NC.5.3 gives

$$r \cdot y = (d - 1)c - zh, \qquad a \cdot y = g - zc.$$

Hence

$$\gamma_y \cdot y = S((3-n)c^2 - hg),$$

while

$$\operatorname{tr} B = (n-3)Sc^2.$$

Thus $D_w(q) = -2Shg$.

For g, use

$$\nabla g = d(a, c), \qquad \operatorname{Hess}(g) = d\begin{pmatrix} J & r \\ r^T & h \end{pmatrix}.$$

After division by d, the expression $D_w(g)/d$ equals

$$\gamma_y \cdot a + \gamma_z c + B : J + 2b \cdot r + \delta h.$$

Direct substitution cancels all terms except

$$2c\left[-c\, w \cdot (Jy + zr) + T(y \cdot r + zh)\right].$$

By Lemma NC.5.3, this is

$$2c\left[-c(d-1)T + T(d-1)c\right] = 0.$$

□

Proposition NC.5.6 (Ideal preservation)**.** *The operator D_w preserves (q, g):*

$$\boxed{D_w((q, g)) \subseteq (q, g).}$$

Hence it descends to an element, again denoted D_w, of $D^2_{A/k}$. Moreover $D_w(1) = 0$.

Proof. For a second-order operator with symmetric principal matrix M, the Leibniz expansion gives, for every $f, H \in P$,

$$D_w(fH) - fD_w(H) - HD_w(f) = 2(M\nabla f) \cdot \nabla H.$$

For $f = q, g$, Lemma NC.5.4 puts $M\nabla f$ in $(q, g)P^n$, while Lemma NC.5.5 puts $D_w(f)$ in (q, g). Therefore $D_w(qH), D_w(gH) \in (q, g)$ for all H, proving ideal preservation. The formula for D_w has no zeroth-order term, so $D_w(1) = 0$. □

NC.5.6 Critical internal degree

Lemma NC.5.7 (Critical degree of the explicit operator)**.** *Every entry of M has degree $2d-1$, every entry of γ has degree $2d-2$, and therefore*

$$\boxed{\deg D_w = 2d - 3.}$$

For a type-$(2, d)$ complete intersection,

$$m = 2 + d - 3 = d - 1,$$

so

$$\boxed{\deg D_w = 2m - 1 < 2m.}$$

Proof. The degree statements follow immediately from

$$\deg a = \deg c = d - 1, \qquad \deg J = \deg r = \deg h = d - 2,$$

and the displayed formulas. A second derivative lowers degree by two and a first derivative lowers degree by one. The formula for m is the NC.3 complete-intersection degree formula. □

NC.5.7 Generic-chart certificate for a nonzero projective symbol

Lemma NC.5.8 (The coefficient c is nonzero)**.** *Under the standing smoothness assumptions, the class of*

$$c = \frac{1}{d} g_z$$

in A is nonzero.

Proof. The singular locus of the corank-one quadric q is the projective point $[0 : \cdots : 0 : 1]$. Smoothness of X forces this point not to lie on $V(g)$; hence the coefficient of z^d in g is nonzero. Therefore c contains a nonzero pure z^{d-1} term. Since $\deg c = d - 1 < d$, an identity $c \in (q, g)$ would force $c \in (q)$, but a multiple of $q = \sum y_i^2$ cannot contain a pure z^{d-1} monomial. Thus $c \neq 0$ in A. □

Lemma NC.5.9 (Nonzero projective second symbol)**.** *Assume A is a domain and $n \geq 4$. For every nonzero $w \in k^{n-1}$, the descended operator D_w has nonzero projective second symbol on the chart $D_+(z)$:*

$$\boxed{\sigma^X_{2,t}(D_w) \neq 0}$$

for every twist t on which the homogeneous operator is sheafified.

Proof. Because (q, g) contains no nonzero linear form, the classes of z and $S = w \cdot y$ in A are nonzero. By Lemma NC.5.8, $c \neq 0$. Since A is a domain,

$$zSc \neq 0.$$

Work over the fraction field $K = \mathrm{Frac}(A)$, hence on the generic point of the nonempty chart $D_+(zSc) \subseteq D_+(z)$.

In the K-vector space K^{n-1} with bilinear form $u \cdot v$, the relations $q = g = 0$ give

$$y \cdot y = 0, \qquad y \cdot a = -zc.$$

Thus the Gram determinant of y, a is

$$\det \begin{pmatrix} 0 & -zc \\ -zc & R \end{pmatrix} = -z^2 c^2 \neq 0.$$

Hence $U := \mathrm{span}_K\{y, a\}$ is nondegenerate. Its orthogonal complement

$$W := \{u \in K^{n-1} : y \cdot u = a \cdot u = 0\}$$

is therefore nondegenerate of dimension $n - 3 \geq 1$. Choose $u, v \in W$ with $u \cdot v \neq 0$.

On the projective chart $D_+(z)$, write $Y_i := y_i/z$. For $u \in W$, consider the local cotangent covector

$$\lambda_u := \sum_i u_i \, dY_i.$$

At $z = 1$ its ambient representative is

$$(u, -y \cdot u) = (u, 0),$$

because $y \cdot u = 0$. Moreover $(u, 0)$ is orthogonal to the two conormal vectors $(y, 0)$ and (a, c), since $u \in W$. As $\mathrm{span}\{y, a\}$ is nondegenerate, a nonzero $u \in W$ gives a nonzero cotangent class on X. The same holds for v.

The local second symbol evaluated on these cotangent classes is therefore

$$\sigma^X_{2,t}(D_w)(\lambda_u, \lambda_v) = (u, 0)^T M (v, 0) = u^T B v = Sc^2 (u \cdot v),$$

because all terms containing $y \cdot u$ or $y \cdot v$ vanish. The right hand side is nonzero in K. Hence the projective principal symbol cannot vanish identically on $D_+(z)$. □

NC.5.8 Strong escape for the corank-one branch

Theorem NC.5.10 (Corank-one type-$(2, d)$ strong escape)**.** *Under the standing assumptions of NC.5 and the graded derivation-generation hypothesis of NC.3, for every nonzero $w \in k^{n-1}$ the operator D_w descends to*

$$D_w \in D^2_{A/k,\,2d-3}$$

and satisfies

$$\boxed{D_w \notin \Delta_A.}$$

Consequently,

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.}$$

Proof. Proposition NC.5.6 gives the actual quotient differential operator. Lemma NC.5.7 gives

$$\deg D_w = 2m - 1 < 2m.$$

Lemma NC.5.9 gives a nonzero projective second symbol. Apply the second-order strong-escape interface, Corollary NC.4.14. □

Corollary NC.5.11 (Critical-degree witness)**.** *Within the same scope,*

$$\boxed{\exists D \in D^2_{A/k,\,2m-1} \setminus \Delta_A.}$$

Moreover the witness may be chosen with $D(1) = 0$.

NC.5.9 Adversarial audit and remaining all-$(2, d)$ obligation

Proposition NC.5.12 (What has been closed in the corank-one branch)**.** *The proof of Theorem NC.5.10 contains independent checks of all five required edges:*

(i) the principal matrix is tangent to both defining equations modulo the ideal;

(ii) the lower-order correction sends both defining equations into the ideal;

(iii) the full second-order operator preserves the complete-intersection ideal;

(iv) the operator lies exactly at the critical internal degree $2m - 1$;

(v) its projective second symbol is nonzero on an explicit nonempty chart.

No dimension subtraction, universal T^1-vanishing, or projective-to-cone surjectivity theorem is used.

Audit Obligation NC.5.13 (Nondegenerate quadric branch)**.** *To promote the historical "all smooth type-$(2, d)$" claim to an NC theorem, one must still provide and independently audit the nondegenerate-quadric branch*

$$\operatorname{rank} q = n.$$

The current NC source corpus does not contain a complete independently replayable formula for that branch. Therefore Theorem NC.5.10 shall not be cited as an all-$(2, d)$ theorem.

Remark NC.5.14 (Evidence status)**.** The algebraic identities in NC.5 are C0 once checked from the displayed formulas. The corank-one strong-escape theorem remains C2-high until the imported Kersken-type source binding, base-field conventions, and independent symbolic replay are frozen. The historical all-$(2, d)$ claim remains C2/open at the nondegenerate branch.

Remark NC.5.15 (Exact symbolic replay artifact)**.** The finite polynomial identities of Lemmas NC.5.4 and NC.5.5 admit direct computer-algebra replay without entering the proof as an axiom. The construction package includes the companion script AK_NC5_corank_one_replay.py, which tests representative homogeneous polynomials in multiple dimensions and degrees using exact symbolic arithmetic. Successful replay is C3 evidence for implementation fidelity; the displayed algebraic proofs remain the theorem evidence.

Declaration NC.5.16 (NC.5 closure)**.** NC.5 closes the corank-one type-$(2, d)$ branch as a complete internal proof chain:

$$\boxed{\text{explicit } D_w \Longrightarrow D_w(q,g) \subset (q,g) \Longrightarrow \deg D_w = 2m-1 \Longrightarrow \sigma_2^X(D_w) \neq 0 \Longrightarrow D_w \notin \Delta_A.}$$

The next NC stage may either reconstruct the nondegenerate $(2, d)$ branch or proceed to the Quantized Stress calculus for broader complete-intersection families. No all-$(2, d)$ promotion occurs until Audit Obligation NC.5.13 is discharged.

NC.6 The Nondegenerate Type $(2, d)$ Branch and Full Smooth Closure

NC.6.1 Purpose

NC.6 discharges Audit Obligation NC.5.13. The principal difficulty is the one-degree mismatch already visible in the naive Gram projector: the ordinary projector onto the common tangent space of q and g has coefficient degree $2d$, while the NC.4 critical window requires coefficient degree $2d - 1$, equivalently operator internal degree $2d - 3$. The construction below replaces that projector by a contracted cofactor tensor depending on a constant vector w. Its degree is exactly one lower, but it retains the two principal tangency identities needed for descent.

NC.6.2 Coordinate-free setup for a nondegenerate quadric

Let

$$P = k[x_1, \dots, x_n], \qquad A = P/(q,g), \qquad \deg q = 2, \qquad \deg g = d \geq 2,$$

with $n \geq 4$. Assume q is nondegenerate. Write

$$q(x) = x^T Q x$$

for a symmetric invertible matrix $Q \in \mathrm{GL}_n(k)$, and set

$$\langle u, v \rangle_q := u^T Q v.$$

Define the q-gradient vector and normalized Hessian endomorphism

$$a := \frac{1}{d} Q^{-1} \nabla g, \qquad H := \frac{1}{d} Q^{-1} \operatorname{Hess}(g).$$

Then H is self-adjoint for $\langle\ ,\ \rangle_q$, and Euler homogeneity gives

$$\boxed{\langle x, a \rangle_q = g, \qquad Hx = (d-1)a.} \tag{NC.6.1}$$

Put

$$R := \langle a, a \rangle_q, \qquad \tau := \operatorname{tr} H.$$

Fix a nonzero constant vector $w \in k^n$, and define

$$S := \langle w, x \rangle_q, \qquad T := \langle w, a \rangle_q.$$

NC.6.3 The one-degree-lower cofactor tensor

Let $v^\flat$ denote the q-covector

$$v^\flat(u) := \langle v, u \rangle_q.$$

Define the q-self-adjoint endomorphism

$$\boxed{\begin{aligned} L_w := {} & R\big(SI - x \otimes w^\flat - w \otimes x^\flat\big) \\ & - S\, a \otimes a^\flat + T\big(x \otimes a^\flat + a \otimes x^\flat\big). \end{aligned}} \tag{NC.6.2}$$

Since L_w is q-self-adjoint, the matrix

$$\boxed{M_w := L_w Q^{-1}} \tag{NC.6.3}$$

is symmetric in the ordinary coordinate basis.

Define also

$$U := Ta - Rw, \qquad V := Tx - Sa.$$

Lemma NC.6.1 (Nondegenerate principal tangency identities)**.** *The cofactor tensor satisfies the exact polynomial identities*

$$\boxed{L_w x = q\,U + g\,V,} \tag{NC.6.4}$$

and

$$\boxed{L_w a = g\,U.} \tag{NC.6.5}$$

Consequently,

$$M_w \nabla q \in (q, g)P^n, \qquad M_w \nabla g \in (q, g)P^n.$$

Proof. Using

$$\langle x, x\rangle_q = q, \qquad \langle x, a\rangle_q = g, \qquad \langle w, x\rangle_q = S, \qquad \langle w, a\rangle_q = T,$$

one obtains

$$\begin{aligned} L_w x &= R(Sx - Sx - qw) - Sga + T(gx + qa) \\ &= q(Ta - Rw) + g(Tx - Sa). \end{aligned}$$

Similarly,

$$\begin{aligned} L_w a &= R(Sa - Tx - gw) - SRa + T(Rx + ga) \\ &= g(Ta - Rw). \end{aligned}$$

Finally,

$$\nabla q = 2Qx, \qquad \nabla g = dQa,$$

so

$$M_w \nabla q = 2L_w x, \qquad M_w \nabla g = dL_w a,$$

which proves principal tangency modulo (q, g). □

NC.6.4 Trace and Hessian contractions

Define

$$\rho := \langle a, Ha\rangle_q.$$

Lemma NC.6.2 (Trace identities)**.** *The following exact identities hold:*

$$\boxed{\operatorname{tr} L_w = (n-3)SR + 2Tg,} \tag{NC.6.6}$$

and

$$\boxed{\operatorname{tr}(L_w H) = S(R\tau - \rho).} \tag{NC.6.7}$$

Proof. For a rank-one endomorphism $u \otimes v^\flat$,

$$\operatorname{tr}(u \otimes v^\flat) = \langle v, u\rangle_q.$$

Applying this to (NC.6.2) yields

$$\operatorname{tr} L_w = nSR - 2SR - SR + 2Tg = (n-3)SR + 2Tg.$$

For the second identity, use self-adjointness of H and $Hx = (d-1)a$. Then

$$\operatorname{tr}\big((x \otimes w^\flat)H\big) = \langle w, Hx\rangle_q = (d-1)T,$$

$$\operatorname{tr}\big((a \otimes a^\flat)H\big) = \langle a, Ha\rangle_q = \rho,$$

and

$$\operatorname{tr}\big((x \otimes a^\flat)H\big) = \langle a, Hx\rangle_q = (d-1)R.$$

The two terms proportional to TR cancel, leaving

$$\operatorname{tr}(L_w H) = S(R\tau - \rho).$$

□

NC.6.5 Lower-order correction

Define

$$\boxed{\gamma_w := S(Ha - \tau a) + (n + d - 4)(Ta - Rw).} \tag{NC.6.8}$$

Lemma NC.6.3 (Correction pairings)**.** *One has*

$$\boxed{\langle \gamma_w, x\rangle_q = -(n-3)SR + \big((n + d - 4)T - S\tau\big)g,} \tag{NC.6.9}$$

and

$$\boxed{\langle \gamma_w, a\rangle_q = -S(R\tau - \rho).} \tag{NC.6.10}$$

Proof. By self-adjointness and (NC.6.1),

$$\langle Ha, x\rangle_q = \langle a, Hx\rangle_q = (d-1)R.$$

Also

$$\langle Ta - Rw, x\rangle_q = Tg - RS.$$

Substitution in (NC.6.8) gives

$$\begin{aligned}\langle \gamma_w, x\rangle_q &= S((d-1)R - \tau g) + (n + d - 4)(Tg - RS)\\ &= -(n-3)SR + \big((n + d - 4)T - S\tau\big)g.\end{aligned}$$

For the second identity,

$$\langle Ha - \tau a, a\rangle_q = \rho - \tau R,$$

whereas

$$\langle Ta - Rw, a\rangle_q = TR - RT = 0.$$

Hence

$$\langle \gamma_w, a\rangle_q = -S(R\tau - \rho).$$

□

NC.6.6 The actual ambient differential operator

Define

$$\boxed{\widetilde{D}_w := \gamma_w \cdot \nabla + \sum_{i,j=1}^{n} (M_w)_{ij}\partial_i\partial_j.} \tag{NC.6.11}$$

Proposition NC.6.4 (Defining-equation action)**.** *The operator $\widetilde{D}_w$ satisfies*

$$\boxed{\widetilde{D}_w(q) = 2\big((n + d - 2)T - S\tau\big)g,} \tag{NC.6.12}$$

and

$$\boxed{\widetilde{D}_w(g) = 0.} \tag{NC.6.13}$$

Proof. Since

$$\mathrm{Hess}(q) = 2Q,$$

the second-order contribution to $\widetilde{D}_w(q)$ is

$$2\,\mathrm{tr}(M_wQ) = 2\,\mathrm{tr}\,L_w.$$

The first-order contribution is

$$\gamma_w \cdot \nabla q = 2\langle \gamma_w, x\rangle_q.$$

Using Lemmas NC.6.2 and NC.6.3 gives

$$\widetilde{D}_w(q) = 2\big((n + d - 2)T - S\tau\big)g.$$

Next,

$$\mathrm{Hess}(g) = dQH.$$

Since $M_w = L_w Q^{-1}$,

$$\sum_{i,j} (M_w)_{ij} \partial_i \partial_j g = d \ \mathrm{tr}(L_w H).$$

Also

$$\gamma_w \cdot \nabla g = d\langle \gamma_w, a \rangle_q.$$

The two contributions cancel by (NC.6.7) and (NC.6.10). □

Proposition NC.6.5 (Ideal preservation and quotient descent)**.** *The operator $\widetilde{D}_w$ preserves the ideal (q, g):*

$$\widetilde{D}_w((q, g)) \subseteq (q, g).$$

Hence it descends to an operator

$$D_w \in D^2_{A/k}$$

with $D_w(1) = 0$.

Proof. For any polynomial f and h,

$$\widetilde{D}_w(fh) = f\widetilde{D}_w(h) + h\widetilde{D}_w(f) + 2(M_w \nabla f) \cdot \nabla h.$$

For $f = q, g$, Lemma NC.6.1 puts $M_w \nabla f$ in $(q, g)P^n$, and Proposition NC.6.4 puts $\widetilde{D}_w(f)$ in (q, g). Therefore $\widetilde{D}_w(qh)$ and $\widetilde{D}_w(gh)$ belong to (q, g) for every h. The displayed formula has no zeroth-order term, so $D_w(1) = 0$. □

NC.6.7 Critical degree

Lemma NC.6.6 (Nondegenerate branch lies exactly in the critical window)**.** *Every entry of M_w has degree $2d - 1$, every entry of γ_w has degree $2d - 2$, and*

$$\boxed{\deg D_w = 2d - 3.}$$

For type $(2, d)$,

$$m = d - 1,$$

so

$$\boxed{\deg D_w = 2m - 1 < 2m.}$$

Proof. One has

$$\deg a = d - 1, \qquad \deg H = d - 2, \qquad \deg R = 2d - 2, \qquad \deg S = 1, \qquad \deg T = d - 1.$$

Every summand in L_w, hence in M_w, has degree $2d - 1$. Every summand in γ_w has degree $2d - 2$. Subtracting the differential order gives internal degree $2d - 3$. The formula $m = d - 1$ follows from the NC.3 complete-intersection threshold. □

NC.6.8 Nonzero projective symbol

Lemma NC.6.7 (Generic nonvanishing of the nondegenerate symbol)**.** *Assume A is a domain, $X = \mathrm{Proj}\, A$ is smooth, and $w \neq 0$. Then*

$$\boxed{\sigma^X_{2,t}(D_w) \neq 0}$$

on a nonempty open subset of X.

Proof. Because Q is nondegenerate and $w \neq 0$,

$$S = \langle w, x \rangle_q$$

is a nonzero linear form. Since (q, g) contains no nonzero linear form, its class in the domain A is nonzero; hence S is nonzero at the generic point of X.

Work over the function field $K = k(X)$. The nondegenerate form $\langle\ ,\ \rangle_q$ on K^n satisfies

$$\langle x, x \rangle_q = 0.$$

Therefore it induces a nondegenerate form on the projective tangent space of the quadric,

$$E := x^{\perp}/\langle x \rangle.$$

The class

$$\bar{a} \in E$$

is nonzero because smoothness of the complete intersection is exactly the independence of dq and dg, equivalently $a \notin Kx$. The projective tangent space of X is

$$H_X := \bar{a}^{\perp} \subset E.$$

Modulo every tensor carrying the radial factor x, the tensor (NC.6.2) induces on E

$$\boxed{S\big(R\, I_E - \bar{a} \otimes \bar{a}^{\flat}\big).} \tag{NC.6.14}$$

It annihilates the conormal $\bar{a}^{\flat}$, so it descends to the second symbol on H_X.

If $R \neq 0$, then on $\bar{a}^{\perp}$ the second term vanishes upon pairing with tangent cotangent directions, while the first term is SR times the nondegenerate inverse quadric form. Since $\dim H_X = n - 3 \geq 1$, the induced tensor is nonzero.

If $R = 0$, then $\bar{a}$ is isotropic and therefore $\bar{a} \in H_X$. The induced tensor becomes

$$-S\, \bar{a} \otimes \bar{a},$$

which is nonzero because $S \neq 0$, $\bar{a} \neq 0$, and $\operatorname{char} K = 0$. Thus in both cases the projective second symbol is nonzero. □

NC.6.9 Strong escape in the nondegenerate branch

Theorem NC.6.8 (Nondegenerate type-$(2, d)$ strong escape)**.** *Assume:*

(i) $\operatorname{char} k = 0$*;*

(ii) $n \geq 4$*;*

(iii) q *is nondegenerate;*

(iv) $A = P/(q, g)$ *is a normal domain and* $X = \operatorname{Proj} A$ *is smooth; and*

(v) the graded derivation-generation hypothesis of NC.3 applies.

Then for every nonzero $w \in k^n$*,*

$$D_w \in D^2_{A/k,\, 2d-3}$$

and

$$\boxed{D_w \notin \Delta_A.}$$

Consequently,

$$\boxed{D_{A/k} \neq A\langle \operatorname{Der}_k(A) \rangle.}$$

Proof. Proposition NC.6.5 gives an actual quotient differential operator. Lemma NC.6.6 places it at

$$2d - 3 = 2m - 1 < 2m.$$

Lemma NC.6.7 gives a nonzero projective second symbol. Corollary NC.4.14 therefore gives $D_w \notin \Delta_A$. □

NC.6.10 Faithfully flat field-extension detection

Lemma NC.6.9 (Finite-order differential operators commute with field extension)**.** *Let A be a finitely generated k-algebra and let K/k be a field extension. Put $A_K := A \otimes_k K$. For every $s \geq 0$,*

$$\boxed{D^s_{A/k} \otimes_k K \cong D^s_{A_K/K}.} \tag{NC.6.15}$$

Moreover,

$$\boxed{\mathrm{Der}_k(A) \otimes_k K \cong \mathrm{Der}_K(A_K).} \tag{NC.6.16}$$

Proof. Let

$$I_\Delta := \ker(A \otimes_k A \to A)$$

be the diagonal ideal and

$$\mathcal{P}^s_{A/k} := (A \otimes_k A)/I_\Delta^{s+1}$$

the module of principal parts. Because A is of finite type over a field, $\mathcal{P}^s_{A/k}$ is a finitely presented left A-module and

$$D^s_{A/k} \cong \mathrm{Hom}_A(\mathcal{P}^s_{A/k}, A).$$

Flat base change gives

$$\mathcal{P}^s_{A_K/K} \cong \mathcal{P}^s_{A/k} \otimes_k K.$$

Finite presentation then gives

$$\mathrm{Hom}_A(\mathcal{P}^s_{A/k}, A) \otimes_k K \cong \mathrm{Hom}_{A_K}(\mathcal{P}^s_{A_K/K}, A_K),$$

which proves (NC.6.15). The derivation statement follows in the same way from the finitely presented module of Kähler differentials,

$$\Omega^1_{A_K/K} \cong \Omega^1_{A/k} \otimes_k K.$$

□

Proposition NC.6.10 (Generated algebra commutes with field extension)**.** *Under the same hypotheses,*

$$\boxed{\Delta_A \otimes_k K \cong \Delta_{A_K}.}$$

Consequently,

$$D_{A_K/K} \neq \Delta_{A_K} \quad \Longrightarrow \quad D_{A/k} \neq \Delta_A.$$

Proof. The algebra Δ_A is generated by A and $\mathrm{Der}_k(A)$. Lemma NC.6.9 identifies their scalar extensions with A_K and $\mathrm{Der}_K(A_K)$, respectively, so the generated subalgebra is exactly Δ_{A_K}.

If $D_{A/k} = \Delta_A$, then after tensoring with the faithfully flat field extension K/k, and using

$$D_{A/k} = \bigcup_{s \geq 0} D^s_{A/k},$$

Lemma NC.6.9 would give

$$D_{A_K/K} = \Delta_{A_K}.$$

The contrapositive proves the final implication. □

NC.6.11 Closure of the full smooth type-$(2, d)$ branch

Theorem NC.6.11 (Smooth type-$(2, d)$ strong non-generation)**.** *Let k be a characteristic-zero field and*

$$A = k[x_1, \ldots, x_n]/(q, g), \qquad n \geq 4, \qquad \deg q = 2, \qquad \deg g = d \geq 2.$$

Assume:

(i) q, g form a homogeneous regular sequence;

(ii) *A is a normal domain;*

(iii) *$X = \operatorname{Proj} A$ is smooth; and*

(iv) *after scalar extension to an algebraic closure, the graded Kersken-type derivation-generation conclusion used in NC.3 applies.*

Then

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.}$$

Equivalently, this entire smooth type-$(2,d)$ class satisfies the strong non-generation conclusion required by the contrapositive Nakai strategy.

Proof. Extend scalars to an algebraic closure $\bar{k}$. Smoothness is preserved by field extension. By Lemma NC.5.1, the quadric $q_{\bar{k}}$ has corank at most one.

If $q_{\bar{k}}$ is nondegenerate, apply Theorem NC.6.8 over $\bar{k}$.

If $q_{\bar{k}}$ has corank one, a linear coordinate change over $\bar{k}$ puts it in the split corank-one normal form used in NC.5. Theorem NC.5.10 then gives

$$D_{A_{\bar{k}}/\bar{k}} \neq \Delta_{A_{\bar{k}}}.$$

In either case, Proposition NC.6.10 descends the non-generation conclusion to k. □

Corollary NC.6.12 (Discharge of the NC.5 nondegenerate audit obligation)**.** *Audit Obligation NC.5.13 is discharged. Under the hypotheses of Theorem NC.6.11, the historical all-smooth-type-$(2,d)$ claim is promoted from an open branchwise candidate to an NC internal theorem candidate.*

NC.6.12 Adversarial status and nonclaim ceiling

Proposition NC.6.13 (What NC.6 does and does not prove)**.** *NC.6 proves, subject to the explicitly imported derivation-generation theorem, a complete internal proof chain for the smooth type-$(2,d)$ class:*

$$\boxed{\begin{aligned}&\operatorname{corank} q \leq 1\\ &\Longrightarrow \left[\textit{nondegenerate explicit escape or corank-one explicit escape}\right]\\ &\Longrightarrow D_{A/k} \neq \Delta_A.\end{aligned}}$$

It does not prove the Nakai conjecture for arbitrary codimension-two (a,b), arbitrary quadratic complete intersections of codimension $c > 2$, all homogeneous ICIS, or arbitrary singular finite-type k-algebras.

Remark NC.6.14 (Evidence status)**.** The identities defining L_w, M_w, γ_w, the descent calculation, and the field-extension lemmas are C0 once the displayed algebra is checked. The projective-symbol nonvanishing and the assembled strong-escape theorem are classified C2-high pending the final source binding for the Kersken theorem, independent proof audit, and prior-art audit. The companion exact symbolic replay of the nondegenerate formulas is C3 evidence and is not used as a replacement for the displayed proof.

Declaration NC.6.15 (NC.6 closure)**.** NC.6 closes the previously missing nondegenerate branch and, together with NC.5, produces the first full smooth family theorem of Appendix NC:

$$\boxed{\text{smooth type-}(2,d) \Longrightarrow D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle}$$

under the exact NC standing hypotheses. The next construction stage should no longer spend effort on the type-$(2,d)$ branch except for independent audit and literature priority. The mathematically higher-value successor is the Quantized Stress calculus for higher-codimension quadratic complete intersections.

NC.7 Jacobian Stress, Hessian Obstruction, and the Second-Order Quantization Gate

NC.7.1 Purpose and presentation level

NC.7 isolates the second-order quantization problem for a general homogeneous complete intersection. The objective is deliberately narrower than a classification of all operators in $D^2_{A/k}$: starting from a symmetric polynomial principal-symbol matrix, NC.7 gives necessary and sufficient equations for an *ambient polynomial* second-order operator of the prescribed form to preserve the defining ideal and therefore descend to A.

This distinction is essential. No statement in NC.7 asserts that every order-two differential operator on A admits an ambient polynomial lift of this form. The resulting construction is therefore an exact quantization criterion for the declared ambient presentation, not a global surjectivity theorem for $D^2_{A/k}$.

Let

$$P := k[x_1, \dots, x_n], \qquad I := (f_1, \dots, f_c), \qquad A := P/I,$$

where $f_1, \dots, f_c$ is a homogeneous regular sequence with

$$\deg f_\alpha = d_\alpha \geq 2.$$

Write

$$J = \left(\frac{\partial f_\alpha}{\partial x_i} \right)_{\alpha,i}$$

for the $c \times n$ Jacobian matrix. All degree-m pieces A_m with $m < 0$ are understood to be zero.

NC.7.2 The degreewise stress and correction spaces

Fix an internal degree $r \in \mathbb{Z}$.

Definition NC.7.1 (Second-order coefficient space)**.** Define

$$\mathcal{B}_r(A) := \left\{ B = (B_{ij}) \in M_n(A) : B^T = B,\ B_{ij} \in A_{r+2} \right\}.$$

Define also the first-order correction space

$$\Gamma_r(A) := A^n_{r+1}.$$

Definition NC.7.2 (Jacobian tangency map)**.** Define

$$\mathsf{Tan}_r : \mathcal{B}_r(A) \longrightarrow \bigoplus_{\alpha=1}^{c} A^n_{r+d_\alpha+1}$$

by

$$\boxed{\mathsf{Tan}_r(B) := JB.} \tag{NC.7.1}$$

Equivalently, the α-th row is

$$\nabla f_\alpha^T B.$$

The *second-order Jacobian stress space* is

$$\boxed{\mathcal{S}_{2,r}(A) := \ker \mathsf{Tan}_r.} \tag{NC.7.2}$$

Remark NC.7.3. Because B is symmetric, $JB = 0$ is equivalent to

$$B\,J^T = 0.$$

Both forms will be used according to whether row or column notation is more convenient.

Definition NC.7.4 (First-order Jacobian correction map)**.** Define

$$\mathsf{j}_r : \Gamma_r(A) \longrightarrow \bigoplus_{\alpha=1}^{c} A_{r+d_\alpha}$$

by

$$\boxed{\mathsf{j}_r(\gamma) := J\gamma = (\nabla f_\alpha \cdot \gamma)_{\alpha=1}^{c}.} \tag{NC.7.3}$$

Definition NC.7.5 (Hessian contraction map)**.** For $B \in \mathcal{B}_r(A)$, define

$$\boxed{\mathsf{h}_r(B) := \frac{1}{2}\,(B : \mathrm{Hess}(f_\alpha))_{\alpha=1}^{c} \in \bigoplus_{\alpha=1}^{c} A_{r+d_\alpha},} \tag{NC.7.4}$$

where

$$B : \mathrm{Hess}(f_\alpha) := \sum_{i,j=1}^{n} B_{ij}\,\frac{\partial^2 f_\alpha}{\partial x_i \partial x_j}.$$

Remark NC.7.6 (Degree check)**.** The three defining maps are homogeneous of degree zero relative to the declared degree slices:

$$\deg B_{ij} = r + 2, \quad \deg \partial_i f_\alpha = d_\alpha - 1, \quad \deg \partial_i \partial_j f_\alpha = d_\alpha - 2.$$

Hence

$$\deg\big((JB)_{\alpha j}\big) = r + d_\alpha + 1,$$

whereas

$$\deg\big(\mathsf{j}_r(\gamma)_\alpha\big) = \deg\big(\mathsf{h}_r(B)_\alpha\big) = r + d_\alpha.$$

Thus no hidden grading shift is being suppressed in (NC.7.1)–(NC.7.4).

NC.7.3 The exact ambient Leibniz identity

Choose homogeneous polynomial lifts of $B \in \mathcal{B}_r(A)$ and $\gamma \in \Gamma_r(A)$ to P, and define

$$\boxed{\widetilde{D}_{B,\gamma} := \frac{1}{2}\sum_{i,j=1}^{n} B_{ij}\partial_i\partial_j + \sum_{i=1}^{n} \gamma_i \partial_i.} \tag{NC.7.5}$$

Lemma NC.7.7 (Second-order Leibniz identity)**.** *For all $f, h \in P$,*

$$\boxed{\widetilde{D}_{B,\gamma}(fh) - f\,\widetilde{D}_{B,\gamma}(h) = (B\nabla f)\cdot\nabla h + \left(\gamma\cdot\nabla f + \frac{1}{2}B : \mathrm{Hess}(f)\right) h.} \tag{NC.7.6}$$

Proof. Expand

$$\partial_i\partial_j(fh) = f\,\partial_i\partial_j h + (\partial_i f)(\partial_j h) + (\partial_j f)(\partial_i h) + h\,\partial_i\partial_j f.$$

Since B is symmetric, the two mixed terms combine to

$$(B\nabla f)\cdot\nabla h.$$

The first-order part contributes

$$f\,\gamma\cdot\nabla h + h\,\gamma\cdot\nabla f.$$

Subtracting $f\widetilde{D}_{B,\gamma}(h)$ gives (NC.7.6). □

Corollary NC.7.8 (Generator action)**.** *For every defining equation f_α,*

$$\boxed{\widetilde{D}_{B,\gamma}(f_\alpha) = \mathsf{j}_r(\gamma)_\alpha + \mathsf{h}_r(B)_\alpha \quad \text{in } A.} \tag{NC.7.7}$$

NC.7.4 Exact descent criterion

Theorem NC.7.9 (Ambient second-order descent criterion)**.** *Let $B \in \mathcal{B}_r(A)$ and $\gamma \in \Gamma_r(A)$, and choose arbitrary homogeneous polynomial lifts in P. Then the ambient operator $\widetilde{D}_{B,\gamma}$ preserves I,*

$$\widetilde{D}_{B,\gamma}(I) \subseteq I,$$

if and only if

$$\boxed{\mathsf{Tan}_r(B) = 0} \tag{NC.7.8}$$

and

$$\boxed{\mathsf{j}_r(\gamma) + \mathsf{h}_r(B) = 0.} \tag{NC.7.9}$$

When these conditions hold, $\widetilde{D}_{B,\gamma}$ descends to

$$D_{B,\gamma} \in D^2_{A/k,r}.$$

The descended operator is independent of the chosen polynomial lifts of the coefficient classes B_{ij}, γ_i.

Proof. Assume first that (NC.7.8)–(NC.7.9) hold. For every $h \in P$, Lemma NC.7.7 gives

$$\widetilde{D}_{B,\gamma}(f_\alpha h) = f_\alpha \widetilde{D}_{B,\gamma}(h) + (B\nabla f_\alpha)\cdot\nabla h + \widetilde{D}_{B,\gamma}(f_\alpha)h.$$

The first term lies in I. Equation (NC.7.8) says

$$B\nabla f_\alpha \in I\,P^n,$$

and (NC.7.9) says

$$\widetilde{D}_{B,\gamma}(f_\alpha) \in I.$$

Hence every displayed term lies in I, so

$$\widetilde{D}_{B,\gamma}(I) \subseteq I.$$

Conversely, suppose $\widetilde{D}_{B,\gamma}(I) \subseteq I$. Taking $h = 1$ gives

$$\widetilde{D}_{B,\gamma}(f_\alpha) \in I,$$

which is exactly (NC.7.9). Subtract this term from (NC.7.6) and choose $h = x_j$. Then

$$(B\nabla f_\alpha)_j \in I$$

for every j, proving (NC.7.8).

Finally, replacing any coefficient lift by another lift changes that coefficient by an element of I. Every resulting term of the ambient operator has a coefficient in I, hence acts trivially after passage to A. Therefore the descended operator depends only on the classes of B and γ in A. □

Remark NC.7.10 (What the theorem does not classify)**.** Theorem NC.7.9 characterizes descent for operators presented in the ambient form (NC.7.5). It does not assert that every element of $D^2_{A/k,r}$ has such a presentation. Accordingly, NC.7 provides an exact construction mechanism and obstruction theory, not an exhaustion theorem for the full ring of differential operators.

NC.7.5 Cotangent-cohomology target

Definition NC.7.11 (Degree-r first cotangent-cohomology slice)**.** Define

$$\boxed{\mathcal{T}^1_r(A) := \operatorname{coker}\left(\mathsf{j}_r : A^n_{r+1} \longrightarrow \bigoplus_{\alpha=1}^{c} A_{r+d_\alpha}\right).} \tag{NC.7.10}$$

Proposition NC.7.12 (Identification with the graded T^1-piece)**.** *For the complete-intersection presentation above,*

$$\boxed{\mathcal{T}_r^{\,1}(A) \cong \big(T^1_{A/k}\big)_r,} \tag{NC.7.11}$$

where

$$T^1_{A/k} := H^1\big(\mathrm{Hom}_A(L_{A/k}, A)\big)$$

is the first cotangent-cohomology module.

Proof. For a complete intersection, the cotangent complex is represented by the two-term complex

$$L_{A/k} \simeq \Big[I/I^2 \longrightarrow \Omega^1_{P/k} \otimes_P A\Big]$$

in cohomological degrees $-1, 0$. The regular sequence gives

$$I/I^2 \cong \bigoplus_{\alpha=1}^{c} A(-d_\alpha), \qquad \Omega^1_{P/k} \otimes_P A \cong A(-1)^n.$$

After applying $\mathrm{Hom}_A(-, A)$, the relevant differential is

$$A(1)^n \xrightarrow{J} \bigoplus_{\alpha=1}^{c} A(d_\alpha).$$

Its degree-r component is precisely

$$A^n_{r+1} \xrightarrow{\mathsf{j}_r} \bigoplus_{\alpha=1}^{c} A_{r+d_\alpha}.$$

Taking the cokernel proves (NC.7.11). □

Remark NC.7.13. This proposition repairs a common shorthand. The expression "$T^1 = \mathrm{coker}\, J$" is safe only after the grading shifts and the chosen complete-intersection presentation have been made explicit. NC.7 will always use the degreewise form (NC.7.10).

NC.7.6 The Hessian obstruction map

Definition NC.7.14 (Hessian obstruction)**.** For

$$B \in \mathcal{S}_{2,r}(A),$$

define

$$\boxed{\omega_r(B) := [\mathsf{h}_r(B)] \in \mathcal{T}_r^{\,1}(A).} \tag{NC.7.12}$$

This is the *second-order Hessian obstruction map*

$$\boxed{\omega_r : \mathcal{S}_{2,r}(A) \longrightarrow \mathcal{T}_r^{\,1}(A).} \tag{NC.7.13}$$

Theorem NC.7.15 (Second-order Quantization Gate)**.** *For a Jacobian stress*

$$B \in \mathcal{S}_{2,r}(A),$$

the following are equivalent:

(i) there exists $\gamma \in \Gamma_r(A)$ such that $\widetilde{D}_{B,\gamma}$ preserves I;

(ii) there exists $\gamma \in \Gamma_r(A)$ such that

$$\mathsf{j}_r(\gamma) = -\mathsf{h}_r(B);$$

(iii)

$$\boxed{\omega_r(B) = 0.}$$

Hence the ambient-quantizable second-order stress symbols are exactly

$$\boxed{\mathcal{L}^{\mathrm{amb}}_{2,r}(A) := \ker \omega_r.} \tag{NC.7.14}$$

Proof. For $B \in \mathcal{S}_{2,r}(A)$, the principal tangency equation (NC.7.8) already holds. Theorem NC.7.9 therefore reduces descent exactly to

$$\mathsf{j}_r(\gamma) + \mathsf{h}_r(B) = 0.$$

Such a γ exists if and only if

$$\mathsf{h}_r(B) \in \operatorname{im} \mathsf{j}_r,$$

which is equivalent to

$$[\mathsf{h}_r(B)] = 0 \quad \text{in} \quad \mathcal{T}^1_r(A).$$

□

Corollary NC.7.16 (Universal T^1-vanishing is unnecessary)**.** *To quantize a particular stress B, it is not necessary to prove*

$$\mathcal{T}^1_r(A) = 0.$$

It is necessary and sufficient to prove only

$$\boxed{\omega_r(B) = 0.}$$

Remark NC.7.17. This is the precise repair of the earlier discarded universal-T^1-vanishing route. The obstruction module may be large and nonzero while a distinguished critical stress class still lies in $\ker \omega_r$.

NC.7.7 Parameter-space exact sequence

Definition NC.7.18 (Ambient second-order quantization pairs)**.** Define

$$Q^{\mathrm{amb}}_{2,r}(A) := \{(B,\gamma) \in \mathcal{B}_r(A) \times \Gamma_r(A) : \mathsf{Tan}_r(B) = 0,\ \mathsf{j}_r(\gamma) + \mathsf{h}_r(B) = 0\}.$$

Proposition NC.7.19 (Quantization-pair exact sequence)**.** *There is a natural short exact sequence of k-vector spaces*

$$\boxed{0 \longrightarrow \mathrm{Der}_k(A)_r \longrightarrow Q^{\mathrm{amb}}_{2,r}(A) \xrightarrow{\pi_B} \ker \omega_r \longrightarrow 0,} \tag{NC.7.15}$$

where

$$\pi_B(B,\gamma) = B.$$

Proof. By the dualized cotangent presentation used in Proposition NC.7.12,

$$\mathrm{Der}_k(A)_r = \ker\left(\mathsf{j}_r : A^n_{r+1} \to \bigoplus_\alpha A_{r+d_\alpha}\right).$$

Thus the kernel of π_B consists exactly of pairs

$$(0,\gamma) \quad \text{with} \quad \gamma \in \mathrm{Der}_k(A)_r.$$

Theorem NC.7.15 says that the image of π_B is precisely $\ker \omega_r$. □

Remark NC.7.20 (Parameter exactness versus operator exactness)**.** Sequence (NC.7.15) is an exact sequence of *ambient coefficient presentations*. It is not asserted to be an exact sequence inside the order filtration of $D_{A/k}$. A nonzero B may in principle yield an operator whose actual order drops after passage to A. The later NC escape test must therefore verify a nonzero second principal symbol; that verification is logically independent of $\omega_r(B) = 0$.

NC.7.8 Presentation dependence and invariance ceiling

Policy 8 (Fixed-presentation status)**.** *The spaces*

$$\mathcal{S}_{2,r}(A), \qquad \mathcal{L}^{\mathrm{amb}}_{2,r}(A), \qquad Q^{\mathrm{amb}}_{2,r}(A)$$

are defined in NC.7 relative to the fixed homogeneous presentation

$$P \twoheadrightarrow A, \qquad I = (f_1, \dots, f_c).$$

Although the obstruction target $\mathcal{T}^1_r(A) \cong (T^1_{A/k})_r$ is intrinsic, NC.7 does not yet claim presentation independence of the entire stress space or of the concrete Hessian representative $\mathsf{h}_r(B)$. Any later intrinsic reformulation must prove its change-of-presentation law.

Remark NC.7.21 (Constant change of defining generators)**.** If the defining regular sequence is replaced by a constant invertible linear combination among equations of the same degree, then J, the Hessian vector, and the correction target transform by the same constant matrix. Hence the vanishing condition

$$\omega_r(B) = 0$$

is unchanged. More general homogeneous changes of regular-sequence generators are intentionally left outside the present claim.

NC.7.9 Diagonal quadratic collapse of the obstruction

The new formalism becomes especially simple for simultaneously diagonal quadrics.

Proposition NC.7.22 (Zero-diagonal stresses have zero Hessian obstruction)**.** *Assume*

$$d_\alpha = 2 \quad \textit{for all } \alpha,$$

and in the fixed coordinates

$$f_\alpha = \sum_{i=1}^{n} a_{\alpha i} x_i^2.$$

Let

$$B \in \mathcal{S}_{2,r}(A)$$

be a symmetric stress with

$$B_{ii} = 0 \quad \textit{for every } i.$$

Then

$$\boxed{\mathsf{h}_r(B) = 0, \qquad \omega_r(B) = 0.} \tag{NC.7.16}$$

Consequently B is ambient-quantizable with the choice

$$\gamma = 0.$$

Proof. Each Hessian is diagonal:

$$\mathrm{Hess}(f_\alpha) = 2\ \mathrm{diag}(a_{\alpha 1}, \dots, a_{\alpha n}).$$

Therefore

$$\frac{1}{2} B : \mathrm{Hess}(f_\alpha) = \sum_{i=1}^{n} a_{\alpha i} B_{ii} = 0.$$

Hence $\mathsf{h}_r(B) = 0$, and the remaining descent condition is exactly the stress equation $JB = 0$. □

Corollary NC.7.23 (Diagonal quadratic Quantization Gate collapses to stress)**.** *For the zero-diagonal family in Proposition NC.7.22,*

$$\boxed{\textit{ambient quantization} \iff JB = 0.}$$

Thus the second-order Quantization Gate contains no additional Hessian obstruction on this family.

NC.7.10 Finite linear-algebra character

Proposition NC.7.24 (Finite computability at fixed degree)**.** *For fixed r, once finite k-bases of the graded pieces*

$$A_{r+2}, \quad A_{r+1}, \quad A_{r+d_\alpha+1}, \quad A_{r+d_\alpha}$$

are chosen, the maps

$$\mathsf{Tan}_r, \qquad \mathsf{j}_r, \qquad \mathsf{h}_r, \qquad \omega_r$$

are represented by finite matrices over k. Therefore

$$\dim_k \mathcal{S}_{2,r}(A), \qquad \dim_k \ker \omega_r, \qquad \dim_k \mathcal{T}_r^1(A)$$

are exact finite linear-algebra invariants of the declared presentation and degree slice.

Proof. Every displayed source and target is a finite-dimensional graded piece of the finitely generated standard-graded algebra A. Multiplication by the finitely many partial derivatives and Hessian entries is k-linear on these graded pieces. Thus all defining maps become finite matrices after basis selection. □

Policy 9 (Computation is evidence, not automatic theorem globalization)**.** *Exact rank computation of the NC.7 matrices at one algebra A proves the corresponding finite-dimensional statement for that algebra, subject to a replayable exact quotient basis. It does not prove a family theorem for all complete intersections of the same multidegree without a separate symbolic or geometric globalization argument. Modular rank agreement is recorded at C4 unless an independent lifting theorem promotes it.*

NC.7.11 NC.7 closure

Theorem NC.7.25 (NC.7 Quantization-Gate reduction)**.** *For every homogeneous complete-intersection presentation*

$$A = P/(f_1, \dots, f_c)$$

and every internal degree r, the second-order ambient quantization problem for a symmetric coefficient matrix B reduces exactly to:

$$\boxed{\begin{aligned} &JB = 0, \\ &\omega_r(B) = 0 \quad \textit{in} \quad (T^1_{A/k})_r. \end{aligned}} \tag{NC.7.17}$$

Equivalently,

$$\boxed{\mathcal{L}^{\mathrm{amb}}_{2,r}(A) = \ker\left(\omega_r : \mathcal{S}_{2,r}(A) \to (T^1_{A/k})_r\right).} \tag{NC.7.18}$$

For simultaneously diagonal quadrics and zero-diagonal stresses,

$$\omega_r = 0$$

identically on that stress family.

Proof. The first two statements are Theorem NC.7.15 together with Proposition NC.7.12. The diagonal quadratic statement is Proposition NC.7.22. □

Declaration NC.7.26 (NC.7 status)**.** NC.7 converts the former abstract Quantization Gate into a degreewise algebraic obstruction problem:

$$\boxed{\text{Jacobian stress} \longrightarrow \text{Hessian class in } T^1 \longrightarrow \text{quantizable stress kernel.}}$$

The descent identity, grading, T^1-identification, and kernel criterion are C0 under the fixed complete-intersection presentation. No projective-symbol nonvanishing or strong escape is inferred merely from $\ker \omega_r \neq 0$. The next stage, NC.8, will quotient the quantizable stress space by the projectively generated/radial sector and define the critical Quantized Stress defect that interfaces directly with the NC.4 Critical Window.

NC.8 Critical Quantized Stress Defect and the NC Master Escape Bridge

NC.8.1 Purpose

NC.7 identifies the ambient-quantizable second-order stress space

$$\mathcal{L}^{\mathrm{amb}}_{2,r}(A) = \ker \omega_r.$$

Quantizability alone is not enough for Nakai escape: a quantized stress may have vanishing projective second symbol, or its projective symbol may already occur in the algebra generated by first-order derivations. NC.8 separates these two issues.

The construction is intentionally made at the projective-symbol level. It does not identify the kernel of projectivization with a single informal "radial" formula. Instead it defines the entire projectively invisible sector intrinsically as a kernel, then compares the visible quantized-stress sector with the actual projective-symbol sector of Δ_A.

Throughout NC.8 assume that $X = \operatorname{Proj} A$ is smooth on the declared source scope and that the NC.4 projectivization of homogeneous differential operators is available.

NC.8.2 The projective symbol map for quantized stresses

Definition NC.8.1 (Zero-twist second principal symbol)**.** Let

$$D \in D^2_{A/k,r}$$

be homogeneous of internal degree r. Its projectivization on the zero twist is a differential operator

$$\widetilde{D}_0 : \mathcal{O}_X \longrightarrow \mathcal{O}_X(r).$$

Its order-two principal symbol is denoted

$$\boxed{\sigma^X_{2,0}(D) \in H^0\Big(X, \operatorname{Sym}^2 T_X(r)\Big).} \tag{NC.8.1}$$

Remark NC.8.2. The zero twist is fixed only to remove unnecessary bookkeeping. NC.4 proves the Critical Window for every twist, hence in particular for $t = 0$. A nonzero zero-twist symbol is therefore sufficient for strong escape. NC.8 does not require an identification of principal symbols across all twists.

Lemma NC.8.3 (Second symbol depends only on the stress matrix)**.** *Let*

$$B \in \ker \omega_r$$

and let

$$\gamma, \gamma' \in \Gamma_r(A)$$

be two corrections satisfying

$$\mathsf{j}_r(\gamma) = \mathsf{j}_r(\gamma') = -\mathsf{h}_r(B).$$

Then

$$\boxed{\sigma^X_{2,0}(D_{B,\gamma}) = \sigma^X_{2,0}(D_{B,\gamma'}).} \tag{NC.8.2}$$

Proof. The difference is

$$D_{B,\gamma} - D_{B,\gamma'} = (\gamma - \gamma') \cdot \nabla.$$

By Proposition NC.7.19,

$$\gamma - \gamma' \in \operatorname{Der}_k(A)_r.$$

Thus the difference has differential order at most one, so its order-two projective principal symbol vanishes. □

Definition NC.8.4 (Quantized-stress projective symbol map)**.** Define

$$\boxed{\Psi_{2,r} : \ker \omega_r \longrightarrow H^0\Big(X, \operatorname{Sym}^2 T_X(r)\Big)} \tag{NC.8.3}$$

by choosing any correction γ supplied by Theorem NC.7.15 and setting

$$\boxed{\Psi_{2,r}(B) := \sigma^X_{2,0}(D_{B,\gamma}).} \tag{NC.8.4}$$

Lemma NC.8.3 makes the definition independent of the chosen correction.

Proposition NC.8.5 (Linearity)**.** *The map $\Psi_{2,r}$ is k-linear.*

Proof. Let $B_1, B_2 \in \ker \omega_r$ and choose corrections γ_1, γ_2. For $a, b \in k$,

$$a\gamma_1 + b\gamma_2$$

is a correction for

$$aB_1 + bB_2$$

because both j_r and h_r are linear. Principal symbols are linear in differential operators, so

$$\Psi_{2,r}(aB_1 + bB_2) = a\Psi_{2,r}(B_1) + b\Psi_{2,r}(B_2).$$

□

NC.8.3 Projectively invisible and visible quantized stresses

Definition NC.8.6 (Projectively invisible stress sector)**.** Define

$$\boxed{\mathcal{R}^{\mathrm{amb}}_{2,r}(A) := \ker \Psi_{2,r}.} \tag{NC.8.5}$$

Elements of $\mathcal{R}^{\mathrm{amb}}_{2,r}(A)$ are quantizable ambient stresses whose zero-twist projective second symbol vanishes.

Policy 10 (Radial terminology ceiling)**.** *The notation $\mathcal{R}^{\mathrm{amb}}_{2,r}(A)$ records* projective invisibility. *It may contain familiar Euler/radial symbols, but NC.8 does not identify it a priori with the subspace generated by tensors of the form*

$$E \odot \delta.$$

Any equality between the full kernel and a simpler radial formula requires a separate theorem.

Definition NC.8.7 (Visible quantized-stress space)**.** Define

$$\boxed{\mathcal{Z}_{2,r}(A) := \operatorname{im} \Psi_{2,r} \subseteq H^0\Big(X, \operatorname{Sym}^2 T_X(r)\Big).} \tag{NC.8.6}$$

This is the *projectively visible ambient quantized-stress space*.

Proposition NC.8.8 (Quantized-stress visibility quotient)**.** *There is a natural short exact sequence*

$$\boxed{0 \longrightarrow \mathcal{R}^{\mathrm{amb}}_{2,r}(A) \longrightarrow \ker \omega_r \xrightarrow{\Psi_{2,r}} \mathcal{Z}_{2,r}(A) \longrightarrow 0.} \tag{NC.8.7}$$

Consequently,

$$\boxed{\mathcal{Z}_{2,r}(A) \cong \ker \omega_r / \mathcal{R}^{\mathrm{amb}}_{2,r}(A).} \tag{NC.8.8}$$

Proof. This is the first isomorphism theorem applied to the linear map $\Psi_{2,r}$. □

NC.8.4 The generated projective-symbol sector

Definition NC.8.9 (Generated second-symbol sector)**.** Define

$$\boxed{\mathcal{G}^X_{2,r}(A) := \left\{\sigma^X_{2,0}(P) : P \in (\Delta_A)_r \cap D^2_{A/k}\right\}.} \tag{NC.8.9}$$

It is a k-linear subspace of

$$H^0\Big(X, \operatorname{Sym}^2 T_X(r)\Big).$$

Remark NC.8.10. The definition uses the *full* generated algebra $\Delta_A = A\langle \mathrm{Der}_k(A)\rangle$, not merely words of length at most two. Hence the long-word/order-drop issue is already included in $\mathcal{G}^X_{2,r}(A)$. NC.4 controls this full sector in the Critical Window.

Proposition NC.8.11 (Critical-window vanishing of the generated sector)**.** *Assume the hypotheses of the NC.4 Critical-Window Theorem and $m > 0$. If*

$$r < 2m,$$

then

$$\boxed{\mathcal{G}^X_{2,r}(A) = 0.} \tag{NC.8.10}$$

Proof. Corollary NC.4.12 gives

$$\sigma^X_{2,0}(P) = 0$$

for every

$$P \in (\Delta_A)_r \cap D^2_{A/k}.$$

□

NC.8.5 The Quantized Stress escape defect

Definition NC.8.12 (Quantized Stress escape defect)**.** Define

$$\boxed{\mathfrak{Q}_{2,r}(A) := \frac{\mathcal{Z}_{2,r}(A) + \mathcal{G}^X_{2,r}(A)}{\mathcal{G}^X_{2,r}(A)}.} \tag{NC.8.11}$$

Equivalently, by the second isomorphism theorem,

$$\boxed{\mathfrak{Q}_{2,r}(A) \cong \frac{\mathcal{Z}_{2,r}(A)}{\mathcal{Z}_{2,r}(A) \cap \mathcal{G}^X_{2,r}(A)}.} \tag{NC.8.12}$$

Remark NC.8.13 (Why this quotient is the correct defect)**.** A nonzero class in $\mathfrak{Q}_{2,r}(A)$ is represented by a projectively visible symbol of an *actual quantized stress operator* that does not belong to the projective second-symbol sector of the full generated algebra. Thus the quotient removes exactly the generated ambiguity relevant to strong non-generation. No dimension subtraction or unproved H^1-vanishing is used.

Theorem NC.8.14 (Quantized-Stress escape theorem)**.** *If*

$$\boxed{\mathfrak{Q}_{2,r}(A) \neq 0,}$$

then

$$\boxed{D_{A/k} \neq \Delta_A.}$$

More precisely, there exists

$$D \in D^2_{A/k,r} \setminus \Delta_A$$

arising from an ambient quantized stress.

Proof. Choose a nonzero class in $\mathfrak{Q}_{2,r}(A)$. By (NC.8.12), it has a representative

$$s \in \mathcal{Z}_{2,r}(A)$$

with

$$s \notin \mathcal{G}^X_{2,r}(A).$$

By definition of $\mathcal{Z}_{2,r}(A)$, there exists

$$B \in \ker \omega_r$$

and a correction γ such that

$$D := D_{B,\gamma} \in D^2_{A/k,r}$$

and

$$\sigma^X_{2,0}(D) = s.$$

If $D \in \Delta_A$, then by definition

$$s \in \mathcal{G}^X_{2,r}(A),$$

a contradiction. □

NC.8.6 Critical Quantized Stress defect

Definition NC.8.15 (Critical second-order Quantized Stress defect)**.** Assume $m > 0$. Define

$$\boxed{\mathfrak{Q}^{\mathrm{crit}}_2(A) := \mathfrak{Q}_{2,\,2m-1}(A).} \tag{NC.8.13}$$

Theorem NC.8.16 (Critical-window collapse of the defect)**.** *Assume the hypotheses of NC.4. At the critical degree*

$$r = 2m - 1,$$

one has

$$\boxed{\mathcal{G}^X_{2,\,2m-1}(A) = 0,}$$

and therefore canonical identifications

$$\boxed{\mathfrak{Q}^{\mathrm{crit}}_2(A) \cong \mathcal{Z}_{2,\,2m-1}(A) \cong \frac{\ker \omega_{2m-1}}{\mathcal{R}^{\mathrm{amb}}_{2,\,2m-1}(A)}.} \tag{NC.8.14}$$

Proof. Since

$$2m - 1 < 2m,$$

Proposition NC.8.11 gives

$$\mathcal{G}^X_{2,2m-1}(A) = 0.$$

The first identification follows immediately from Definition NC.8.11; the second is Proposition NC.8.8. □

Corollary NC.8.17 (Critical Quantized-Stress strong escape)**.** *Under the same hypotheses,*

$$\boxed{\mathfrak{Q}^{\mathrm{crit}}_2(A) \neq 0 \quad \Longrightarrow \quad D_{A/k} \neq \Delta_A.} \tag{NC.8.15}$$

Equivalently, it is enough to find

$$B \in \ker \omega_{2m-1}$$

such that

$$\boxed{\Psi_{2,\,2m-1}(B) \neq 0.} \tag{NC.8.16}$$

Proof. At the critical degree the generated projective-symbol sector vanishes by Theorem NC.8.16. Thus every nonzero element of $\mathcal{Z}_{2,2m-1}(A)$ gives a nonzero class in $\mathfrak{Q}^{\mathrm{crit}}_2(A)$, and Theorem NC.8.14 applies. □

NC.8.7 The new NC Master Bridge

Theorem NC.8.18 (Second-order NC Master Bridge)**.** *Let A be a standard-graded homogeneous ICIS in the NC source class. Assume:*

(i) the graded Kersken-type derivation structure used in NC.3 holds;

(ii) $m > 0$*;*

(iii) $X = \operatorname{Proj} A$ *is smooth on the declared projective locus; and*

(iv) there exists a critical ambient stress

$$B \in \mathcal{S}_{2,\,2m-1}(A)$$

such that

$$\omega_{2m-1}(B) = 0$$

and

$$\Psi_{2,\,2m-1}(B) \neq 0.$$

Then there exists

$$D \in D^2_{A/k,\,2m-1} \setminus A\langle \mathrm{Der}_k(A)\rangle.$$

Hence

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.8.17}$$

Proof. The equation

$$\omega_{2m-1}(B) = 0$$

quantizes B by Theorem NC.7.15. The nonzero projective symbol gives a nonzero element of $\mathcal{Z}_{2,2m-1}(A)$. The critical-window generated sector vanishes by Theorem NC.8.16. Corollary NC.8.17 then gives the strong escape. □

Remark NC.8.19 (Compression of the earlier proof DAG)**.** Theorem NC.8.18 compresses the earlier two-stage route

$$\text{projective critical tensor} \longrightarrow \text{quantization} \longrightarrow \text{escape}$$

into the direct criterion

$$\boxed{B \in \ker \omega_{2m-1} \setminus \mathcal{R}^{\mathrm{amb}}_{2,2m-1}(A).}$$

The direct route searches from the outset for a symbol that is both quantizable and projectively visible. Projective tensor existence remains useful as an independent discovery route, but it is no longer logically prior to every second-order NC proof.

NC.8.8 Diagonal quadrics as the first natural source class

Corollary NC.8.20 (Diagonal quadratic reduction)**.** *Assume all f_α are simultaneously diagonal quadrics and let*

$$B \in \mathcal{S}_{2,\,2m-1}(A)$$

be zero-diagonal. Then

$$\omega_{2m-1}(B) = 0$$

automatically. Therefore

$$\boxed{\Psi_{2,\,2m-1}(B) \neq 0}$$

is the only remaining NC.8 condition needed for strong escape.

Proof. Proposition NC.7.22 gives

$$\omega_{2m-1}(B) = 0.$$

Apply Theorem NC.8.18. □

NC.8.9 A concrete horizontal witness for the canonical $(2,2)$ control

Consider

$$A = \mathbb{Q}[x_1, \dots, x_6]/(q_0, q_1),$$

where

$$q_0 = \sum_{i=1}^{6} x_i^2, \qquad q_1 = \sum_{i=1}^{6} i\, x_i^2.$$

Let B be the zero-diagonal symmetric matrix used in the NC.7 exact replay, with in particular

$$B_{12} = x_3 x_4 x_5.$$

Define the homogeneous one-form on the cone

$$\alpha := x_2\, dx_1 - x_1\, dx_2.$$

Lemma NC.8.21 (Horizontal nonzero pairing)**.** *For the above stress,*

$$\boxed{\iota_E \alpha = 0,} \tag{NC.8.18}$$

and the symmetric symbol pairing satisfies

$$\boxed{B(\alpha, \alpha) = -2x_1 x_2 x_3 x_4 x_5.} \tag{NC.8.19}$$

If A is a domain, the right-hand side is nonzero.

Proof. Since

$$E = \sum_i x_i \partial_i,$$

one has

$$\iota_E \alpha = x_2 x_1 - x_1 x_2 = 0.$$

Because $B_{11} = B_{22} = 0$,

$$B(\alpha, \alpha) = 2B_{12}(x_2)(-x_1) = -2x_1 x_2 x_3 x_4 x_5.$$

□

Remark NC.8.22 (Use of the control)**.** Lemma NC.8.21 is an affine-horizontal certificate supporting projective visibility of the canonical $(2,2)$ stress. NC.8 does not use this one example to prove a family theorem. The higher-codimension family construction is deferred to NC.9, where the stress-kernel dimension bound and a projective nonvanishing certificate are proved uniformly.

NC.8.10 Finite computation and evidence discipline

Proposition NC.8.23 (Finite critical-stress computation)**.** *At fixed internal degree r, the spaces*

$$\ker \omega_r, \qquad \mathcal{R}^{\mathrm{amb}}_{2,r}(A), \qquad \mathcal{Z}_{2,r}(A),$$

and, after a finite presentation of the projective-symbol target,

$$\mathfrak{Q}_{2,r}(A),$$

are finite-dimensional k-vector spaces. Thus an exact quotient-basis implementation can compute the critical defect by finite linear algebra.

Proof. NC.7 proves finite-dimensionality of $\ker \omega_r$. The projective target

$$H^0(X, \mathrm{Sym}^2 T_X(r))$$

is finite-dimensional because X is projective. Kernels, images, intersections, sums, and quotients of finite-dimensional vector spaces are finite-dimensional. □

Policy 11 (No dimension-subtraction shortcut)**.** *The identity*

$$\dim \mathfrak{Q}_{2,r} = \dim \mathcal{Z}_{2,r} - \dim(\mathcal{Z}_{2,r} \cap \mathcal{G}^X_{2,r})$$

may be used only after both spaces and their intersection have been represented by valid linear maps. No dimension formula inferred from a nonexact global-sections sequence, no unproved H^1-vanishing, and no mere count of source and target dimensions is admissible as a substitute.

NC.8.11 NC.8 closure

Theorem NC.8.24 (NC.8 closure theorem)**.** *The second-order NC problem now has a single critical defect:*

$$\boxed{\mathfrak{Q}^{\mathrm{crit}}_2(A) \cong \frac{\ker \omega_{2m-1}}{\mathcal{R}^{\mathrm{amb}}_{2,2m-1}(A)}}$$

under the NC.4 Critical Window. If this space is nonzero, then

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.}$$

For simultaneously diagonal quadrics, zero-diagonal stresses satisfy

$$\omega_{2m-1} = 0$$

automatically, so the remaining problem is the construction of a projectively visible critical stress.

Proof. Combine Theorem NC.8.16, Corollary NC.8.17, and Proposition NC.7.22. □

Declaration NC.8.25 (NC.8 status and successor obligation)**.** NC.8 establishes the conceptual bridge

$$\boxed{\text{critical Jacobian stress} \longrightarrow \text{Hessian-obstruction kernel} \longrightarrow \text{projective visibility} \longrightarrow \text{strong Nakai escape.}}$$

The quotient construction and the implication

$$\mathfrak{Q}^{\mathrm{crit}}_2(A) \neq 0 \Longrightarrow D_{A/k} \neq \Delta_A$$

are C0–C2 consequences of NC.4 and NC.7. NC.8 does not prove that the critical defect is nonzero for every ICIS. The next stage, NC.9, supplies a uniform nonzero critical stress for a broad higher-codimension family of diagonal quadratic complete intersections and thereby tests the full bridge on an infinite non-hypersurface source class.

NC.9 Higher-Codimension Diagonal Quadratic Stress Escape

NC.9.1 Scope and objective

NC.9 applies the NC.7–NC.8 Quantized Stress calculus to complete intersections of simultaneously diagonal quadrics of arbitrary codimension. The principal result is a uniform sufficient dimension condition under which a critical zero-diagonal stress exists, is automatically quantizable, has nonzero projective second symbol, and therefore escapes the full algebra generated by first-order derivations.

Let

$$P = k[x_1, \ldots, x_n], \qquad A = P/(f_1, \ldots, f_c), \qquad c \geq 2,$$

where

$$\boxed{f_\alpha = \sum_{i=1}^{n} a_{\alpha i} x_i^2 \qquad (1 \leq \alpha \leq c).} \tag{NC.9.1}$$

Write

$$\mathbf{A} := (a_{\alpha i}) \in M_{c\times n}(k)$$

for the coefficient matrix, and set

$$X := \mathrm{Proj}\, A, \qquad p := \dim X = n - c - 1.$$

NC.9.2 Normal-domain closure from projective smoothness

Lemma NC.9.1 (Smooth projective complete-intersection cone is normal)**.** *Assume $f_1, \dots, f_c$ is a homogeneous regular sequence, $p \geq 1$, and $X = \operatorname{Proj} A$ is smooth. Then the affine cone A is a normal domain and its only possible singular point is the homogeneous vertex*

$$\mathfrak{m} = A_+.$$

Proof. Because A is a complete intersection, it is Cohen–Macaulay and therefore satisfies Serre's condition S_2. At every nonzero point of the affine cone, smoothness of X gives Jacobian rank c, hence the punctured cone is smooth. Thus the singular locus is contained in the vertex.

Since

$$\dim A = p + 1 \geq 2,$$

the vertex has codimension at least two. Therefore every prime of height at most one is regular, so A satisfies R_1. Serre's criterion gives normality.

A normal Noetherian ring is a finite product of normal domains along its connected components. The standard grading with $A_0 = k$ admits no nontrivial idempotent: normality gives reducedness, and comparison of the highest homogeneous term in $e^2 = e$ forces every idempotent e to lie in $A_0 = k$. Hence $\operatorname{Spec} A$ is connected, so only one normal-domain factor occurs. Thus A is a domain. □

Remark NC.9.2. The NC.9 dimension inequality below forces $p \geq 2$, so the hypotheses used in Lemma NC.9.1 are automatically in the safe codimension range for the vertex.

NC.9.3 Coefficient rank

Lemma NC.9.3 (Full row rank of the diagonal coefficient matrix)**.** *If $f_1, \dots, f_c$ is a regular sequence of quadrics as in (NC.9.1), then*

$$\boxed{\operatorname{rank} \mathbf{A} = c.} \tag{NC.9.2}$$

Proof. If the rows were linearly dependent over k, after reordering one defining quadric would be a constant k-linear combination of the others. Its class modulo the preceding generators would then be zero, contradicting the regular-sequence requirement that the next generator be a nonzerodivisor on a nonzero quotient. □

NC.9.4 The finite stress kernel

Fix a subset

$$I \subseteq \{1, \dots, n\}, \qquad |I| = N,$$

such that the column submatrix

$$\mathbf{A}_I \in M_{c \times N}(k)$$

has rank c.

Definition NC.9.4 (Constant zero-diagonal stress space)**.** Let

$$C_I := \left\{ C \in M_N(k) : C^T = C,\ C_{ii} = 0 \right\}.$$

Define

$$\boxed{\Phi_I : C_I \longrightarrow M_{N \times c}(k), \qquad C \longmapsto C\mathbf{A}_I^T.} \tag{NC.9.3}$$

A matrix

$$0 \neq C \in \ker \Phi_I$$

is called a constant diagonal-quadratic stress on I.

Lemma NC.9.5 (Symmetry constraint on the image)**.** *For every $C \in C_I$, if*

$$Y = C\mathbf{A}_I^T,$$

then

$$\boxed{\mathbf{A}_I Y = \mathbf{A}_I C \mathbf{A}_I^T}$$

is symmetric.

Proof. The matrix C is symmetric, so

$$(\mathbf{A}_I C \mathbf{A}_I^T)^T = \mathbf{A}_I C^T \mathbf{A}_I^T = \mathbf{A}_I C \mathbf{A}_I^T.$$

□

Lemma NC.9.6 (Codimension of the symmetry constraint)**.** *Let*

$$\Theta : M_{N\times c}(k) \longrightarrow \mathrm{Alt}_c(k)$$

be

$$\Theta(Y) := \mathbf{A}_I Y - (\mathbf{A}_I Y)^T.$$

Then Θ is surjective and

$$\boxed{\dim \ker \Theta = Nc - \binom{c}{2}.} \tag{NC.9.4}$$

Proof. Because $\operatorname{rank} \mathbf{A}_I = c$, choose a right inverse

$$R \in M_{N\times c}(k), \qquad \mathbf{A}_I R = I_c.$$

For any alternating matrix $K \in \mathrm{Alt}_c(k)$, set

$$Y := \frac{1}{2} RK.$$

Characteristic zero gives

$$\Theta(Y) = \frac{1}{2}K - \frac{1}{2}K^T = K.$$

Thus Θ is surjective. Since

$$\dim M_{N\times c}(k) = Nc$$

and

$$\dim \mathrm{Alt}_c(k) = \binom{c}{2},$$

the dimension formula follows. □

Theorem NC.9.7 (Stress-kernel lower bound)**.** *One has*

$$\boxed{\dim_k \ker \Phi_I \geq \binom{N}{2} - Nc + \binom{c}{2} = \binom{N-c}{2} - c.} \tag{NC.9.5}$$

Proof. By Lemma NC.9.5,

$$\operatorname{im} \Phi_I \subseteq \ker \Theta.$$

Therefore

$$\operatorname{rank} \Phi_I \leq Nc - \binom{c}{2}.$$

Since

$$\dim C_I = \binom{N}{2},$$

rank–nullity gives

$$\dim \ker \Phi_I \geq \binom{N}{2} - Nc + \binom{c}{2}.$$

A direct simplification yields

$$\binom{N}{2} - Nc + \binom{c}{2} = \binom{N-c}{2} - c.$$

□

NC.9.5 The dimension criterion

Theorem NC.9.8 (Uniform existence of a nonzero constant stress)**.** *Assume*

$$\boxed{p(p+1) > 2c.} \tag{NC.9.6}$$

Set

$$N := \min\{n,\ 2c+1\}.$$

Then there exists a subset I of cardinality N with

$$\operatorname{rank} \mathbf{A}_I = c$$

and

$$\boxed{0 \neq C \in \ker \Phi_I.}$$

Proof. By Lemma NC.9.3, the full coefficient matrix has row rank c. Choose c independent columns and enlarge them to a subset of cardinality N; this gives $\operatorname{rank} \mathbf{A}_I = c$.

If

$$n \geq 2c+1,$$

then $N = 2c+1$, so Theorem NC.9.7 gives

$$\dim \ker \Phi_I \geq \binom{c+1}{2} - c = \frac{c(c-1)}{2} > 0.$$

If

$$n < 2c+1,$$

then $N = n$ and

$$N - c = n - c = p + 1.$$

Hence

$$\dim \ker \Phi_I \geq \binom{p+1}{2} - c = \frac{p(p+1) - 2c}{2} > 0$$

by (NC.9.6). □

NC.9.6 Critical polynomial stress

Fix

$$0 \neq C \in \ker \Phi_I$$

as above. Define

$$\delta := 2c + 1 - N \geq 0.$$

If $\delta = 0$, put $h := 1$. If $\delta > 0$, choose any coordinate x_s and put

$$h := x_s^{\delta}.$$

By Lemma NC.9.1 and the fact that the ideal is generated in degree two, $h \neq 0$ in A.

Definition NC.9.9 (Critical polynomial stress matrix)**.** Define a symmetric $n \times n$ matrix B^C by

$$B^C_{ii} := 0,$$

and, for $i \neq j$,

$$\boxed{B^C_{ij} := \begin{cases} h\, C_{ij} \displaystyle\prod_{\ell \in I \setminus \{i,j\}} x_\ell, & i, j \in I, \\ 0, & \text{otherwise.} \end{cases}} \tag{NC.9.7}$$

Lemma NC.9.10 (Critical coefficient degree)**.** *Every nonzero entry of B^C has degree*

$$\boxed{2c-1.} \tag{NC.9.8}$$

Consequently the associated second-order operator has internal degree

$$\boxed{r=2c-3.} \tag{NC.9.9}$$

For a codimension-c quadratic complete intersection,

$$m=c-1,$$

so

$$\boxed{r=2m-1<2m.} \tag{NC.9.10}$$

Proof. By construction,

$$\deg B^C_{ij}=\delta+(N-2)=(2c+1-N)+(N-2)=2c-1.$$

Subtracting differential order two gives internal degree $2c-3$. Finally,

$$m=\sum_{\alpha=1}^{c}2-(c+1)=c-1.$$

□

NC.9.7 Jacobian stress and automatic quantization

Theorem NC.9.11 (Critical Jacobian stress identity)**.** *The matrix B^C satisfies*

$$\boxed{J(B^C)=0 \quad \text{in } A.} \tag{NC.9.11}$$

Equivalently,

$$B^C J^T=0.$$

Thus

$$B^C\in\mathcal{S}_{2,\,2c-3}(A).$$

Proof. For $i\notin I$, the i-th column of JB^C is zero. For $i\in I$, using

$$\partial_j f_\alpha=2a_{\alpha j}x_j,$$

one gets

$$\begin{aligned}(B^C\nabla f_\alpha)_i&=2\sum_{j\in I}hC_{ij}\left(\prod_{\ell\in I\setminus\{i,j\}}x_\ell\right)a_{\alpha j}x_j\\&=2h\left(\prod_{\ell\in I\setminus\{i\}}x_\ell\right)\sum_{j\in I}C_{ij}a_{\alpha j}.\end{aligned}$$

The final sum is zero because

$$C\mathbf{A}_I^T=0.$$

□

Corollary NC.9.12 (Hessian obstruction vanishes)**.** *The stress B^C satisfies*

$$\boxed{\omega_{2c-3}(B^C)=0.} \tag{NC.9.12}$$

Hence the ambient operator

$$\boxed{D_C:=\frac{1}{2}\sum_{i,j=1}^{n}B^C_{ij}\partial_i\partial_j} \tag{NC.9.13}$$

descends to

$$D_C\in D^2_{A/k,\,2c-3}$$

with no first-order correction.

Proof. The matrix B^C has zero diagonal. Proposition NC.7.22 therefore gives

$$\mathsf{h}_{2c-3}(B^C) = 0$$

and hence

$$\omega_{2c-3}(B^C) = 0.$$

The NC.7 Quantization Gate supplies the descended operator with $\gamma = 0$. □

NC.9.8 Projective visibility

Choose $i \neq j$ in I with

$$C_{ij} \neq 0.$$

Such a pair exists because $C \neq 0$ and C has zero diagonal. On $\mathbf{P}^{n-1}$, the homogeneous one-form

$$\alpha_{ij} := x_j\, dx_i - x_i\, dx_j$$

defines a global section

$$\alpha_{ij} \in H^0\Big(\mathbf{P}^{n-1}, \Omega^1_{\mathbf{P}^{n-1}}(2)\Big)$$

by the Euler sequence. Write

$$\bar{\alpha}_{ij} \in H^0\Big(X, \Omega^1_X(2)\Big)$$

for its image after restriction to X.

Lemma NC.9.13 (Horizontal critical pairing)**.** *For the quantized stress B^C,*

$$\boxed{\Psi_{2,\,2c-3}(B^C)(\bar{\alpha}_{ij}, \bar{\alpha}_{ij}) = -2C_{ij}\, h \prod_{\ell \in I} x_\ell} \tag{NC.9.14}$$

as a section of

$$\mathcal{O}_X(2c+1).$$

Proof. The order-two principal symbol of (NC.9.13) is the symmetric matrix B^C. The coefficient vector of α_{ij} has only two nonzero entries: x_j in position i and $-x_i$ in position j. Since $B^C_{ii} = B^C_{jj} = 0$,

$$B^C(\alpha_{ij}, \alpha_{ij}) = -2x_i x_j B^C_{ij}.$$

Substituting (NC.9.7) gives

$$-2C_{ij} h \prod_{\ell \in I} x_\ell.$$

Tangency of B^C to every defining equation, proved in Theorem NC.9.11, ensures that the pairing factors through Ω^1_X, hence computes the projective symbol pairing. □

Theorem NC.9.14 (Projective visibility of every nonzero constant stress)**.** *Under the hypotheses above,*

$$\boxed{\Psi_{2,\,2c-3}(B^C) \neq 0.} \tag{NC.9.15}$$

Proof. By Lemma NC.9.1, A is a domain. The defining ideal contains no nonzero linear form, so every coordinate class x_ℓ is nonzero in A. The chosen h is also nonzero. Hence

$$C_{ij}\, h \prod_{\ell \in I} x_\ell \neq 0 \quad \text{in } A.$$

Lemma NC.9.13 therefore gives a nonzero projective-symbol evaluation, proving

$$\Psi_{2,\,2c-3}(B^C) \neq 0.$$

□

NC.9.9 Higher-codimension diagonal quadratic theorem

Theorem NC.9.15 (Higher-codimension diagonal quadratic strong escape)**.** *Let k be a field of characteristic zero and let*

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_c), \qquad c \geq 2,$$

where $f_1, \dots, f_c$ is a homogeneous regular sequence of simultaneously diagonal quadrics as in (NC.9.1). Assume $X = \operatorname{Proj} A$ is smooth and

$$\boxed{p(p+1) > 2c, \qquad p = n - c - 1.}$$

Assume also the graded Kersken-type derivation-generation theorem used in NC.3 at its exact source strength. Then there exists

$$\boxed{D_C \in D^2_{A/k,\,2c-3} \setminus A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.9.16}$$

Consequently,

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.9.17}$$

Proof. Theorem NC.9.8 supplies

$$0 \neq C \in \ker \Phi_I.$$

Theorem NC.9.11 constructs the critical stress

$$B^C \in \mathcal{S}_{2,\,2c-3}(A).$$

Corollary NC.9.12 gives

$$\omega_{2c-3}(B^C) = 0,$$

and Theorem NC.9.14 gives

$$\Psi_{2,\,2c-3}(B^C) \neq 0.$$

By Lemma NC.9.10,

$$2c - 3 = 2m - 1.$$

The NC.8 Master Bridge therefore gives the strong escape (NC.9.16). □

Corollary NC.9.16 (Critical defect nonvanishing)**.** *Under the hypotheses of Theorem NC.9.15,*

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0.} \tag{NC.9.18}$$

Remark NC.9.17 (Asymptotic range)**.** The sufficient condition

$$p(p+1) > 2c$$

is equivalent to

$$p > \frac{-1 + \sqrt{1 + 8c}}{2}.$$

Thus the required projective dimension grows only on the order of $\sqrt{2c}$ as the codimension increases. NC.9 does not claim that this sufficient bound is optimal.

NC.9.10 Explicit Vandermonde family

NC.9 now gives an abstract stress-existence theorem. There is also a canonical family in which the constant stress has a closed formula.

Fix $c \geq 2$, put

$$N := 2c + 1,$$

and choose pairwise distinct scalars

$$\lambda_1, \dots, \lambda_N \in k.$$

For

$$a = 0, \dots, c - 1,$$

define

$$\boxed{f_a := \sum_{i=1}^{N} \lambda_i^a x_i^2.} \tag{NC.9.19}$$

Lemma NC.9.18 (Smoothness of the Vandermonde family)**.** *The projective complete intersection*

$$X_c := V(f_0, \dots, f_{c-1}) \subset \mathbf{P}^{2c}$$

is smooth of dimension

$$p = c.$$

Proof. We first verify nonemptiness and the expected codimension after extension to an algebraic closure $\bar{k}$. Let

$$V = (\lambda_i^a)_{\substack{0 \le a \le c-1 \\ 1 \le i \le N}}$$

be the $c \times N$ Vandermonde coefficient matrix. Every $c \times c$ minor obtained from distinct columns is nonzero, hence

$$\operatorname{rank} V = c, \qquad \dim_{\bar{k}} \ker V = N - c = c + 1.$$

For each coordinate i, deleting the i-th column still leaves rank c, so the coordinate projection

$$\ker V \longrightarrow \bar{k}, \qquad y \longmapsto y_i$$

is nonzero. Thus each coordinate hyperplane cuts $\ker V$ in a proper linear subspace. Because $\bar{k}$ is infinite, the finite union of those proper subspaces does not cover $\ker V$. Choose

$$y = (y_1, \dots, y_N) \in \ker V$$

with every $y_i \neq 0$, and choose $x_i \in \bar{k}$ with $x_i^2 = y_i$. Then $[x] \in \mathbf{P}^{2c}_{\bar{k}}$ is a common zero of all f_a. Hence the projective zero locus is nonempty.

Now the Jacobian matrix has rows

$$(2\lambda_i^a x_i)_{i=1}^N.$$

Suppose its rank is $< c$ at a projective common zero x. Then there exists a nonzero polynomial

$$H(t) = \sum_{a=0}^{c-1} \nu_a t^a$$

of degree at most $c - 1$ such that

$$H(\lambda_i) x_i = 0 \qquad \text{for every } i.$$

Thus the support

$$S_x := \{i : x_i \neq 0\}$$

is contained in the root set of H, so

$$|S_x| \le c - 1.$$

Set $y_i = x_i^2$. The equations $f_a(x) = 0$ give

$$\sum_{i \in S_x} \lambda_i^a y_i = 0 \qquad (a = 0, \dots, c-1).$$

The corresponding $c \times |S_x|$ Vandermonde matrix has full column rank because the λ_i are distinct and $|S_x| \le c - 1 < c$. Hence every $y_i = 0$, so every $x_i = 0$, contradicting projectivity. Therefore the Jacobian rank is c at every geometric point of the common zero locus.

The Jacobian criterion now shows that the zero locus is geometrically smooth of codimension c, hence of dimension

$$2c - c = c.$$

Since the polynomial ring is Cohen–Macaulay and the ideal has height c while being generated by c elements, $f_0, \dots, f_{c-1}$ is a regular sequence. □

Let

$$P_\lambda(t) := \prod_{i=1}^{N}(t-\lambda_i), \qquad \mu_i := \frac{1}{P'_\lambda(\lambda_i)}.$$

Lemma NC.9.19 (Barycentric moment cancellation)**.** *For every integer*

$$0 \le r \le N-2,$$

one has

$$\boxed{\sum_{i=1}^{N} \mu_i \lambda_i^r = 0.} \tag{NC.9.20}$$

Proof. Apply Lagrange interpolation to the polynomial t^r:

$$t^r = \sum_{i=1}^{N} \lambda_i^r \frac{P_\lambda(t)}{(t-\lambda_i)P'_\lambda(\lambda_i)}.$$

Since $r \le N-2$, the coefficient of t^{N-1} on the left is zero. The coefficient of t^{N-1} in the i-th Lagrange basis polynomial is

$$\frac{1}{P'_\lambda(\lambda_i)} = \mu_i.$$

Comparing coefficients proves (NC.9.20). □

Definition NC.9.20 (Vandermonde stress matrix)**.** Define

$$C^{\mathrm{V}}_{ii} := 0$$

and, for $i \ne j$,

$$\boxed{C^{\mathrm{V}}_{ij} := \mu_i \mu_j (\lambda_i - \lambda_j)^2.} \tag{NC.9.21}$$

Theorem NC.9.21 (Closed-form Vandermonde stress)**.** *Let*

$$\mathbf{A}_{\mathrm{V}} = (\lambda_i^a)_{\substack{0 \le a \le c-1 \\ 1 \le i \le N}}.$$

Then

$$\boxed{C^{\mathrm{V}} \mathbf{A}_{\mathrm{V}}^T = 0.} \tag{NC.9.22}$$

In particular $C^{\mathrm{V}} \ne 0$ is an explicit constant stress.

Proof. Fix i and $0 \le a \le c-1$. Since the $j = i$ term vanishes,

$$\begin{aligned}\sum_{j=1}^{N} C^{\mathrm{V}}_{ij} \lambda_j^a &= \mu_i \sum_{j=1}^{N} \mu_j (\lambda_i - \lambda_j)^2 \lambda_j^a \\ &= \mu_i \left[\lambda_i^2 \sum_j \mu_j \lambda_j^a - 2\lambda_i \sum_j \mu_j \lambda_j^{a+1} + \sum_j \mu_j \lambda_j^{a+2} \right].\end{aligned}$$

Here

$$a+2 \le c+1 \le 2c-1 = N-2$$

because $c \ge 2$. All three sums therefore vanish by Lemma NC.9.19. Thus every matrix entry of $C^{\mathrm{V}} \mathbf{A}_{\mathrm{V}}^T$ is zero. Since the λ_i are pairwise distinct and every $\mu_i \ne 0$, the off-diagonal entries are nonzero, so $C^{\mathrm{V}} \ne 0$. □

Corollary NC.9.22 (Explicit infinite family of strong escapes)**.** *For every integer $c \geq 2$, the smooth codimension-c quadratic complete intersection*

$$A_c = k[x_1, \ldots, x_{2c+1}]/(f_0, \ldots, f_{c-1})$$

of (NC.9.19) *has*

$$p = c$$

and satisfies

$$p(p+1) > 2c.$$

Assuming the NC.3 graded derivation-generation theorem at its exact source strength,

$$\boxed{D_{A_c/k} \neq A_c\langle \mathrm{Der}_k(A_c)\rangle.} \tag{NC.9.23}$$

Moreover a witnessing critical operator is obtained explicitly from

$$B_{ij} = C^{\mathrm{V}}_{ij} \prod_{\ell \neq i,j} x_\ell, \qquad B_{ii} = 0,$$

by

$$\boxed{D_{\mathrm{V}} = \frac{1}{2}\sum_{i,j} B_{ij}\partial_i\partial_j.} \tag{NC.9.24}$$

Proof. Lemma NC.9.18 gives

$$p = (2c+1) - c - 1 = c.$$

For $c \geq 2$,

$$p(p+1) = c(c+1) > 2c.$$

Apply Theorem NC.9.15. Since $N = 2c+1$, the degree-correction factor h equals 1, and Theorem NC.9.21 supplies the displayed closed-form stress. □

NC.9.11 Evidence status and novelty ceiling

Declaration NC.9.23 (NC.9 claim status)**.** The stress-kernel dimension bound, polynomial-stress construction, Jacobian identity, vanishing Hessian obstruction, critical-degree calculation, and Vandermonde cancellation are C0 algebraic statements. The assembled strong-escape theorem is classified

$$\boxed{\text{C2-high}}$$

until the imported Kersken theorem is source-bound at the exact invoked hypotheses, an independent proof audit is completed, and the relevant complete-intersection/Nakai literature is exhaustively checked.

NC.9 does not claim that the numerical condition

$$p(p+1) > 2c$$

is necessary or optimal. It does not claim strong escape for every quadratic complete intersection: simultaneous diagonalization is an actual hypothesis of the main theorem. It does not claim that the Vandermonde family is new in the literature before the prior-art audit.

Theorem NC.9.24 (NC.9 closure theorem)**.** *Within the simultaneously diagonal quadratic source class, the second-order Nakai problem is closed by the finite stress inequality*

$$\boxed{p(p+1) > 2c.}$$

Under this inequality, NC.9 constructs a critical stress

$$B^C \in \ker \omega_{2m-1}$$

with

$$\Psi_{2,2m-1}(B^C) \neq 0,$$

and hence

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0 \quad \Longrightarrow \quad D_{A/k} \neq \Delta_A.}$$

The Vandermonde family supplies an explicit such example in every codimension $c \geq 2$.

Proof. This is the combination of Theorems NC.9.8, NC.9.11, NC.9.14, NC.9.15, and Corollary NC.9.22. □

Declaration NC.9.25 (Successor obligation)**.** NC.9 is the first Appendix-NC theorem family whose codimension is unbounded. The next mathematical obstruction is no longer the existence of a higher-codimension example. The next target is to remove the simultaneous-diagonal hypothesis. For general quadratic forms

$$f_\alpha = x^T Q_\alpha x,$$

the Jacobian-stress equation persists, but the Hessian obstruction becomes

$$\mathrm{tr}(BQ_\alpha),$$

which need not vanish. NC.10 should therefore study whether the NC.7 obstruction map

$$\omega_{2m-1}$$

has a nonzero projectively visible kernel for arbitrary smooth quadratic complete intersections.

NC.10 General Quadratic ICIS: Exact-Stress Divergence Quantization and the One-Degree-Lower Barrier

NC.10.1 Purpose and current boundary

NC.9 proves strong Nakai escape for a broad infinite family of *simultaneously diagonal* quadratic complete intersections. NC.10 removes that coordinate hypothesis from the obstruction formalism and determines exactly which part of the argument survives for arbitrary quadratic forms.

Let

$$P = k[x_1, \ldots, x_n], \qquad A = P/(f_1, \ldots, f_c), \qquad c \geq 2,$$

where

$$\boxed{f_\alpha = x^T Q_\alpha x, \qquad Q_\alpha = Q_\alpha^T \in M_n(k).} \tag{NC.10.1}$$

Then

$$\nabla f_\alpha = 2Q_\alpha x, \qquad \mathrm{Hess}(f_\alpha) = 2Q_\alpha.$$

For a quadratic complete intersection,

$$m = c - 1, \qquad r_{\mathrm{crit}} = 2m - 1 = 2c - 3.$$

The main conclusion of NC.10 is deliberately twofold.

First, every *exact* symmetric polynomial stress

$$BQ_\alpha x = 0 \quad \text{in } P^n$$

is automatically quantizable by a canonical divergence correction. Thus the Hessian obstruction disappears on the exact-stress lane, without any diagonal hypothesis.

Second, there is a universal Gram-cofactor exact stress of degree $2c$, hence operator degree $2c - 2 = 2m$. This lies exactly one degree *above* the NC.4 escape window. Therefore the general quadratic problem is reduced to a precise one-degree-lower question rather than solved by the universal Gram construction.

NC.10.2 Exact quadratic stresses

Definition NC.10.1 (Exact quadratic stress module)**.** For an integer r, define

$$\boxed{\mathcal{E}^{\mathrm{quad}}_{2,r} := \left\{B = B^T \in M_n(P_{r+2}) : BQ_\alpha x = 0 \text{ in } P^n \text{ for all } \alpha\right\}.} \tag{NC.10.2}$$

This is the *exact* stress module at internal degree r.

Remark NC.10.2. The condition in (NC.10.2) is stronger than the NC.7 Jacobian-stress equation

$$JB = 0 \quad \text{in } A.$$

Every exact stress defines a Jacobian stress after reduction modulo I, but the converse need not hold.

Definition NC.10.3 (Matrix divergence)**.** For a polynomial matrix $B = (B_{ij})$, define

$$\boxed{(\operatorname{div} B)_j := \sum_{i=1}^{n} \partial_i B_{ij}.} \tag{NC.10.3}$$

Thus

$$\operatorname{div} B \in P^n.$$

NC.10.3 Divergence identity and automatic quantization

Lemma NC.10.4 (Exact-stress divergence identity)**.** *Let $B = B^T$ be a polynomial matrix and let $Q = Q^T$ be a constant symmetric matrix. If*

$$BQx = 0 \quad \text{in } P^n,$$

then

$$\boxed{\operatorname{tr}(BQ) = -(\operatorname{div} B) \cdot Qx.} \tag{NC.10.4}$$

Proof. Take the ordinary divergence of the zero vector BQx:

$$0 = \sum_{i=1}^{n} \partial_i \left(\sum_{j=1}^{n} B_{ij} (Qx)_j \right).$$

Expanding,

$$0 = \sum_{i,j} (\partial_i B_{ij})(Qx)_j + \sum_{i,j} B_{ij} Q_{ji}.$$

The first term is

$$(\operatorname{div} B) \cdot Qx,$$

and the second is

$$\operatorname{tr}(BQ)$$

because Q is symmetric. □

Theorem NC.10.5 (Exact-Stress Divergence Quantization)**.** *Let*

$$B \in \mathcal{E}^{\mathrm{quad}}_{2,r}.$$

Define

$$\boxed{\gamma_B := \frac{1}{2} \operatorname{div} B} \tag{NC.10.5}$$

and

$$\boxed{\widetilde{\mathscr{D}}_B := \frac{1}{2} \sum_{i,j=1}^{n} B_{ij} \partial_i \partial_j + \gamma_B \cdot \nabla.} \tag{NC.10.6}$$

Then, for every α,

$$\boxed{\widetilde{\mathscr{D}}_B(f_\alpha) = 0} \tag{NC.10.7}$$

and

$$\boxed{\widetilde{\mathscr{D}}_B(f_\alpha h) = f_\alpha \widetilde{\mathscr{D}}_B(h) \qquad (h \in P).} \tag{NC.10.8}$$

Consequently

$$\widetilde{\mathscr{D}}_B(I) \subseteq I$$

and the operator descends to

$$\boxed{\mathscr{D}_B \in D^2_{A/k,r}.} \tag{NC.10.9}$$

Proof. For a quadratic $f_\alpha = x^T Q_\alpha x$, the NC.7 Hessian contraction is

$$\frac{1}{2} B : \mathrm{Hess}(f_\alpha) = \mathrm{tr}(BQ_\alpha).$$

The first-order contribution is

$$\gamma_B \cdot \nabla f_\alpha = (\mathrm{div}\, B) \cdot Q_\alpha x.$$

Lemma NC.10.4 therefore gives

$$\widetilde{\mathscr{D}}_B(f_\alpha) = 0.$$

Moreover, the mixed term in the NC.7 Leibniz identity is

$$(B\nabla f_\alpha) \cdot \nabla h = 2(BQ_\alpha x) \cdot \nabla h = 0.$$

Hence

$$\widetilde{\mathscr{D}}_B(f_\alpha h) = f_\alpha \widetilde{\mathscr{D}}_B(h).$$

This proves ideal preservation and descent. □

Corollary NC.10.6 (Exact stresses lie in the Hessian-obstruction kernel)**.** *Reduction modulo I sends*

$$\mathcal{E}^{\mathrm{quad}}_{2,r}$$

into

$$\ker \omega_r.$$

More precisely,

$$\boxed{\mathsf{h}_r(B) = -\mathsf{j}_r\left(\frac{1}{2}\,\mathrm{div}\, B\right)} \tag{NC.10.10}$$

for every exact stress B.

Proof. Apply Lemma NC.10.4 to each Q_α and compare with the definitions of h_r and j_r. □

Remark NC.10.7 (Conceptual consequence)**.** The Hessian obstruction in a general quadratic ICIS is not intrinsically an obstruction to every stress. It is an obstruction only to stresses that are tangent modulo the defining ideal but fail to possess an exact ambient representative. On the exact-stress lane, quantization is canonical.

NC.10.4 Divergence-form interpretation

Proposition NC.10.8 (Divergence-form operator)**.** *For symmetric B,*

$$\boxed{\widetilde{\mathscr{D}}_B = \frac{1}{2}\sum_{j=1}^{n} \partial_j \left(\sum_{i=1}^{n} B_{ij}\partial_i\right).} \tag{NC.10.11}$$

Proof. Expand the right-hand side:

$$\frac{1}{2}\sum_{i,j}(\partial_j B_{ij})\partial_i + \frac{1}{2}\sum_{i,j} B_{ij}\partial_j\partial_i.$$

Because B is symmetric, relabeling indices in the first term gives

$$\frac{1}{2}(\operatorname{div} B)\cdot\nabla,$$

while the second term is the stated symmetric second-order part. □

NC.10.5 Critical exact-stress escape criterion

Definition NC.10.9 (Critical exact-stress sector)**.** For a codimension-c quadratic complete intersection, define

$$\boxed{\mathcal{E}_2^{\mathrm{crit}}(A) := \mathcal{E}^{\mathrm{quad}}_{2,\,2c-3}.} \tag{NC.10.12}$$

Theorem NC.10.10 (Critical exact-stress escape)**.** *Assume the NC.4 hypotheses and let*

$$0 \neq B \in \mathcal{E}_2^{\mathrm{crit}}(A).$$

If the projective second symbol of the descended divergence operator is nonzero,

$$\boxed{\sigma^X_{2,0}(\mathscr{D}_B) \neq 0,} \tag{NC.10.13}$$

then

$$\boxed{\mathscr{D}_B \notin A\langle \mathrm{Der}_k(A)\rangle}$$

and therefore

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.10.14}$$

Proof. By Theorem NC.10.5,

$$\mathscr{D}_B \in D^2_{A/k,\,2c-3}.$$

Since

$$m = c - 1,$$

one has

$$2c - 3 = 2m - 1 < 2m.$$

The nonzero projective symbol therefore triggers Corollary NC.4.12 and the strong-symbol escape theorem of NC.4. □

Corollary NC.10.11 (NC.9 lies entirely in the exact-stress lane)**.** *Every NC.9 stress matrix B^C satisfies*

$$B^C Q_\alpha x = 0 \quad \text{in } P^n$$

for every α. Hence the NC.9 operators may equivalently be quantized by Theorem NC.10.5. For the zero-diagonal diagonal-family stresses, the divergence correction and the NC.7 correction differ, if at all, only by a derivation of A.

Proof. The proof of Theorem NC.9.11 establishes the vanishing as a polynomial identity before reduction modulo the ideal. Both the NC.7 correction and the divergence correction quantize the same principal matrix. Lemma NC.8.3 shows that any two such corrections differ by a first-order derivation. □

NC.10.6 A universal exact stress at the threshold

The exact-stress lane exists for every collection of quadratic forms, but the universal construction naturally appears one degree too late.

Fix any constant invertible symmetric matrix

$$H = H^T \in \mathrm{GL}_n(k).$$

Let

$$V(x) := \begin{pmatrix} Q_1 x & \cdots & Q_c x \end{pmatrix} \in M_{n\times c}(P_1)$$

and define the $c \times c$ Gram matrix

$$\boxed{G_H(x) := V(x)^T H V(x).} \tag{NC.10.15}$$

Definition NC.10.12 (Gram-cofactor stress)**.** Define

$$\boxed{\mathcal{P}_H := \det(G_H)\, H - HV\ \mathrm{adj}(G_H)\, V^T H.} \tag{NC.10.16}$$

Lemma NC.10.13 (Universal exact tangency)**.** *The matrix $\mathcal{P}_H$ is symmetric and satisfies*

$$\boxed{\mathcal{P}_H V = 0.} \tag{NC.10.17}$$

Equivalently,

$$\mathcal{P}_H Q_\alpha x = 0 \quad (\alpha = 1, \dots, c).$$

Proof. Symmetry follows from symmetry of H, G_H, and $\mathrm{adj}(G_H)$. Moreover,

$$\begin{aligned} \mathcal{P}_H V &= \det(G_H) HV - HV\, \mathrm{adj}(G_H) V^T HV \\ &= HV\, (\det(G_H) I_c - \mathrm{adj}(G_H) G_H) = 0 \end{aligned}$$

by the adjugate identity. □

Lemma NC.10.14 (Degree of the universal Gram stress)**.** *Every entry of G_H has degree* 2*. Hence every entry of $\mathcal{P}_H$ has degree*

$$\boxed{2c.} \tag{NC.10.18}$$

The divergence-quantized operator

$$\mathscr{D}_{\mathcal{P}_H}$$

therefore has internal degree

$$\boxed{2c - 2 = 2m.} \tag{NC.10.19}$$

Proof. The determinant of the $c \times c$ quadratic matrix G_H has degree $2c$. Each entry of

$$V\, \mathrm{adj}(G_H) V^T$$

has degree

$$1 + 2(c-1) + 1 = 2c.$$

Subtracting differential order two gives internal degree $2c - 2$. Since $m = c - 1$, this equals $2m$. □

Theorem NC.10.15 (Universal Gram benchmark)**.** *For every quadratic complete-intersection presentation* (NC.10.1)*, the Gram-cofactor matrix $\mathcal{P}_H$ is an exact stress and therefore admits the canonical divergence quantization of Theorem NC.10.5. However its internal degree is*

$$2m,$$

so the NC.4 Critical Window does not force its escape from Δ_A.

Proof. Exact tangency is Lemma NC.10.13; quantization is Theorem NC.10.5; the degree statement is Lemma NC.10.14. NC.4 applies only for

$$r < 2m,$$

whereas the Gram operator has

$$r = 2m.$$

□

Remark NC.10.16 (The one-degree-lower barrier)**.** Theorem NC.10.15 gives a canonical explanation of the earlier Gram-projector failure. The universal exact tangent tensor exists, but it has coefficient degree

$$2c$$

rather than the critical

$$2c - 1.$$

Thus the unresolved general quadratic problem is not the absence of a tangent tensor. It is the absence, in general, of a *one-degree-lower* projectively visible quantizable tensor.

NC.10.7 Two-lane architecture for arbitrary quadrics

Definition NC.10.17 (Exact lane and modulo-ideal lane)**.** For quadratic complete intersections define two logically distinct search lanes.

(i) The *exact lane* searches for

$$B \in \mathcal{E}_2^{\mathrm{crit}}(A)$$

with nonzero projective symbol. Quantization is then automatic by divergence.

(ii) The *modulo-ideal lane* searches for

$$B \in \mathcal{S}_{2,\,2c-3}(A)$$

that need not satisfy exact ambient tangency, but does satisfy

$$\omega_{2c-3}(B) = 0$$

and has nonzero projective symbol. This is the full NC.7–NC.8 route.

Theorem NC.10.18 (Quadratic two-lane escape criterion)**.** *Assume the NC.4 hypotheses for a smooth quadratic ICIS. Strong escape follows if either:*

$$\boxed{\exists B \in \mathcal{E}_2^{\mathrm{crit}}(A) : \sigma_{2,0}^{X}(\mathscr{D}_B) \neq 0,} \tag{NC.10.20}$$

or

$$\boxed{\exists B \in \mathcal{S}_{2,\,2c-3}(A) : \omega_{2c-3}(B) = 0,\ \Psi_{2,\,2c-3}(B) \neq 0.} \tag{NC.10.21}$$

Proof. The first implication is Theorem NC.10.10. The second is the NC.8 Master Bridge. □

NC.10.8 What is already solved inside the quadratic class

Proposition NC.10.19 (Codimension-two quadratic class is already closed)**.** *For $c = 2$, the smooth quadratic complete-intersection class is already covered by the NC.6 type-$(2, d)$ theorem with $d = 2$. Thus NC.10 introduces no new open branch in codimension two.*

Proof. A codimension-two quadratic complete intersection has type $(2, 2)$. Theorem NC.6.11 applies with $d = 2$. □

Proposition NC.10.20 (Higher codimension is genuinely different)**.** *For $c \geq 3$, simultaneous diagonalization is not a formal consequence of smoothness or of the complete-intersection condition. Therefore NC.9 cannot be promoted to all smooth quadratic ICIS merely by a coordinate change.*

Proof. After one nondegenerate quadratic form is normalized, simultaneous diagonalization of the remaining forms requires simultaneous diagonalizability of the associated self-adjoint endomorphisms, in particular a common eigenbasis; a generic tuple of self-adjoint endomorphisms need not commute and hence need not admit such a basis. Smoothness imposes a Jacobian-rank condition on the common zero locus, not pairwise commutativity of these endomorphisms. Thus no implication from smoothness to simultaneous diagonalization is available. □

NC.10.9 The exact remaining obstruction for all quadratic ICIS

Definition NC.10.21 (Critical quadratic quantization kernel)**.** Define

$$\boxed{\mathcal{K}^{\mathrm{quad}}_{\mathrm{crit}}(A) := \ker\left(\omega_{2c-3} : \mathcal{S}_{2,\,2c-3}(A) \to (T^1_{A/k})_{2c-3}\right).} \tag{NC.10.22}$$

Theorem NC.10.22 (All-quadratic reduction)**.** *Let A be a smooth-projective homogeneous quadratic ICIS satisfying the NC.3 derivation-generation hypotheses. Then the strong Nakai conclusion*

$$D_{A/k} \neq \Delta_A$$

follows as soon as

$$\boxed{\mathcal{K}^{\mathrm{quad}}_{\mathrm{crit}}(A) \not\subseteq \mathcal{R}^{\mathrm{amb}}_{2,\,2c-3}(A).} \tag{NC.10.23}$$

Equivalently,

$$\boxed{\mathfrak{Q}^{\mathrm{crit}}_2(A) \neq 0.} \tag{NC.10.24}$$

Proof. This is Theorem NC.8.16 specialized to quadratic codimension c, where

$$m = c - 1.$$

□

Remark NC.10.23 (What remains genuinely open)**.** Theorem NC.10.22 is a reduction theorem, not a nonvanishing theorem. For $c \geq 3$, NC.10 does not yet prove that every smooth quadratic ICIS has

$$\mathfrak{Q}^{\mathrm{crit}}_2(A) \neq 0.$$

That nonvanishing is the precise unresolved step required to promote the NC.9 diagonal family to the full quadratic class.

NC.10.10 Research consequences of the Gram benchmark

The Gram benchmark provides two useful search diagnostics.

Proposition NC.10.24 (Critical factor-extraction criterion)**.** *Suppose a nonzero homogeneous polynomial ℓ divides every entry of $\mathcal{P}_H$:*

$$\mathcal{P}_H = \ell B,$$

with

$$\deg \ell = 1.$$

Then

$$B \in \mathcal{E}^{\mathrm{crit}}_2(A).$$

If its projective symbol is nonzero, strong escape follows.

Proof. Because P is a domain on the polynomial level,

$$0 = \mathcal{P}_H Q_\alpha x = \ell\, B Q_\alpha x$$

implies

$$B Q_\alpha x = 0.$$

The coefficient degree drops from $2c$ to $2c - 1$, hence the operator internal degree is $2c - 3$. Apply Theorem NC.10.10. □

Remark NC.10.25. NC.10 does not claim that such a common linear factor exists generically. The proposition identifies one concrete mechanism by which the universal post-threshold stress can fall into the critical window. The diagonal NC.9 construction should be viewed as a more structured one-degree-lower phenomenon, not as evidence that generic Gram cofactors possess such a divisor.

NC.10.11 NC.10 closure

Theorem NC.10.26 (NC.10 closure theorem)**.** *For arbitrary homogeneous quadratic complete intersections:*

(i) every exact symmetric polynomial stress admits canonical divergence quantization;

(ii) the Hessian obstruction vanishes automatically on the exact-stress lane;

(iii) a universal Gram-cofactor exact stress always exists at operator degree

$$2m;$$

(iv) NC.4 strong escape requires a projectively visible quantized stress at degree

$$2m - 1;$$

(v) therefore the general quadratic problem is reduced to a precise one-degree-lower nonvanishing problem, expressible equivalently by

$$\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0.$$

Proof. Items (i)–(ii) are Theorem NC.10.5 and Corollary NC.10.6. Item (iii) is Theorem NC.10.15. Item (iv) is NC.4. Item (v) is Theorem NC.10.22. □

Declaration NC.10.27 (NC.10 status and successor obligation)**.** NC.10 does *not* claim the all-quadratic theorem for $c \geq 3$. Instead it closes the algebraic architecture of the obstruction and identifies the unique remaining proof target:

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0 \quad \text{for every smooth quadratic ICIS.}}$$

The next stage should attack this nonvanishing by one or more of:

(a) critical exact-stress syzygies of degree $2c - 1$,
(b) modulo-ideal stresses with nontrivial Hessian correction,
(c) Buchsbaum–Rim/Eagon–Northcott or Jacobian–Koszul duality,
(d) local duality or socle control of $\ker \omega_{2c-3}$,
(e) exact computation on genuinely non-simultaneously-diagonal controls.

The universal Gram stress is a boundary object at degree $2m$, not a solution of the critical nonvanishing problem.

NC.11 Cofactor Syzygies, Buchsbaum–Rim Symmetrization, and the Radial Trap

NC.11.1 Purpose

NC.10 reduces the arbitrary quadratic problem to the construction of a projectively visible critical stress of coefficient degree

$$2c - 1.$$

NC.11 analyzes the first syzygy layer of the linear matrix

$$V(x) = \begin{pmatrix} Q_1 x & \cdots & Q_c x \end{pmatrix}$$

and explains both why the universal cofactor machinery is relevant and why its most immediate one-degree-lower construction still does not solve the Nakai problem.

The key distinction is:

$$\boxed{\text{critical stress existence} \neq \text{critical visible stress existence.}}$$

The first Buchsbaum–Rim/cofactor syzygies produce natural critical tensors, but their cheapest one-degree-lower symmetrization carries an Euler factor and hence lies in the generated/projectively invisible sector. The genuine next target is a *nonradial symmetrization* of the cofactor syzygy module.

NC.11.2 The linear-gradient matrix

Retain the NC.10 quadratic setup

$$f_\alpha = x^T Q_\alpha x, \qquad Q_\alpha = Q_\alpha^T,$$

and put

$$\boxed{V = \begin{pmatrix} v_1 & \cdots & v_c \end{pmatrix}, \qquad v_\alpha := Q_\alpha x.} \tag{NC.11.1}$$

Thus

$$\nabla f_\alpha = 2 v_\alpha.$$

Consider the graded map

$$\boxed{\varphi : P(-1)^n \longrightarrow P^c, \qquad \varphi = V^T.} \tag{NC.11.2}$$

Its kernel is the exact linear-gradient syzygy module

$$\boxed{\mathcal{K}_V := \ker(V^T).} \tag{NC.11.3}$$

NC.11.3 Cofactor syzygies

Let

$$J = \{j_0, \dots, j_c\} \subseteq \{1, \dots, n\}$$

be an ordered subset of cardinality $c + 1$. For $0 \le t \le c$, write

$$V^T_{J \setminus \{j_t\}}$$

for the $c \times c$ submatrix of V^T with the listed columns.

Definition NC.11.1 (Cofactor syzygy)**.** Define

$$z_J \in P_c^n$$

by

$$\boxed{(z_J)_{j_t} = (-1)^t \det\left(V^T_{J \setminus \{j_t\}}\right), \qquad (z_J)_i = 0 \text{ for } i \notin J.} \tag{NC.11.4}$$

Lemma NC.11.2 (Cofactor identity)**.** *For every J,*

$$\boxed{V^T z_J = 0} \tag{NC.11.5}$$

as an exact polynomial identity.

Proof. Fix a row α of V^T. The α-component of $V^T z_J$ is the alternating expansion obtained by adjoining the α-th row of V_J^T as a repeated row to the $c \times (c+1)$ matrix V_J^T. Equivalently, it is the Laplace expansion of a determinant with two equal rows. Hence it vanishes. □

Corollary NC.11.3 (Cofactor derivations)**.** *Define*

$$\boxed{\delta_J := \sum_{i=1}^{n} (z_J)_i \partial_i.} \tag{NC.11.6}$$

Then

$$\delta_J(f_\alpha) = 0 \qquad (\alpha = 1, \dots, c),$$

so δ_J descends to a derivation of A. Its internal degree is

$$\boxed{c - 1 = m.} \tag{NC.11.7}$$

Proof. Since

$$\nabla f_\alpha = 2v_\alpha,$$

Lemma NC.11.2 gives

$$\delta_J(f_\alpha) = 2z_J^T v_\alpha = 0.$$

The coefficients of z_J have degree c, so the derivation has internal degree $c-1$, which equals the quadratic ICIS threshold m. □

Remark NC.11.4. The cofactor derivations are the quadratic specialization of the familiar "trivial derivation" construction by Jacobian minors. NC.11 proves only the displayed exact identities at this stage; completeness of this family as generators of $\ker(V^T)$ requires the Buchsbaum–Rim exactness hypothesis introduced below.

NC.11.4 Raw dimension count cannot force a critical exact stress

For $d \geq 0$, let

$$N_d := \dim_k P_d = \binom{n+d-1}{d}.$$

Consider the exact-stress map

$$\boxed{\Xi_d : \mathrm{Sym}_n(P_d) \longrightarrow M_{n\times c}(P_{d+1}), \qquad B \longmapsto BV.} \tag{NC.11.8}$$

Theorem NC.11.5 (Critical raw-dimension no-go)**.** *At the critical coefficient degree*

$$d = 2c - 1$$

with $c \geq 2$,

$$\boxed{\dim_k M_{n\times c}(P_{2c}) > \dim_k \mathrm{Sym}_n(P_{2c-1}).} \tag{NC.11.9}$$

More precisely,

$$\boxed{\frac{nc\, N_{2c}}{\binom{n+1}{2} N_{2c-1}} = \frac{n+2c-1}{n+1} > 1.} \tag{NC.11.10}$$

Therefore rank–nullity using only source and target dimensions can never force

$$\ker \Xi_{2c-1} \neq 0.$$

Proof. The standard monomial-count ratio is

$$\frac{N_{2c}}{N_{2c-1}} = \frac{n+2c-1}{2c}.$$

Hence

$$\frac{nc\, N_{2c}}{\binom{n+1}{2} N_{2c-1}} = \frac{2c}{n+1}\,\frac{n+2c-1}{2c} = \frac{n+2c-1}{n+1}.$$

For $c \geq 2$, the numerator exceeds the denominator. □

Remark NC.11.6. The theorem does not say that the critical exact-stress kernel is zero. It says something more methodological: its nonvanishing, when it occurs, must come from structural syzygies of V, not from a naive excess-of-unknowns dimension count.

NC.11.5 The radial cofactor trap

The relation

$$x^T V = (f_1, \dots, f_c) \tag{NC.11.11}$$

holds identically. Thus x is a syzygy of V^T only after reduction modulo the defining ideal.

Fix a cofactor syzygy z_J, and let

$$h \in P_{c-2}$$

be homogeneous.

Definition NC.11.7 (Radial cofactor stress)**.** Define

$$\boxed{B^{\mathrm{rad}}_{J,h} := h\left(x z_J^T + z_J x^T\right).} \tag{NC.11.12}$$

Lemma NC.11.8 (Critical degree and quotient tangency)**.** *The matrix $B^{\mathrm{rad}}_{J,h}$ is symmetric of coefficient degree*

$$2c - 1,$$

and

$$\boxed{B^{\mathrm{rad}}_{J,h} V = h\, z_J(f_1, \dots, f_c).} \tag{NC.11.13}$$

Hence it defines a critical Jacobian stress on A:

$$B^{\mathrm{rad}}_{J,h} \in \mathcal{S}_{2,\,2c-3}(A).$$

Proof. The degree is

$$(c-2) + 1 + c = 2c - 1.$$

Using

$$z_J^T V = 0$$

and (NC.11.11),

$$\begin{aligned} B^{\mathrm{rad}}_{J,h} V &= h\left(x(z_J^T V) + z_J(x^T V)\right) \\ &= h\, z_J(f_1, \dots, f_c). \end{aligned}$$

Thus the product vanishes modulo I. □

Theorem NC.11.9 (Radial cofactor stresses are generated)**.** *Let*

$$E = \sum_i x_i \partial_i$$

be the Euler derivation and let δ_J be the cofactor derivation. Define

$$\boxed{\mathscr{P}_{J,h} := \frac{h}{2}\left(E\delta_J + \delta_J E\right).} \tag{NC.11.14}$$

Then

$$\mathscr{P}_{J,h} \in \Delta_A, \qquad \deg \mathscr{P}_{J,h} = 2c - 3,$$

and its second-order coefficient matrix in the NC.7 convention is exactly

$$\boxed{B^{\mathrm{rad}}_{J,h}.} \tag{NC.11.15}$$

Consequently

$$\boxed{\omega_{2c-3}\left(B^{\mathrm{rad}}_{J,h}\right) = 0}$$

but

$$\boxed{\Psi_{2,\,2c-3}\left(B^{\mathrm{rad}}_{J,h}\right) = 0.} \tag{NC.11.16}$$

Proof. Both E and δ_J are derivations of A, so $\mathscr{P}_{J,h} \in \Delta_A$. Its internal degree is

$$(c-2) + 0 + (c-1) = 2c - 3.$$

The second-order part of

$$\frac{1}{2}(E\delta_J + \delta_J E)$$

is

$$\frac{1}{2}\sum_{i,j}\left(x_i(z_J)_j + (z_J)_i x_j\right)\partial_i\partial_j.$$

Multiplication by h gives precisely the NC.7 coefficient matrix (NC.11.12). Because an actual descended operator with this principal matrix exists, Theorem NC.7.15 gives

$$\omega_{2c-3}(B^{\mathrm{rad}}_{J,h}) = 0.$$

Finally the matrix has an Euler factor. Equivalently, for every covector λ satisfying

$$\lambda(x) = 0,$$

one has

$$\lambda^T B^{\mathrm{rad}}_{J,h}\lambda = 2h\,\lambda(x)\lambda(z_J) = 0.$$

Thus its projective second symbol vanishes. The same conclusion follows directly from NC.4 because $\mathscr{P}_{J,h} \in \Delta_A$ and $2c - 3 < 2m$. □

Corollary NC.11.10 (First one-degree-lower cofactor construction cannot escape)**.** *The universal construction*

$$z_J \longmapsto h(x \odot z_J)$$

does reach the critical coefficient degree $2c-1$*, but every such stress lies in the generated/projectively invisible sector. Therefore first cofactor syzygies plus the universal quotient syzygy* x *do* not *prove the Nakai escape.*

Remark NC.11.11 (Structural interpretation)**.** This is the *radial trap*. The one-degree-lower barrier can be crossed syntactically by pairing a degree-c exact cofactor syzygy with the degree-one Euler vector x, but that same Euler factor is exactly what NC.4 kills after projectivization. A successful general construction must therefore lower the degree without using the radial vector as an unavoidable tensor factor.

NC.11.6 Cofactor matrix and nonradial symmetrization

Let

$$M := \binom{n}{c+1},$$

and form the $n \times M$ matrix

$$\boxed{Z := \left(z_J\right)_{|J|=c+1}.} \tag{NC.11.17}$$

Every entry of Z has degree c, and

$$\boxed{V^T Z = 0.} \tag{NC.11.18}$$

For

$$A(x) \in M_{M\times n}(P_{c-1}),$$

the product

$$ZA$$

has coefficient degree $2c-1$.

Definition NC.11.12 (Cofactor symmetrization map)**.** Define

$$\boxed{\Sigma_{\mathrm{cof}} : M_{M\times n}(P_{c-1}) \longrightarrow \mathrm{Alt}_n(P_{2c-1})} \tag{NC.11.19}$$

by

$$\boxed{\Sigma_{\mathrm{cof}}(A) := ZA - (ZA)^T.} \tag{NC.11.20}$$

Theorem NC.11.13 (Cofactor symmetrization produces critical exact stresses)**.** *If*

$$A \in \ker \Sigma_{\mathrm{cof}},$$

then

$$\boxed{B_A := ZA} \tag{NC.11.21}$$

is symmetric, has coefficient degree $2c-1$*, and satisfies*

$$\boxed{B_A V = 0} \tag{NC.11.22}$$

as an exact polynomial identity. Thus

$$B_A \in \mathcal{E}_2^{\mathrm{crit}}(A).$$

Proof. The kernel condition says exactly that ZA is symmetric. Equation (NC.11.18) gives

$$V^T B_A = V^T ZA = 0.$$

Transposing and using symmetry of B_A,

$$B_A V = 0.$$

The coefficient degree is

$$c + (c-1) = 2c - 1.$$

□

Definition NC.11.14 (Cofactor critical-stress quotient)**.** Define

$$\boxed{\mathfrak{S}^{\mathrm{cof}}_{\mathrm{crit}}(V) := \frac{\ker \Sigma_{\mathrm{cof}}}{\{A : ZA = 0\}}.} \tag{NC.11.23}$$

By Theorem NC.11.13, this quotient identifies with the space of critical exact stresses obtained from cofactor symmetrization.

Corollary NC.11.15 (Visible cofactor-symmetrization escape)**.** *If there exists*

$$A \in \ker \Sigma_{\mathrm{cof}}$$

such that

$$ZA \neq 0$$

and

$$\sigma^X_{2,0}(\mathscr{D}_{ZA}) \neq 0,$$

then

$$\boxed{D_{A/k} \neq \Delta_A.}$$

Proof. Theorem NC.11.13 places ZA in the critical exact-stress sector, and Theorem NC.10.10 applies. □

NC.11.7 Buchsbaum–Rim completeness gate

The cofactor columns always give syzygies. Whether they generate the *entire* exact syzygy module is a separate determinantal-grade statement.

Assumption NC.11.16 (Expected maximal-minor grade)**.** *Assume the ideal of $c \times c$ minors of V^T contains a regular sequence of length*

$$\boxed{n - c + 1.} \tag{NC.11.24}$$

Theorem NC.11.17 (Imported Buchsbaum–Rim first-syzygy theorem)**.** *Under Assumption NC.11.16, the Buchsbaum–Rim complex of*

$$\varphi : P(-1)^n \to P^c$$

is exact at $P(-1)^n$. Consequently the cofactor syzygies z_J generate

$$\ker(V^T)$$

as a graded P-module.

Remark NC.11.18 (Source status)**.** Theorem NC.11.17 is an imported theorem slot. The exactness mechanism is the classical Buchsbaum–Rim complex under the expected maximal-minor grade hypothesis; see [4], with modern commutative-algebra treatments such as [6]. The precise grading convention used here is still audited internally rather than copied from notation in either source. No smoothness hypothesis in NC is silently identified with Assumption NC.11.16.

Theorem NC.11.19 (Conditional completeness of cofactor symmetrization)**.** *Assume Theorem NC.11.17. Then every exact critical stress*

$$B \in \mathcal{E}_2^{\mathrm{crit}}(A)$$

can be written

$$\boxed{B = ZA} \tag{NC.11.25}$$

for some

$$A \in M_{M\times n}(P_{c-1})$$

satisfying

$$\Sigma_{\mathrm{cof}}(A) = 0.$$

Hence

$$\boxed{\mathcal{E}_2^{\mathrm{crit}}(A) = \mathrm{im}\left(\ker \Sigma_{\mathrm{cof}} \xrightarrow{A\mapsto ZA} \mathrm{Sym}_n(P_{2c-1})\right).} \tag{NC.11.26}$$

Proof. Because B is symmetric and

$$BV = 0,$$

one also has

$$V^T B = 0.$$

Thus every column of B lies in $\ker(V^T)$. By Theorem NC.11.17, each column is a P-linear combination of the degree-c cofactor columns z_J. Since every column of B has degree $2c - 1$, the coefficients may be chosen homogeneous of degree $c - 1$. Collecting them gives a matrix

$$A \in M_{M\times n}(P_{c-1})$$

with

$$B = ZA.$$

Symmetry of B is exactly

$$\Sigma_{\mathrm{cof}}(A) = 0.$$

□

Corollary NC.11.20 (Exact-lane all-quadratic reduction under expected grade)**.** *Under Assumption NC.11.16, the exact critical lane for the all-quadratic problem reduces to the finite-degree symmetrization system*

$$\boxed{ZA = (ZA)^T, \qquad A \in M_{M\times n}(P_{c-1}),} \tag{NC.11.27}$$

together with projective visibility of the resulting ZA.

NC.11.8 Why Buchsbaum–Rim alone is not yet the all-quadratic proof

Policy 12 (No syzygy-to-escape promotion)**.** *The facts*

$$V^T z_J = 0$$

and, under expected grade,

$$\ker(V^T) = \langle z_J \rangle_P$$

do not imply

$$\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0.$$

A proof must still produce a symmetric degree-$(2c-1)$ *combination whose projective symbol is nonzero. The radial family of Theorem NC.11.9 is an explicit counterexample to any reasoning that treats "critical syzygy" as synonymous with "escape symbol".*

Definition NC.11.21 (Nonradial cofactor defect)**.** Define the projectively visible cofactor defect

$$\boxed{\mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof,vis}}(V) := \mathrm{im}\left[\mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof}}(V) \longrightarrow H^0\Big(X, \mathrm{Sym}^2 T_X(2c-3)\Big)\right],} \tag{NC.11.28}$$

where the map sends a class represented by A to the projective principal symbol of the divergence-quantized exact stress ZA.

Theorem NC.11.22 (Cofactor-defect escape criterion)**.** *If*

$$\boxed{\mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof,vis}}(V) \neq 0,} \tag{NC.11.29}$$

then

$$\boxed{D_{A/k} \neq \Delta_A.}$$

Under Assumption NC.11.16, this criterion exhausts the exact-stress *lane at critical degree, but not the larger modulo-ideal lane of NC.7–NC.8.*

Proof. A nonzero visible class has a representative

$$B = ZA \in \mathcal{E}_2^{\mathrm{crit}}(A)$$

with nonzero projective second symbol. Theorem NC.10.10 gives strong escape.

Under the Buchsbaum–Rim grade hypothesis, Theorem NC.11.19 shows that every exact critical stress occurs in this form. NC.7 stresses tangent only modulo I need not admit an exact representative, so they remain outside this completeness claim. □

NC.11.9 Finite linear-algebra realization

Proposition NC.11.23 (Finite cofactor-symmetrization matrix)**.** *Fix monomial bases of*

$$P_{c-1} \quad \textit{and} \quad P_{2c-1}.$$

Then

$$\Sigma_{\mathrm{cof}}$$

is represented by a finite matrix over k. Hence

$$\ker \Sigma_{\mathrm{cof}}, \qquad \mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof}}(V),$$

and, after projective-symbol reduction,

$$\mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof,vis}}(V)$$

are accessible by exact finite linear algebra for a fixed quadratic tuple $(Q_1, \ldots, Q_c)$.

Proof. Every entry of Z is a degree-c polynomial and every entry of A has degree $c-1$. The coefficients of

$$ZA - (ZA)^T$$

in the monomial basis of P_{2c-1} depend k-linearly on the coefficients of A. Thus Σ_{cof} is a finite linear map between finite-dimensional graded pieces. □

Policy 13 (Generic-rank evidence ceiling)**.** *A modular or rational rank computation of Σ_{cof} for one quadratic tuple proves only the corresponding finite statement for that tuple unless a separate specialization or semicontinuity argument is stated and audited. Failure to find a nonradial kernel vector in a finite sample does not prove that the all-quadratic theorem is false.*

NC.11.10 NC.11 closure

Theorem NC.11.24 (NC.11 closure theorem)**.** *For arbitrary quadratic complete intersections, the first syzygy layer yields the following exact architecture:*

$$\boxed{\begin{array}{l} \textit{cofactor minors } z_J \Longrightarrow V^T z_J = 0, \\ z_J \odot x \Longrightarrow \textit{critical stress but radial/generated}, \\ ZA = (ZA)^T, \quad \deg A = c-1 \Longrightarrow \textit{critical exact stress}, \\ \textit{visible } ZA \Longrightarrow D_{A/k} \neq \Delta_A. \end{array}} \tag{NC.11.30}$$

Under the Buchsbaum–Rim expected-grade hypothesis, the cofactor-symmetrization system

$$ZA = (ZA)^T$$

exhausts all critical exact stresses. Thus the exact-lane all-quadratic problem becomes a nonradial symmetrization problem*, while the full NC problem still allows the larger modulo-ideal Hessian-obstruction lane.*

Proof. The cofactor identity is Lemma NC.11.2. The radial statement is Theorem NC.11.9. The exact-stress construction is Theorem NC.11.13. Visible exact stresses escape by Theorem NC.10.10. Conditional completeness is Theorem NC.11.19. □

Declaration NC.11.25 (NC.11 status and successor obligation)**.** NC.11 does not prove

$$\mathfrak{S}_{\mathrm{crit}}^{\mathrm{cof,vis}}(V) \neq 0$$

for every smooth quadratic ICIS. Instead it identifies a new negative control and a sharper positive target.

The negative control is the

$$\boxed{\text{Radial Cofactor Trap}}$$

showing that the easiest degree-lowering use of first cofactor syzygies is already generated by Euler and a trivial derivation.

The positive target is

$$\boxed{\ker \Sigma_{\mathrm{cof}} \text{ contains a nonzero projectively visible class.}}$$

If that target fails generically, the remaining all-quadratic route must use stresses tangent only modulo I, where the NC.7 Hessian obstruction and its local-duality structure become essential.

Accordingly, NC.12 should combine:

$$\text{exact rank diagnostics on genuinely non-diagonal controls}$$

with

$$\text{the modulo-ideal obstruction complex}$$

before any attempt to claim the full quadratic theorem.

NC.12 Exact Non-Diagonal Quadratic Controls and Branch Separation

NC.12.1 Purpose

NC.12 tests the NC.7–NC.11 architecture on explicit quadratic complete intersections that are smooth but are not simultaneously diagonalizable. The objective is not merely to find another positive example. The objective is to determine, by exact rational linear algebra, whether the critical defect is already explained by the exact-stress/cofactor lane or whether the larger modulo-ideal Hessian-obstruction lane contributes genuinely new classes.

Two controls are used: a projective curve control with $n = 5, c = 3$, and a projective surface control with $n = 6, c = 3$. In both cases

$$m = c - 1 = 2, \qquad r_{\mathrm{crit}} = 2m - 1 = 3.$$

NC.12.2 Control C_5: a smooth non-simultaneously-diagonal curve

Let

$$A_5 = \mathbb{Q}[x_1, \dots, x_5]/(f_1, f_2, f_3),$$

with

$$\boxed{\begin{aligned} f_1 &= x_1^2 + x_4^2 + x_4x_5, \\ f_2 &= x_2^2 + x_5^2 + 2x_4x_5, \\ f_3 &= x_3^2 + x_4^2 + x_4x_5 + 3x_5^2. \end{aligned}} \tag{NC.12.1}$$

The three leading monomials

$$x_1^2, \quad x_2^2, \quad x_3^2$$

are pairwise coprime for the declared elimination order, so the three displayed quadrics form a homogeneous complete intersection. Thus

$$X_5 := \operatorname{Proj} A_5$$

has expected dimension

$$p = 1.$$

Proposition NC.12.1 (Exact smoothness certificate for C_5)**.** *Let $\mathfrak{s}_5$ be the homogeneous ideal generated by f_1, f_2, f_3 and the 3×3 minors of their Jacobian matrix. An exact rational Gröbner basis of $\mathfrak{s}_5$ has leading monomials containing pure powers of every variable:*

$$x_1^2, \quad x_2^2, \quad x_3^2, \quad x_4^4, \quad x_5^5.$$

Hence $P/\mathfrak{s}_5$ is zero-dimensional. Because $\mathfrak{s}_5$ is homogeneous, its affine zero set contains no nonzero point. Therefore the affine cone is smooth away from the origin and

$$\boxed{X_5 \text{ is smooth.}}$$

Proposition NC.12.2 (Non-simultaneous diagonalization of C_5)**.** *Write*

$$f_\alpha = x^T Q_\alpha x$$

and set

$$Q_\Sigma := Q_1 + Q_2 + Q_3.$$

Then

$$\det Q_\Sigma = 4 \neq 0,$$

but

$$\boxed{[Q_\Sigma^{-1}Q_1,\, Q_\Sigma^{-1}Q_2] \neq 0.} \tag{NC.12.2}$$

Hence the triple (Q_1, Q_2, Q_3) is not simultaneously diagonalizable by congruence over $\overline{\mathbb{Q}}$.

Proof. If all three forms were simultaneously diagonalizable by congruence, then after the same change of basis the nonsingular form Q_Σ would also be diagonal and

$$Q_\Sigma^{-1} Q_\alpha$$

would become diagonal by similarity. These operators would therefore commute. The exact commutator in (NC.12.2) has a nonzero entry $3/8$, giving a contradiction. □

NC.12.3 Control C_6: a smooth non-simultaneously-diagonal surface

Let

$$A_6 = \mathbb{Q}[x_1, \dots, x_6]/(g_1, g_2, g_3),$$

with

$$\boxed{\begin{aligned} g_1 &= x_1^2 + x_4^2 + x_5 x_6, \\ g_2 &= x_2^2 + x_5^2 + x_4 x_6, \\ g_3 &= x_3^2 + x_6^2 + x_4 x_5. \end{aligned}} \tag{NC.12.3}$$

Again the leading monomials x_1^2, x_2^2, x_3^2 are pairwise coprime. Thus

$$X_6 := \operatorname{Proj} A_6$$

is a codimension-three quadratic complete intersection of expected dimension

$$p = 2.$$

Proposition NC.12.3 (Exact smoothness certificate for C_6)**.** *Let $\mathfrak{s}_6$ be generated by g_1, g_2, g_3 and all 3×3 Jacobian minors. An exact rational Gröbner basis of $\mathfrak{s}_6$ has leading monomials containing*

$$x_1^2, \quad x_2^2, \quad x_3^2, \quad x_4^3, \quad x_5^4, \quad x_6^5.$$

Hence the homogeneous singular ideal is zero-dimensional and has only the origin as an affine zero. Therefore

$$\boxed{X_6 \text{ is smooth.}}$$

Proposition NC.12.4 (Non-simultaneous diagonalization of C_6)**.** *For the symmetric matrices of* (NC.12.3)*, let*

$$Q_\Sigma = Q_1 + Q_2 + Q_3.$$

Then

$$\det Q_\Sigma = \frac{1}{2} \neq 0$$

and

$$\boxed{[Q_\Sigma^{-1} Q_1,\ Q_\Sigma^{-1} Q_2] \neq 0.} \tag{NC.12.4}$$

Hence C_6 is genuinely outside the simultaneously diagonal source class of NC.9.

NC.12.4 Exact graded quotient bases

For both controls, the displayed defining equations form a Gröbner basis with leading monomials

$$x_1^2, \quad x_2^2, \quad x_3^2.$$

Therefore an exact monomial basis of each graded piece is given by monomials in which the exponents of x_1, x_2, x_3 are at most one.

For C_5,

$$\boxed{\dim(A_5)_4 = 28, \quad \dim(A_5)_5 = 36, \quad \dim(A_5)_6 = 44, \quad \dim(A_5)_7 = 52.} \tag{NC.12.5}$$

For C_6,

$$\boxed{\dim(A_6)_4 = 66, \quad \dim(A_6)_5 = 102, \quad \dim(A_6)_6 = 146, \quad \dim(A_6)_7 = 198.} \tag{NC.12.6}$$

These bases are used in every rank statement below. No floating-point arithmetic or modular lifting is used in the final ranks.

NC.12.5 The critical quantization matrix

For either control, define

$$\boxed{\mathcal{F}_3(B,\gamma) := (JB,\ J\gamma + \mathsf{h}_3(B))\,.} \tag{NC.12.7}$$

Then

$$\ker \mathcal{F}_3 = Q_{2,3}^{\mathrm{amb}}(A)$$

by NC.7. The exact sequence of NC.7 gives

$$\dim\ker \omega_3 = \dim\ker \mathcal{F}_3 - \dim\ker \mathsf{j}_3. \tag{NC.12.8}$$

Theorem NC.12.5 (Exact critical quantization ranks)**.** *Exact rational row reduction gives:*

	C_5	C_6
$\dim \mathrm{Dom}(\mathcal{F}_3)$	680	2538
$\mathrm{rank}_{\mathbb{Q}}\,\mathcal{F}_3$	611	2336
$\dim\ker \mathcal{F}_3$	69	202
$\mathrm{rank}_{\mathbb{Q}}\,\mathsf{j}_3$	108	306
$\dim\ker \mathsf{j}_3$	32	90
$\boxed{\dim\ker \omega_3}$	$\boxed{37}$	$\boxed{112}$

(NC.12.9)

NC.12.6 Radial sector and a horizontal projective detector

Define

$$\boxed{\rho_3(u) := xu^T + ux^T, \qquad u \in \mathrm{Der}_{\mathbb{Q}}(A)_3.} \tag{NC.12.10}$$

Let

$$\alpha_{12} := x_2\,dx_1 - x_1\,dx_2, \qquad \iota_E\alpha_{12} = 0,$$

and define

$$\boxed{H_{12}(B) := B(\alpha_{12},\alpha_{12}) \in A_7.} \tag{NC.12.11}$$

Lemma NC.12.6 (Radial image is horizontal-invisible)**.** *For every derivation u,*

$$H_{12}(\rho_3(u)) = 0.$$

Hence

$$\mathrm{im}\,\rho_3 \subseteq \mathcal{R}_{2,3}^{\mathrm{amb}}(A) \subseteq \ker\left(H_{12}|_{\ker\omega_3}\right).$$

Proof. For a horizontal covector α with $\alpha(x) = 0$,

$$(xu^T + ux^T)(\alpha,\alpha) = 2\alpha(x)\alpha(u) = 0.$$

Projectively invisible symbols vanish under every horizontal test. □

Theorem NC.12.7 (Exact radial and horizontal ranks)**.** *For both controls the radial map is injective on the critical derivation space. Moreover:*

	C_5	C_6
$\dim \mathrm{im}\,\rho_3$	32	90
$\mathrm{rank}[\mathcal{F}_3; H_{12}]$	616	2358
$\dim\ker(H_{12}|_{\ker\omega_3})$	32	90

(NC.12.12)

Consequently,

$$\boxed{\mathrm{im}\,\rho_3 = \mathcal{R}_{2,3}^{\mathrm{amb}}(A) = \ker(H_{12}|_{\ker\omega_3})} \tag{NC.12.13}$$

for both controls.

Proof. Exact rank computation of

$$[\mathsf{j}_3;\rho_3]$$

gives full column rank 140 for C_5 and 396 for C_6, so ρ_3 is injective on $\ker \mathsf{j}_3$.

For C_5,

$$\dim\ker[\mathcal{F}_3;H_{12}] = 680 - 616 = 64.$$

Subtracting the 32-dimensional correction ambiguity leaves dimension 32 on $\ker\omega_3$.

For C_6,

$$\dim\ker[\mathcal{F}_3;H_{12}] = 2538 - 2358 = 180,$$

and subtracting the 90-dimensional derivation ambiguity leaves dimension 90.

Lemma NC.12.6 gives nested spaces whose outer dimensions are equal, hence equality follows. □

Corollary NC.12.8 (Exact critical Quantized Stress defects)**.** *One has*

$$\boxed{\dim_{\mathbb{Q}}\mathfrak{Q}_2^{\mathrm{crit}}(A_5) = 37 - 32 = 5,} \tag{NC.12.14}$$

and

$$\boxed{\dim_{\mathbb{Q}}\mathfrak{Q}_2^{\mathrm{crit}}(A_6) = 112 - 90 = 22.} \tag{NC.12.15}$$

Therefore both controls satisfy

$$D_{A_i/\mathbb{Q}} \neq A_i\langle \mathrm{Der}_{\mathbb{Q}}(A_i)\rangle.$$

NC.12.7 Exact cofactor lane

Let

$$Z = (z_J)_{|J|=4}$$

be the cofactor-syzygy matrix of NC.11. For $c = 3$, a critical coefficient matrix is

$$A_{\mathrm{cof}} \in M_{\binom{n}{4}\times n}(P_2),$$

and, under the Buchsbaum–Rim expected-grade condition, exact critical stresses are represented by

$$B = ZA_{\mathrm{cof}}, \qquad ZA_{\mathrm{cof}} = (ZA_{\mathrm{cof}})^T.$$

Proposition NC.12.9 (Expected maximal-minor grade in both controls)**.** *For C_5, the ideal of 3×3 minors of V^T has height*

$$3 = n - c + 1.$$

For C_6, it has height

$$4 = n - c + 1.$$

Hence the Buchsbaum–Rim completeness hypothesis of NC.11 holds for both controls.

Proof. Exact Gröbner bases of the maximal-minor ideals have initial monomial ideals whose minimum vertex-cover sizes are respectively 3 and 4. Initial ideals preserve Krull dimension and hence height. □

Theorem NC.12.10 (Exact-lane rank separation)**.** *Exact rational ranks of the NC.11 cofactor symmetrization system are:*

	C_5	C_6
$\dim \mathrm{Dom}(\Sigma_{\mathrm{cof}})$	375	1890
$\mathrm{rank}\,\Sigma_{\mathrm{cof}}$	300	1275
$\dim\ker\Sigma_{\mathrm{cof}}$	75	615
$\dim\ker(A_{\mathrm{cof}} \mapsto ZA_{\mathrm{cof}})$	75	612
$\boxed{\dim \mathcal{E}_2^{\mathrm{crit}}}$	$\boxed{0}$	$\boxed{3}$

(NC.12.16)

For C_6, augmenting Σ_{cof} by the horizontal test $H_{12}(ZA_{\mathrm{cof}})$ increases rank from 1275 to 1278. Hence all three nonzero exact critical stress directions are projectively visible.

Proof. The kernel of

$$A_{\mathrm{cof}} \mapsto ZA_{\mathrm{cof}}$$

is contained in $\ker \Sigma_{\mathrm{cof}}$.

For C_5, both kernels have dimension 75, hence every symmetric critical cofactor combination has $ZA_{\mathrm{cof}} = 0$. Buchsbaum–Rim completeness gives

$$\mathcal{E}_2^{\mathrm{crit}}(A_5) = 0.$$

For C_6, the quotient dimension is

$$615 - 612 = 3.$$

The horizontal augmented rank rises by exactly three, so H_{12} is injective on this three-dimensional exact-stress quotient. □

NC.12.8 The branch-separation theorem

Theorem NC.12.11 (Exact lane versus modulo-ideal lane)**.** *For* C_5*,*

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}} = 5, \qquad \dim \mathcal{E}_2^{\mathrm{crit}} = 0.} \tag{NC.12.17}$$

Thus every visible critical escape class necessarily uses tangency only modulo the defining ideal and the NC.7 correction mechanism.

For C_6*,*

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}} = 22, \qquad \dim \mathcal{E}_2^{\mathrm{crit}} = 3.} \tag{NC.12.18}$$

The exact lane therefore explains only three of the twenty-two critical escape directions. At least

$$\boxed{22 - 3 = 19} \tag{NC.12.19}$$

independent visible critical classes are genuinely outside the exact-stress lane.

Proof. Combine Corollary NC.12.8, Proposition NC.12.9, and Theorem NC.12.10. □

Corollary NC.12.12 (The Hessian-obstruction formalism is essential)**.** *The NC.7 Hessian-obstruction framework is not redundant with the exact-stress divergence lane. Even on the smooth, genuinely non-simultaneously-diagonal surface control* C_6*, the majority of the critical escape space occurs only after allowing Jacobian tangency modulo* I *and lower-order correction.*

NC.12.9 Interpretation and evidence ceiling

Remark NC.12.13 (Why C_6 is the primary control)**.** The curve control C_5 is retained because it cleanly proves that a nonzero critical defect can exist even when the exact critical lane is empty. Its Nakai non-generation conclusion may overlap classical low-dimensional complete-intersection results and therefore carries no novelty claim.

The surface control C_6 has

$$\dim X_6 = 2, \qquad \dim A_6 = 3,$$

is genuinely non-simultaneously diagonal, and exhibits a 22-dimensional critical defect. It is the principal NC.12 positive control for the next all-quadratic stage.

Policy 14 (C3-only globalization ceiling)**.** *Every rank, dimension, smoothness certificate, maximal-minor height, and commutator certificate in NC.12 is an exact rational computation and is therefore classified C3 for the displayed algebra only. No family theorem, generic-rank theorem, moduli statement, or openness statement is inferred merely from these controls.*

Theorem NC.12.14 (NC.12 closure theorem)**.** *There exists an explicit smooth, genuinely non-simultaneously-diagonal codimension-three quadratic surface control A_6 for which*

$$\boxed{\dim_{\mathbb{Q}} \mathfrak{Q}_2^{\mathrm{crit}}(A_6) = 22 > 0.}$$

Hence the NC.4–NC.8 Master Bridge succeeds beyond the simultaneous-diagonal family.

Moreover,

$$\boxed{\dim \mathcal{E}_2^{\mathrm{crit}}(A_6) = 3 < 22,}$$

so the general quadratic problem cannot be reduced, even empirically on this control, to exact polynomial stresses alone. The modulo-ideal Hessian-obstruction lane contributes at least nineteen additional visible critical directions.

Declaration NC.12.15 (NC.12 successor obligation)**.** The primary question is no longer:

"Can a non-diagonal quadratic ICIS have a critical escape?"

The answer is yes.

The next question is:

$$\boxed{\text{Can the nonzero critical defect be forced uniformly over the smooth quadratic locus?}}$$

The next stage should therefore analyze the family-level ranks of

$$\mathcal{F}_{2c-3}, \qquad \mathrm{j}_{2c-3}, \qquad H_{ij}, \qquad \Sigma_{\mathrm{cof}},$$

and determine which inequalities are generic, open, closed, or structurally forced before any "all smooth quadratic ICIS" theorem is claimed.

NC.13 Determinantal Rank Geometry on a Fixed Normal-Form Slice

NC.13.1 Purpose

NC.12 proves that the critical Quantized Stress defect is nonzero on explicit smooth, genuinely non-simultaneously-diagonal quadratic controls. NC.13 upgrades that pointwise computation to a family-level determinantal framework.

The central principle is:

$$\boxed{\text{rank difference} = \text{visible critical directions detected on the quantization kernel.}}$$

This makes the passage from one exact control to a generic theorem a precise question about polynomial matrix ranks.

NC.13 does not yet claim the all-smooth-quadratic theorem. Instead it proves:

(i) a universal finite matrix model on a fixed Gröbner normal-form slice;

(ii) rank-stratified Zariski openness of the positive critical defect;

(iii) a nonempty locally closed family containing the NC.12 surface control on which the defect is at least 22; and

(iv) an exact single-rank promotion criterion whose proof would yield a generic smooth quadratic theorem on that normal-form slice.

NC.13.2 A fixed quadratic Gröbner normal-form slice

Fix integers

$$n > c \geq 2$$

and write

$$y = (x_{c+1}, \ldots, x_n).$$

Let

$$N_y := n - c.$$

Policy 15 (Normal-form-slice ceiling)**.** *The family below is a fixed affine normal-form slice on which the quotient monomial basis is uniform. NC.13 does* not *prove that this slice is a Zariski chart, dominant slice, or moduli atlas for the full parameter space of quadratic complete intersections. Accordingly, "generic" in NC.13 means generic* inside this slice *unless a separate dominance theorem is supplied.*

Define the affine normal-form parameter space

$$\boxed{\mathcal{G}_{n,c} := \bigoplus_{\alpha=1}^{c} \mathrm{Sym}^2(k^{N_y})^\vee.} \tag{NC.13.1}$$

A point

$$u = (q_1, \ldots, q_c) \in \mathcal{G}_{n,c}$$

defines

$$\boxed{f_{\alpha,u} = x_\alpha^2 + q_\alpha(y) \qquad (1 \leq \alpha \leq c).} \tag{NC.13.2}$$

Let

$$A_u = k[x_1, \ldots, x_n]/(f_{1,u}, \ldots, f_{c,u}).$$

Proposition NC.13.1 (Uniform complete-intersection chart)**.** *For every $u \in \mathcal{G}_{n,c}$, the sequence*

$$f_{1,u}, \ldots, f_{c,u}$$

is a homogeneous regular sequence. Moreover, for any term order with

$$\mathrm{LM}(f_{\alpha,u}) = x_\alpha^2,$$

the displayed equations form a Gröbner basis and the standard monomials are exactly those monomials whose exponents in

$$x_1, \ldots, x_c$$

are at most one. Consequently every graded dimension

$$\dim_k (A_u)_d$$

is independent of u.

Proof. The leading monomials

$$x_1^2, \ldots, x_c^2$$

are pairwise coprime. Buchberger's coprime-leading-term criterion therefore makes the displayed sequence a Gröbner basis. The quotient standard monomials are those not divisible by any x_α^2. The quotient has codimension c, while it is generated by c homogeneous elements in the regular ring P; hence the sequence is regular. □

Corollary NC.13.2 (Constant graded bases)**.** *For each degree d, the standard monomials of Proposition NC.13.1 define a basis*

$$\mathcal{B}_d$$

that identifies every fiber $(A_u)_d$ with one fixed vector space

$$W_d.$$

All multiplication maps by the variables have matrices whose entries are polynomial functions on $\mathcal{G}_{n,c}$.

NC.13.3 Universal critical matrices

Set

$$m = c - 1, \qquad r_* := 2m - 1 = 2c - 3.$$

Using the fixed bases $\mathscr{B}_d$, the NC.7 maps become matrices with entries in the coordinate ring

$$R_{n,c} := k[\mathscr{G}_{n,c}].$$

Definition NC.13.3 (Universal quantization matrix)**.** Let

$$\boxed{\mathbf{F}_{n,c}} \tag{NC.13.3}$$

be the universal matrix representing

$$\mathcal{F}_{r_*}(B, \gamma) = \big(JB,\ J\gamma + \mathsf{h}_{r_*}(B)\big)$$

on the fixed graded bases.

Let

$$\boxed{\mathbf{J}_{n,c}} \tag{NC.13.4}$$

represent the critical first-order Jacobian correction map

$$\mathsf{j}_{r_*}.$$

For every pair $i < j$, let

$$\boxed{\mathbf{H}^{ij}_{n,c}} \tag{NC.13.5}$$

represent the horizontal contraction

$$B \longmapsto B(\alpha_{ij}, \alpha_{ij}), \qquad \alpha_{ij} = x_j dx_i - x_i dx_j,$$

extended by zero on the correction coordinates.

Proposition NC.13.4 (Polynomial dependence)**.** *Every entry of*

$$\mathbf{F}_{n,c}, \qquad \mathbf{J}_{n,c}, \qquad \mathbf{H}^{ij}_{n,c}$$

is a polynomial function of the coefficients of the quadrics

$$q_1, \dots, q_c.$$

Proof. In the fixed Gröbner normal-form slice, reduction is generated by the monic rules

$$x_\alpha^2 = -q_\alpha(y).$$

Their coefficients are polynomial in the parameters. The Jacobian entries are linear in the parameters and the Hessian entries are constant-linear in them. Every matrix entry in the NC.7 formulas is therefore obtained by finitely many additions and multiplications in $R_{n,c}$. □

NC.13.4 Rank loci

Let $M(u)$ denote the specialization of a universal matrix M at a parameter point u.

Definition NC.13.5 (Determinantal rank loci)**.** For any polynomial matrix M, define

$$\mathscr{R}_{\le r}(M) := \{u : \operatorname{rank} M(u) \le r\},$$

$$\mathscr{R}_{\ge r}(M) := \{u : \operatorname{rank} M(u) \ge r\},$$

and

$$\mathscr{R}_{=r}(M) := \mathscr{R}_{\le r}(M) \cap \mathscr{R}_{\ge r}(M).$$

Proposition NC.13.6 (Determinantal topology)**.** *The locus*

$$\mathscr{R}_{\le r}(M)$$

is Zariski closed,

$$\mathscr{R}_{\ge r}(M)$$

is Zariski open, and

$$\mathscr{R}_{=r}(M)$$

is locally closed.

Proof. The rank is at most r exactly when all $(r+1)\times(r+1)$ minors vanish. The complement is the union of principal open sets defined by nonzero $(r+1)$-minors. □

NC.13.5 The visibility rank increment

Fix a horizontal detector $H = \mathbf{H}^{ij}_{n,c}$ and define the stacked matrix

$$\boxed{\mathbf{F}^{H}_{n,c} := \begin{bmatrix} \mathbf{F}_{n,c} \\ H \end{bmatrix}.} \tag{NC.13.6}$$

Definition NC.13.7 (Horizontal visibility increment)**.** At $u \in \mathscr{G}_{n,c}$, define

$$\boxed{\varepsilon_H(u) := \operatorname{rank} \mathbf{F}^{H}_{n,c}(u) - \operatorname{rank} \mathbf{F}_{n,c}(u).} \tag{NC.13.7}$$

Lemma NC.13.8 (Rank-increment identity)**.** *Let*

$$F : U \to V, \qquad H : U \to W$$

be linear maps between finite-dimensional vector spaces. Then

$$\boxed{\operatorname{rank} \begin{bmatrix} F \\ H \end{bmatrix} - \operatorname{rank} F = \operatorname{rank}(H|_{\ker F}).} \tag{NC.13.8}$$

Proof. The kernel of the stacked map is

$$\ker F \cap \ker H.$$

By rank–nullity,

$$\begin{aligned} \operatorname{rank}[F;H] - \operatorname{rank} F &= \dim \ker F - \dim(\ker F \cap \ker H) \\ &= \operatorname{rank}(H|_{\ker F}). \end{aligned}$$

□

Theorem NC.13.9 (Rank increment detects strong escape)**.** *Assume the NC.4 hypotheses at a parameter point u. If*

$$\boxed{\varepsilon_H(u) > 0,} \tag{NC.13.9}$$

then

$$\boxed{D_{A_u/k} \neq A_u\langle \mathrm{Der}_k(A_u)\rangle.} \tag{NC.13.10}$$

Proof. By Lemma NC.13.8, there exists a quantization pair

$$(B,\gamma) \in \ker \mathcal{F}_{r_*}$$

with

$$H(B) \neq 0.$$

Thus B is quantizable and its projective second symbol is nonzero, because H is evaluation on the horizontal projective one-form α_{ij}. The internal degree is

$$r_* = 2m - 1 < 2m.$$

The NC.8 Master Bridge gives strong escape. □

Corollary NC.13.10 (Quantitative lower bound)**.** *At every admissible parameter point,*

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}}(A_u) \geq \varepsilon_H(u).} \tag{NC.13.11}$$

Remark NC.13.11. Equality need not hold for one horizontal detector. The detector may miss visible symbols that vanish on the chosen α_{ij}. NC.12 is stronger at its controls because an independent exact dimension comparison proves that H_{12} detects the full projectively visible quotient.

NC.13.6 Rank-stratified openness

Theorem NC.13.12 (Relative openness on a fixed quantization-rank stratum)**.** *Fix an integer r and $s \geq 1$. On the locally closed stratum*

$$\mathscr{R}_{=r}(\mathbf{F}_{n,c}),$$

the locus

$$\boxed{\varepsilon_H \geq s}$$

is Zariski open relative to the stratum. More explicitly,

$$\boxed{\{u \in \mathscr{R}_{=r}(\mathbf{F}_{n,c}) : \varepsilon_H(u) \geq s\} = \mathscr{R}_{=r}(\mathbf{F}_{n,c}) \cap \mathscr{R}_{\geq r+s}(\mathbf{F}_{n,c}^H).} \tag{NC.13.12}$$

Proof. On the fixed stratum,

$$\operatorname{rank} \mathbf{F}_{n,c} = r.$$

Hence

$$\varepsilon_H \geq s \iff \operatorname{rank} \mathbf{F}_{n,c}^H \geq r + s.$$

The latter condition is Zariski open by Proposition NC.13.6. □

Policy 16 (No global semicontinuity shortcut)**.** *The difference*

$$\operatorname{rank} \mathbf{F}^H - \operatorname{rank} \mathbf{F}$$

is not asserted to be lower or upper semicontinuous on the entire parameter space. Both ranks may jump. Therefore one positive specialization does not by itself prove a generic positive defect. The fixed-rank-stratum formulation in Theorem NC.13.12 is the safe statement.

NC.13.7 Smoothness and non-diagonal loci

Let

$$\mathscr{U}_{n,c}^{\mathrm{sm}} \subseteq \mathscr{G}_{n,c}$$

be the locus for which

$$X_u = \operatorname{Proj} A_u$$

is smooth.

Proposition NC.13.13 (Smooth locus is open)**.** *The locus*

$$\mathscr{U}_{n,c}^{\mathrm{sm}}$$

is Zariski open.

Proof. The universal family over the fixed Gröbner chart is flat because the standard monomial basis is constant in every degree. Projective smoothness is an open condition in a finite-type flat family. Equivalently, the complement is the image of the universal Jacobian rank-drop incidence locus under the proper projection to parameter space. □

For the codimension-three surface normal-form slice

$$(n, c) = (6, 3),$$

write

$$Q_{\Sigma,u} = Q_{1,u} + Q_{2,u} + Q_{3,u}.$$

Definition NC.13.14 (Certified non-diagonal open locus)**.** Fix one matrix entry of the commutator

$$C_u = [Q_{\Sigma,u}^{-1} Q_{1,u}, Q_{\Sigma,u}^{-1} Q_{2,u}].$$

Define

$$\mathscr{U}^{\mathrm{nd}}$$

by the simultaneous nonvanishing of

$$\det Q_{\Sigma,u}$$

and the numerator of that chosen commutator entry after clearing the power of $\det Q_{\Sigma,u}$.

Proposition NC.13.15 (Certified non-diagonal locus)**.** *The locus*

$$\mathscr{U}^{\mathrm{nd}}$$

is Zariski open, and every point of it is not simultaneously diagonalizable by congruence.

NC.13.8 The NC.12 surface control as a rank-stratum seed

Let

$$u_6 \in \mathscr{G}_{6,3}$$

be the NC.12 point

$$\begin{aligned} g_1 &= x_1^2 + x_4^2 + x_5 x_6, \\ g_2 &= x_2^2 + x_5^2 + x_4 x_6, \\ g_3 &= x_3^2 + x_6^2 + x_4 x_5. \end{aligned}$$

NC.12 proves

$$\boxed{\operatorname{rank} \mathbf{F}_{6,3}(u_6) = 2336,} \tag{NC.13.13}$$

$$\boxed{\operatorname{rank} \mathbf{F}_{6,3}^{H_{12}}(u_6) = 2358,} \tag{NC.13.14}$$

and

$$\boxed{\operatorname{rank} \mathbf{J}_{6,3}(u_6) = 306,} \tag{NC.13.15}$$

where 306 is the full target dimension of the critical correction map.

Theorem NC.13.16 (Nonempty positive rank-stratum family)**.** *Define*

$$\mathscr{W}_{6,3} := \mathscr{U}_{6,3}^{\mathrm{sm}} \cap \mathscr{U}^{\mathrm{nd}} \cap \mathscr{R}_{=2336}(\mathbf{F}_{6,3}) \cap \mathscr{R}_{\geq 2358}(\mathbf{F}_{6,3}^{H_{12}}) \cap \mathscr{R}_{\geq 306}(\mathbf{J}_{6,3}).$$

Then:

(i) $\mathscr{W}_{6,3}$ *is locally closed in* $\mathscr{G}_{6,3}$*;*

(ii) it contains the NC.12 point u_6*, hence is nonempty;*

(iii) the correction map j_3 *is surjective at every point of* $\mathscr{W}_{6,3}$*;*

(iv) every $u \in \mathscr{W}_{6,3}$ satisfies

$$\boxed{\varepsilon_{H_{12}}(u) \geq 22;} \tag{NC.13.16}$$

and therefore

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}}(A_u) \geq 22}$$

and

$$\boxed{D_{A_u/k} \neq A_u\langle \mathrm{Der}_k(A_u)\rangle.}$$

Proof. Local closedness follows from Propositions NC.13.6, NC.13.13, and NC.13.15. NC.12 places u_6 in every displayed factor.

The target of $\mathbf{J}_{6,3}$ has dimension 306, so rank at least 306 means surjectivity. On the stratum

$$\operatorname{rank} \mathbf{F}_{6,3} = 2336,$$

the stacked-rank condition gives

$$\varepsilon_{H_{12}} \geq 2358 - 2336 = 22.$$

Apply Corollary NC.13.10 and Theorem NC.13.9. □

Remark NC.13.17 (What has been promoted)**.** NC.12 gave one exact non-diagonal surface. Theorem NC.13.16 gives a nonempty algebraic rank-stratified family on which the same strong conclusion holds uniformly. This is a genuine family-level promotion, but it is not yet a statement that the family is Zariski open in the entire smooth quadratic locus.

NC.13.9 The generic promotion gate

The parameter space

$$\mathscr{G}_{6,3} \cong \mathbb{A}^{18}$$

is irreducible.

Theorem NC.13.18 (Conditional rank-2336 promotion criterion on the normal-form slice)**.** *Assume the universal upper bound*

$$\boxed{\operatorname{rank} \mathbf{F}_{6,3}(u) \leq 2336 \qquad \text{for every } u \in \mathscr{G}_{6,3}.} \tag{NC.13.17}$$

Then there exists a nonempty Zariski-open subset

$$\mathscr{U}^{\mathrm{gen}} \subseteq \mathscr{G}_{6,3}$$

such that every smooth point in

$$\mathscr{U}^{\mathrm{gen}}$$

satisfies

$$\boxed{\varepsilon_{H_{12}} \geq 22}$$

and hence

$$\boxed{D_{A_u/k} \neq A_u\langle \mathrm{Der}_k(A_u)\rangle.}$$

In particular, strong Nakai non-generation holds for a generic member of this codimension-three quadratic surface chart.

Proof. At u_6, a 2336×2336 minor of $\mathbf{F}_{6,3}$ is nonzero. Therefore

$$\operatorname{rank} \mathbf{F}_{6,3} \geq 2336$$

on a nonempty Zariski-open set. Under the assumed universal upper bound, equality holds there.

Likewise, the rank 2358 at u_6 produces a nonempty open set on which

$$\operatorname{rank} \mathbf{F}_{6,3}^{H_{12}} \geq 2358.$$

Intersecting the two open sets gives

$$\varepsilon_{H_{12}} \geq 22.$$

Intersect further with the nonempty smooth locus. Theorem NC.13.9 gives the conclusion. □

Corollary NC.13.19 (Equivalent determinantal obligation)**.** *To discharge* (NC.13.17)*, it is enough to prove that every* 2337×2337 *minor of the universal matrix*

$$\mathbf{F}_{6,3}$$

vanishes identically in the parameter ring

$$R_{6,3}.$$

Because NC.12 already supplies a nonzero 2336*-minor, such a proof would determine the generic rank exactly:*

$$\boxed{\operatorname{rank}_{\text{gen}} \mathbf{F}_{6,3} = 2336.} \tag{NC.13.18}$$

Remark NC.13.20 (Preferred proof route)**.** Brute-force expansion of all 2337-minors is not a viable mathematical strategy. The desired upper bound should come from a structural lower bound

$$\dim \ker \mathbf{F}_{6,3} \geq 202$$

over the universal parameter field. Candidates include a universal complex, representation-theoretic decomposition, a Hilbert-series computation of the critical stress/correction module, or a sheaf-cohomological interpretation of the 22-dimensional visible quotient.

NC.13.10 Why generic is still not all

Policy 17 (Generic-to-all nonpromotion)**.** *Even after Theorem NC.13.18 is discharged, the conclusion would hold on a dense open smooth locus, not automatically on every smooth quadratic complete intersection. Special fibers may lie on different rank strata where*

$$\operatorname{rank} \mathbf{F}$$

and

$$\operatorname{rank} \mathbf{F}^H$$

jump simultaneously. The all-smooth theorem requires either:

(i) a structural nonvanishing theorem valid on every smooth rank stratum;

(ii) a geometric identification of the critical defect with a vector bundle or cohomology group whose nonvanishing is uniform; or

(iii) a degeneration argument proving that positive critical classes cannot disappear on smooth specializations.

NC.13.11 NC.13 closure

Theorem NC.13.21 (NC.13 closure theorem)**.** *On the fixed quadratic Gröbner normal-form slice, the critical Nakai problem is governed by polynomial determinantal data.*

For every horizontal detector H*,*

$$\boxed{\varepsilon_H = \operatorname{rank}[\mathbf{F}; H] - \operatorname{rank} \mathbf{F}}$$

is exactly the dimension detected by H *on the critical quantization kernel.*

On each fixed rank stratum of $\mathbf{F}$*, the condition*

$$\varepsilon_H > 0$$

is relatively Zariski open and implies strong escape.

For the NC.12 surface control, there exists a nonempty locally closed family on which

$$\boxed{\varepsilon_{H_{12}} \geq 22.}$$

Finally, the generic codimension-three surface theorem is reduced to the single universal rank bound

$$\boxed{\operatorname{rank} \mathbf{F}_{6,3} \leq 2336.}$$

Declaration NC.13.22 (NC.13 successor obligation — reaudited)**.** The rank-2336 promotion gate above remains a correct conditional statement about the special rank stratum through the NC.12 control. It is no longer the primary route to strong non-generation.

NC.14 proves that every smooth $(2, 2, 2)$ K3 complete intersection already has a nonzero critical projective stress space, with a uniform lower bound 18. Consequently the natural universal rank bound for the full smooth $(2, 2, 2)$ family is weaker than 2336, while the value 2336 records a special cohomological jump at the NC.12 control. The NC.13 determinantal machinery is retained for stratification and jumping questions, not as a prerequisite for the all-smooth $(2, 2, 2)$ theorem.

NC.14 K3 Critical Cohomology and the All-Smooth $(2, 2, 2)$ Theorem

NC.14.1 Global re-audit register before NC.14

NC.14 begins with the corrections forced by a complete re-audit of NC.1–NC.13.

Declaration NC.14.1 (Re-audit conclusions)**.** The following statements are retained.

(i) The NC.4 Critical Window remains the strong-escape engine; its projectivization argument avoids an unproved affine PBW or bounded-word-length assumption.

(ii) The NC.7 Quantization Gate remains necessary. NC.12 exhibits smooth non-diagonal controls in which the majority, and in one control all, of the visible critical classes are not represented by exact polynomial stresses.

(iii) The NC.9 diagonal theorem and the NC.12 pointwise non-diagonal controls remain valid at their declared C2/C3 strength.

(iv) The NC.13 parameter family is a fixed Gröbner *normal-form slice*; no dominance statement over the full quadratic moduli space has been proved.

(v) The number 22 observed in NC.12 is not to be identified with the second Betti number of a K3 surface and is not forced by Riemann–Roch. The intrinsic Euler characteristic relevant to the critical bundle is 18.

(vi) The NC.13 bound

$$\operatorname{rank} \mathbf{F}_{6,3} \leq 2336$$

is therefore not required for strong non-generation. It is a stronger special-stratum condition equivalent to a larger-than-minimal critical cohomology space.

Policy 18 (Novelty ceiling)**.** *NC.14 proves an internal all-smooth theorem for the declared* $(2, 2, 2)$ *source class. It does not claim literature priority. Recent Nakai work located in the current audit includes homogeneous* hypersurface *results, whereas the present source is a codimension-three complete intersection; this difference is not itself a novelty proof. A dedicated complete-intersection prior-art audit remains mandatory before any priority statement.*

NC.14.2 The K3 geometry of a smooth triple quadric

Let

$$X = V(f_1, f_2, f_3) \subset \mathbf{P}^5_k$$

be a smooth complete intersection of three quadrics over a field k of characteristic zero, and let

$$A = k[x_1, \ldots, x_6]/(f_1, f_2, f_3).$$

Write

$$H := c_1(\mathcal{O}_X(1)).$$

Lemma NC.14.2 (K3 numerical data)**.** *One has*

$$\boxed{K_X \cong \mathcal{O}_X, \qquad H^2 = 8, \qquad c_2(T_X) = 3H^2, \qquad \int_X c_2(T_X) = 24.} \tag{NC.14.1}$$

In particular

$$\chi(\mathcal{O}_X) = 2.$$

Proof. Adjunction gives

$$K_X = \mathcal{O}_X(2 + 2 + 2 - 6) = \mathcal{O}_X.$$

The degree is

$$H^2 = 2^3 = 8.$$

The tangent-normal sequence gives

$$c(T_X) = \frac{(1 + H)^6}{(1 + 2H)^3}.$$

Expanding through degree two,

$$(1 + H)^6 = 1 + 6H + 15H^2 + \cdots,$$
$$(1 + 2H)^{-3} = 1 - 6H + 24H^2 + \cdots,$$

so

$$c_1(T_X) = 0, \qquad c_2(T_X) = 3H^2.$$

Thus

$$\int_X c_2(T_X) = 3 \cdot 8 = 24.$$

Noether–Riemann–Roch for $\mathcal{O}_X$ gives

$$\chi(\mathcal{O}_X) = \frac{c_1^2 + c_2}{12} = 2.$$

□

Remark NC.14.3. The variety X is therefore a polarized K3 surface of degree 8. NC.14 uses only the explicit complete-intersection geometry above, not a classification theorem for K3 surfaces.

NC.14.3 Universal vanishing of the critical T^1 slice

For $c = 3$,

$$m = c - 1 = 2, \qquad r_* = 2m - 1 = 3.$$

The NC.7 obstruction target is

$$(T^1_{A/k})_3 = \operatorname{coker}\left(\mathrm{j}_3 : A_4^6 \longrightarrow A_5^3\right).$$

Lemma NC.14.4 (ACM line-bundle vanishing)**.** *For the complete-intersection surface X,*

$$\boxed{H^1(X, \mathcal{O}_X(t)) = 0 \qquad \text{for every } t \in \mathbb{Z}.} \tag{NC.14.2}$$

Proof. A projective complete intersection is arithmetically Cohen–Macaulay. Since $\dim X = 2$, its only intermediate line-bundle cohomology group is H^1, which therefore vanishes for every twist. □

Lemma NC.14.5 (Ambient cotangent vanishing)**.** *One has*

$$\boxed{H^1\left(X, \Omega^1_{\mathbf{P}^5}(3)|_X\right) = 0.} \tag{NC.14.3}$$

Proof. Tensor the Koszul resolution of $\mathcal{O}_X$ by $\Omega^1_{\mathbf{P}^5}(3)$:

$$0 \to \Omega^1_{\mathbf{P}^5}(-3) \to \Omega^1_{\mathbf{P}^5}(-1)^3 \to \Omega^1_{\mathbf{P}^5}(1)^3 \to \Omega^1_{\mathbf{P}^5}(3) \to \Omega^1_{\mathbf{P}^5}(3)|_X \to 0.$$

The Euler sequence on $\mathbf{P}^5$ shows that

$$\Omega^1_{\mathbf{P}^5}(-3), \quad \Omega^1_{\mathbf{P}^5}(-1), \quad \Omega^1_{\mathbf{P}^5}(1)$$

have no cohomology in the degrees contributing to H^1 of the restriction. For $\Omega^1_{\mathbf{P}^5}(3)$, higher cohomology vanishes and the map

$$H^0(\mathcal{O}_{\mathbf{P}^5}(2))^6 \longrightarrow H^0(\mathcal{O}_{\mathbf{P}^5}(3))$$

is surjective, because every cubic is a sum

$$\sum_i x_i q_i.$$

The hypercohomology of the displayed Koszul complex therefore gives (NC.14.3). □

Lemma NC.14.6 (Critical tangent cohomology vanishing)**.** *One has*

$$\boxed{H^1(X, T_X(3)) = 0.} \tag{NC.14.4}$$

Proof. The conormal sequence twisted by 3 is

$$0 \to \mathcal{O}_X(1)^3 \to \Omega^1_{\mathbf{P}^5}(3)|_X \to \Omega^1_X(3) \to 0.$$

By Lemma NC.14.5, the middle H^1 vanishes. Serre duality and $K_X \cong \mathcal{O}_X$ give

$$H^2(X, \mathcal{O}_X(1)) \cong H^0(X, \mathcal{O}_X(-1))^\vee = 0.$$

Hence

$$H^1(X, \Omega^1_X(3)) = 0.$$

For a rank-two bundle E,

$$E^\vee \cong E \otimes (\det E)^{-1}.$$

Applying this to $E = \Omega^1_X$ and using $\det \Omega^1_X = K_X \cong \mathcal{O}_X$ gives

$$T_X \cong \Omega^1_X.$$

Thus (NC.14.4) follows. □

Theorem NC.14.7 (Universal critical T^1-vanishing for triples of quadrics)**.** *For every smooth complete intersection of three quadrics in* $\mathbf{P}^5$*,*

$$\boxed{(T^1_{A/k})_3 = 0.} \tag{NC.14.5}$$

Equivalently,

$$\boxed{\mathrm{j}_3 : A^6_4 \twoheadrightarrow A^3_5.} \tag{NC.14.6}$$

Proof. The Euler sequence restricted to X, twisted by 3, is

$$0 \to \mathcal{O}_X(3) \to \mathcal{O}_X(4)^6 \to T_{\mathbf{P}^5}|_X(3) \to 0.$$

By Lemma NC.14.4,

$$H^1(X, \mathcal{O}_X(3)) = 0,$$

so

$$H^0(X, \mathcal{O}_X(4))^6 \twoheadrightarrow H^0(X, T_{\mathbf{P}^5}|_X(3)).$$

The normal sequence twisted by 3 is

$$0 \to T_X(3) \to T_{\mathbf{P}^5}|_X(3) \to \mathcal{O}_X(5)^3 \to 0.$$

Lemma NC.14.6 gives surjectivity

$$H^0(X, T_{\mathbf{P}^5}|_X(3)) \twoheadrightarrow H^0(X, \mathcal{O}_X(5))^3.$$

The composite is exactly the graded Jacobian map

$$\mathrm{j}_3 : A_4^6 \to A_5^3.$$

NC.7 identifies its cokernel with $(T^1_{A/k})_3$. □

Corollary NC.14.8 (Every critical stress is quantizable)**.** *For every smooth triple quadric,*

$$\boxed{\ker \omega_3 = \mathcal{S}_{2,3}(A).} \tag{NC.14.7}$$

Thus every critical Jacobian stress admits a first-order correction.

NC.14.4 The critical stress/projective-tensor exact sequence

Let

$$\mathcal{V}_X := \mathrm{Sym}^2 T_X(3).$$

Definition NC.14.9 (Universal radial map)**.** Define

$$\boxed{\rho_3 : \mathrm{Der}_k(A)_3 \longrightarrow \mathcal{S}_{2,3}(A), \qquad u \longmapsto xu^T + ux^T.} \tag{NC.14.8}$$

Theorem NC.14.10 (Critical stress exact sequence)**.** *For every smooth complete intersection of three quadrics in* $\mathbf{P}^5$*, there is a natural short exact sequence*

$$\boxed{0 \longrightarrow \mathrm{Der}_k(A)_3 \xrightarrow{\rho_3} \mathcal{S}_{2,3}(A) \xrightarrow{\Psi_{2,3}} H^0(X, \mathcal{V}_X) \longrightarrow 0.} \tag{NC.14.9}$$

Proof. The Euler sequence

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(1)^6 \to T_{\mathbf{P}^5}|_X \to 0$$

has a line subbundle on the left. Taking the symmetric square and twisting by 3 gives

$$0 \to \mathcal{O}_X(4)^6 \to \mathcal{O}_X(5)^{21} \to \mathrm{Sym}^2 T_{\mathbf{P}^5}|_X(3) \to 0. \tag{NC.14.10}$$

Lemma NC.14.4 gives

$$H^1(X, \mathcal{O}_X(4)) = 0,$$

so every global ambient projective symmetric tensor has a coefficient-matrix lift

$$B_0 \in \mathrm{Sym}_6(A_5).$$

Take

$$s \in H^0(X, \mathrm{Sym}^2 T_X(3)) \subset H^0(X, \mathrm{Sym}^2 T_{\mathbf{P}^5}|_X(3))$$

and choose such a lift B_0. Because s is tangent to X, contraction with each conormal df_α vanishes projectively. The Euler sequence twisted by 5 therefore gives

$$B_0 \nabla f_\alpha = x\, \lambda_\alpha$$

for some

$$\lambda_\alpha \in A_5.$$

By Theorem NC.14.7, choose

$$u \in A_4^6$$

with

$$\mathrm{j}_3(u) = -(\lambda_1, \lambda_2, \lambda_3).$$

Set

$$B = B_0 + xu^T + ux^T.$$

Since

$$x \cdot \nabla f_\alpha = 2f_\alpha = 0 \quad \text{in } A,$$

one has

$$B\nabla f_\alpha = x\big(\lambda_\alpha + u \cdot \nabla f_\alpha\big) = 0.$$

Thus

$$B \in \mathcal{S}_{2,3}(A),$$

and the radial correction does not change the projective symbol. Hence $\Psi_{2,3}$ is surjective.

Now suppose

$$B \in \mathcal{S}_{2,3}(A)$$

has zero projective symbol. Because its ambient projective tensor is tangent to X, zero in $\mathrm{Sym}^2 T_X(3)$ means zero already in the ambient tangent tensor along X. Exactness of (NC.14.10) gives

$$B = xu^T + ux^T$$

for some

$$u \in A_4^6.$$

Tangency gives

$$0 = B\nabla f_\alpha = x\,(u \cdot \nabla f_\alpha).$$

The affine cone is a domain by the normality argument used earlier in NC, so

$$u \cdot \nabla f_\alpha = 0$$

for every α. Thus

$$u \in \mathrm{Der}_k(A)_3.$$

Conversely every $\rho_3(u)$ is projectively radial and hence invisible.

Finally, if

$$xu^T + ux^T = 0,$$

its diagonal entries give

$$2x_i u_i = 0.$$

The domain property and $x_i \neq 0$ imply $u_i = 0$ for all i, so ρ_3 is injective. □

Corollary NC.14.11 (Intrinsic form of the critical defect)**.** *Assume the NC.4 graded derivation-generation hypotheses. Then for every smooth triple quadric,*

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0\Big(X, \mathrm{Sym}^2 T_X(3)\Big).} \tag{NC.14.11}$$

Proof. By Corollary NC.14.8,

$$\ker \omega_3 = \mathcal{S}_{2,3}(A).$$

By Theorem NC.14.10, the projectively invisible sector is exactly the radial image of $\mathrm{Der}_k(A)_3$. NC.4 kills the generated projective second-symbol sector at the critical degree $3 < 2m = 4$. Now apply the NC.8 critical-defect identification. □

NC.14.5 Riemann–Roch for the critical K3 tensor bundle

Lemma NC.14.12 (Chern classes of the critical bundle)**.** *For*

$$\mathcal{V}_X = \operatorname{Sym}^2 T_X(3),$$

one has

$$\boxed{\operatorname{rank} \mathcal{V}_X = 3, \qquad c_1(\mathcal{V}_X) = 9H, \qquad c_2(\mathcal{V}_X) = 39H^2.} \tag{NC.14.12}$$

In particular,

$$\int_X c_2(\mathcal{V}_X) = 312.$$

Proof. Let the Chern roots of T_X be a, b. Since

$$a + b = c_1(T_X) = 0, \qquad ab = c_2(T_X) = 3H^2,$$

the roots of $\operatorname{Sym}^2 T_X$ are

$$2a, \quad a + b, \quad 2b.$$

Therefore

$$c_1(\operatorname{Sym}^2 T_X) = 0, \qquad c_2(\operatorname{Sym}^2 T_X) = 4ab = 12H^2.$$

Twisting a rank-three bundle by $\mathcal{O}_X(3)$ adds

$$3(3H) = 9H$$

to c_1, and since the untwisted first Chern class is zero,

$$c_2(\mathcal{V}_X) = 12H^2 + \binom{3}{2}(3H)^2 = 39H^2.$$

Because $H^2 = 8$,

$$\int_X 39H^2 = 312.$$

□

Theorem NC.14.13 (Critical Euler characteristic)**.** *For every smooth triple quadric,*

$$\boxed{\chi(X, \mathcal{V}_X) = 18.} \tag{NC.14.13}$$

Proof. On a K3 surface, Riemann–Roch for a rank-r vector bundle E gives

$$\chi(E) = 2r + \frac{1}{2}c_1(E)^2 - c_2(E).$$

Using Lemma NC.14.12,

$$\begin{aligned} \chi(\mathcal{V}_X) &= 6 + \frac{1}{2}(9H)^2 - 312 \\ &= 6 + \frac{1}{2} \cdot 81 \cdot 8 - 312 \\ &= 18. \end{aligned}$$

□

NC.14.6 Vanishing of the top cohomology

Lemma NC.14.14 (Negative ambient symmetric tensors have no sections)**.** *One has*

$$\boxed{H^0\Big(X, \operatorname{Sym}^2 T_{\mathbf{P}^5}|_X(-3)\Big) = 0.} \tag{NC.14.14}$$

Proof. Symmetrizing the Euler sequence and twisting by -3 gives

$$0 \to \mathcal{O}_X(-2)^6 \to \mathcal{O}_X(-1)^{21} \to \operatorname{Sym}^2 T_{\mathbf{P}^5}|_X(-3) \to 0.$$

Now

$$H^0(X, \mathcal{O}_X(-1)) = 0$$

and, by Lemma NC.14.4,

$$H^1(X, \mathcal{O}_X(-2)) = 0.$$

The long exact sequence gives the result. □

Lemma NC.14.15 (Top critical cohomology vanishes)**.** *One has*

$$\boxed{H^2(X, \mathcal{V}_X) = 0.} \tag{NC.14.15}$$

Proof. The tangent inclusion

$$T_X \hookrightarrow T_{\mathbf{P}^5}|_X$$

is a subbundle inclusion. Taking symmetric squares and twisting by -3 gives

$$\operatorname{Sym}^2 T_X(-3) \hookrightarrow \operatorname{Sym}^2 T_{\mathbf{P}^5}|_X(-3).$$

Lemma NC.14.14 therefore implies

$$H^0(X, \operatorname{Sym}^2 T_X(-3)) = 0.$$

Since $K_X \cong \mathcal{O}_X$ and $T_X \cong \Omega^1_X$,

$$\mathcal{V}_X^\vee = \operatorname{Sym}^2 \Omega^1_X(-3) \cong \operatorname{Sym}^2 T_X(-3).$$

Serre duality gives

$$H^2(X, \mathcal{V}_X)^\vee \cong H^0(X, \mathcal{V}_X^\vee) = 0.$$

□

Theorem NC.14.16 (Uniform nonvanishing and quantitative lower bound)**.** *For every smooth complete intersection of three quadrics in* $\mathbf{P}^5$*,*

$$\boxed{h^0\Big(X, \operatorname{Sym}^2 T_X(3)\Big) = 18 + h^1\Big(X, \operatorname{Sym}^2 T_X(3)\Big) \geq 18.} \tag{NC.14.16}$$

In particular,

$$\boxed{H^0\Big(X, \operatorname{Sym}^2 T_X(3)\Big) \neq 0.}$$

Proof. Combine Theorem NC.14.13 with Lemma NC.14.15:

$$18 = h^0 - h^1 + h^2 = h^0 - h^1.$$

□

NC.14.7 The all-smooth triple-quadric theorem

Theorem NC.14.17 (All-smooth $(2,2,2)$ strong Nakai escape)**.** *Let k be a characteristic-zero field and let*

$$A = k[x_1, \dots, x_6]/(f_1, f_2, f_3)$$

be a homogeneous complete intersection of three quadrics such that

$$X = \operatorname{Proj} A$$

is smooth. Assume the graded Kersken-type derivation-generation conclusion of NC.3 at its exact source strength. Then

$$\boxed{\dim_k \mathfrak{Q}_2^{\mathrm{crit}}(A) \geq 18,} \tag{NC.14.17}$$

and there exists

$$\boxed{D \in D^2_{A/k,3} \setminus A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.14.18}$$

Consequently

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.14.19}$$

Proof. Corollary NC.14.11 identifies the critical Quantized Stress defect with

$$H^0(X, \operatorname{Sym}^2 T_X(3)).$$

Theorem NC.14.16 gives dimension at least 18. Thus the critical defect is nonzero. NC.8 then produces an actual order-two critical operator outside Δ_A. □

Corollary NC.14.18 (No diagonal or generic hypothesis is needed)**.** *Theorem NC.14.17 applies to every smooth triple quadric in $\mathbf{P}^5$. It requires neither simultaneous diagonalization nor membership in the NC.13 normal-form slice nor a generic-rank assumption.*

NC.14.8 Reinterpretation of the NC.12 number 22

For a smooth triple quadric, the Hilbert series is

$$\frac{(1-t^2)^3}{(1-t)^6}.$$

Hence

$$\dim A_4 = 66, \qquad \dim A_5 = 102.$$

Corollary NC.14.19 (Universal critical derivation dimension)**.** *For every smooth triple quadric,*

$$\boxed{\dim_k \mathrm{Der}_k(A)_3 = 6 \cdot 66 - 3 \cdot 102 = 90.} \tag{NC.14.20}$$

Proof. Theorem NC.14.7 makes

$$\mathrm{j}_3 : A_4^6 \to A_5^3$$

surjective. Its kernel is $\mathrm{Der}_k(A)_3$. □

Corollary NC.14.20 (Universal stress and quantization-pair dimensions)**.** *Write*

$$h_X := h^0(X, \operatorname{Sym}^2 T_X(3)).$$

Then

$$\boxed{\dim \mathcal{S}_{2,3}(A) = 90 + h_X,} \tag{NC.14.21}$$

and

$$\boxed{\dim \mathcal{Q}^{\mathrm{amb}}_{2,3}(A) = 180 + h_X.} \tag{NC.14.22}$$

In the NC.12 matrix presentation with domain dimension 2538,

$$\boxed{\operatorname{rank} \mathcal{F}_3 = 2358 - h_X \leq 2340.} \tag{NC.14.23}$$

Proof. Theorem NC.14.10 and Corollary NC.14.19 give

$$\dim \mathcal{S}_{2,3} = 90 + h_X.$$

NC.7 gives the exact sequence

$$0 \to \mathrm{Der}_k(A)_3 \to Q^{\mathrm{amb}}_{2,3}(A) \to \ker \omega_3 \to 0.$$

By Corollary NC.14.8,

$$\ker \omega_3 = \mathcal{S}_{2,3},$$

giving the second formula. The final rank formula is rank–nullity. Theorem NC.14.16 gives

$$h_X \geq 18.$$

□

Corollary NC.14.21 (The NC.12 control is a cohomology-jump fiber)**.** *For the NC.12 surface control* C_6*,*

$$\dim \mathfrak{Q}^{\mathrm{crit}}_2 = 22$$

implies

$$\boxed{h^0(X_6, \mathcal{V}_{X_6}) = 22, \qquad h^1(X_6, \mathcal{V}_{X_6}) = 4.} \tag{NC.14.24}$$

Its matrix rank

$$2336$$

is therefore explained by

$$2358 - 22 = 2336.$$

Proof. Use Corollary NC.14.11, Theorem NC.14.16, and Corollary NC.14.20. □

Remark NC.14.22 (Why 22 was misleading)**.** The numerical coincidence

$$h^0(\mathcal{V}_{X_6}) = 22 = b_2(\mathrm{K3})$$

does not identify the two vector spaces and does not supply a topological explanation. Riemann–Roch gives the invariant quantity

$$\chi(\mathcal{V}_X) = 18,$$

while the extra four sections at the NC.12 control arise from

$$h^1(\mathcal{V}_{X_6}) = 4.$$

Thus 22 is a cohomological jump value, not the universal K3 count relevant to the NC proof.

NC.14.9 Relation to explicit symmetric-cotangent cohomology

Remark NC.14.23 (External consistency)**.** The bundle

$$\mathrm{Sym}^2 T_X(3)$$

is, on the K3 surface X, isomorphic after choosing $K_X \cong \mathcal{O}_X$ to

$$\mathrm{Sym}^2 \Omega^1_X(3).$$

Explicit cohomology of twisted symmetric powers of cotangent bundles on smooth complete intersections is studied by Etesse [5]. NC.14 does not import a numerical theorem from that paper: the vanishing, Chern-class calculation, and lower bound 18 above are proved internally. The reference is retained for comparison and prior-art audit.

NC.14.10 Architectural consequences

Theorem NC.14.24 (NC.14 architectural closure)**.** *For smooth* $(2,2,2)$ *complete intersections in* $\mathbf{P}^5$*, the second-order NC architecture collapses to the intrinsic projective bundle*

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0(X, \mathrm{Sym}^2 T_X(3)),}$$

and this vector space has dimension at least 18*.*

Thus:

$$\boxed{\textit{all smooth } (2,2,2) \Longrightarrow \textit{critical visible stress} \Longrightarrow \textit{strong Nakai escape}.}$$

*The NC.13 rank-*2336 *problem is thereby demoted from a proof gate to a cohomology-jump/stratification problem. The universal smooth-fiber matrix bound supplied by NC.14 is*

$$\boxed{\mathrm{rank}\, \mathcal{F}_3 \leq 2340,}$$

with equality exactly when

$$h^1(X, \mathrm{Sym}^2 T_X(3)) = 0.$$

Declaration NC.14.25 (NC.14 status and successor obligation)**.** NC.14 supplies the first non-diagonal, non-generic, all-smooth quadratic family theorem of Appendix NC:

$$\boxed{\text{every smooth triple quadric } X \subset \mathbf{P}^5 \text{ has strong order-two Nakai escape.}}$$

The internal geometry and cohomology argument is C0–C2. The final strong-escape implication still consumes the NC.3 imported derivation-generation theorem, so publication status remains C2-high until that source/interface audit and the complete-intersection prior-art audit are closed.

The next mathematical target is no longer the $(2,2,2)$ K3 family. The successor problem is to determine which parts of

$$(T^1)_{2m-1} = 0, \qquad \mathcal{S}_{2,2m-1}/\mathrm{radial} \cong H^0(\mathrm{Sym}^2 T_X(2m-1)),$$

and the Riemann–Roch nonvanishing mechanism persist for:

higher-dimensional intersections of three quadrics, higher-codimension quadratic complete intersections, and eventually general homogeneous ICIS.

NC.15 Quadratic Completion: Universal Critical Quantization and Intrinsic Cofactor Escape

Integrated NC.15–NC.16 source abstract

This standalone XeLaTeX source gives the re-audited construction draft of Appendix NC.15 and NC.16.

NC.15 upgrades the NC.14 triple-quadric theorem to a conditional all-smooth quadratic complete-intersection theorem. The proof architecture has two independent gates: universal critical T^1-vanishing and nonzero intrinsic critical projective tensors. The first gate uses an exact Vandermonde diagonal control together with the imported Brückmann twisted-Hodge numerical theorem; the second is proved geometrically from the conormal Gauss map and the adjugate of the second fundamental form.

NC.16 removes the quadratic-degree hypothesis at the architectural level. It proves the degree-independent identity

$$\operatorname{Sym}^2 T_X(2m-1) \cong \operatorname{Sym}^2 \Omega_X^{d-1}(2d-1),$$

closes all smooth positive-dimensional homogeneous complete-intersection curves under the NC.3 derivation-generation interface, and identifies the equidegree order-two matching index

$$\delta_2(a,d) = (a-2)d - a + 1.$$

The exact support bound has been re-audited and corrected to

$$s_{\max} = (c-1)(2a-2) + (d+2)(a-2).$$

The first order-two boundary is therefore the cubic-surface line $(a,d) = (3,2)$.

No literature-priority claim is made. Imported interfaces remain explicitly marked, and the assembled publication status is kept at C2-high.

NC.15.1 Purpose, hypotheses, and claim ceiling

Let

$$P = k[x_0,\dots,x_N], \qquad A = P/(f_1,\dots,f_c),$$

where k is a field of characteristic zero,

$$c \geq 2, \qquad f_1,\dots,f_c \in P_2$$

form a homogeneous regular sequence, and

$$X := \operatorname{Proj} A \subset \mathbf{P}_k^N$$

is smooth of positive dimension

$$d := \dim X = N - c \geq 1.$$

The NC.3 graded derivation-generation conclusion is consumed only at the final strong-escape step. Until the Kersken local-to-graded interface and the complete-intersection prior-art audit are fully source-bound, the final assembled theorem is classified C2-high.

Policy 19 (No hypersurface folding)**.** *The present section assumes $c \geq 2$. For $c = 1$, the NC threshold m and the present critical-window normalization degenerate. The homogeneous hypersurface case is therefore kept separate.*

Policy 20 (No novelty claim)**.** *NC.15 asserts an internal theorem architecture and proof candidate. It does not assert literature priority for any all-quadratic conclusion.*

NC.15.2 Critical numerical data

For a complete intersection of c quadrics, the NC.3 threshold is

$$m = \sum_{\alpha=1}^{c} 2 - (c+1) = c - 1.$$

Hence the maximal order-two NC.4 escape degree is

$$\boxed{r_{\mathrm{crit}} = 2m - 1 = 2c - 3.} \tag{NC.15.1}$$

Adjunction gives

$$K_X \cong O_X(2c - N - 1) = O_X(c - d - 1).$$

Since $m = c - 1$,

$$\boxed{K_X \cong O_X(m - d).} \tag{NC.15.2}$$

For a smooth d-fold,

$$T_X \cong \Omega_X^{d-1} \otimes K_X^{-1}.$$

Consequently

$$\begin{aligned} \mathrm{Sym}^2 T_X(2c-3) &\cong \mathrm{Sym}^2 \Omega_X^{d-1} \otimes K_X^{-2} \otimes O_X(2c-3) \\ &\cong \mathrm{Sym}^2 \Omega_X^{d-1}(2d-1). \end{aligned}$$

Thus

$$\boxed{\mathrm{Sym}^2 T_X(2c-3) \cong \mathrm{Sym}^2 \Omega_X^{d-1}(2d-1).} \tag{NC.15.3}$$

Remark NC.15.1 (First structural cancellation)**.** The right-hand twist $2d-1$ is independent of the codimension c. This cancellation will be shown in NC.16 to hold for arbitrary homogeneous complete intersections, not only quadrics.

NC.15.3 Gate Q: the critical T^1 slice

Write

$$r := 2c - 3.$$

The graded Jacobian map at the order-two critical degree is

$$\mathrm{j}_r : A_{r+1}^{N+1} \longrightarrow A_{r+2}^{c}.$$

By the NC.7 identification,

$$(T^1_{A/k})_r \cong \mathrm{coker}(\mathrm{j}_r).$$

We first compute this slice on one smooth control in every pair (c, d).

Definition NC.15.2 (Vandermonde quadratic control)**.** After a harmless characteristic-zero field extension, choose pairwise distinct

$$a_0, \ldots, a_N$$

and put

$$\lambda_j := (1, a_j, \ldots, a_j^{c-1})^T \in k^c.$$

Define

$$f_\alpha := \sum_{j=0}^{N} a_j^{\alpha-1} x_j^2, \qquad 1 \le \alpha \le c.$$

Every c of the vectors λ_j are linearly independent.

Lemma NC.15.3 (Smoothness of the Vandermonde control)**.** *The projective complete intersection*

$$X_{\mathrm{V}} = V(f_1, \ldots, f_c) \subset \mathbf{P}^N$$

is smooth of dimension $d = N - c$.

Proof. At a projective point x, the j-th Jacobian column equals

$$2x_j\lambda_j.$$

If the Jacobian rank were $< c$, then the support

$$S := \{j : x_j \neq 0\}$$

would contain at most $c-1$ indices, because any c of the λ_j are linearly independent.

The defining equations are the vector equation

$$\sum_{j\in S} \lambda_j x_j^2 = 0.$$

The vectors indexed by S are independent, so $x_j^2 = 0$ for all $j \in S$, a contradiction. □

For the diagonal control set

$$M := A^c \Big/ \sum_{j=0}^{N} A\, x_j\lambda_j.$$

Then

$$M_{2c-1} \cong (T^1_{A/k})_{2c-3}.$$

Theorem NC.15.4 (Quadratic support-saturation theorem)**.** *For the Vandermonde control,*

$$\boxed{M_s = 0 \qquad \text{for every } s \geq 2c-1.} \tag{NC.15.4}$$

In particular,

$$\boxed{(T^1_{A/k})_{2c-3} = 0.} \tag{NC.15.5}$$

Proof. Consider a monomial vector $x^\mu v$, $v \in k^c$.

If the support of x^μ contains at least c indices, choose c supported indices $j_1, \ldots, j_c$. Then $\lambda_{j_1}, \ldots, \lambda_{j_c}$ form a basis of k^c. Writing

$$v = \sum_{\nu=1}^{c} b_\nu \lambda_{j_\nu},$$

we obtain

$$x^\mu v = \sum_{\nu=1}^{c} b_\nu \frac{x^\mu}{x_{j_\nu}}(x_{j_\nu}\lambda_{j_\nu}) = 0 \quad \text{in } M.$$

Assume now that the support has cardinality $t < c$. If every supported exponent were at most 2, then

$$\deg x^\mu \leq 2t \leq 2c-2,$$

contrary to $s \geq 2c-1$. Hence some supported variable x_i has exponent at least 3.

Extend the support to a set B of cardinality c. Since the λ_j, $j \in B$, form a basis, the quadratic vector relations solve

$$x_i^2 = \sum_{\ell\notin B} b_{i\ell} x_\ell^2 \quad \text{in } A.$$

For every $\ell \notin B$, the coefficient of λ_i in the basis expansion of λ_ℓ is nonzero: otherwise λ_ℓ would lie in the span of the remaining $c-1$ basis vectors, contradicting the full-spark property.

Substituting the displayed relation decreases the exponent of x_i by 2, so x_i remains in the support, while each nonzero term acquires a new supported variable outside B. Thus the support strictly increases and total degree is unchanged.

If the new support still has size $< c$, the same degree argument again forces an exponent at least 3. Iteration reaches support size c, where the class vanishes by the first part. □

NC.15.4 Intrinsic interpretation and numerical transport

Lemma NC.15.5 (Critical T^1/tangent-cohomology identification)**.** *For every smooth quadratic complete intersection in the declared scope,*

$$\boxed{(T^1_{A/k})_{2c-3} \cong H^1(X, T_X(2c-3))\,.} \tag{NC.15.6}$$

By Serre duality,

$$\boxed{(T^1_{A/k})^\vee_{2c-3} \cong H^{d-1}\Big(X, \Omega^1_X(2-N)\Big)\,.} \tag{NC.15.7}$$

Proof. Twist the Euler sequence by $r = 2c-3$:

$$0 \to \mathcal{O}_X(r) \to \mathcal{O}_X(r+1)^{N+1} \to T_{\mathbf{P}^N}|_X(r) \to 0.$$

Twist the normal sequence by r:

$$0 \to T_X(r) \to T_{\mathbf{P}^N}|_X(r) \to \mathcal{O}_X(r+2)^c \to 0.$$

A projective complete intersection is arithmetically Cohen–Macaulay. For $d \geq 3$, the needed H^1 and H^2 line-bundle groups vanish as intermediate cohomology. For $d = 2$, the only additional endpoint is

$$H^2(X, \mathcal{O}_X(r)) \cong H^0(X, K_X(-r))^\vee = H^0(X, \mathcal{O}_X(-c))^\vee = 0.$$

For $d = 1$,

$$K_X = \mathcal{O}_X(c-2)$$

and

$$H^1(X, \mathcal{O}_X(r)) = H^1(X, \mathcal{O}_X(r+1)) = 0$$

by Serre duality and $c \geq 2$.

Thus the cokernel of the global Jacobian map is exactly $H^1(X, T_X(r))$, proving the first statement.

For the second,

$$H^1(X, T_X(r))^\vee \cong H^{d-1}(X, \Omega^1_X \otimes K_X(-r)).$$

Using $K_X = \mathcal{O}_X(c-d-1)$ and $r = 2c-3$,

$$c - d - 1 - (2c-3) = 2 - (c+d) = 2 - N.$$

□

Theorem NC.15.6 (Twisted-Hodge numerical transport; imported)**.** *For a smooth projective complete intersection over characteristic zero, the dimensions*

$$h^q(X, \Omega^p_X(t))$$

are determined by the numerical complete-intersection data

$$(d, c; d_1, \ldots, d_c; p, q, t).$$

In particular, for fixed ambient dimension and fixed multidegree they are constant across the smooth locus of the parameter space.

Declaration NC.15.7 (Source binding)**.** Theorem NC.15.6 is imported. Brückmann's 1976–1977 papers compute the twisted Hodge groups of smooth complete intersections; later Brückmann–Rackwitz work gives a purely algebraic positive-characteristic extension and explicitly states dependence only on the numerical complete-intersection data and the characteristic. The exact theorem-number binding to the notation used here remains a publication-stage obligation.

Theorem NC.15.8 (Universal quadratic critical T^1-vanishing)**.** *Assume Theorem NC.15.6 at the declared source strength. Then every smooth positive-dimensional complete intersection of $c \geq 2$ quadrics satisfies*

$$\boxed{(T^1_{A/k})_{2c-3} = 0.} \tag{NC.15.8}$$

Proof. By Lemma NC.15.5, the dimension of the critical T^1 slice equals

$$h^{d-1}\Big(X, \Omega^1_X(2-N)\Big),$$

which, by Theorem NC.15.6, depends only on the complete-intersection type $(2, \dots, 2)$. The smooth Vandermonde member of the same type has zero critical T^1 by Theorem NC.15.4. Hence every smooth member has zero critical T^1. □

Corollary NC.15.9 (Collapse of the quadratic Quantization Gate)**.** *At critical degree $r = 2c - 3$,*

$$\boxed{\ker \omega_r = \mathcal{S}_{2,r}(A).} \tag{NC.15.9}$$

Thus every critical quadratic Jacobian stress is quantizable.

NC.15.5 Gate N: intrinsic adjugate tensors

Let

$$V := \mathrm{Span}_k\{f_1, \dots, f_c\}.$$

For $q \in V$, the second fundamental form in the normal direction defined by q is a symmetric morphism

$$h_q : T_X \longrightarrow \Omega^1_X(2).$$

Taking the adjugate gives an intrinsic polynomial map in q

$$\alpha_X : \mathrm{Sym}^{d-1} V \longrightarrow H^0\Big(X, \mathrm{Sym}^2 T_X(2c-4)\Big). \tag{NC.15.10}$$

Locally, $\alpha_X(q^{d-1})$ is represented by the cofactor matrix of the symmetric $d \times d$ matrix representing h_q.

Policy 21 (Separation from the NC.11 radial trap)**.** *The construction α_X is not the NC.11 affine cofactor symmetrization*

$$z_J \longmapsto x \odot z_J.$$

The latter contains an Euler/radial factor and is projectively invisible. The present adjugate is formed intrinsically on T_X and is already a projective tangent tensor.

Lemma NC.15.10 (Conormal Gauss map)**.** *After base change to an algebraic closure, define*

$$\gamma : X \times \mathbf{P}(V) \longrightarrow (\mathbf{P}^N)^\vee$$

by

$$(x, [q]) \longmapsto [dq(x)].$$

This is a morphism and

$$\gamma^* O_{(\mathbf{P}^N)^\vee}(1) \cong O_X(1) \boxtimes O_{\mathbf{P}(V)}(1).$$

Moreover

$$\int_{X \times \mathbf{P}^{c-1}} c_1(\gamma^* O(1))^{N-1} = \boxed{2^c \binom{d+c-1}{c-1}} > 0. \tag{NC.15.11}$$

Hence γ is generically finite onto an $(N-1)$-dimensional image.

Proof. Smoothness of X makes the conormal vectors

$$df_1(x), \dots, df_c(x)$$

linearly independent at every x, so no nonzero $q \in V$ has $dq(x) = 0$. Thus γ is everywhere defined.

For quadrics, the coordinates of $dq(x)$ are bihomogeneous of degree $(1,1)$ in (x, q), giving the line-bundle identity.

Since

$$\dim(X \times \mathbf{P}^{c-1}) = d + c - 1 = N - 1,$$

the top self-intersection is

$$\binom{N-1}{d}\left(\int_X H^d\right)\left(\int_{\mathbf{P}^{c-1}} L^{c-1}\right).$$

Here

$$\int_X H^d = 2^c, \qquad \int_{\mathbf{P}^{c-1}} L^{c-1} = 1,$$

which proves the formula. A morphism whose pulled-back hyperplane class has positive top self-intersection has image of full source dimension. □

Theorem NC.15.11 (Universal nonvanishing of the quadratic adjugate)**.** *For every smooth quadratic complete intersection in the declared scope,*

$$\boxed{\alpha_X \neq 0.} \tag{NC.15.12}$$

Consequently

$$\boxed{H^0\Big(X, \operatorname{Sym}^2 T_X(2c-4)\Big) \neq 0.} \tag{NC.15.13}$$

Proof. By Lemma NC.15.10, the conormal Gauss map is generically finite. In characteristic zero it is generically separable, so its differential has full rank at a general conormal pair $(x, [q])$.

The vertical differential has rank $c - 1$. Modulo the vertical directions, the horizontal differential is the second fundamental form

$$h_q(x) : T_x X \to T_x^* X(2).$$

Therefore $h_q(x)$ has full rank d at a general conormal pair. Its adjugate is then nonzero.

The assertion descends from the algebraic closure because coherent H^0 commutes with field extension and the map α_X is defined over the original field. □

Corollary NC.15.12 (Universal critical intrinsic tensor)**.** *For every smooth quadratic complete intersection,*

$$\boxed{H^0\Big(X, \operatorname{Sym}^2 T_X(2c-3)\Big) \neq 0.} \tag{NC.15.14}$$

Proof. Choose

$$0 \neq s \in H^0\Big(X, \operatorname{Sym}^2 T_X(2c-4)\Big)$$

and a nonzero

$$\ell \in H^0(X, \mathcal{O}_X(1)).$$

A smooth positive-dimensional complete intersection is geometrically integral, hence multiplication by the nonzero section ℓ cannot annihilate s. Thus

$$0 \neq \ell s \in H^0\Big(X, \operatorname{Sym}^2 T_X(2c-3)\Big).$$

□

NC.15.6 Universal quadratic stress/projective-tensor sequence

Set $r = 2c - 3$. Define

$$\rho_r : \operatorname{Der}_k(A)_r \longrightarrow \mathcal{S}_{2,r}(A), \qquad u \longmapsto xu^T + ux^T.$$

Theorem NC.15.13 (Quadratic stress/projective-tensor exact sequence)**.** *Assume the critical T^1-vanishing*

$$(T^1_{A/k})_r = 0.$$

Then

$$\boxed{0 \longrightarrow \operatorname{Der}_k(A)_r \xrightarrow{\rho_r} \mathcal{S}_{2,r}(A) \xrightarrow{\Psi_{2,r}} H^0\Big(X, \operatorname{Sym}^2 T_X(r)\Big) \longrightarrow 0.} \tag{NC.15.15}$$

Proof. Restrict the Euler sequence to X and take its symmetric square. Since the Euler subbundle has rank one, after twisting by r one gets

$$0 \to O_X(r+1)^{N+1} \to O_X(r+2)^{\binom{N+2}{2}} \to \operatorname{Sym}^2 T_{\mathbf{P}^N}|_X(r) \to 0.$$

The required $H^1(X, O_X(r+1))$ vanishes: for $d \geq 2$ by the ACM property, and for $d = 1$ by Serre duality and positivity of $r+1$. Therefore every global ambient projective symmetric tensor admits a coefficient-matrix lift

$$B_0 \in \operatorname{Sym}_{N+1}(A_{r+2}).$$

Take

$$s \in H^0(X, \operatorname{Sym}^2 T_X(r))$$

and choose such a lift B_0. Tangency of s implies

$$B_0 \nabla f_\alpha = x\lambda_\alpha$$

for suitable

$$\lambda_\alpha \in A_{r+2}.$$

The critical T^1-vanishing gives

$$u \in A_{r+1}^{N+1}$$

such that

$$u \cdot \nabla f_\alpha = -\lambda_\alpha \qquad (1 \leq \alpha \leq c).$$

Set

$$B := B_0 + xu^T + ux^T.$$

Since

$$x \cdot \nabla f_\alpha = 2f_\alpha = 0 \quad \text{in } A,$$

we get

$$B \nabla f_\alpha = 0.$$

Thus B is a critical stress with projective symbol s, proving surjectivity.

If a stress B has zero projective symbol, the symmetric Euler sequence gives

$$B = xu^T + ux^T$$

for some $u \in A_{r+1}^{N+1}$. Tangency yields

$$0 = B \nabla f_\alpha = x(u \cdot \nabla f_\alpha).$$

The homogeneous coordinate ring of a smooth positive-dimensional complete intersection is a domain, so

$$u \cdot \nabla f_\alpha = 0$$

for every α, hence

$$u \in \operatorname{Der}_k(A)_r.$$

Finally,

$$xu^T + ux^T = 0$$

forces $u = 0$ by examining diagonal entries and using $\operatorname{char} k = 0$. □

Corollary NC.15.14 (Intrinsic quadratic critical defect)**.** *Assume the NC.3 graded Kersken-type derivation-generation conclusion. Then*

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0\Big(X, \operatorname{Sym}^2 T_X(2c-3)\Big).} \tag{NC.15.16}$$

In particular,

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \neq 0.} \tag{NC.15.17}$$

NC.15.7 All-quadratic strong escape

Theorem NC.15.15 (All smooth quadratic complete intersections: strong order-two escape)**.** *Assume:*

(i) k *has characteristic zero;*

(ii) $c \geq 2$*;*

(iii) $A = k[x_0, \dots, x_N]/(f_1, \dots, f_c)$ *is a homogeneous complete intersection of quadrics;*

(iv) $X = \operatorname{Proj} A$ *is smooth and* $\dim X \geq 1$*;*

(v) the imported Brückmann numerical transport theorem is available at the strength stated in Theorem NC.15.6;

(vi) the NC.3 Kersken-type graded derivation-generation theorem is available at its exact invoked source strength.

Then there exists

$$\boxed{D \in D^2_{A/k,\,2c-3} \setminus A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.15.18}$$

Consequently

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.15.19}$$

Proof. Theorem NC.15.8 closes Gate Q. Corollary NC.15.12 closes Gate N. Theorem NC.15.13 lifts the nonzero critical projective tensor to a quantizable critical stress.

Since

$$2c - 3 < 2(c-1) = 2m,$$

the NC.4 Critical Window forces the projective second symbol of every order-two element generated by first-order derivations to vanish at this degree. The lifted operator has nonzero projective second symbol and therefore escapes the full derivation-generated algebra. □

NC.15.8 Low-dimensional compatibility checks

Corollary NC.15.16 (Quadratic complete-intersection curves)**.** *If* $d = 1$*, then*

$$\mathrm{Sym}^2 T_X(2c-3) \cong O_X(1),$$

and

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}}(A) = h^0(X, O_X(1)) = c + 2.} \tag{NC.15.20}$$

For $c = 3$*, the defect dimension is* 5*.*

Remark NC.15.17 (Recovery of NC.14)**.** For $c = 3$ and $d = 2$,

$$r_{\mathrm{crit}} = 3, \qquad \mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0(X, \mathrm{Sym}^2 T_X(3)).$$

NC.14 supplies the stronger numerical estimate

$$h^0(X, \mathrm{Sym}^2 T_X(3)) \geq 18.$$

Thus NC.14 is retained as the refined K3 numerical layer inside the quadratic completion theorem.

NC.15.9 NC.15 status freeze

Declaration NC.15.18 (Re-audited status)**.** The following parts are internal proof candidates:

(i) the Vandermonde smooth control;

(ii) the quadratic support-saturation theorem;

(iii) the critical T^1/tangent-cohomology identification;

(iv) the conormal top-intersection calculation;

(v) nonvanishing of the intrinsic adjugate;

(vi) the critical stress/projective-tensor exact sequence;

(vii) the NC.4/NC.7/NC.8 assembly.

The following interfaces remain imported:

(a) Brückmann twisted-Hodge numerical transport;

(b) Kersken derivation generation and the NC.3 local-to-graded transfer.

Accordingly the assembled theorem is classified

$$\boxed{\text{C2-high}}$$

until source binding and prior-art review are closed.

NC.16 General-Degree Complete Intersections: Intrinsic Critical Bundles, Curve Closure, and the Order-Selection Frontier

NC.16.1 General homogeneous setup

Let

$$P = k[x_0, \dots, x_N], \qquad A = P/(f_1, \dots, f_c),$$

where

$$c \geq 2, \qquad \deg f_i = d_i \geq 2,$$

and

$$X = \operatorname{Proj} A \subset \mathbf{P}^N$$

is smooth of positive dimension

$$d = N - c.$$

Put

$$S := \sum_{i=1}^{c} d_i.$$

The NC.3 threshold is

$$\boxed{m = S - (c+1).} \tag{NC.16.1}$$

Hence the order-two critical degree is

$$\boxed{r_2 = 2m - 1.} \tag{NC.16.2}$$

Adjunction gives

$$K_X \cong \mathcal{O}_X(S - N - 1).$$

Since $N = d + c$,

$$S - N - 1 = S - (c+1) - d = m - d.$$

Thus

$$\boxed{K_X \cong \mathcal{O}_X(m - d).} \tag{NC.16.3}$$

NC.16.2 Universal critical-bundle identity

Theorem NC.16.1 (Degree-independent intrinsic critical bundle)**.** *For every smooth homogeneous complete intersection in the declared scope,*

$$\boxed{\operatorname{Sym}^2 T_X(2m-1) \cong \operatorname{Sym}^2 \Omega_X^{d-1}(2d-1).} \tag{NC.16.4}$$

Proof. Using

$$T_X \cong \Omega_X^{d-1} \otimes K_X^{-1}$$

and (NC.16.3),

$$\begin{aligned} \operatorname{Sym}^2 T_X(2m-1) &\cong \operatorname{Sym}^2 \Omega_X^{d-1} \otimes O_X(-2(m-d)+2m-1) \\ &= \operatorname{Sym}^2 \Omega_X^{d-1}(2d-1). \end{aligned}$$

□

Remark NC.16.2 (Second structural cancellation)**.** The intrinsic critical projective bundle depends on the projective dimension d, but its twist is independent of the codimension and of the multidegree. Degree dependence survives only in the quantization gate and in the construction of nonzero sections.

NC.16.3 General critical reduction

Lemma NC.16.3 (General critical T^1/tangent-cohomology identification)**.** *For every smooth homogeneous complete intersection with $c \geq 2$, $d \geq 1$, and $d_i \geq 2$,*

$$\boxed{(T^1_{A/k})_{2m-1} \cong H^1(X, T_X(2m-1)).} \tag{NC.16.5}$$

Proof. The Euler and normal sequences are twisted by $r_2 = 2m-1$. Because

$$m = \sum_i d_i - (c+1) \geq 2c - (c+1) = c - 1 \geq 1,$$

we have $r_2 \geq 1$.

For $d \geq 3$, the necessary H^1 and H^2 line-bundle terms vanish by the ACM property. For $d = 2$,

$$H^2(X, O_X(r_2)) \cong H^0(X, K_X(-r_2))^\vee = H^0(X, O_X(-m-2))^\vee = 0.$$

For $d = 1$, the corresponding H^1 terms vanish by Serre duality because all relevant dual twists are negative. The cokernel of the graded Jacobian map is therefore $H^1(X, T_X(r_2))$. □

Theorem NC.16.4 (General order-two completion criterion)**.** *Assume*

$$\boxed{(T^1_{A/k})_{2m-1} = 0.} \tag{NC.16.6}$$

Then

$$\boxed{0 \to \operatorname{Der}_k(A)_{2m-1} \to \mathcal{S}_{2,2m-1}(A) \to H^0\Big(X, \operatorname{Sym}^2 T_X(2m-1)\Big) \to 0.} \tag{NC.16.7}$$

Under the NC.3 and NC.4 hypotheses,

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0\Big(X, \operatorname{Sym}^2 \Omega_X^{d-1}(2d-1)\Big).} \tag{NC.16.8}$$

Consequently,

$$(T^1)_{2m-1} = 0 \quad \textit{and} \quad H^0\Big(X, \operatorname{Sym}^2 \Omega_X^{d-1}(2d-1)\Big) \neq 0$$

imply strong order-two escape.

Proof. The proof is the degree-sensitive version of Theorem NC.15.13. For the equation f_i of degree d_i, projective tangency gives

$$B_0 \nabla f_i = x\lambda_i, \qquad \lambda_i \in A_{2m-1+d_i}.$$

The vanishing of the critical T^1 slice produces

$$u \in A_{2m}^{N+1}$$

with

$$u \cdot \nabla f_i = -\lambda_i.$$

The radial correction

$$B = B_0 + xu^T + ux^T$$

then kills all normal contractions because

$$x \cdot \nabla f_i = d_i f_i = 0 \quad \text{in } A.$$

The kernel is again the radial image. Finally use Theorem NC.16.1 and the NC.4 Critical Window. □

NC.16.4 Complete closure in projective dimension one

Assume $d = 1$. Then

$$K_X = O_X(m-1), \qquad T_X = K_X^{-1} = O_X(1-m).$$

Therefore

$$T_X(2m-1) = O_X(m).$$

Theorem NC.16.5 (Universal critical T^1-vanishing for CI curves)**.** *For every smooth homogeneous complete-intersection curve with $c \geq 2$ and $d_i \geq 2$,*

$$\boxed{(T^1_{A/k})_{2m-1} = 0.} \tag{NC.16.9}$$

Proof. By Lemma NC.16.3,

$$(T^1_{A/k})_{2m-1} \cong H^1(X, T_X(2m-1)) = H^1(X, O_X(m)).$$

Serre duality gives

$$H^1(X, O_X(m))^\vee \cong H^0(X, K_X(-m)) = H^0(X, O_X(-1)) = 0.$$

□

For $d = 1$, Theorem NC.16.1 gives

$$\operatorname{Sym}^2 T_X(2m-1) \cong O_X(1).$$

Because the ideal contains no linear form,

$$h^0(X, O_X(1)) = N + 1 = c + 2.$$

Theorem NC.16.6 (All smooth homogeneous complete-intersection curves: strong order-two escape)**.** *Assume the NC.3 graded Kersken-type derivation-generation conclusion. Then every smooth homogeneous complete-intersection curve*

$$X \subset \mathbf{P}^{c+1}$$

of arbitrary multidegree

$$(d_1, \ldots, d_c), \qquad d_i \geq 2,$$

satisfies

$$\boxed{\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0(X, O_X(1)),} \tag{NC.16.10}$$

and

$$\boxed{\dim \mathfrak{Q}_2^{\mathrm{crit}}(A) = c + 2.} \tag{NC.16.11}$$

Hence there exists

$$\boxed{D \in D^2_{A/k,\,2m-1} \setminus A\langle \mathrm{Der}_k(A)\rangle,} \tag{NC.16.12}$$

so

$$\boxed{D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.16.13}$$

NC.16.5 Equidegree adjugates and the matching index

Assume temporarily

$$d_1 = \cdots = d_c = a \geq 2.$$

Then

$$m = c(a-1) - 1, \qquad r_2 = 2c(a-1) - 3. \tag{NC.16.14}$$

Let

$$V = \mathrm{Span}\{f_1, \ldots, f_c\}.$$

For $q \in V$, the second fundamental form is

$$h_q : T_X \longrightarrow \Omega^1_X(a).$$

Its adjugate defines

$$\alpha_{X,a} : \mathrm{Sym}^{d-1} V \longrightarrow H^0\Big(X, \mathrm{Sym}^2 T_X(t_{\mathrm{cof}})\Big),$$

where

$$\boxed{t_{\mathrm{cof}} = a(d+2c-1) - 2(d+c+1).} \tag{NC.16.15}$$

Derivation of the twist. The symmetric form h_q is a section of

$$\mathrm{Sym}^2 \Omega^1_X \otimes O_X(a).$$

Its adjugate lies in

$$\mathrm{Sym}^2 T_X \otimes K_X^2 \otimes O_X(a(d-1)).$$

For an equidegree-a complete intersection,

$$K_X = O_X(ca - d - c - 1).$$

Thus the total twist is

$$2(ca - d - c - 1) + a(d-1),$$

which is (NC.16.15). □

Lemma NC.16.7 (Equidegree conormal nonvanishing)**.** *For every smooth equidegree complete intersection,*

$$\boxed{\alpha_{X,a} \neq 0.} \tag{NC.16.16}$$

Proof. The conormal Gauss map

$$\gamma_a : X \times \mathbf{P}^{c-1} \to (\mathbf{P}^N)^\vee$$

is defined by derivatives of bidegree $(a-1, 1)$. Therefore

$$\gamma_a^* O(1) = O_X(a-1) \boxtimes O_{\mathbf{P}^{c-1}}(1).$$

Its top self-intersection equals

$$\binom{N-1}{d}(a-1)^d \deg(X) = \binom{d+c-1}{c-1}(a-1)^d a^c > 0.$$

Hence γ_a is generically finite. As in the quadratic case, generic separability in characteristic zero forces a general scalar second fundamental form h_q to have full rank, so its adjugate is nonzero. □

Definition NC.16.8 (Order-two Cofactor–Critical Matching Index)**.** Define

$$\boxed{\delta_2(a,d) := t_{\mathrm{cof}} - r_2.} \tag{NC.16.17}$$

Then

$$\boxed{\delta_2(a,d) = (a-2)d - a + 1.} \tag{NC.16.18}$$

Proof. Subtract

$$r_2 = 2c(a-1) - 3$$

from (NC.16.15). The codimension cancels:

$$\delta_2 = ad - a - 2d + 1 = (a-2)d - a + 1.$$

□

Proposition NC.16.9 (Sign classification)**.** *For $a \geq 2$, $d \geq 1$,*

$$\boxed{\begin{aligned} a = 2 &\Longrightarrow \delta_2 = -1 \quad \textit{for every } d, \\ d = 1 &\Longrightarrow \delta_2 = -1 \quad \textit{for every } a, \\ (a,d) = (3,2) &\Longrightarrow \delta_2 = 0, \\ a \geq 3,\ d \geq 2,\ (a,d) \neq (3,2) &\Longrightarrow \delta_2 > 0. \end{aligned}} \tag{NC.16.19}$$

Remark NC.16.10 (Architectural meaning)**.** If $\delta_2 < 0$, the adjugate is born below the NC.4 order-two critical ceiling and can be moved to the critical degree by multiplying with a nonzero section of $O_X(-\delta_2)$. If $\delta_2 = 0$, it is born exactly at the ceiling. If $\delta_2 > 0$, this specific second-order construction appears too late for the NC.4 order-two route.

NC.16.6 The same index on the diagonal Quantization Gate

For an equidegree diagonal control put

$$f_\alpha = \sum_{j=0}^{N} (\lambda_j)_\alpha x_j^a,$$

where every c of the $\lambda_j \in k^c$ are linearly independent. The same support argument as in NC.15 proves smoothness.

At critical internal degree r_2, the Jacobian output degree is

$$\boxed{s = r_2 + a = 2c(a-1) + a - 3.} \tag{NC.16.20}$$

Set

$$M = A^c \Big/ \sum_{j=0}^{N} A\, x_j^{a-1} \lambda_j.$$

Call an index j *heavy* for a monomial if its exponent is at least $a-1$.

If a monomial has at least c heavy indices, its vector class in M vanishes.

Suppose it has at most $c-1$ heavy indices and no heavy exponent is at least $2a-1$. Then a heavy exponent is at most $2a-2$, while a nonheavy exponent is at most $a-2$. Since there are $N+1 = d+c+1$ variables, the maximal degree under those constraints is

$$\boxed{s_{\max} = (c-1)(2a-2) + (d+2)(a-2).} \tag{NC.16.21}$$

A direct calculation gives

$$\boxed{s - s_{\max} = -ad + a + 2d - 1 = -\delta_2(a,d).} \tag{NC.16.22}$$

Theorem NC.16.11 (Matching-index support criterion)**.** *If*

$$\delta_2(a,d) < 0,$$

then the diagonal control satisfies

$$\boxed{(T^1_{A/k})_{r_2} = 0.} \tag{NC.16.23}$$

Assuming the imported Brückmann numerical transport theorem, the same vanishing holds for every smooth equidegree complete intersection of the same numerical type.

Proof. By (NC.16.22),

$$\delta_2 < 0 \quad \Longrightarrow \quad s > s_{\max}.$$

Hence every degree-s monomial vector with fewer than c heavy indices must contain a heavy variable x_i with exponent at least $2a - 1$.

Choose a c-element basis set B containing all current heavy indices and i. The vector equations

$$\sum_j \lambda_j x_j^a = 0$$

solve

$$x_i^a = \sum_{\ell \notin B} b_{i\ell} x_\ell^a.$$

After substitution, the exponent of x_i remains at least $a - 1$, while each nonzero term acquires a new heavy variable outside B. Thus the number of heavy indices strictly increases without changing total degree.

Iteration reaches c heavy variables, where the coefficient vector is killed by the c corresponding basis vectors λ_j. Hence the critical cokernel vanishes.

Numerical transport then propagates the zero dimension across the smooth equidegree family. □

Remark NC.16.12 (Re-audit correction)**.** The correct support ceiling is

$$(c - 1)(2a - 2) + (d + 2)(a - 2),$$

not

$$a(c - 1) + (N - c + 2)(a - 2).$$

The corrected bound is essential because it exactly preserves the identity

$$s - s_{\max} = -\delta_2(a, d).$$

NC.16.7 First post-quadratic boundary: cubic surfaces

By Proposition NC.16.9, the unique equidegree point with

$$\delta_2 = 0$$

is

$$\boxed{(a, d) = (3, 2).} \tag{NC.16.24}$$

At this boundary:

$$t_{\text{cof}} = r_2,$$

so the intrinsic adjugate is already born at the order-two critical degree. However

$$s = s_{\max},$$

so the support-saturation proof no longer forces the critical T^1 slice to vanish.

Policy 22 (No boundary overclaim)**.** *NC.16 does not claim*

$$(T^1_{A/k})_{r_2} \neq 0$$

for cubic complete-intersection surfaces. It only records that the universal support proof reaches equality and therefore ceases to prove vanishing.

The correct next object is the actual obstruction class

$$\omega_{r_2}(\alpha_{X,3}),$$

not the entire T^1 space. Even if $(T^1)_{r_2}$ is nonzero, the adjugate may still lie in the kernel of the Hessian obstruction.

NC.16.8 Beyond the order-two wall

For general order $q \geq 2$, NC.4 supplies the ceiling

$$\boxed{r_q^{\mathrm{crit}} = qm - 1.} \tag{NC.16.25}$$

For a genuine order-q intrinsic tensor construction of internal degree t_q, define

$$\boxed{\delta_q := t_q - (qm - 1).} \tag{NC.16.26}$$

A viable higher-order route requires three separate gates:

$$\boxed{\begin{array}{ll} \mathsf{G}_q & \delta_q \leq 0 \quad \text{(degree gate)}, \\ \mathsf{Q}_q & \text{the projective } q\text{-symbol quantizes}, \\ \mathsf{N}_q & \text{the resulting projective } q\text{-symbol is nonzero.} \end{array}} \tag{NC.16.27}$$

Policy 23 (No order-relabeling shortcut)**.** *An order-two tensor lying above the order-two ceiling cannot be reclassified as an order-q symbol merely because* $qm - 1$ *is larger. A genuine order-q symbol and a genuine order-q quantization theorem are required.*

NC.16.9 Mixed multidegrees

For a non-equidegree regular sequence

$$(d_1, \ldots, d_c),$$

partition the equations into equal-degree blocks

$$V_e := \mathrm{Span}\{f_i : \deg f_i = e\}.$$

Second-fundamental-form adjugates may be formed inside a block V_e, but different degree blocks carry different twists. They cannot be added without an explicit degree correction.

The general multidegree frontier therefore separates:

$$\boxed{\begin{array}{ll} \mathsf{B}_1 & \text{single-degree-block adjugates}, \\ \mathsf{B}_2 & \text{mixed-degree Schur/tensor constructions}, \\ \mathsf{B}_3 & \text{higher-order intrinsic tensors}, \\ \mathsf{B}_4 & \text{order-}q\text{ quantization complexes.} \end{array}} \tag{NC.16.28}$$

NC.16.10 NC.16 architectural closure

Theorem NC.16.13 (NC.16 closure)**.** *Under the imported interfaces already declared, the current homogeneous-complete-intersection landscape is:*

source class	*status*
$c \geq 2,\ d = 1,$ *arbitrary multidegree*	*strong order-two escape*
$c \geq 2,\ d \geq 1,\ (d_1, \ldots, d_c) = (2, \ldots, 2)$	*NC.15 strong order-two escape*
$c \geq 2,\ d \geq 2,\ \max d_i \geq 3$	*remaining internal frontier.*

(NC.16.29)

The first equidegree order-two boundary is

$$\boxed{\textit{smooth complete-intersection surfaces of cubics.}}$$

Declaration NC.16.14 (Successor obligations)**.** The next mathematical order is:

(1) compute the critical T^1 slice for smooth complete intersections of cubics in projective dimension two;

(2) compute the specific obstruction

$$\omega_{r_2}(\alpha_{X,3});$$

(3) determine whether the boundary adjugate quantizes even when the entire critical T^1 slice does not vanish;

(4) construct the order-q analogue of the NC.7 Quantization Gate;

(5) search for higher-adjugate or Schur-functor tensors with

$$t_q < qm$$

in the region $\delta_2 > 0$.

Declaration NC.16.15 (NC.16 claim ceiling)**.** NC.16 does not claim the general Nakai conjecture. It closes all smooth homogeneous complete-intersection curves under the NC.3 interface and gives a re-audited order-selection architecture for the remaining homogeneous-ICIS problem. Any literature-priority statement remains prohibited until the dedicated complete-intersection prior-art audit is finished.

NC.15–NC.16 re-audit notes

1. The previous informal attribution of a general complete-intersection cofactor theorem to Beauville–Etesse–Höring–Liu–Voisin was too broad. Their 2024 paper treats intersections of two quadrics. The general equidegree adjugate used here is therefore proved internally from the conormal Gauss map; the external paper is retained only as related $c = 2$ context.

2. The equidegree support ceiling has been corrected to

$$s_{\max} = (c-1)(2a-2) + (d+2)(a-2).$$

With this correction, the identity

$$s - s_{\max} = -\delta_2(a,d)$$

is exact.

3. Universal quadratic T^1-vanishing is not presented as a purely internal theorem. Its propagation from the exact Vandermonde control to all smooth members consumes the imported Brückmann numerical twisted-Hodge theorem.

4. The final escape implication remains conditional on the exact NC.3 Kersken local-to-graded interface.

5. No general-Nakai or literature-novelty claim is made.

NC.17 General Homogeneous-ICIS Completion: Strong Escape Master Theorem

Source abstract

This integrated Appendix NC consolidates the re-audited NC.1–NC.14 foundation, the post-NC.14 finite-order refinements, the threefold/fourfold asymptotic completion, and the subsequent general-dimensional proof architecture into one standalone theorem package.

Let

$$A = k[x_0, \dots, x_N]/(f_1, \dots, f_c), \qquad c \geq 2, \qquad \deg f_i = d_i \geq 2,$$

be a standard homogeneous complete intersection over a characteristic-zero field. Assume that the equations form a homogeneous regular sequence, that A is a domain, and that

$$X := \operatorname{Proj} A \subset \mathbf{P}^N_k$$

is smooth of positive dimension

$$d := N - c \geq 1.$$

Put

$$m := \sum_{i=1}^{c} d_i - (c+1).$$

Then $m \geq 1$, the punctured affine cone is smooth, and the cone is a normal isolated complete-intersection singularity at its vertex.

The proof has four independent layers.

First, the Kersken Euler-plus-trivial derivation theorem is transferred from the local ICIS at the vertex to the global standard-graded module by localization and a graded detection lemma. Every basic trivial derivation has internal degree exactly m. Projectivization then scalarizes every Euler factor. Consequently arbitrary long words in first-order derivations have projective order bounded solely by the number of trivial factors. This gives the robust Critical Window

$$r < qm \quad \Longrightarrow \quad \sigma_q^X(P) = 0 \qquad (P \in \Delta_A \cap D^q_{A/k} \text{ homogeneous of degree } r),$$

with maximal critical degree $qm - 1$. No bounded-word-length or affine PBW hypothesis is used.

Second, for

$$E_d := T_X(m) \cong \Omega_X^{d-1}(d),$$

the bundle E_d is globally generated. For the quotient projectivization

$$\pi : Z := \mathbf{P}_X(E_d) \to X, \qquad L := \mathcal{O}_Z(1),$$

we fix the convention $\pi_* L^q = \operatorname{Sym}^q E_d$. If

$$e_i = d_i - 1, \qquad S = \sum_i e_i,$$

then the complete-symmetric generating series is

$$\sum_{j \geq 0} h_j(E_d) t^j = \frac{(1-(S-1)Ht)\prod_i (1-(S+e_i)Ht)}{(1-SHt)^{c+d+1}}, \qquad H = c_1(\mathcal{O}_X(1)).$$

The coefficient $h_d(E_d)$ admits an exact two-layer decomposition

$$h_d(E_d) = (A_d + B_{d-1})H^d$$

in which every monomial coefficient of A_d is nonnegative and every monomial coefficient of B_{d-1} is strictly positive. Therefore

$$\int_X h_d(E_d) > 0$$

for all $d \geq 1$, $c \geq 2$, and all multidegrees $d_i \geq 2$. Since L is nef, this proves that L is big.

Third, for $d \geq 2$, the critical projective symbol space satisfies

$$h^0(X, \operatorname{Sym}^q T_X(qm-1)) = \frac{L^{2d-1}}{(2d-1)!} q^{2d-1} + O(q^{2d-2}).$$

All lower symbol layers at distance at least $d+2$ from the top vanish by Kodaira. The remaining $d+1$ boundary layers are each $O(q^{2d-2})$ by the nef higher-cohomology estimate. These estimates propagate through the cone-tangent filtration and the principal-symbol filtration on the punctured cone. Thus the total quantization obstruction is subleading, and a nonzero projectively visible quantizable critical symbol survives for all sufficiently large q.

Fourth, normality and Hartogs extension extend the resulting operator across the cone vertex. The Critical Window then forces strong escape:

$$\boxed{D_{A/k} \neq A\langle \operatorname{Der}_k(A) \rangle.}$$

For curves, a separate exact order-two proof is given. Hence the result holds for every smooth positive-dimensional standard homogeneous complete intersection of codimension at least two in the stated source class.

This is a scoped Nakai theorem for homogeneous complete intersections, not a proof of the unrestricted Nakai conjecture for arbitrary finite-type characteristic-zero algebras. The assembled result remains labelled C2-high pending independent line-by-line audit and a complete prior-art review. No literature-priority claim is made.

Declaration NC.17.1 (Re-audited ownership status)**.** NC.17 is retained as the strongest AK/NC internal positive Escape theorem candidate. Its mathematical claim is independent of the Family 048 collapse witness. Its publication ceiling remains C2-high pending independent line-by-line refereeing, formal verification where feasible, and a dedicated prior-art audit. This edition makes no priority claim beyond the cited literature.

NC.17.1 Scope, authority, and integration policy

NC.17.1.1 Target and source class

Definition NC.17.2 (Nakai target)**.** For a finitely generated k-algebra A, let

$$D_{A/k} := \bigcup_{q\geq 0} D^q_{A/k}, \qquad \Delta_A := A\langle \mathrm{Der}_k(A)\rangle \subseteq D_{A/k}.$$

The Nakai implication asks whether

$$D_{A/k} = \Delta_A \quad \Longrightarrow \quad A \text{ is regular.}$$

For a declared singular source class, its contrapositive target is

$$A \text{ singular} \quad \Longrightarrow \quad D_{A/k} \neq \Delta_A.$$

Definition NC.17.3 (Integrated NC source class)**.** Fix a field k of characteristic zero and

$$P = k[x_0, \ldots, x_N]$$

with the standard grading. Let

$$I = (f_1, \ldots, f_c), \qquad c \geq 2,$$

be generated by a homogeneous regular sequence with

$$\deg f_i = d_i \geq 2,$$

and set

$$A = P/I, \qquad X = \mathrm{Proj}\, A, \qquad d = \dim X = N - c \geq 1.$$

Throughout the master theorem we assume that A is a domain and X is smooth over k.

Policy 24 (Frozen-v21 boundary)**.** *Appendix NC is a post-v21 research supplement. Nothing proved here silently modifies AK-HPDST v21.0.0, its Claim Register, or its stable Core. Promotion requires a separate successor-release admission step.*

Policy 25 (Evidence and claim ceiling)**.** *Algebraic identities proved internally are used at their displayed strength. Imported theorems are consumed only at the hypotheses explicitly bound here. Finite computation, numerical replay, or successful XeLaTeX compilation never substitutes for a missing theorem edge. The integrated master theorem is reported as* C2-high *until independent publication-grade audit and prior-art review are complete.*

NC.17.1.2 Strong escape versus order-wise escape

Definition NC.17.4 (Strong escape)**.** A strong escape is an actual operator

$$D \in D_{A/k} \setminus \Delta_A.$$

This is stronger than exhibiting an operator outside a sector generated by at most a bounded number of first-order factors.

Policy 26 (No weak-to-strong promotion)**.** *No argument may identify word length with differential order. In particular, arbitrarily long words whose affine differential order drops after commutation or cancellation must be excluded. The Critical Window below is designed precisely to close this issue after projectivization.*

NC.17.1.3 Legacy-to-integrated map

The integrated proof retains the earlier research as follows.

Legacy layer	Retained mathematical role	Integrated status
NC.1–NC.4	Governance, graded derivation structure, projectivization, robust Critical Window	Core proof gate
NC.5–NC.13	Explicit type-$(2, d)$, Quantized Stress, diagonal/non-diagonal controls, cofactor and determinantal lanes	Refinements / controls
NC.14	All-smooth $(2, 2, 2)$ K3 order-two theorem; $h^0 \geq 18$	Quantitative refinement
NC.15	All smooth positive-dimensional quadratic CI, order two	Finite-order refinement
NC.16 curves / surfaces	Exact or finite-order closure and negative cubic order-two control	Refinements / diagnostics
NC.16 threefold/fourfold	First asymptotic q^{2d-1} versus q^{2d-2} proofs	Special cases of master proof
NC.17 candidate	General top positivity and general asymptotic quantization	Re-audited and integrated here

The earlier family-specific routes are not erased. They provide independent checks, low-order witnesses, and failure data. They are simply no longer proof gates for the all-dimensional homogeneous-CI theorem.

NC.17.2 Graded ICIS derivations and the robust Critical Window

NC.17.2.1 The derivation threshold

Define

$$\boxed{m := \sum_{i=1}^{c} d_i - (c + 1).} \tag{NC.2.1}$$

Because $c \geq 2$ and $d_i \geq 2$,

$$m \geq c - 1 \geq 1.$$

Let

$$E := \sum_{j=0}^{N} x_j \partial_{x_j}$$

be the Euler derivation.

For each $(c + 1)$-tuple of distinct coordinate indices $\nu = (\nu_0, \ldots, \nu_c)$, define the standard trivial derivation by expanding the determinant

$$\delta_\nu = \det \begin{pmatrix} \partial_{\nu_0} & \cdots & \partial_{\nu_c} \\ \partial_{\nu_0} f_1 & \cdots & \partial_{\nu_c} f_1 \\ \vdots & & \vdots \\ \partial_{\nu_0} f_c & \cdots & \partial_{\nu_c} f_c \end{pmatrix}.$$

Lemma NC.17.5 (Degree of trivial derivations)**.** *Every nonzero basic trivial derivation is homogeneous of internal degree*

$$\boxed{\deg \delta_\nu = m.} \tag{NC.2.2}$$

Proof. Each coefficient is a $c \times c$ Jacobian minor of degree

$$\sum_{i=1}^{c} (d_i - 1) = \sum_i d_i - c.$$

The factor ∂_{x_j} has internal degree -1, so every summand has degree

$$\sum_i d_i - c - 1 = m.$$

□

NC.17.2.2 Imported local theorem and exact graded transfer

Theorem NC.17.6 (Euler-plus-trivial generation; imported local interface)**.** *Let $(R, \mathfrak{m})$ be a quasihomogeneous normal isolated complete intersection singularity over a characteristic-zero field in the Kersken regime. Then*

$$\mathrm{Der}_k(R)$$

is generated as an R-module by an Euler derivation and the trivial Jacobian derivations.

Remark NC.17.7 (Source binding)**.** This is Kersken's theorem as explicitly restated by Granger–Schulze: for a quasihomogeneous normal ICIS, the module of derivations is generated by the Euler derivation and the trivial derivations. Herzog–Martsinkovsky, Corollary 4.3 and Remark 4.4, give an algebraic characteristic-zero formulation in which the trivial derivations together with the Euler derivation minimally generate the derivation module. The local theorem is not silently identified with the global graded module; that transfer is proved next.

Let

$$\mathfrak{m} := A_+ = \bigoplus_{n>0} A_n.$$

Lemma NC.17.8 (Localization of derivations)**.** *For any multiplicative set $S \subset A$,*

$$S^{-1} \mathrm{Der}_k(A) \cong \mathrm{Der}_k(S^{-1}A).$$

In particular,

$$\mathrm{Der}_k(A)_{\mathfrak{m}} \cong \mathrm{Der}_k(A_{\mathfrak{m}}).$$

Proof. Because A is finite type over k, $\Omega^1_{A/k}$ is finitely presented and

$$\mathrm{Der}_k(A) = \mathrm{Hom}_A(\Omega^1_{A/k}, A).$$

Kähler differentials commute with localization, and localization of Hom out of a finitely presented module gives the claim. □

Lemma NC.17.9 (The irrelevant maximal ideal detects graded modules)**.** *If M is a graded A-module and*

$$M_{\mathfrak{m}} = 0,$$

then $M = 0$.

Proof. Suppose that $0 \neq u \in M$ is homogeneous of degree a. Since $u/1 = 0$ in $M_{\mathfrak{m}}$, some $s \in A \setminus \mathfrak{m}$ satisfies $su = 0$. Write

$$s = s_0 + s_+, \qquad s_0 \in k^\times, \qquad s_+ \in \mathfrak{m}.$$

The degree-a part of su is $s_0 u$, hence $u = 0$, a contradiction. □

Proposition NC.17.10 (Graded Euler-plus-trivial generation)**.** *Assume the local ring $A_{\mathfrak{m}}$ satisfies Theorem NC.17.6. Let $T \subset \mathrm{Der}_k(A)$ be the graded A-submodule generated by the global trivial derivations. Then*

$$\boxed{\mathrm{Der}_k(A) = AE + T.} \tag{NC.2.3}$$

Every basic trivial generator has internal degree m.

Proof. After localization at $\mathfrak{m}$, the imported theorem gives

$$(\mathrm{Der}_k(A)/(AE + T))_{\mathfrak{m}} = 0.$$

The quotient is graded, so the preceding detection lemma forces the quotient to vanish globally. The degree assertion was proved above. □

NC.17.2.3 Internal-degree factor budget

Lemma NC.17.11 (Homogeneous generator reduction)**.** *Every homogeneous element of Δ_A is a finite k-linear combination of homogeneous words in homogeneous multiplication operators, Euler factors, and basic trivial derivations, each word having the same total internal degree as the element represented.*

Proof. Expand every first-order derivation using $\mathrm{Der}_k(A) = AE+T$, distribute products, decompose coefficients into homogeneous parts, and retain the required homogeneous component. □

Lemma NC.17.12 (Non-Euler factor budget)**.** *If a homogeneous generator word has internal degree r and contains s trivial-derivation factors, then*

$$\boxed{r \geq sm.} \tag{NC.2.4}$$

Consequently

$$r < qm \quad \Longrightarrow \quad s \leq q - 1.$$

Proof. Every trivial factor contributes degree m, every Euler factor degree zero, and every homogeneous multiplication coefficient has nonnegative degree. Internal degrees add under composition. □

NC.17.2.4 Projectivization and Euler scalarization

For a finitely generated algebra B, let

$$\mathcal{P}^q_{B/k} := (B \otimes_k B)/I_\Delta^{q+1}$$

be the module of principal parts.

Lemma NC.17.13 (Localization of finite-order differential operators)**.** *If $D \in D^q_{B/k}$ and $S \subset B$ is multiplicative, then D localizes uniquely to an element of $D^q_{S^{-1}B/k}$, compatibly with composition. Homogeneous internal degree is preserved in the graded case.*

Proof. An order-q operator is equivalently a left B-linear map $\mathcal{P}^q_{B/k} \to B$. Localizing principal parts identifies

$$S^{-1}\mathcal{P}^q_{B/k} \cong \mathcal{P}^q_{S^{-1}B/k};$$

indeed $1 \otimes s$ differs from $s \otimes 1$ by a nilpotent element modulo I_Δ^{q+1}, hence is invertible after localizing the left factor. The asserted operator follows by localization of the representing map. □

Let $D \in D^q_{A/k,r}$ be homogeneous. For every integer t, its localized degree-t pieces glue to a sheaf differential operator

$$\widetilde{D}_t : \mathcal{O}_X(t) \longrightarrow \mathcal{O}_X(t + r).$$

If X is smooth, its order-q projective principal symbol lies in

$$\sigma^X_{q,t}(D) \in H^0(X, \mathrm{Sym}^q T_X(r)) .$$

Lemma NC.17.14 (Euler scalarization)**.** *On every twist $\mathcal{O}_X(t)$, the projectivized Euler derivation is multiplication by the scalar t. Hence it has projective differential order zero.*

Proof. For a homogeneous fraction $u = a/f^N$ of degree t,

$$E(u) = (\deg a - N \deg f)u = tu.$$

□

Lemma NC.17.15 (Word projective-order bound)**.** *If a homogeneous generator word contains exactly s trivial derivations, then on every twist its projectivization has differential order at most s.*

Proof. Read the composition factor by factor. Multiplications and every Euler factor are order zero after projectivization. Each trivial derivation has order at most one. Differential order is subadditive under composition. □

Theorem NC.17.16 (Projective order budget)**.** *Let $P \in (\Delta_A)_r$ be homogeneous, with $r \geq 0$. Then for every twist t,*

$$\boxed{\operatorname{ord}(\widetilde{P}_t) \leq \left\lfloor \frac{r}{m} \right\rfloor.} \tag{NC.2.5}$$

Proof. Expand P into homogeneous generator words. A word with s trivial factors satisfies $s \leq \lfloor r/m \rfloor$ by Lemma NC.17.12, and has projective order at most s by Lemma NC.17.15. Finite sums preserve the order bound. □

Theorem NC.17.17 (Robust Critical Window)**.** *Let $q \geq 2$. If*

$$\boxed{r < qm,} \tag{NC.2.6}$$

then every homogeneous $P \in (\Delta_A)_r$ satisfies

$$\operatorname{ord}(\widetilde{P}_t) \leq q - 1$$

for every twist t. Consequently, if additionally $P \in D^q_{A/k,r}$, then

$$\boxed{\sigma^X_{q,t}(P) = 0 \quad \textit{for every } t.} \tag{NC.2.7}$$

The largest integral degree in the window is

$$\boxed{r_q^{\mathrm{crit}} = qm - 1.} \tag{NC.2.8}$$

Proof. The inequality $r < qm$ gives $\lfloor r/m \rfloor \leq q - 1$. Apply Theorem NC.17.16. An operator of order at most $q - 1$ has zero order-q principal symbol. □

Corollary NC.17.18 (Critical-symbol strong escape)**.** *Let*

$$D \in D^q_{A/k,r}, \qquad r < qm,$$

be homogeneous. If for some twist t,

$$\sigma^X_{q,t}(D) \neq 0,$$

then

$$\boxed{D \notin \Delta_A.} \tag{NC.2.9}$$

Proof. Membership in Δ_A would contradict Theorem NC.17.17. □

Remark NC.17.19 (Long-word/order-drop immunity)**.** The proof never bounds affine word length, never identifies word length with differential order, never assumes absence of cancellation, and never invokes an affine PBW theorem. Arbitrarily many Euler factors and multiplication factors are harmless because they are projective order zero. This is the version of the Critical Window consumed by the master theorem below.

NC.17.3 The universal critical bundle and projective convention

Adjunction gives

$$K_X \cong O_X\left(\sum_i d_i - N - 1\right) = \boxed{O_X(m-d).} \tag{NC.3.1}$$

Since $\operatorname{rk} T_X = d$,

$$T_X \cong \Omega_X^{d-1} \otimes K_X^{-1}.$$

Therefore

$$\boxed{E_d := T_X(m) \cong \Omega_X^{d-1}(d).} \tag{NC.3.2}$$

For every $q \geq 0$,

$$\boxed{\operatorname{Sym}^q T_X(qm-1) \cong \operatorname{Sym}^q E_d(-1).} \tag{NC.3.3}$$

Lemma NC.17.20 (Global generation of the universal critical bundle)**.** *The rank-d bundle E_d is globally generated.*

Proof. For $p = d-1$, the exterior Euler/Koszul sequence gives a surjection

$$\bigwedge^{p+1} H^0(\mathbf{P}^N, O(1))^* \otimes O_{\mathbf{P}^N} \twoheadrightarrow \Omega^p_{\mathbf{P}^N}(p+1),$$

so $\Omega^{d-1}_{\mathbf{P}^N}(d)$ is globally generated. The conormal surjection

$$\Omega^1_{\mathbf{P}^N}|_X \twoheadrightarrow \Omega^1_X$$

induces after taking $(d-1)$-st exterior powers a surjection

$$\Omega^{d-1}_{\mathbf{P}^N}(d)|_X \twoheadrightarrow \Omega^{d-1}_X(d) = E_d.$$

A quotient of a globally generated bundle is globally generated. □

NC.17.3.1 Projectivization convention

We use the *quotient* projectivization

$$\pi : Z := \mathbf{P}_X(E_d) := \operatorname{Proj}_X(\operatorname{Sym} E_d) \longrightarrow X$$

and the tautological quotient line bundle

$$\boxed{L := O_Z(1).} \tag{NC.3.4}$$

With this convention,

$$\boxed{\pi_* L^q = \operatorname{Sym}^q E_d, \qquad R^i\pi_* L^q = 0 \quad (i > 0, q \geq 0).} \tag{NC.3.5}$$

Because E_d is globally generated, L is globally generated and therefore nef. Also

$$\boxed{\dim Z = 2d-1.} \tag{NC.3.6}$$

To avoid sign ambiguity in the word "Segre", define $h_j(E_d)$ by

$$\boxed{\sum_{j\geq 0} h_j(E_d)t^j := \frac{1}{c_{-t}(E_d)}.} \tag{NC.3.7}$$

Thus h_j is the complete symmetric polynomial in the Chern roots. In the quotient convention,

$$\boxed{\pi_*\left(c_1(L)^{d-1+j}\right) = h_j(E_d).} \tag{NC.3.8}$$

In particular,

$$L^{2d-1} = \int_X h_d(E_d).$$

NC.17.4 Universal positivity of the top complete-symmetric class

Set

$$e_i := d_i - 1 > 0, \qquad S := \sum_{i=1}^{c} e_i. \tag{NC.4.1}$$

Then

$$m = S - 1.$$

NC.17.4.1 K-theory and generating function

The tangent-normal and Euler sequences give in $K^0(X)$

$$[T_X] = (c + d + 1)[\mathcal{O}_X(1)] - [\mathcal{O}_X] - \sum_i [\mathcal{O}_X(d_i)].$$

Twisting by $\mathcal{O}_X(m) = \mathcal{O}_X(S - 1)$ yields

$$\boxed{[E_d] = (c + d + 1)[\mathcal{O}_X(S)] - [\mathcal{O}_X(S - 1)] - \sum_i [\mathcal{O}_X(S + e_i)].} \tag{NC.4.2}$$

Lemma NC.17.21 (Complete-symmetric generating function)**.** *Writing $H = c_1(\mathcal{O}_X(1))$,*

$$\boxed{\sum_{j\geq 0} h_j(E_d)t^j = \frac{(1 - (S - 1)Ht)\prod_{i=1}^{c}(1 - (S + e_i)Ht)}{(1 - SHt)^{c+d+1}}.} \tag{NC.4.3}$$

Proof. Apply multiplicativity of Chern polynomials to (NC.4.2), replace t by $-t$, and invert according to (NC.4.3). □

Let $\sigma_k(e)$ denote the elementary symmetric polynomial of degree k in $e_1, \ldots, e_c$, with $\sigma_0 = 1$.

NC.17.4.2 Exact two-layer decomposition

Theorem NC.17.22 (Two-layer decomposition)**.** *There are symmetric polynomials A_d and B_{d-1}, homogeneous of degrees d and $d - 1$ in the variables e_i, such that*

$$\boxed{h_d(E_d) = \big(A_d(e) + B_{d-1}(e)\big)H^d.} \tag{NC.4.4}$$

Explicitly,

$$\boxed{A_d = \sum_{k=0}^{\min(c,d)} (-1)^k \binom{2d - 1}{d - k} \sigma_k(e) S^{d-k},} \tag{NC.4.5}$$

and

$$\boxed{B_{d-1} = \sum_{k=0}^{\min(c,d-1)} (-1)^k \binom{2d - 1}{d - 1 - k} \sigma_k(e) S^{d-1-k}.} \tag{NC.4.6}$$

Proof. Suppress H in the coefficient extraction and use

$$1 - (S - 1)t = (1 - St) + t$$

and

$$1 - (S + e_i)t = (1 - St)\left(1 - e_i \frac{t}{1 - St}\right).$$

Then (NC.4.3) becomes

$$(1 - St)^{-d} \prod_i \left(1 - e_i \frac{t}{1 - St}\right) + t(1 - St)^{-(d+1)} \prod_i \left(1 - e_i \frac{t}{1 - St}\right).$$

Taking the coefficient of t^d in the first summand gives (NC.4.5); taking the coefficient of t^{d-1} after removing the leading t from the second gives (NC.4.6). □

NC.17.4.3 The coefficient inequality

Lemma NC.17.23 (Elementary-symmetric ratio bound)**.** *Let*

$$\alpha = (\alpha_1, \dots, \alpha_s), \qquad \alpha_i \in \mathbf{Z}_{>0}, \qquad n := \sum_i \alpha_i,$$

and write $\varepsilon_k(\alpha)$ for the elementary symmetric polynomial of degree k in the positive integers α_i. Then

$$\boxed{(k+1)\varepsilon_{k+1}(\alpha) \le n\varepsilon_k(\alpha).} \tag{NC.4.7}$$

Proof. The exact identity

$$(k+1)\varepsilon_{k+1}(\alpha) = \sum_{i=1}^{s} \alpha_i \, \varepsilon_k(\alpha_1, \dots, \widehat{\alpha_i}, \dots, \alpha_s)$$

holds by counting each $(k+1)$-fold product once for each chosen factor. Every omitted-variable elementary symmetric polynomial is at most $\varepsilon_k(\alpha)$. Sum the coefficients α_i. □

Theorem NC.17.24 (Coefficient positivity)**.** *Every monomial coefficient of A_d is nonnegative. Every monomial coefficient of B_{d-1} is strictly positive. Hence for all $e_i > 0$,*

$$\boxed{A_d(e) \ge 0, \qquad B_{d-1}(e) > 0.} \tag{NC.4.8}$$

Proof. Let

$$e^\beta = e_1^{\beta_1} \cdots e_c^{\beta_c}, \qquad \beta_i \ge 0, \qquad \beta! := \prod_i \beta_i!,$$

and let s be the size of the support of β.

First assume $|\beta| = d$. Multinomial extraction from (NC.4.5) gives

$$[e^\beta] A_d = \frac{(2d-1)!}{\beta!} \sum_{k=0}^{s} (-1)^k \frac{\varepsilon_k(\beta)}{(d+k-1)!}. \tag{NC.4.9}$$

Pair terms of indices k and $k+1$ for even k:

$$\frac{\varepsilon_k}{(d+k-1)!} - \frac{\varepsilon_{k+1}}{(d+k)!} = \frac{1}{(d+k-1)!}\left(\varepsilon_k - \frac{\varepsilon_{k+1}}{d+k}\right).$$

Lemma NC.17.23, applied with $n = d$, gives

$$\frac{\varepsilon_{k+1}}{d+k} \le \frac{d}{(k+1)(d+k)}\varepsilon_k \le \varepsilon_k.$$

Thus every pair is nonnegative. If a final term is unpaired, its index is even and the term is positive. Hence $[e^\beta] A_d \ge 0$.

Now assume $|\beta| = d-1$. From (NC.4.6),

$$[e^\beta] B_{d-1} = \frac{(2d-1)!}{\beta!} \sum_{k=0}^{s} (-1)^k \frac{\varepsilon_k(\beta)}{(d+k)!}. \tag{NC.4.10}$$

Pairing even k with $k+1$ yields

$$\frac{\varepsilon_k}{(d+k)!} - \frac{\varepsilon_{k+1}}{(d+k+1)!} = \frac{1}{(d+k)!}\left(\varepsilon_k - \frac{\varepsilon_{k+1}}{d+k+1}\right).$$

Now $n = d-1$, so

$$\frac{\varepsilon_{k+1}}{d+k+1} \le \frac{d-1}{(k+1)(d+k+1)}\varepsilon_k < \varepsilon_k.$$

Every pair is strictly positive, and a possible final unpaired even term is positive. Therefore every degree-$(d-1)$ monomial coefficient of B_{d-1} is strictly positive. For $d = 1$, the statement is simply $B_0 = 1$. □

Theorem NC.17.25 (Universal top complete-symmetric positivity)**.** *For every member of the integrated NC source class,*

$$\boxed{\int_X h_d(E_d) > 0.} \tag{NC.4.11}$$

Equivalently,

$$\boxed{L^{2d-1} > 0.} \tag{NC.4.12}$$

Since L is nef, L is big.

Proof. Theorems NC.17.22 and NC.17.24 give

$$A_d(e) + B_{d-1}(e) > 0$$

for all $e_i = d_i - 1 > 0$. Also

$$\int_X H^d = \deg X = \prod_i d_i > 0.$$

Hence (NC.4.11) follows. By the fixed quotient-projective convention,

$$\int_Z c_1(L)^{2d-1} = \int_X h_d(E_d) > 0.$$

A nef line bundle on a projective variety is big exactly when its top self-intersection is positive. □

Remark NC.17.26 (Effect of the theorem)**.** This single positivity theorem replaces the dimension-by-dimension top-class computations previously required in dimensions three and four. Those formulas remain useful numerical checks but are no longer logical dependencies.

NC.17.5 Asymptotic abundance of critical projective symbols

Let

$$H_Z := \pi^* O_X(1).$$

Using (NC.3.3) and the projective-bundle convention,

$$\begin{aligned} H^0(X, \operatorname{Sym}^q T_X(qm-1)) &= H^0(X, \operatorname{Sym}^q E_d(-1)) \\ &= H^0(Z, L^q \otimes H_Z^{-1}). \end{aligned} \tag{NC.5.1}$$

Theorem NC.17.27 (Critical-symbol asymptotics)**.** *Assume $d \geq 2$. Then, as $q \to \infty$,*

$$\boxed{h^0(X, \operatorname{Sym}^q T_X(qm-1)) = \frac{L^{2d-1}}{(2d-1)!} q^{2d-1} + O(q^{2d-2}).} \tag{NC.5.2}$$

The leading coefficient is strictly positive.

Proof. The variety Z has dimension $2d-1$ and L is nef and big. Asymptotic Riemann–Roch gives

$$\chi(Z, H_Z^{-1} \otimes L^q) = \frac{L^{2d-1}}{(2d-1)!} q^{2d-1} + O(q^{2d-2}).$$

For a nef line bundle L and fixed coherent sheaf $\mathcal{F}$, the standard higher-cohomology estimate gives

$$h^i(Z, \mathcal{F} \otimes L^q) = O(q^{2d-1-i}) \qquad (i > 0).$$

Taking $\mathcal{F} = H_Z^{-1}$, all higher cohomology contributes at most $O(q^{2d-2})$. Therefore h^0 has the asserted asymptotic. □

NC.17.6 Lower symbol layers: deep vanishing and boundary growth

Fix q and set the critical internal degree

$$\boxed{r := qm - 1.} \tag{NC.6.1}$$

For a lower symbol order

$$p = q - j, \qquad j \geq 1,$$

we have

$$\boxed{\mathrm{Sym}^p T_X(r) \cong \mathrm{Sym}^p E_d(jm - 1).} \tag{NC.6.2}$$

Hence

$$\boxed{H^1(X, \mathrm{Sym}^p T_X(r)) \cong H^1(Z, L^p \otimes H_Z^{jm-1}).} \tag{NC.6.3}$$

NC.17.6.1 Canonical bundle of the projective bundle

Lemma NC.17.28 (Canonical bundle)**.** *One has*

$$\boxed{K_Z = -dL + dmH_Z.} \tag{NC.6.4}$$

Proof. For the quotient projectivization of a rank-d bundle,

$$K_Z = -dL + \pi^*(K_X + \det E_d).$$

Now

$$K_X = \mathcal{O}_X(m - d)$$

and

$$\det E_d = \det T_X \otimes \mathcal{O}_X(dm) = K_X^{-1} \otimes \mathcal{O}_X(dm) = \mathcal{O}_X(d + (d-1)m).$$

Thus $K_X + \det E_d = \mathcal{O}_X(dm)$. □

Lemma NC.17.29 (Relative-plus-horizontal ampleness)**.** *For every real $a, b > 0$, the divisor class*

$$aL + bH_Z$$

is ample on Z.

Proof. The tautological L is relatively ample over X and nef on Z, while H_Z is the pullback of an ample class on the base. Standard relative ampleness gives an integer $n \gg 0$ such that

$$L + nH_Z$$

is ample. Choose $n > b/a$. Then

$$aL + bH_Z = \left(a - \frac{b}{n}\right) L + \frac{b}{n}(L + nH_Z).$$

The first summand is nef and the second is ample, so their sum is ample. □

Remark NC.17.30 (Correction of the fourfold draft)**.** The earlier shortcut "two linearly independent nef classes imply every positive combination is ample" is not used. Linear independence inside the nef cone is insufficient. Lemma NC.17.29 supplies the correct projective-bundle argument.

Theorem NC.17.31 (Deep lower-layer vanishing)**.** *Assume $d \geq 2$. If*

$$j \geq d + 2, \qquad q \geq j,$$

then

$$\boxed{H^1\big(X, \mathrm{Sym}^{q-j} T_X(qm - 1)\big) = 0.} \tag{NC.6.5}$$

Proof. Let

$$D_{p,j} := pL + (jm-1)H_Z, \qquad p = q-j.$$

Using Lemma NC.17.28,

$$D_{p,j} - K_Z = (p+d)L + \big((j-d)m - 1\big)H_Z.$$

Because $m \geq 1$ and $j \geq d+2$,

$$(j-d)m - 1 \geq 1.$$

Both coefficients are positive, so the difference is ample by Lemma NC.17.29. Since Z is smooth, Kodaira vanishing gives

$$H^1(Z, \mathcal{O}_Z(D_{p,j})) = 0.$$

Apply (NC.6.3). □

Theorem NC.17.32 (Boundary-layer growth)**.** *For each fixed*

$$j \in \{1, \dots, d+1\},$$

one has

$$\boxed{h^1\big(X, \operatorname{Sym}^{q-j} T_X(qm-1)\big) = O(q^{2d-2}).} \tag{NC.6.6}$$

Proof. By (NC.6.3), the group is

$$H^1\Big(Z, L^{q-j} \otimes H_Z^{jm-1}\Big).$$

For fixed j, rewrite it as

$$H^1\big(Z, \mathcal{F}_j \otimes L^q\big), \qquad \mathcal{F}_j := L^{-j} \otimes H_Z^{jm-1},$$

where $\mathcal{F}_j$ is fixed. Since L is nef on the $(2d-1)$-fold Z, the nef higher-cohomology estimate gives

$$h^1 = O(q^{(2d-1)-1}) = O(q^{2d-2}).$$

□

NC.17.7 The cone-tangent filtration and projective lifting

Define the rank-$(d+1)$ bundle

$$\boxed{\mathcal{A}_X := \ker\Big(\mathcal{O}_X(1)^{N+1} \longrightarrow \bigoplus_{i=1}^{c} \mathcal{O}_X(d_i)\Big),} \tag{NC.7.1}$$

where the map is the projective Jacobian map. Combining the Euler and tangent normal sequences gives

$$\boxed{0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{A}_X \longrightarrow T_X \longrightarrow 0.} \tag{NC.7.2}$$

Because the kernel line is trivial, for each $p \geq 1$, local splitting gives

$$\boxed{0 \longrightarrow \operatorname{Sym}^{p-1} \mathcal{A}_X \longrightarrow \operatorname{Sym}^p \mathcal{A}_X \longrightarrow \operatorname{Sym}^p T_X \longrightarrow 0.} \tag{NC.7.3}$$

A projective complete intersection is arithmetically Cohen–Macaulay. Thus, for $d \geq 2$,

$$\boxed{H^1(X, \mathcal{O}_X(t)) = 0 \quad \text{for all } t \in \mathbf{Z}.} \tag{NC.7.4}$$

Theorem NC.17.33 (Cone-symbol obstruction bound)**.** *Assume $d \geq 2$ and $r = qm - 1$. Then*

$$\boxed{H^1(X, \operatorname{Sym}^p \mathcal{A}_X(r)) = 0 \quad (0 \leq p \leq q-d-2),} \tag{NC.7.5}$$

and

$$\boxed{h^1(X, \operatorname{Sym}^{q-1} \mathcal{A}_X(r)) = O(q^{2d-2}).} \tag{NC.7.6}$$

Proof. Start with $p = 0$ and (NC.7.2). Induct through (NC.7.3). If $p \le q - d - 2$, then

$$j = q - p \ge d + 2,$$

so the tangent-bundle H^1 term vanishes by Theorem NC.17.31. This kills all deep cone-symbol layers.

Only

$$p = q - d - 1, \dots, q - 1$$

remain. There are exactly $d + 1$ such boundary layers, and each associated tangent-bundle H^1 is $O(q^{2d-2})$ by Theorem NC.17.32. Repeated long exact sequences preserve the same exponent because the number of layers is independent of q. □

Corollary NC.17.34 (Asymptotically full projective lift)**.** *Let*

$$V_q := H^0(X, \mathrm{Sym}^q T_X(qm - 1))$$

and

$$W_q^{\mathrm{proj}} := im\left[H^0(X, \mathrm{Sym}^q \mathcal{A}_X(qm - 1)) \to V_q\right].$$

Then

$$\boxed{\dim W_q^{\mathrm{proj}} = \frac{L^{2d-1}}{(2d-1)!} q^{2d-1} + O(q^{2d-2}).} \tag{NC.7.7}$$

Proof. The cokernel of the global-section map injects into

$$H^1(X, \mathrm{Sym}^{q-1} \mathcal{A}_X(qm - 1)),$$

which is $O(q^{2d-2})$. Subtract this from the asymptotic dimension of V_q in Theorem NC.17.27. □

NC.17.8 Quantization on the punctured cone

Let

$$Y := \mathrm{Spec}\, A, \qquad U := Y \setminus \{\mathfrak{m}\}.$$

Because X is smooth, U is the principal $\mathbf{G}_m$-bundle associated with $\mathcal{O}_X(-1)$, hence smooth.

NC.17.8.1 Equivariant tangent identification

Lemma NC.17.35 (Cone tangent and weight decomposition)**.** *The $\mathbf{G}_m$-equivariant tangent bundle satisfies*

$$T_U \cong \rho^* \mathcal{A}_X,$$

where $\rho : U \to X$. Consequently, for every $p \ge 0$, integer s, and $i \ge 0$,

$$\boxed{H^i(U, \mathrm{Sym}^p T_U)_s \cong H^i(X, \mathrm{Sym}^p \mathcal{A}_X(s)).} \tag{NC.8.1}$$

Proof. The pullback of the Jacobian kernel (NC.7.1) to the nonzero cone is exactly the tangent subbundle cut out by the equations $df_i = 0$, including the radial Euler direction; hence it identifies $\mathbf{G}_m$-equivariantly with T_U. The torsor ρ is affine and

$$(\rho_* \mathcal{O}_U)_s \cong \mathcal{O}_X(s)$$

for each homogeneous weight s. Apply projection formula to symmetric powers and take weight-s components. Affineness of ρ eliminates higher direct images. □

NC.17.8.2 Principal-symbol filtration

Let $\mathcal{D}_U^{\le p}$ be the sheaf of k-linear differential operators on smooth U of order at most p. In characteristic zero, smoothness gives

$$\boxed{0 \longrightarrow \mathcal{D}_U^{\le p-1} \longrightarrow \mathcal{D}_U^{\le p} \xrightarrow{\sigma_p} \mathrm{Sym}^p T_U \longrightarrow 0.} \tag{NC.8.2}$$

Theorem NC.17.36 (Subleading differential-operator obstruction)**.** *Assume $d \ge 2$ and $r = qm - 1$. Then*

$$\boxed{h^1(U, \mathcal{D}_U^{\le q-1})_r = O(q^{2d-2}).} \tag{NC.8.3}$$

Proof. By Lemma NC.17.35 and Theorem NC.17.33, all symbol-layer H^1 groups vanish in the deep range; only the fixed $d+1$ boundary layers remain, each $O(q^{2d-2})$. Start from

$$\mathcal{D}_U^{\le 0} = \mathcal{O}_U.$$

At weight r,

$$H^1(U, \mathcal{O}_U)_r \cong H^1(X, \mathcal{O}_X(r)) = 0.$$

Induct through (NC.8.2). Each long exact sequence can increase the H^1-dimension only by the corresponding symbol-layer H^1, so the total remains $O(q^{2d-2})$. □

Corollary NC.17.37 (Quantizable cone symbols have subleading codimension)**.** *Let*

$$Q_q := im\left[H^0(U, \mathcal{D}_U^{\le q})_r \longrightarrow H^0(U, \mathrm{Sym}^q T_U)_r\right], \qquad r = qm - 1.$$

Then

$$\boxed{\operatorname{codim} Q_q = O(q^{2d-2}).} \tag{NC.8.4}$$

Proof. The cokernel of the order-q symbol map injects into $H^1(U, \mathcal{D}_U^{\le q-1})_r$. Apply Theorem NC.17.36. □

Theorem NC.17.38 (Asymptotic visible quantization)**.** *Assume $d \ge 2$. For every sufficiently large q, there exists a quantizable cone symbol whose projective image in*

$$H^0(X, \mathrm{Sym}^q T_X(qm - 1))$$

is nonzero. More precisely, the projective image of the quantizable sector has dimension

$$\boxed{\frac{L^{2d-1}}{(2d-1)!} q^{2d-1} + O(q^{2d-2}).} \tag{NC.8.5}$$

Proof. The cone-liftable projective sector has the positive leading term

$$C q^{2d-1} + O(q^{2d-2}), \qquad C = \frac{L^{2d-1}}{(2d-1)!} > 0,$$

by Corollary NC.17.34. Inside the full cone-symbol space, quantizable symbols have codimension $O(q^{2d-2})$ by Corollary NC.17.37.

For a linear map $\varphi : C \to V$ and a subspace $Q \subset C$,

$$\operatorname{codim}_{\varphi(C)} \varphi(Q) \le \operatorname{codim}_C Q.$$

Therefore passage to the projective image can remove only another $O(q^{2d-2})$ dimensions. The strictly positive q^{2d-1} leading term cannot be annihilated. □

NC.17.9 Extension across the cone vertex

Lemma NC.17.39 (Normality and Hartogs)**.** *The affine cone $Y = \operatorname{Spec} A$ is normal and*

$$\boxed{\Gamma(U, \mathcal{O}_U) = A.} \tag{NC.9.1}$$

Proof. A complete intersection is Cohen–Macaulay, hence satisfies S_2. The punctured cone U is smooth. The vertex has codimension

$$\dim A = d + 1 \geq 2,$$

so A is regular in codimension one. Serre's criterion gives normality. Hartogs extension on the normal affine scheme across codimension at least two gives $\Gamma(U, \mathcal{O}_U) = A$. □

Lemma NC.17.40 (Extension of homogeneous differential operators)**.** *Restriction induces an isomorphism*

$$\boxed{D^{\leq q}_{A/k,s} \cong H^0(U, \mathcal{D}_U^{\leq q})_s} \tag{NC.9.2}$$

for every $q \geq 0$ and every homogeneous weight s.

Proof. Injectivity follows because U is dense. For surjectivity, induct on operator order. At order zero use Lemma NC.17.39. Let D be an order-q operator on U. For $a \in A$, the function $D(a|_U)$ lies in $\Gamma(U, \mathcal{O}_U) = A$, so D defines a homogeneous k-linear map $A \to A$. For every $a \in A$, the commutator $[D, a]$ has order at most $q - 1$ and extends by induction. Grothendieck's recursive definition of differential operators then shows that the extended map has order at most q. □

NC.17.10 The curve case: exact order-two closure

Assume $d = 1$. Then

$$K_X = \mathcal{O}_X(m - 1), \qquad T_X = \mathcal{O}_X(1 - m).$$

Set

$$q = 2, \qquad r = 2m - 1.$$

Theorem NC.17.41 (Curve critical symbol and quantization)**.** *For every smooth homogeneous complete-intersection curve in the source class,*

$$\boxed{\operatorname{Sym}^2 T_X(2m - 1) \cong \mathcal{O}_X(1),} \tag{NC.10.1}$$

so

$$H^0(X, \operatorname{Sym}^2 T_X(2m - 1)) \neq 0.$$

Every such projective critical symbol lifts to a cone symbol and quantizes to an order-two homogeneous differential operator on A.

Proof. The bundle identity is immediate from $T_X = \mathcal{O}_X(1 - m)$. For the cone-tangent sequence twisted by r, Serre duality gives

$$H^1(X, \mathcal{O}_X(r))^\vee \cong H^0(X, \mathcal{O}_X(-m)) = 0$$

and

$$H^1(X, T_X(r))^\vee = H^1(X, \mathcal{O}_X(m))^\vee \cong H^0(X, \mathcal{O}_X(-1)) = 0.$$

Thus $H^1(X, \mathcal{A}_X(r)) = 0$. The exact sequence

$$0 \to \mathcal{A}_X(r) \to \operatorname{Sym}^2 \mathcal{A}_X(r) \to \operatorname{Sym}^2 T_X(r) \to 0$$

therefore lifts every projective second symbol to a cone second symbol.

On U,

$$H^1(U, \mathcal{O}_U)_r = 0, \qquad H^1(U, T_U)_r = 0.$$

The order-one principal-symbol sequence implies

$$H^1(U, \mathcal{D}_U^{\leq 1})_r = 0,$$

so every cone second symbol quantizes. Lemma NC.17.40 extends the operator across the vertex. □

Corollary NC.17.42 (Strong order-two escape for curves)**.** *Every smooth homogeneous complete-intersection curve in the source class has*

$$\boxed{D \in D^2_{A/k,2m-1} \setminus \Delta_A.} \tag{NC.10.2}$$

Proof. Theorem NC.17.41 gives an actual operator with nonzero projective second symbol, and

$$2m - 1 < 2m.$$

Apply Corollary NC.17.18. □

NC.17.11 Master theorem: homogeneous complete intersections

Theorem NC.17.43 (Integrated Homogeneous-ICIS Strong Escape)**.** *Let*

$$A = k[x_0, \dots, x_N]/(f_1, \dots, f_c), \qquad c \geq 2, \qquad \deg f_i = d_i \geq 2,$$

be a standard homogeneous complete intersection over a field of characteristic zero. Assume A is a domain and

$$X = \operatorname{Proj} A$$

is smooth of positive dimension $d = N - c \geq 1$. Then

$$\boxed{D_{A/k} \neq A\langle \operatorname{Der}_k(A) \rangle.} \tag{NC.11.1}$$

More precisely:

(i) If $d = 1$, there is a strong escape at order two:

$$D \in D^2_{A/k,2m-1} \setminus \Delta_A.$$

(ii) If $d \geq 2$, there exists q_0 such that for every $q \geq q_0$ there is

$$\boxed{D_q \in D^q_{A/k,qm-1} \setminus \Delta_A.} \tag{NC.11.2}$$

Proof. The case $d = 1$ is the preceding corollary.

Assume $d \geq 2$. By Theorem NC.17.38, for every sufficiently large q there exists a quantizable cone order-q symbol of weight

$$r = qm - 1$$

whose projective image

$$s_q \in H^0(X, \operatorname{Sym}^q T_X(qm - 1))$$

is nonzero. Choose a differential operator on U realizing this symbol. Lemma NC.17.40 extends it across the vertex to

$$D_q \in D^q_{A/k,qm-1}.$$

Its projective order-q symbol is $s_q \neq 0$. Finally,

$$qm - 1 < qm.$$

The robust Critical Window says every degree-$(qm - 1)$ element of Δ_A has zero projective order-q symbol. Hence

$$D_q \notin \Delta_A.$$

□

Corollary NC.17.44 (Arbitrarily high-order critical escape)**.** *If $d \geq 2$, then*

$$\boxed{q \gg 0 \quad \Longrightarrow \quad \exists D_q \in D^q_{A/k,qm-1} \setminus \Delta_A.} \tag{NC.11.3}$$

Corollary NC.17.45 (Scoped Nakai theorem)**.** *Within the declared source class of positive-dimensional standard homogeneous complete intersections of codimension at least two, a singular affine cone cannot have its full ring of differential operators generated by first-order derivations. Thus the contrapositive Nakai implication holds on this source class.*

Remark NC.17.46 (What this theorem does not prove)**.** The theorem does not address arbitrary finite-type algebras, arbitrary non-complete-intersection singularities, nonisolated sources, or the full unrestricted Nakai conjecture. The codimension-one hypersurface case is also not consumed by this proof; it has a separate literature route.

NC.17.12 Legacy refinements, negative controls, and dependency reduction

NC.17.12.1 NC.14: the K3 quantitative refinement

For a smooth triple quadric

$$X = (2,2,2) \subset \mathbf{P}^5,$$

legacy NC.14 proves the exact order-two identification

$$\mathfrak{Q}_2^{\mathrm{crit}}(A) \cong H^0(X, \mathrm{Sym}^2 T_X(3))$$

and the Riemann–Roch lower bound

$$\boxed{h^0(X, \mathrm{Sym}^2 T_X(3)) \geq 18.}$$

This remains stronger quantitatively than the asymptotic master theorem at that specific source. Its former number 22 is retained only as a cohomology-jump value for a special control, not a universal K3 count.

NC.17.12.2 NC.15: quadrics

The finite-order quadratic theorem remains a refinement:

$$\text{smooth positive-dimensional quadratic CI} \Longrightarrow \text{strong order-two escape.}$$

The master theorem above no longer requires Brückmann-type numerical transport to prove existence, but that machinery remains relevant to explicit low-order cohomology and family uniformity.

NC.17.12.3 NC.16: surfaces and the cubic order-two failure

The cubic-surface finite computation established that a universal order-two route is false: there are smooth cubic controls for which the critical second-order defect vanishes. This negative result is retained as a genuine architectural constraint. The successor finite-order constructions at order three (cubic), order four (quartic), and matched/safe orders on general surfaces remain independent explicit witnesses.

The general theorem explains why failure at one low order is not a failure of the Nakai target: sufficiently high critical orders possess a symbol-dimension surplus over all quantization obstructions.

NC.17.12.4 Threefold and fourfold asymptotics

The earlier dimension-three and dimension-four arguments are exactly the first special cases of the universal mechanism

$$\boxed{q^{2d-1} \quad \text{candidate symbols} \qquad \text{versus} \qquad O(q^{2d-2}) \quad \text{quantization loss.}}$$

Their explicit top-class formulas remain replay controls for Theorem NC.17.25.

Theorem NC.17.47 (Dependency reduction)**.** *The proof of Theorem NC.17.43 does not require any of the following as logical gates:*

(i) universal quadratic critical T^1-vanishing;

(ii) Brückmann transport across the quadratic family;

(iii) surface Riemann–Roch positivity at one fixed order;

(iv) Aubin–Yau/Donaldson–Uhlenbeck–Yau polystability used in an earlier surface route;

(v) dimension-by-dimension explicit top-class expansions;

(vi) finite-control globalization from a selected diagonal or non-diagonal matrix example.

They remain refinements, diagnostics, and independent consistency checks.

NC.17.13 Audit register, imported edges, and publication posture

NC.17.13.1 Imported edges actually consumed

The master theorem consumes the following external mathematical results at substantive strength:

(1) Kersken's Euler-plus-trivial generation theorem for quasihomogeneous normal ICIS, together with its modern explicit restatement;

(2) standard smooth-projective complete-intersection facts, including adjunction and arithmetically Cohen–Macaulay line-bundle cohomology;

(3) projective-bundle intersection theory in the fixed quotient convention;

(4) the criterion that a nef line bundle is big iff its top self-intersection is positive;

(5) Kodaira vanishing;

(6) asymptotic Riemann–Roch and the standard nef higher-cohomology estimate

$$h^i(Z, \mathcal{F} \otimes L^q) = O(q^{\dim Z - i});$$

(7) the principal-symbol exact sequence for differential operators on a smooth scheme in characteristic zero;

(8) Serre normality and Hartogs extension across codimension at least two.

NC.17.13.2 Internally proved edges

The following are internal components of the present integrated proof:

(1) the localization/graded-detection transfer from local Kersken generation to $\mathrm{Der}_k(A) = AE + T$;

(2) the long-word-safe projective Critical Window;

(3) the identity $E_d = T_X(m) \cong \Omega_X^{d-1}(d)$ and its global generation;

(4) the general generating function (NC.4.3);

(5) the two-layer decomposition and coefficientwise positivity proof;

(6) the exponent-gap architecture q^{2d-1} versus q^{2d-2};

(7) propagation through the cone-tangent and differential-operator filtrations;

(8) extension of finite-order operators across the normal cone vertex by commutator induction;

(9) the final Critical-Window escape step.

NC.17.13.3 Adversarial checks retained

Policy 27 (No symbol/quantization conflation)**.** *A nonzero projective class is not by itself an operator on the cone. In particular, the existence of*

$$H^0(X, \mathrm{Sym}^q T_X(r)) \neq 0$$

does not close quantization. The proof uses two separate codimension estimates: first for lifting to cone symbols through $\mathcal{A}_X$, then for quantization through $\mathcal{D}_U^{\leq p}$.

Policy 28 (Fixed number of boundary layers)**.** *The exponent-gap proof requires that the number of surviving boundary layers be independent of q. Here it is exactly $d+1$, while d is fixed for the source variety. Therefore a sum of boundary losses remains $O(q^{2d-2})$.*

Policy 29 (No RR-only nonvanishing shortcut)**.** *The critical symbol nonvanishing does not follow merely from a positive Euler characteristic. The fixed coherent twist is handled by asymptotic Riemann–Roch together with the higher-cohomology estimate, yielding an actual positive leading term for h^0.*

Policy 30 (Projective convention lock)**.** *All pushforward and top-intersection formulas use the quotient projectivization convention $\pi_* \mathcal{O}_Z(q) = \mathrm{Sym}^q E_d$. The class denoted $h_d(E_d)$ is explicitly defined by $1/c_{-t}(E_d)$; no opposite-sign Segre convention is imported implicitly.*

NC.17.13.4 Prior-art ceiling

Current literature contains substantial progress for hypersurface and plane curve classes, including recent homogeneous/weighted-homogeneous hypersurface results and an effective homogeneous plane-curve result. Those source classes and proof mechanisms are different from the codimension-$c \geq 2$ theorem assembled here. This difference is not itself a novelty proof.

Declaration NC.17.48 (Publication posture)**.** The present integrated theorem is mathematically stronger than the previous NC status freeze inside the declared homogeneous-complete-intersection source class. Nevertheless it remains

C2-high

until all of the following are complete:

(a) an independent line-by-line audit of every internal proof;

(b) exact theorem-level source binding for each imported positivity and operator-sheaf result;

(c) a dedicated complete-intersection/Nakai prior-art search across journals, preprints, theses, and older differential-operator literature;

(d) independent reproduction of the coefficient algebra and selected low-dimensional specializations.

No literature-priority claim is made in this draft.

NC.17.14 Research consequences and next frontier

Declaration NC.17.49 (What is closed inside the present source class)**.** Subject to the imported interfaces listed above, one theorem now covers

all smooth positive-dimensional standard homogeneous complete intersections of codimension $c \geq 2$.

The earlier curve, surface, threefold, fourfold, and quadratic results are retained as refinements rather than prerequisites.

Declaration NC.17.50 (What remains genuinely open)**.** The next frontier is no longer "dimension $d \geq 5$" inside the same homogeneous-CI class. The unresolved directions require changing the source class, for example:

(1) weighted-homogeneous complete intersections beyond the standard graded case;

(2) nonhomogeneous complete intersections;

(3) nonisolated complete-intersection singularities;

(4) singularities that are not complete intersections;

(5) arbitrary finite-type characteristic-zero algebras in the unrestricted Nakai conjecture.

Remark NC.17.51 (Architectural interpretation)**.** The decisive change is that NC no longer needs to construct a specially tuned low-order stress on each geometric family. The master proof is a *dimension-surplus theorem*: a positive-volume projective symbol sector grows in dimension $2d-1$, whereas every quantization boundary obstruction lives one exponent lower. NC.4 then converts the surviving symbol into an absolute algebra-generation obstruction. This separates the proof into three independent reusable engines:

Critical Window + positivity / symbol volume + subleading quantization loss

This architecture is the natural starting point for any successor attempt beyond complete intersections.

.1 Specialization checks in dimensions three and four

The general positivity theorem specializes to the previously obtained threefold and fourfold formulas. These are retained as audit controls rather than proof inputs.

For $d = 3$, writing

$$S = \sum e_i, \qquad R = \sum e_i^2, \qquad T = \sum e_i^3,$$

one recovers

$$L^5 = \deg X \, \frac{14S^3 + 33S^2 - 12SR - 3R - 2T}{6} > 0.$$

For $d = 4$, with

$$U = \sum e_i^4,$$

one recovers

$$L^7 = \deg X \, \frac{3R^2 - 174RS^2 - 72RS + 225S^4 + 416S^3 - 48ST - 8T - 6U}{24} > 0.$$

The general coefficientwise proof removes the need for the dimension-four majorization estimate, but the equality is a valuable regression test.

.2 Claim register for the integrated edition

Claim ID	Statement	Status
NC-I.1	Graded Euler-plus-trivial generation after local-to-graded transfer	C1 input + C0/C2 transfer
NC-I.2	Robust Critical Window $r < qm$ for full Δ_A, arbitrary word length	C2-high
NC-I.3	$E_d = T_X(m) \cong \Omega_X^{d-1}(d)$ globally generated	C0 + standard Euler sequence
NC-I.4	Universal positivity $\int_X h_d(E_d) > 0$	C2-high internal proof
NC-I.5	Critical symbol growth $Cq^{2d-1} + O(q^{2d-2})$	C1 asymptotics + C2 assembly
NC-I.6	Total projective/cone quantization loss $O(q^{2d-2})$	C2-high

NC-I.7	Vertex extension of finite-order operators	C2-high / standard normality
NC-I.8	All $d \geq 2$ strong escape for all sufficiently large q	C2-high
NC-I.9	All CI curves strong order-two escape	C2-high
NC-I.10	Master scoped theorem for all positive-dimensional homogeneous CI, $c \geq 2$	C2-high

.3 References and exact source notes

A General Nakai Reduction and Exact Rees Quantization Gate

Source abstract

Appendix NC.17 closed the smooth-projectivization homogeneous complete- intersection source class by combining the robust Critical Window with a positive-volume symbol sector and a subleading quantization loss. NC.18 begins the passage from that scoped theorem to the unrestricted Nakai problem.

The purpose of NC.18 is not to claim the general Nakai conjecture. Its purpose is to replace the informal slogan "degenerate a singularity to its tangent cone and reuse the graded proof" by an exact bridge theorem with a visible obstruction module.

Let k be a field of characteristic zero and let A be a finitely generated k-algebra. Write

$$D_{A/k} = \bigcup_{q \geq 0} D^q_{A/k}, \qquad \Delta_A = A\langle \mathrm{Der}_k(A) \rangle.$$

First, finite-presentation of principal-parts modules implies localization of finite-order differential operators. Hence any counterexample to the Nakai implication localizes to a singular local ring at a prime minimal in the singular locus. Such a localization has an isolated singular point. In characteristic zero the singular locus is invariant under derivations; thus, for the isolated local core $(R, \mathfrak{m})$,

$$\delta(\mathfrak{m}) \subseteq \mathfrak{m} \qquad (\delta \in \mathrm{Der}_k(R)).$$

Consequently every element of Δ_R preserves every power $\mathfrak{m}^n$.

Second, let

$$\mathcal{R} = R[\mathfrak{m}t, t^{-1}] \subset R[t, t^{-1}], \qquad u = t^{-1}, \qquad B = k[u].$$

Then

$$\mathcal{R}/u\mathcal{R} \cong G := \mathrm{gr}_{\mathfrak{m}} R, \qquad \mathcal{R}[u^{-1}] \cong R[t, t^{-1}].$$

For the relative principal-parts module

$$P^q = P^q_{\mathcal{R}/B}, \qquad \mathcal{D}^q = \mathrm{Diff}^q_{\mathcal{R}/B} = \mathrm{Hom}_{\mathcal{R}}(P^q, \mathcal{R}),$$

the special-fiber base change is

$$P^q/uP^q \cong P^q_{G/k}.$$

Because multiplication by u is injective on $\mathcal{R}$, applying $\mathrm{Hom}_{\mathcal{R}}(P^q, -)$ to

$$0 \to \mathcal{R} \xrightarrow{u} \mathcal{R} \to G \to 0$$

gives the exact Rees Quantization Gate

$$0 \longrightarrow \mathcal{D}^q/u\mathcal{D}^q \xrightarrow{\mathrm{sp}_q} D^q_{G/k} \longrightarrow \mathrm{Ext}^1_{\mathcal{R}}(P^q, \mathcal{R})[u] \longrightarrow 0.$$

Thus the failure of a tangent-cone operator to lift is measured exactly by the u-torsion of an Ext^1 module; no flatness shortcut is used.

Third, all derivations of the isolated local core lift canonically to B-linear derivations of $\mathcal{R}$. Let $\widetilde{\Delta}$ be the subalgebra of $\mathrm{Diff}_{\mathcal{R}/B}$ generated by $\mathcal{R}$ and these lifted derivations, and let

$$\mathrm{Sat}_u(\widetilde{\Delta}) = \mathrm{Diff}_{\mathcal{R}/B} \cap \widetilde{\Delta}[u^{-1}].$$

The special fiber of this saturation is the *saturated derivation shadow*. NC.18 proves the exact escape-transfer theorem: if a tangent-cone operator lies in the image of sp_q but outside the saturated derivation shadow, then some finite-order differential operator on R lies outside Δ_R.

Finally, in every finite-dimensional graded sector this yields a quantitative criterion. If the tangent-cone operator sector has dimension $Cq^a + O(q^{a-1})$ while both the Rees lifting obstruction and the saturated derivation shadow are $O(q^{a-1})$, then strong Nakai escape follows for all sufficiently large q. This is the exact general form of the dimension-surplus mechanism that closed NC.17.

NC.18 therefore removes two major ambiguities: the unrestricted problem may be reduced to isolated local singularities, and the Rees/tangent-cone passage has an exact obstruction theory. What remains is a single genuinely new universal problem: construct, on the tangent cone of an arbitrary isolated singularity, a sufficiently large liftable operator sector whose saturated derivation shadow is asymptotically smaller. This remaining statement is isolated as the *Universal Shadow-Defect Gate*; it is not asserted here as proved.

Declaration A.1 (Re-audited role)**.** NC.18 is retained as a transfer/quantization bridge. It is not promoted to a proof of the unrestricted Nakai conjecture. The universal nonvanishing ambition recorded historically inside NC.18 is superseded by the existence of the Collapse region established externally through Family 048 and the classical Becker–Rego implication.

A.1 General Nakai reduction and Rees quantization bridge

A.1.1 Scope, target, and relation to NC.17

Definition A.2 (Nakai generation algebra)**.** Let k be a field and A a commutative k-algebra. Set

$$D_{A/k} := \bigcup_{q\geq 0} D^q_{A/k}, \qquad \Delta_A := A\langle \mathrm{Der}_k(A)\rangle \subseteq D_{A/k}.$$

A *strong escape* is an element

$$D \in D_{A/k} \setminus \Delta_A.$$

Policy 31 (No promotion from NC.17)**.** *The homogeneous-complete-intersection theorem of NC.17 is a scoped source-class result. NC.18 does not infer the unrestricted Nakai conjecture from it. Every new passage from an arbitrary singularity to a graded degeneration must be proved independently.*

Declaration A.3 (NC.18 objective)**.** NC.18 has four proof obligations:

(i) reduce a hypothetical global Nakai counterexample to an isolated local counterexample;

(ii) prove that first-order derivations preserve the maximal-ideal filtration of that isolated local core;

(iii) derive an exact obstruction theory for lifting differential operators from the tangent cone through the extended Rees algebra;

(iv) formulate an exact escape criterion that transports failure of first-order generation back to the original local ring.

Only after these four obligations are closed is it legitimate to ask for a universal asymptotic construction on tangent cones.

A.1.2 Principal parts and localization

Let A be a finitely generated k-algebra and let

$$I_A := \ker(A \otimes_k A \xrightarrow{\mu} A).$$

For $q \geq 0$ define

$$P^q_{A/k} := (A \otimes_k A)/I_A^{q+1},$$

viewed as a left A-module through the first tensor factor. There is the standard identification

$$D^q_{A/k} \cong \mathrm{Hom}_A(P^q_{A/k}, A).$$

Since A is of finite presentation over the field k, $P^q_{A/k}$ is a finitely presented left A-module for every finite q.

Theorem A.4 (Localization of finite-order operators)**.** *Let $S \subset A$ be multiplicatively closed. For every $q \geq 0$, restriction induces a natural isomorphism of left $S^{-1}A$-modules*

$$S^{-1}D^q_{A/k} \xrightarrow{\sim} D^q_{S^{-1}A/k}.$$

Moreover

$$S^{-1}\operatorname{Der}_k(A) \xrightarrow{\sim} \operatorname{Der}_k(S^{-1}A), \qquad S^{-1}\Delta_A = \Delta_{S^{-1}A}.$$

Consequently

$$D_{A/k} = \Delta_A \quad \Longrightarrow \quad D_{S^{-1}A/k} = \Delta_{S^{-1}A}.$$

Proof. The sheaf of differential operators of order at most q associated with a morphism locally of finite presentation is quasi-coherent because it is the $\mathcal{O}$-dual of the finite-presentation principal-parts sheaf. On the affine scheme Spec A, restriction to Spec $S^{-1}A$ therefore equals localization. Equivalently, one may use

$$P^q_{S^{-1}A/k} \cong S^{-1}P^q_{A/k}$$

together with finite presentation to commute Hom with localization.

For derivations,

$$\operatorname{Der}_k(A) = \operatorname{Hom}_A(\Omega_{A/k}, A),$$

and $\Omega_{A/k}$ is finitely presented. Hence derivations localize. Since localization respects finite sums and finite compositions, the algebra generated by coefficients and derivations also localizes. The final implication follows order by order and then after taking the filtered union. □

Remark A.5 (Why this is stronger than mere extension)**.** It is not enough to know that a differential operator on A extends to a localization. The reduction argument requires surjectivity: every finite-order operator on the localization must come from the localized finite-order module. Finite presentation of principal parts is the decisive input.

A.1.3 Reduction to an isolated local singularity

Theorem A.6 (Isolated-core reduction)**.** *Assume the Nakai implication fails for some finitely generated characteristic-zero k-algebra A. Then there exists a local k-algebra*

$$(R, \mathfrak{m}) = (A_{\mathfrak{p}}, \mathfrak{p}A_{\mathfrak{p}})$$

essentially of finite type over k such that

$$R \text{ is singular}, \qquad \operatorname{Sing}(\operatorname{Spec} R) = \{\mathfrak{m}\}, \qquad D_{R/k} = \Delta_R.$$

Thus a counterexample exists only if an isolated local counterexample exists.

Proof. For a finite-type algebra over a field the regular locus is open, hence the singular locus is closed. Choose a prime $\mathfrak{p}$ minimal among the primes in $\operatorname{Sing}(\operatorname{Spec} A)$. Put $R = A_{\mathfrak{p}}$. The ring R is singular because $A_{\mathfrak{p}}$ is singular. A nonmaximal prime of R is of the form $\mathfrak{q}A_{\mathfrak{p}}$ for some $\mathfrak{q} \subsetneq \mathfrak{p}$. Minimality of $\mathfrak{p}$ in the singular locus forces $A_{\mathfrak{q}}$ to be regular. Hence every nonmaximal localization of R is regular, so the singular locus of Spec R consists only of its maximal ideal.

If $D_{A/k} = \Delta_A$, the localization theorem gives

$$D_{R/k} = \Delta_R.$$

□

Corollary A.7 (Correct global strategy)**.** *To prove the Nakai implication for all finitely generated k-algebras, it is sufficient to rule out equality $D_{R/k} = \Delta_R$ for every singular local k-algebra essentially of finite type whose singular locus consists only of its maximal ideal.*

Remark A.8 (Domain and reducedness)**.** The isolated-core reduction itself does not require a domain hypothesis. Some later geometric constructions may impose reducedness, equidimensionality, or a domain condition. Such assumptions must then be recorded as additional source restrictions rather than silently inserted into the general Nakai statement.

A.1.4 Derivation invariance of the isolated maximal ideal

Theorem A.9 (Derivation invariance of the isolated core)**.** *Let* $(R, \mathfrak{m})$ *be essentially of finite type over a characteristic-zero field and assume*

$$\operatorname{Sing}(\operatorname{Spec} R) = \{\mathfrak{m}\}.$$

Then

$$\delta(\mathfrak{m}) \subseteq \mathfrak{m} \qquad \text{for every } \delta \in \operatorname{Der}_k(R).$$

Consequently

$$\delta(\mathfrak{m}^n) \subseteq \mathfrak{m}^n \qquad (n \geq 0).$$

Proof. The first assertion is an imported characteristic-zero invariance theorem for the singular locus. In algebraic form, derivations preserve the relevant Fitting ideals of the Kähler differential module; in characteristic zero the radical ideal defining the set-theoretic singular locus is derivation-stable. For an isolated local singularity that radical ideal is precisely $\mathfrak{m}$. Hence $\delta(\mathfrak{m}) \subseteq \mathfrak{m}$. The power statement follows from the Leibniz rule. □

Corollary A.10 (Filtration rigidity of the derivation-generated algebra)**.** *Every element* $P \in \Delta_R$ *preserves the* $\mathfrak{m}$*-adic filtration:*

$$P(\mathfrak{m}^n) \subseteq \mathfrak{m}^n \qquad (n \geq 0).$$

Therefore, if there exists any finite-order differential operator $D \in D_{R/k}$ *and any n for which*

$$D(\mathfrak{m}^n) \not\subseteq \mathfrak{m}^n,$$

then $D \notin \Delta_R$ *and strong Nakai escape is already proved for R.*

Proof. Multiplication by an element of R preserves every ideal $\mathfrak{m}^n$, and every first-order derivation preserves every $\mathfrak{m}^n$. Stability is preserved by finite sums and compositions. □

Declaration A.11 (Counterexample rigidity)**.** A hypothetical isolated local Nakai counterexample must satisfy the much stronger condition

$$D_{R/k}(\mathfrak{m}^n) \subseteq \mathfrak{m}^n \qquad \text{for every } D \in D_{R/k} \text{ and every } n \geq 0,$$

because equality $D_{R/k} = \Delta_R$ forces every differential operator to preserve the maximal-ideal filtration.

A.1.5 Extended Rees algebra and tangent cone

Fix from now on an isolated local source $(R, \mathfrak{m})$ as above. Define the extended Rees algebra

$$\mathcal{R} := R[\mathfrak{m}t, t^{-1}] \subseteq R[t, t^{-1}].$$

Put

$$u := t^{-1}, \qquad B := k[u].$$

The algebra $\mathcal{R}$ is naturally a B-algebra. Since R is essentially of finite type over k and $\mathfrak{m}$ is finitely generated, $\mathcal{R}$ is essentially of finite presentation over B; in particular it is Noetherian.

Lemma A.12 (Generic and special fibers)**.** *There are canonical isomorphisms*

$$\mathcal{R}[u^{-1}] \cong R[t, t^{-1}], \qquad \mathcal{R}/u\mathcal{R} \cong G := \operatorname{gr}_{\mathfrak{m}} R.$$

Moreover multiplication by u on $\mathcal{R}$ *is injective.*

Proof. After inverting $u = t^{-1}$, the element $t = u^{-1}$ becomes available and the extended Rees algebra becomes the Laurent extension of R. The quotient by t^{-1} identifies the class of $\mathfrak{m}^n t^n$ modulo the preceding filtration layer with $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, giving the associated graded ring. Finally $\mathcal{R}$ is a subring of $R[t, t^{-1}]$, where t^{-1} is a unit; hence multiplication by u is injective even when R has zero divisors. □

Remark A.13 (Why the extended Rees algebra is used)**.** The ordinary Rees algebra does not contain the degeneration parameter whose zero fiber is literally the associated graded ring. The extended Rees algebra contains $u = t^{-1}$, and the exact sequence

$$0 \to \mathcal{R} \xrightarrow{u} \mathcal{R} \to G \to 0$$

is the algebraic source of the quantization obstruction below.

A.1.6 Principal parts under the Rees degeneration

For $q \geq 0$ define relative principal parts over $B = k[u]$ by

$$P^q := P^q_{\mathcal{R}/B} = (\mathcal{R} \otimes_B \mathcal{R})/I^{q+1},$$

where I is the diagonal ideal. Set

$$\mathcal{D}^q := \mathrm{Diff}^q_{\mathcal{R}/B} \cong \mathrm{Hom}_{\mathcal{R}}(P^q, \mathcal{R}).$$

Lemma A.14 (Special-fiber base change for principal parts)**.** *For every $q \geq 0$ there is a canonical isomorphism of left G-modules*

$$P^q/uP^q \cong P^q_{G/k}.$$

Proof. Principal parts are defined by a tensor square followed by quotient by a power of the diagonal ideal. Base change $B \to B/(u) = k$ sends

$$\mathcal{R} \otimes_B \mathcal{R} \quad \text{to} \quad G \otimes_k G,$$

and sends the diagonal ideal to the diagonal ideal of G/k. The image of I^{q+1} is the $(q+1)$-st power of the image ideal. Right exactness of tensor product therefore gives the displayed quotient isomorphism. No flatness of P^q over B is assumed. □

Lemma A.15 (Generic-fiber identification)**.** *For every $q \geq 0$,*

$$\mathcal{D}^q[u^{-1}] \cong D^q_{R/k} \otimes_k k[t, t^{-1}].$$

Proof. Localization of finite-order relative differential operators applies because $\mathcal{R}$ is essentially of finite presentation over the Noetherian ring B. After inverting u, the relative algebra is

$$R[t, t^{-1}]/k[t, t^{-1}],$$

and the Laurent variable belongs to the base. Relative principal parts are therefore obtained from those of R/k by Laurent scalar extension. □

A.1.7 The exact Rees Quantization Gate

For a $\mathcal{R}$-module M define its first u-torsion layer by

$$M[u] := \ker(u : M \to M).$$

Theorem A.16 (Exact Rees Quantization Gate)**.** *For every finite order $q \geq 0$, specialization gives a natural exact sequence*

$$0 \longrightarrow \mathcal{D}^q/u\mathcal{D}^q \xrightarrow{\mathrm{sp}_q} D^q_{G/k} \xrightarrow{\partial_q} \mathrm{Ext}^1_{\mathcal{R}}(P^q, \mathcal{R})[u] \longrightarrow 0.$$

In particular, a tangent-cone operator

$$\overline{D} \in D^q_{G/k}$$

lifts to a relative operator in $\mathcal{D}^q$ if and only if

$$\partial_q(\overline{D}) = 0.$$

Proof. Use the injective multiplication map from the extended Rees exact sequence

$$0 \longrightarrow \mathcal{R} \xrightarrow{u} \mathcal{R} \longrightarrow G \longrightarrow 0.$$

Apply the left exact functor $\mathrm{Hom}_{\mathcal{R}}(P^q, -)$. The beginning of the long exact sequence is

$$0 \to \mathrm{Hom}(P^q, \mathcal{R}) \xrightarrow{u} \mathrm{Hom}(P^q, \mathcal{R}) \to \mathrm{Hom}(P^q, G) \to \mathrm{Ext}^1(P^q, \mathcal{R}) \xrightarrow{u} \mathrm{Ext}^1(P^q, \mathcal{R}).$$

Thus the cokernel of reduction modulo u on $\mathrm{Hom}(P^q, \mathcal{R})$ is precisely the kernel of multiplication by u on the Ext^1 module.

Since u acts trivially on G,

$$\mathrm{Hom}_{\mathcal{R}}(P^q, G) \cong \mathrm{Hom}_G(P^q/uP^q, G),$$

and the special-fiber principal-parts lemma identifies the latter with

$$\mathrm{Hom}_G(P^q_{G/k}, G) = D^q_{G/k}.$$

Substitution yields the asserted exact sequence. □

Definition A.17 (Rees lifting obstruction)**.** Set

$$\mathfrak{O}_q(R) := \mathrm{Ext}^1_{\mathcal{R}}(P^q_{\mathcal{R}/B}, \mathcal{R})[u].$$

This is the exact order-q obstruction space for lifting tangent-cone operators through the Rees degeneration.

Policy 32 (No flatness shortcut)**.** *The statement*

$$D^q_{G/k} \stackrel{?}{=} \mathcal{D}^q/u\mathcal{D}^q$$

is not used unless $\mathfrak{O}_q(R) = 0$ has been proved. The obstruction module is part of the theorem data, not an error term that may be silently discarded.

Remark A.18 (Finite generation)**.** Because $\mathcal{R}$ is Noetherian and P^q is finitely presented, the module $\mathrm{Ext}^1_{\mathcal{R}}(P^q, \mathcal{R})$ is finitely generated. Hence its u-torsion is a finite $\mathcal{R}$-module killed by some power of u. This does not prove that the obstruction is small in the order parameter q; it does show that the specialization defect is algebraically localized on the degeneration fiber.

A.1.8 Lifting first-order derivations

Lemma A.19 (Canonical Rees lift of derivations)**.** *For every $\delta \in \mathrm{Der}_k(R)$ there is a canonical B-linear derivation*

$$\widetilde{\delta} \in \mathrm{Der}_B(\mathcal{R})$$

defined on homogeneous Rees terms by

$$\widetilde{\delta}(at^n) = \delta(a)t^n.$$

After inverting u, it becomes the scalar extension of δ to $R[t, t^{-1}]$ with $\delta(t) = 0$.

Proof. Derivation invariance gives

$$\delta(\mathfrak{m}^n) \subseteq \mathfrak{m}^n$$

for all $n \geq 0$, so the formula preserves the extended Rees algebra. It is a derivation by the Leibniz rule and it kills $u = t^{-1}$, hence is B-linear. The generic-fiber statement is immediate. □

Definition A.20 (Rees derivation algebra)**.** Let

$$\widetilde{\Delta} \subseteq \mathrm{Diff}_{\mathcal{R}/B}$$

be the B-relative differential-operator subalgebra generated by multiplication by elements of $\mathcal{R}$ and all lifted derivations $\widetilde{\delta}$ arising from $\delta \in \mathrm{Der}_k(R)$.

Lemma A.21 (Generic fiber of the Rees derivation algebra)**.** *There is an equality inside* $\mathrm{Diff}_{\mathcal{R}/B}[u^{-1}]$*:*

$$\widetilde{\Delta}[u^{-1}] = \Delta_R \otimes_k k[t, t^{-1}].$$

Proof. Every generator of $\widetilde{\Delta}$ specializes generically to either a Laurent coefficient or a scalar-extended derivation of R. Conversely every generator of Δ_R has the canonical Rees lift. Finite sums and finite compositions give equality of the generated algebras. □

A.1.9 Saturation and the derivation shadow

Definition A.22 (u-saturation)**.** Inside $\mathrm{Diff}_{\mathcal{R}/B}[u^{-1}]$ define

$$\mathrm{Sat}_u(\widetilde{\Delta}) := \mathrm{Diff}_{\mathcal{R}/B} \cap \widetilde{\Delta}[u^{-1}].$$

Equivalently,

$$\mathrm{Sat}_u(\widetilde{\Delta}) = \left\{ P \in \mathrm{Diff}_{\mathcal{R}/B} : u^N P \in \widetilde{\Delta} \text{ for some } N \geq 0 \right\}.$$

Definition A.23 (Order-q saturated derivation shadow)**.** Let

$$\mathfrak{S}_q(R) := \mathrm{sp}_q\left(\frac{\mathrm{Sat}_u(\widetilde{\Delta}) \cap \mathcal{D}^q + u\mathcal{D}^q}{u\mathcal{D}^q} \right) \subseteq D^q_{G/k}.$$

This is the special-fiber shadow of all relative operators whose generic fiber belongs to the derivation-generated algebra.

Remark A.24 (Why saturation is mandatory)**.** An operator may fail to lie in $\widetilde{\Delta}$ before specialization while becoming derivation-generated after the parameter u is inverted. Such an operator cannot witness strong escape on R. Saturation removes exactly this false-positive sector. Replacing $\mathrm{Sat}_u(\widetilde{\Delta})$ by $\widetilde{\Delta}$ would therefore be unsound.

A.1.10 Exact escape transfer

Theorem A.25 (Rees strong-escape transfer)**.** *Suppose that for some $q \geq 0$ there exists*

$$\overline{D} \in D^q_{G/k}$$

such that

$$\overline{D} \in \mathrm{im}(\mathrm{sp}_q) \qquad \textit{and} \qquad \overline{D} \notin \mathfrak{S}_q(R).$$

Then

$$D_{R/k} \neq \Delta_R.$$

More precisely, at least one finite-order coefficient of a generic Rees lift of $\overline{D}$ is an operator on R outside Δ_R.

Proof. Choose $P \in \mathcal{D}^q$ with

$$\mathrm{sp}_q(P \bmod u) = \overline{D}.$$

Assume for contradiction that the generic fiber

$$P[u^{-1}] \in D^q_{R/k} \otimes_k k[t, t^{-1}]$$

belongs to

$$\Delta_R \otimes_k k[t, t^{-1}].$$

By the generic-fiber lemma for $\widetilde{\Delta}$,

$$P[u^{-1}] \in \widetilde{\Delta}[u^{-1}],$$

so

$$P \in \mathrm{Diff}_{\mathcal{R}/B} \cap \widetilde{\Delta}[u^{-1}] = \mathrm{Sat}_u(\widetilde{\Delta}).$$

Its special fiber would then lie in $\mathfrak{S}_q(R)$, contradicting the hypothesis.

Thus $P[u^{-1}]$ does not lie in $\Delta_R \otimes_k k[t, t^{-1}]$. Write it as a finite Laurent sum with coefficients in $D^q_{R/k}$. If every coefficient belonged to Δ_R, the whole Laurent sum would belong to the scalar extension of Δ_R. Therefore at least one coefficient lies in $D^q_{R/k} \setminus \Delta_R$. □

Corollary A.26 (Necessary special-fiber condition for a Nakai counterexample)**.** *If*

$$D_{R/k} = \Delta_R,$$

then for every $q \geq 0$,

$$\mathrm{im}(\mathrm{sp}_q) \subseteq \mathfrak{S}_q(R).$$

Since the reverse inclusion is tautological from the definition,

$$\boxed{\mathrm{im}(\mathrm{sp}_q) = \mathfrak{S}_q(R)}$$

for every q. Thus an isolated local Nakai counterexample would force every Rees-liftable tangent-cone operator to be explained by the saturated shadow of first-order derivations.

A.1.11 The Rees defect quotient

Definition A.27 (Quantizable sector)**.** Set

$$\mathfrak{Q}_q(R) := \mathrm{im}(\mathrm{sp}_q) \subseteq D^q_{G/k}.$$

By the exact Rees Quantization Gate,

$$D^q_{G/k}/\mathfrak{Q}_q(R) \cong \mathfrak{O}_q(R).$$

Definition A.28 (Rees–Nakai defect)**.** Define

$$\mathfrak{N}_q(R) := \mathfrak{Q}_q(R)/\mathfrak{S}_q(R).$$

Theorem A.29 (Defect criterion)**.** *For an isolated local singularity* $(R, \mathfrak{m})$*,*

$$\mathfrak{N}_q(R) \neq 0 \quad \Longrightarrow \quad D_{R/k} \neq \Delta_R.$$

Consequently, proving

$$\exists q \geq 0 : \ \mathfrak{N}_q(R) \neq 0$$

for every isolated local singularity essentially of finite type over a characteristic-zero field proves the unrestricted Nakai implication.

Proof. A nonzero class in $\mathfrak{N}_q(R)$ has a representative in the quantizable sector but outside the saturated derivation shadow, so the strong-escape transfer theorem applies. The second assertion follows from the isolated-core reduction. □

Remark A.30 (NC.18 compression of the general conjecture)**.** The unrestricted Nakai problem is therefore reduced to a single local defect statement:

$$\boxed{R \text{ isolated singular} \quad \Longrightarrow \quad \mathfrak{N}_q(R) \neq 0 \text{ for some finite } q.}$$

This is an exact reformulation-by-sufficient-reduction, not yet a proof of the nonvanishing.

A.1.12 Graded sectors and a dimension-surplus theorem

The tangent cone

$$G = \mathrm{gr}_{\mathfrak{m}} R$$

is a standard nonnegatively graded algebra over its canonical degree-zero field

$$K := G_0 = R/\mathfrak{m}.$$

The extension K/k may be transcendental when the isolated core is obtained by localizing at the generic point of a positive-dimensional singular component. Therefore a count using all k-linear operators in a homogeneous degree need not be finite-dimensional over k.

The correct universal finite-dimensional sector is obtained by imposing K-linearity only on the *special fiber*. Since

$$k \longrightarrow K \longrightarrow G,$$

every K-linear differential operator is also k-linear, and there are natural inclusions

$$D^q_{G/K} \subseteq D^q_{G/k}.$$

For fixed q and internal degree r set

$$V^K_{q,r} := (D^q_{G/K})_r.$$

Because G is a finitely generated standard graded K-algebra and $P^q_{G/K}$ is a finite graded G-module, $V^K_{q,r}$ is finite-dimensional over K.

Restrict the Rees obstruction map

$$\partial_q : D^q_{G/k} \longrightarrow \mathfrak{O}_q(R)$$

to $V^K_{q,r}$ and define

$$O^K_{q,r} := \operatorname{im}\left(\partial_q|_{V^K_{q,r}}\right), \qquad Q^K_{q,r} := V^K_{q,r} \cap \mathfrak{Q}_q(R), \qquad S^K_{q,r} := V^K_{q,r} \cap \mathfrak{S}_q(R).$$

The map ∂_q is G-linear, hence its restriction is K-linear. All three displayed spaces are therefore finite-dimensional K-vector spaces.

Proposition A.31 (Exact dimension of the K-linear liftable sector)**.** *For every q, r,*

$$\dim_K Q^K_{q,r} = \dim_K V^K_{q,r} - \dim_K O^K_{q,r}.$$

Proof. By exactness of the Rees Quantization Gate,

$$\mathfrak{Q}_q(R) = \ker \partial_q.$$

Intersect with $V^K_{q,r}$. Thus

$$Q^K_{q,r} = \ker(\partial_q|_{V^K_{q,r}}),$$

and rank–nullity over K gives the formula. □

Theorem A.32 (General dimension-surplus escape)**.** *If for some q, r,*

$$\dim_K V^K_{q,r} - \dim_K O^K_{q,r} > \dim_K S^K_{q,r},$$

then

$$D_{R/k} \neq \Delta_R.$$

Proof. The left-hand side before comparison with $S^K_{q,r}$ is exactly $\dim_K Q^K_{q,r}$. Since

$$S^K_{q,r} \subseteq Q^K_{q,r},$$

strict inequality produces an element of

$$Q^K_{q,r} \setminus S^K_{q,r} \subseteq \mathfrak{Q}_q(R) \setminus \mathfrak{S}_q(R).$$

Apply the Rees strong-escape transfer theorem. □

Corollary A.33 (Asymptotic NC mechanism)**.** *Suppose there exist an integer-valued degree choice $r(q)$, a real number $C > 0$, and an exponent $a > 0$ such that*

$$\dim_K V^K_{q,r(q)} = Cq^a + O(q^{a-1}),$$

while

$$\dim_K O^K_{q,r(q)} = O(q^{a-1}), \qquad \dim_K S^K_{q,r(q)} = O(q^{a-1}).$$

Then

$$D_{R/k} \neq \Delta_R$$

for all sufficiently large q through the selected sectors.

Remark A.34 (Residue-field problem closed at the counting interface)**.** No coefficient field inside R and no completion theorem are required for this dimension count. The field $K = R/\mathfrak{m}$ exists canonically on the special fiber, and $D_{G/K}$ is a canonical subspace of $D_{G/k}$. The Rees lifting obstruction is still computed relative to the original base field k. Thus the isolated-core reduction and the finite-dimensional asymptotic strategy are compatible even when K/k is transcendental.

Remark A.35 (Exact relation to NC.17)**.** NC.17 has precisely this architecture after projectivization: the principal symbol supply has one higher asymptotic exponent than the accumulated quantization boundary, while the robust Critical Window annihilates the first-order-generated top symbol in the selected critical degree. NC.18 shows that the same dimension-surplus principle survives the passage to an arbitrary local singularity provided two new quantities are controlled:

restricted Rees lift loss $O^K_{q,r}$ and restricted saturated shadow $S^K_{q,r}$.

A.1.13 Why the old Critical Window does not automatically globalize

Policy 33 (No false positive-gap theorem)**.** *From*

$$\delta(\mathfrak{m}^n) \subseteq \mathfrak{m}^n$$

one may conclude only that derivations have nonnegative filtration shift. One may not conclude that every non-Euler derivation has a strictly positive uniform shift. The tangent cone may possess many degree-zero derivations. Therefore the homogeneous-ICIS threshold

$$r < qm$$

has no automatic analogue for an arbitrary tangent cone.

Remark A.36 (The actual replacement)**.** The correct general object is not a guessed scalar threshold but the saturated shadow

$$S_{q,r} \subseteq V_{q,r}.$$

For homogeneous ICIS, the Critical Window computes this shadow sharply: its top projective symbol is zero in the critical sector. In general, NC must estimate the shadow directly or construct a quotient in which it becomes subleading. This is the precise content of "Critical Window 2.0."

A.1.14 The universal Shadow-Defect Gate

Gate A.37 (Universal Shadow-Defect Gate)**.** For every singular isolated local k-algebra $(R, \mathfrak{m})$ essentially of finite type in characteristic zero, prove that

$$\boxed{\mathfrak{N}_q(R) \neq 0 \text{ for some finite } q.}$$

Equivalently, it is sufficient to prove the K-linear dimension-surplus inequality of NC.18.12 for some q, r, where $K = R/\mathfrak{m}$. The defect nonvanishing form remains the invariant statement used in the unrestricted reduction.

Theorem A.38 (Conditional completion theorem)**.** *If the Universal Shadow-Defect Gate holds for every isolated local singularity essentially of finite type over characteristic zero fields, then the Nakai conjecture holds for every finitely generated characteristic-zero algebra.*

Proof. Assume a global counterexample exists. By the isolated-core reduction there is an isolated local counterexample $(R, \mathfrak{m})$ with $D_{R/k} = \Delta_R$. The Universal Shadow-Defect Gate produces $\mathfrak{N}_q(R) \neq 0$ for some q. The defect criterion then gives $D_{R/k} \neq \Delta_R$, contradiction. □

Declaration A.39 (Status)**.** The conditional completion theorem is proved. The Universal Shadow-Defect Gate is not proved in NC.18. Therefore NC.18 does *not* claim a proof of the general Nakai conjecture.

A.1.15 Three attack routes for the remaining gate

Declaration A.40 (Route A: tangent-cone symbol abundance)**.** Construct a functorial finite-dimensional sector

$$W_{q,r} \subseteq (D^q_{G/K})_r, \qquad K = R/\mathfrak{m},$$

whose K-dimension has a positive leading term of order q^a, preferably by a projective or blow-up compactification of Spec G. This is the direct successor of the NC.17 symbol-volume engine.

Declaration A.41 (Route B: Rees obstruction codimension)**.** Prove that

$$\dim_K O^K_{q,r} = O(q^{a-1})$$

in the same sector. Since

$$O^K_{q,r} = \mathrm{im}\Big(\partial_q|_{(D^q_{G/K})_r}\Big),$$

this is now a concrete homological-growth problem. The expected mechanism is support on the degeneration fiber and a one-dimension loss relative to the main symbol sector; this expectation requires proof and is not assumed.

Declaration A.42 (Route C: saturated-shadow compression)**.** Prove

$$\dim_K S^K_{q,r} = O(q^{a-1})$$

or construct a quotient of $V^K_{q,r}$ in which the image of $S^K_{q,r}$ is subleading. A likely intrinsic input is the cotangent-complex defect of the singularity. At first order one has

$$\mathrm{Der}_k(R) = \mathrm{Hom}_R(\Omega_{R/k}, R),$$

while the full cotangent complex $L_{R/k}$ records higher singularity data not visible in the locally free smooth case, and higher differential operators retain information from higher principal-parts neighborhoods. NC.18 records this as a research direction, not a theorem.

A.1.16 A sharper contradiction template

Theorem A.43 (Counterexample forcing package)**.** *Suppose* $(R, \mathfrak{m})$ *is an isolated local singularity and assume, for contradiction,*

$$D_{R/k} = \Delta_R.$$

Then all of the following hold simultaneously:

(1) every differential operator preserves every $\mathfrak{m}^n$*;*

(2) every first-order derivation admits a canonical Rees lift;

(3) every Rees-liftable tangent-cone operator belongs to the saturated derivation shadow:

$$\mathfrak{Q}_q(R) = \mathfrak{S}_q(R) \quad \forall q;$$

(4) every canonical K*-linear graded sector satisfies*

$$\dim_K V^K_{q,r} - \dim_K O^K_{q,r} \le \dim_K S^K_{q,r}.$$

Thus a proof of the general Nakai conjecture may proceed by violating any one of these forced inequalities for every singular isolated local ring.

Proof. Item (1) is filtration rigidity. Item (2) is the Rees lifting lemma. Item (3) is the necessary special-fiber condition for a Nakai counterexample. Item (4) follows by taking dimensions in item (3) and using the exact quantization gate. □

A.1.17 Audit: what NC.18 proves and what it does not

Claim ID	Statement	Status
NC18.1	Finite-order differential operators localize for finite-type algebras	C1 imported / C0 assembly
NC18.2	A global Nakai counterexample yields an isolated local counterexample	C2 internal
NC18.3	The maximal ideal of an isolated local singularity is derivation-stable in characteristic zero	C1 imported
NC18.4	Δ_R preserves the full $\mathfrak{m}$-adic filtration	C0 internal
NC18.5	Extended Rees generic/special fibers are $R[t, t^{-1}]$ and $\mathrm{gr}_{\mathfrak{m}} R$	C0 standard
NC18.6	$P^q_{\mathcal{R}/B}/u \cong P^q_{G/k}$	C2 internal
NC18.7	Exact Rees Quantization Gate with obstruction $\mathrm{Ext}^1(P^q, \mathcal{R})[u]$	C2-high internal
NC18.8	Every derivation of the isolated core has a canonical Rees lift	C2 internal

NC18.9	Saturated-shadow strong-escape transfer	C2-high internal
NC18.10	Rees–Nakai defect nonzero implies strong Nakai escape	C2-high internal
NC18.11	Canonical $K = R/\mathfrak{m}$ finite-dimensional sector and dimension-surplus criterion	C2-high internal
NC18.12	Residue-field compatibility via $D_{G/K} \subset D_{G/k}$	C2 internal
NC18.13	Universal Shadow-Defect Gate for every isolated singularity	OPEN / next gate
NC18.14	General Nakai conjecture	NOT CLAIMED

Policy 34 (Current publication ceiling)**.** *NC.18 is a bridge theorem package. It materially narrows the general problem, but it does not by itself settle the unrestricted Nakai conjecture. The correct research status is therefore:*

$$\boxed{\textit{general Nakai reduced to an exact isolated Rees-shadow defect problem.}}$$

Any stronger statement is prohibited until the Universal Shadow-Defect Gate is proved.

A.1.18 Immediate NC continuation

Declaration A.44 (NC.19 target)**.** The next NC section should attack the Universal Shadow-Defect Gate itself, not add another isolated example. The preferred order is:

(1) define an intrinsic tangent-cone/projectivized operator sector that exists without the complete-intersection hypothesis;

(2) estimate the u-torsion Ext^1 obstruction in that sector;

(3) identify the associated graded image of the saturated derivation algebra;

(4) prove a one-exponent dimension gap, or produce a finite-order defect when asymptotics are unavailable;

(5) only then promote the conditional completion theorem.

Remark A.45 (Engineering view of the proof architecture)**.** NC.17 solved a family by manufacturing more valid high-order symbols than all quantization losses could absorb. NC.18 shows that the unrestricted problem has exactly the same engineering shape after the correct Rees interface is inserted:

$$\boxed{\text{operator supply} - \text{Rees lift loss} - \text{saturated first-order shadow} > 0.}$$

The unresolved mathematics has therefore been compressed into two measurable terms rather than an informal "generalize the homogeneous argument" request.

References and source bindings

B Free-Tangent Collapse and Final-Gate Bifurcation

Source abstract

NC.18 reduced the unrestricted Nakai problem to the Universal Shadow-Defect Gate: for every isolated singular local algebra R essentially of finite type in characteristic zero, show $\mathfrak{R}_q(R) \neq 0$ for some finite q. NC.19 performs an adversarial audit of that last gate.

The main internal result of NC.19 is a free-tangent collapse theorem. If A is a normal integral algebra essentially of finite type over a characteristic-zero field and $\mathrm{Der}_k(A)$ is free of rank $\dim A$, then every finite-order k-linear differential operator is generated by multiplication and first-order derivations:

$$\boxed{D_{A/k} = A\langle \mathrm{Der}_k(A) \rangle.}$$

The proof uses only codimension-one smoothness, the smooth principal-symbol theorem, reflexivity of dual modules over a normal domain, and the codimension-one reconstruction of reflexive modules. This is the algebraic mechanism underlying the classical Becker–Rego observation that the Nakai conjecture implies the Zariski–Lipman conjecture.

Consequently, any singular normal affine characteristic-zero variety with free tangent sheaf is automatically a Nakai counterexample and, after localization at a minimal singular point, a counterexample to the Universal Shadow-Defect Gate of NC.18.

A public preprint posted on 23 September 2026 claims precisely such an example: a singular normal affine complex surface with free rank-two tangent sheaf. This manuscript has not been independently certified inside Appendix NC; current public indices list no Lean formalization and no independent endorsement. Therefore NC.19 records a bifurcated status. If that external construction survives audit, the unrestricted Nakai conjecture is false and the NC.18 universal gate is refuted. If it fails, the positive Universal Shadow-Defect Gate remains open. In neither branch is the homogeneous-ICIS theorem of NC.17 affected.

Declaration B.1 (Re-audited ownership status)**.** The implication "normal finite-type characteristic-zero algebra with projective/free tangent module $\Rightarrow$ $D = \Delta$" is not claimed as an AK priority result. NC.19 provides an internal reconstruction and reflexive-extension formulation of the classical Becker–Rego mechanism. It is therefore classified as AK reconstruction of externally established mathematics, not as the independent solution of Nakai.

B.1 NC.19 scope and the final-gate question

NC.18 defined, for an isolated local singularity $(R, \mathfrak{m})$, a quantizable special-fiber sector $\mathfrak{Q}_q(R)$, a saturated first-order derivation shadow $\mathfrak{S}_q(R)$, and the Rees–Nakai defect

$$\mathfrak{R}_q(R) = \mathfrak{Q}_q(R)/\mathfrak{S}_q(R).$$

The remaining universal gate was

$$\boxed{R \text{ singular isolated} \Longrightarrow \exists q \geq 0 : \ \mathfrak{R}_q(R) \neq 0.} \tag{NC.19.1}$$

NC.18 proved that (NC.19.1), if valid for every isolated local singularity essentially of finite type in characteristic zero, implies the unrestricted Nakai conjecture.

Policy 35 (Adversarial direction)**.** *A universal gate must be attacked from both sides. NC.19 therefore does not assume that (NC.19.1) is true. It first asks whether a structurally forced class with $D_{R/k} = \Delta_R$ can exist.*

B.2 Differential operators, the generated algebra, and localization

Let A be an integral k-algebra essentially of finite type. Write

$$D^q_{A/k} := \mathrm{Diff}^q_{A/k}(A, A), \qquad D_{A/k} := \bigcup_{q \geq 0} D^q_{A/k},$$

and

$$\Delta_A := A\langle \mathrm{Der}_k(A) \rangle \subseteq D_{A/k}.$$

For a finite order q, let $\Delta_A^{\leq q}$ denote the A-submodule spanned by compositions of at most q first-order derivations.

The module of principal parts $P^q_{A/k}$ is finite because A is essentially of finite type, and

$$D^q_{A/k} \cong \mathrm{Hom}_A(P^q_{A/k}, A). \tag{NC.19.2}$$

Formation of finite-order differential operators commutes with localization.

B.3 Reflexive extension package

Lemma B.2 (Duals are reflexive over a normal domain)**.** *Let A be a noetherian normal domain and M a finite A-module. Then*

$$M^* := \mathrm{Hom}_A(M, A)$$

is reflexive.

Proof. Replace M by its torsion-free quotient; this does not change M^* because A is a domain. For a finite torsion-free module and a reflexive target, Hom is reflexive. Since A is reflexive over itself, the assertion follows. □

Corollary B.3 (Finite-order differential operators are reflexive)**.** *If A is a noetherian normal domain essentially of finite type over k, then each $D^q_{A/k}$ is a finite reflexive A-module.*

Proof. Apply the lemma to (NC.19.2). □

Lemma B.4 (Codimension-one reconstruction)**.** *Let A be a noetherian normal domain with fraction field K. If M, N are finite reflexive A-submodules of the same finite-dimensional K-vector space and*

$$M_{\mathfrak{p}} = N_{\mathfrak{p}} \qquad \text{for every height-one prime } \mathfrak{p},$$

then $M = N$.

Proof. For a finite reflexive module over a normal domain,

$$M = \bigcap_{\operatorname{ht}\mathfrak{p}=1} M_{\mathfrak{p}}$$

inside $M \otimes_A K$. Apply the same identity to N. □

B.4 The free-tangent collapse theorem

Theorem B.5 (NC.19 Free-Tangent Collapse)**.** *Let k be a characteristic-zero field and let A be a normal integral k-algebra essentially of finite type. Put $d = \dim A$. Assume*

$$\operatorname{Der}_k(A) \cong A^{\oplus d}.$$

Then, for every $q \geq 0$,

$$\boxed{D^q_{A/k} = \Delta_A^{\leq q}.} \tag{NC.19.3}$$

Consequently

$$\boxed{D_{A/k} = A\langle \operatorname{Der}_k(A)\rangle.} \tag{NC.19.4}$$

Proof. Choose an A-basis

$$\theta_1, \dots, \theta_d$$

of $\operatorname{Der}_k(A)$. For each multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbf{N}^d$ define the ordered operator

$$\theta^\alpha := \theta_1^{\alpha_1} \cdots \theta_d^{\alpha_d}.$$

Let F_q be the A-span of the θ^α with $|\alpha| \leq q$. Clearly

$$F_q \subseteq \Delta_A^{\leq q} \subseteq D^q_{A/k}.$$

At the generic point, $\operatorname{Frac}(A)$ is a finitely generated separable field extension of k. The principal symbols of the operators θ^α are the usual monomials in a basis of the tangent space. Filtering a putative A-linear relation by operator order shows successively that every coefficient is zero. Hence F_q is a free A-module.

Now let $\mathfrak{p}$ be a height-one prime of A. Normality gives that $A_{\mathfrak{p}}$ is regular. Since k is perfect and A is essentially of finite type, this localization is smooth over k. The restrictions of $\theta_1, \dots, \theta_d$ form a basis of $\operatorname{Der}_k(A_{\mathfrak{p}})$. On a smooth characteristic-zero algebra the associated graded of the order filtration is

$$\operatorname{gr} D_{A_{\mathfrak{p}}/k} \cong \operatorname{Sym}_{A_{\mathfrak{p}}} \operatorname{Der}_k(A_{\mathfrak{p}}).$$

Therefore the principal symbols of the ordered monomials θ^α with $|\alpha| = j$ form a basis of the jth graded piece. Induction on j gives

$$(F_q)_{\mathfrak{p}} = D^q_{A_{\mathfrak{p}}/k} = (D^q_{A/k})_{\mathfrak{p}}. \tag{NC.19.5}$$

The module F_q is free, hence reflexive. The module $D^q_{A/k}$ is reflexive by the previous section. Equation (NC.19.5) gives equality at every height-one prime. The codimension-one reconstruction lemma now yields

$$F_q = D^q_{A/k}.$$

Since $F_q \subseteq \Delta_A^{\leq q} \subseteq D^q_{A/k}$, all three modules are equal. Taking the union over q proves (NC.19.4). □

Remark B.6 (Noncommuting frames cause no PBW shortcut)**.** The proof does not assume that the basis θ_i commutes. Ordered monomials are controlled through the order filtration: their highest symbols are the ordinary symmetric monomials. Lower-order commutator terms cannot create a relation in the highest graded piece. Thus no global PBW theorem on the singular ring is being imported.

Remark B.7 (Relation to Becker–Rego)**.** Theorem B.5 is the precise NC form of the classical mechanism behind the Becker–Rego observation that Nakai implies Zariski–Lipman. Rego's 1977 proof uses the same two ingredients: monomials in a free derivation frame generate differential operators on the smooth locus, and normality extends the equality across codimension at least two.

B.5 Immediate negative criterion for Nakai

Corollary B.8 (Free-tangent Nakai counterexample criterion)**.** *Let A satisfy the hypotheses of Theorem B.5 and suppose A is singular. Then A is a counterexample to the unrestricted Nakai implication:*

$$\boxed{A \text{ singular} \quad \text{and} \quad D_{A/k} = A\langle \mathrm{Der}_k(A)\rangle.} \tag{NC.19.6}$$

Proof. The equality is Theorem B.5; singularity contradicts the proposed regularity conclusion. □

Corollary B.9 (Local surface form)**.** *Let A be a normal affine integral surface over a characteristic-zero field, and assume $\mathrm{Der}_k(A)$ is free of rank two. If $x \in \mathrm{Spec}\, A$ is singular and $R = A_{\mathfrak{m}_x}$, then*

$$R \text{ is an isolated normal local surface singularity}, \qquad D_{R/k} = R\langle \mathrm{Der}_k(R)\rangle. \tag{NC.19.7}$$

Proof. A normal surface is regular in codimension one, hence its singular locus has codimension at least two; after localizing at a singular closed point, the singularity is isolated. Freeness of derivations and the equality (NC.19.4) localize. □

B.6 Impact on the NC.18 Universal Shadow-Defect Gate

Theorem B.10 (Free tangent forces total shadow collapse)**.** *Let $(R, \mathfrak{m})$ be an isolated normal local singularity essentially of finite type over a characteristic-zero field. If*

$$\mathrm{Der}_k(R) \cong R^{\oplus \dim R},$$

then

$$D_{R/k} = \Delta_R.$$

Consequently the necessary condition proved in NC.18 gives, for every q,

$$\boxed{\mathfrak{Q}_q(R) = \mathfrak{S}_q(R), \qquad \mathfrak{R}_q(R) = 0.} \tag{NC.19.8}$$

Therefore the Universal Shadow-Defect Gate is false on any such singularity.

Proof. The first assertion is Theorem B.5. NC.18 proved that $D_{R/k} = \Delta_R$ forces every Rees-liftable special-fiber operator into the saturated derivation shadow, while the reverse inclusion is tautological. Thus $\mathfrak{Q}_q(R) = \mathfrak{S}_q(R)$ and the quotient defect vanishes. □

Declaration B.11 (What has been killed internally)**.** NC.18's universal nonvanishing gate cannot be accepted as a formal principle independent of the geometry of tangent freeness. Theorem B.10 identifies an exact fatal class: any singular normal free-tangent germ forces the defect to vanish at every order, not merely at low order.

B.7 The September 2026 free-tangent surface claim

On 23 September 2026 a public OpenAI mathematics preprint entitled *A singular normal affine surface with free tangent sheaf* was posted. Its public catalogue statement asserts the existence of a singular normal affine complex surface X with

$$\mathcal{T}_X \cong \mathcal{O}_X^{\oplus 2}. \tag{NC.19.9}$$

Because X is affine, (NC.19.9) would imply for $A = \Gamma(X, \mathcal{O}_X)$ that

$$\mathrm{Der}_{\mathbf{C}}(A) \cong A^{\oplus 2}.$$

Theorem B.12 (Conditional negative resolution from the 2026 surface)**.** *Assume the construction claimed in the September 2026 preprint is mathematically valid at the stated strength: $X = \operatorname{Spec} A$ is a singular normal affine complex surface and $\mathcal{T}_X$ is free of rank two. Then*

$$\boxed{D_{A/\mathbf{C}} = A\langle \mathrm{Der}_{\mathbf{C}}(A)\rangle \quad \textit{while } A \textit{ is singular.}} \tag{NC.19.10}$$

Hence:

(i) the unrestricted characteristic-zero Nakai conjecture is false;

(ii) the characteristic-zero Zariski–Lipman conjecture is false;

(iii) after localization at a singular point, NC.18's Universal Shadow-Defect Gate is false and $\mathfrak{N}_q(R) = 0$ for every q.

Proof. Apply Corollary B.8. The local statement follows from the local surface corollary and Theorem B.10. □

Policy 36 (External validation ceiling)**.** *Theorem B.12 is a genuine theorem with one explicit external premise. Appendix NC does* not *presently certify that premise. Public indices available on 17 October 2026 list the manuscript as a 37-page preprint; no Lean formalization is listed, and a public problem index records the claim without independent endorsement. Therefore NC must not replace "if the construction is valid" by an unqualified statement that Nakai has been refuted.*

B.8 Final-gate bifurcation

Theorem B.13 (NC.19 Final-Gate Bifurcation)**.** *Exactly the following research alternatives remain compatible with the present audit.*

Branch A. *The September 2026 singular free-tangent surface is valid. Then the unrestricted Nakai conjecture is false by Theorem B.12, and the NC.18 Universal Shadow-Defect Gate is definitively false.*

Branch B. *The September 2026 construction fails one of the required properties (normality, singularity, finite type, or tangent freeness). Then NC.19 supplies no counterexample, and the NC.18 positive gate remains an open problem requiring a new restriction that excludes free-tangent collapse or a different global strategy.*

Remark B.14 (There is no third logical outcome)**.** If a singular normal affine variety in characteristic zero has a free tangent module, the differential-operator conclusion is no longer open. Theorem B.5 forces generation by first-order derivations. The only unresolved point in Branch A is the validity of the external geometric construction itself.

B.9 What survives regardless of the external fork

The negative interface does not invalidate the positive NC results on homogeneous complete intersections. NC.17 proves, subject to its declared imported interfaces, that for the standard homogeneous complete-intersection source class of codimension at least two and smooth positive-dimensional projectivization,

$$D_{A/k} \neq A\langle \mathrm{Der}_k(A)\rangle.$$

The two statements are compatible: a free-tangent Nakai counterexample, if valid, lies outside the structural regime in which NC.17's Critical Window and symbol-surplus theorem force strong escape.

Declaration B.15 (Architecture lesson)**.** The attempted universalization of the Critical Window has discovered its exact boundary. The obstruction is not merely that arbitrary tangent cones have degree-zero derivations; it is stronger: there may exist singular normal spaces whose entire tangent module is free, in which case first-order derivations already generate all finite-order differential operators.

B.10 Audit register

Item	Statement	Status
NC19.1	$D^q_{A/k}$ is reflexive for normal A	proved internally / standard dual-reflexive lemma
NC19.2	Free-tangent collapse $D_{A/k} = A\langle \mathrm{Der}_k(A)\rangle$	proved internally
NC19.3	Singular normal free-tangent A is a Nakai counterexample	proved conditionally only on existence of such A
NC19.4	Such a local singularity has $\mathfrak{N}_q = 0$ for all q	proved from NC.18 + NC19.2
NC19.5	September 2026 preprint really constructs the required surface	external claim; not independently certified in NC
NC19.6	Unrestricted Nakai is false	conditional on NC19.5
NC19.7	Positive Universal Shadow-Defect Gate	cannot be promoted; refuted if NC19.5 validates

B.11 Required next audit and claim ceiling

To convert Branch A from conditional to certified negative resolution, no further asymptotic symbol calculation is required. The remaining task is a line-by-line audit of the external surface construction sufficient to certify four properties:

$$\boxed{A \text{ finite type over } \mathbf{C}, \quad A \text{ normal}, \quad A \text{ singular}, \quad \mathrm{Der}_{\mathbf{C}}(A) \cong A^2.} \tag{NC.19.11}$$

Once these four are certified, Theorem B.5 supplies the Nakai counterexample automatically.

Declaration B.16 (NC.19 status freeze)**.** The internal free-tangent kill switch is closed. The unrestricted Nakai conjecture is not claimed as positively proved. A recent explicit negative-resolution candidate exists, but its geometric construction remains externally unaudited. The correct present label is

NC.19: internal kill-switch proved; unrestricted Nakai status conditional-negative.

References and source bindings

C Family 048 Audit and Negative Nakai Transfer

Source abstract

This appendix performs a source-level audit of the September 23, 2026 manuscript *A singular normal affine surface with free tangent sheaf* and connects its main theorem to the Nakai conjecture through the classical Becker–Rego implication and the independent NC.19 free-tangent collapse theorem. The external manuscript supplies a finitely generated normal integral complex algebra A of dimension two, a singular maximal localization, and $\mathrm{Der}_{\mathbf{C}}(A) \simeq A^{\oplus 2}$. NC.20 rechecks the construction in five layers: curve/residue data, low-order jet

obstruction cancellation, analytic contraction and Hartogs extension, algebraization/faithfully-flat descent, and the Nakai transfer.

No fatal gap was found in the audited proof chain. In particular, the order-two residue matrix was replayed independently and its coefficients agree with the manuscript. Subject to the standard cited Grauert, Artin-approximation, formal-functions, and completion theorems, the audited source proves a characteristic-zero counterexample to Zariski–Lipman. Since Becker–Rego prove that Nakai implies Zariski–Lipman, and NC.19 independently proves the corresponding differential-operator collapse for normal free-tangent algebras, the unrestricted characteristic-zero Nakai conjecture is therefore false. The publication status remains conservative: the source is not peer reviewed and no Lean formalization is currently listed, so NC records the result as a proof-level C2-high negative resolution rather than a formal-certification claim.

Declaration C.1 (Resolution-ownership correction)**.** The unrestricted Nakai conjecture is classified in this final edition as **Class III: externally resolved**. The negative conclusion does not mathematically require an AK-specific theorem. The external Family 048 singular normal affine complex surface with free tangent module, together with the classical Becker–Rego implication from Nakai to Zariski–Lipman (equivalently its contrapositive at the conjectural level), already forces the unrestricted Nakai conjecture to be false, conditional only on the correctness of the Family 048 preprint. NC.19–NC.20 remain valuable as an AK reconstruction, transfer audit, and source-consistency audit; they are not the ownership basis of the negative resolution.

C.1 Exact logical target

For an essentially finite-type characteristic-zero algebra R, write

$$D_{R/k} = \bigcup_{q\geq 0} \mathrm{Diff}^q_{R/k}(R,R), \qquad \Delta_R = R\langle \mathrm{Der}_k(R)\rangle.$$

The Nakai implication in the form used throughout Appendix NC is

$$D_{R/k} = \Delta_R \quad \Longrightarrow \quad R \text{ regular}. \tag{C.1}$$

A single singular essentially finite-type R with $D_{R/k} = \Delta_R$ refutes the unrestricted conjecture.

C.2 The external witness theorem

The September 23, 2026 source states the following theorem.

Theorem C.2 (Free-tangent surface theorem; audited external input)**.** *There exist a finitely generated normal integral* $\mathbf{C}$*-algebra* A *of Krull dimension two and a maximal ideal* $\mathfrak{m} \subset A$ *such that*

$$\mathrm{Der}_{\mathbf{C}}(A) \simeq A^{\oplus 2}, \qquad A_{\mathfrak{m}} \text{ is not regular}.$$

The affine surface may be chosen smooth away from the marked point.

The rest of this appendix records the independent NC audit of the proof of this theorem.

C.3 Audit layer I: curve and residue engine

The source constructs an elliptic curve $\Gamma \subset \mathbf{P}^1 \times \mathbf{P}^1$ of bidegree $(2,2)$, a curve C of bidegree $(8,4)$, a reduced six-point divisor Z, and a line bundle

$$L = H + Z, \qquad \deg H = 4, \qquad \deg L = 10, \qquad L^{\otimes 4} \simeq K_C.$$

Adjunction gives

$$\deg K_C = (6,2)\cdot(8,4) = 40, \qquad g(C) = 21,$$

and all numerical identities used later are consistent.

The key section counts were rederived from Riemann–Roch, Serre duality, and the fourth-order contact with Γ:

$$h^0(C,L) = 2, \quad h^0(C,2L) = 3, \quad h^0(C,O_C(Z)) = 1, \quad h^0(C,2L+H) = 6.$$

The evaluation maps

$$H^0(C,(k+1)L)\to H^0(Z,(k+1)L|_Z)$$

are surjective for $k\ge 2$ as claimed. The residue construction yields six local rows satisfying

$$p_j(0)\wedge R_j(0)=1,\qquad W_j=R_j(0)\wedge R_j'(0),\qquad \sum_{j=1}^{6}(W_j-1)=-9\deg L=-90. \tag{C.2}$$

No inconsistency was found in the trace-pairing normalization or the three-direction determination of the binary quadratic form.

C.4 Audit layer II: low-order jet obstruction

The first-order transition is generated by

$$D=t\left(\frac{m'}{3}E_t+m\partial_z\right),\qquad E_t=t\partial_t,$$

and sends the disk row (p,mr) to the outside row $(P,0)$ because

$$\Delta A_1=mr',\qquad \Delta B_1=-mr,\qquad P=p+mr'.$$

At order two, the difficult obstruction is an element of $H^1(C,L^3)^{\oplus 2}$. The manuscript computes its Serre-duality residues through

$$\Psi_{ij}=\left(r_iA_{2,j}+\frac{1}{4}r_i'U_j+\frac{3}{4}r_j'U_i\right)dz.$$

The coefficient calculation was independently replayed. With $m=c/z$, $R=mr$, one obtains exactly

$$\begin{aligned}\Delta A_2&=-\frac{cR}{z^3}+\frac{cR'}{3z^2}+\frac{cR''}{2z}-\frac{cp}{3z^2}+\frac{cp'}{z},\\ \Delta U&=-\frac{cp}{z}-\frac{cR}{3z^2},\end{aligned}$$

and therefore

$$\begin{aligned}F_{ij}&=-\frac{1}{3}R_ip_j-\frac{2}{3}R_iR_j'-R_i'R_j,\\ H_{ij}&=-R_ip_j-\frac{1}{3}R_iR_j'-\frac{2}{3}R_i'R_j.\end{aligned}$$

The symmetric and alternating pieces of the resulting residue matrix are

$$\begin{aligned}T_{ij}^{\mathrm{sym}}&=-\frac{2}{3}\big(R_ip_j+R_jp_i+2R_iR_j'+2R_i'R_j\big),\\ T_{ij}^{\mathrm{alt}}&=\frac{1}{12}\epsilon_{ij}\big((R\wedge p)(0)+(R\wedge R')(0)\big).\end{aligned}$$

The symmetric sum vanishes by the global residue theorem applied to $P_iP_j\,dz$. Using (C.2), the alternating total is

$$\frac{1}{12}\sum_{j=1}^{6}(W_j-1)=-\frac{90}{12}=-\frac{3}{4}\deg L,$$

which exactly equals the outside obstruction. Thus the order-two class vanishes as claimed.

For $k\ge 4$, the vertical obstruction vanishes because

$$H^1(C,L^{k+1})\simeq H^0(C,L^{3-k})^*=0.$$

The exceptional $k=3$ class lies in $H^1(C,K_C)^{\oplus 2}$ and is killed by the six available overlap parameters because the rows $r'(q_j)$ span $\mathbf{C}^2$. The determinant-target obstruction is zero at $k=4$ by exactness of dh and for $k>4$ by

$$H^1(C,L^k)\simeq H^0(C,L^{4-k})^*=0.$$

Thus the finite-jet induction through index seven is coherent.

C.5 Audit layer III: analytic realization

The finite overlap flows are actual holomorphic germs on annuli and define, after shrinking, a Hausdorff smooth surface neighborhood Y of C. The conormal bundle remains L, so

$$C^2 = -\deg L = -10.$$

Let $\mathcal{I} = \mathcal{O}_Y(-C)$ and $\mathcal{F} = T_Y(-\log C)$. The finite jets define two sections of $\mathcal{F}/\mathcal{I}^9\mathcal{F}$ whose ordinary determinant has leading form

$$t^5(1 + O(t))\,\partial_t \wedge \partial_z.$$

The lifting argument uses

$$0 \to \mathcal{O}_C \to \mathcal{F}|_C \to T_C \to 0$$

and therefore, for $n \geq 9$,

$$H^1(C, \mathcal{F}|_C \otimes L^n) = 0$$

because the two outer terms are L^n and L^{n-4} and Serre duality reduces them to $H^0(L^{4-n})$ and $H^0(L^{8-n})$.

Grauert contraction applies to the negative intersection matrix (-10) and contracts C to a point x of a normal surface X. Formal functions, together with the cofinal $\mathcal{I}$-adic and $\mathfrak{m}_x\mathcal{O}_Y$-adic filtrations, gives

$$R^1\pi_*(\mathcal{I}^9\mathcal{F}) = 0,$$

so the finite jets lift to holomorphic vector fields. Their determinant is nonzero on $Y \setminus C$, hence they frame $T_{X\setminus\{x\}}$.

Normality gives Hartogs extension across the codimension-two point. Extending the coefficients of the frame and of an arbitrary derivation proves

$$\mathcal{T}_X \simeq \mathcal{O}_X^{\oplus 2}.$$

The contracted point is singular: if it were smooth, pulling back a nonvanishing local two-form would give $K_Y \cdot C \leq 0$, while adjunction gives

$$K_Y \cdot C = 2g(C) - 2 - C^2 = 40 + 10 = 50,$$

a contradiction.

C.6 Audit layer IV: algebraization and descent

The analytic singularity is algebraized by approximating both defining equations and a finite relation matrix. The source proves a finite-determinacy lemma: if g and a relation matrix B are sufficiently close to f and D, with $gB = 0$ exactly, then

$$\mathbf{C}[[x_1, \ldots, x_e]]/(g) \simeq \mathbf{C}[[x_1, \ldots, x_e]]/(f).$$

The proof uses Artin–Rees to control the relation module and the finite-length module

$$T^1 = \mathrm{Hom}_R(I/I^2, R)/J.$$

For an isolated singularity this T^1 is supported only at the marked point, hence has finite length.

Artin approximation over the henselization produces algebraic equations and relations with the required finite precision and exact relation $gB = 0$. Standard approximation for henselian G-rings supplies the needed solution in an algebraic/étale neighborhood. Completion then recovers the analytic complete local ring.

The tangent module descends correctly. Finite presentation of Ω^1 and flatness of completion give

$$\mathrm{Der}_{\mathbf{C}}(T) \otimes_T \widehat{T} \simeq \mathrm{Der}^{\mathrm{cont}}_{\mathbf{C}}(\widehat{T}, \widehat{T}) \simeq \widehat{T}^{\oplus 2}.$$

Nakayama plus faithful flatness then yields

$$\mathrm{Der}_{\mathbf{C}}(T) \simeq T^{\oplus 2}.$$

Normality and nonregularity are preserved/descent-controlled through completion. Shrinking to a principal affine neighborhood globalizes the free basis and yields a finitely generated normal integral algebra A with

$$\dim A = 2, \qquad \mathrm{Der}_{\mathbf{C}}(A) \simeq A^{\oplus 2}, \qquad A_{\mathfrak{m}} \text{ singular.}$$

C.7 Audit layer V: Nakai transfer

The required transfer is classical.

Theorem C.3 (Becker–Rego / NC.19)**.** *Let R be a normal integral essentially finite-type algebra over a characteristic-zero field, of dimension d. If*

$$\mathrm{Der}_k(R) \simeq R^{\oplus d},$$

then

$$\boxed{D_{R/k} = R\langle \mathrm{Der}_k(R)\rangle.}$$

NC.19 independently proves this by comparing finite-order differential operators and ordered monomials in a free derivation frame at every height-one prime and then using reflexive reconstruction over the normal domain. This is the same mechanism underlying Rego's classical proposition that an affirmative Nakai conjecture would imply Zariski–Lipman.

Corollary C.4 (Negative resolution of unrestricted Nakai)**.** *Let A be the audited algebra above and set $R = A_{\mathfrak{m}}$. Then*

$$R \text{ is singular}, \qquad \mathrm{Der}_{\mathbf{C}}(R) \simeq R^{\oplus 2}, \qquad D_{R/\mathbf{C}} = R\langle \mathrm{Der}_{\mathbf{C}}(R)\rangle.$$

Hence the unrestricted characteristic-zero Nakai conjecture is false.

C.8 Claim register and status

Claim	NC audit	Comment
Curve/canonical-root data	PASS	Numerical and cohomological identities replayed.
Trace/residue normalization	PASS	Bundle degrees, principal-part dimensions and residue identities consistent.
Order-2 jet cancellation	PASS	Coefficients independently replayed; symmetric and alternating totals agree exactly.
Higher-order jet induction	PASS	Remaining obstructions vanish by the stated Serre-duality groups; $k = 3, 4$ exceptions are separately controlled.
Analytic contraction and tangent freeness	PASS	Grauert contraction, formal-functions lifting and Hartogs extension fit the stated hypotheses.
Algebraization/descent	PASS	Artin approximation, finite determinacy, Nakayama and faithful flatness form a coherent chain.
Becker–Rego / NC.19 transfer	PASS	Classical implication plus independent NC.19 reflexive proof.
Unrestricted Nakai	**NEGATIVE**	Proof-level consequence of the audited free-tangent surface.
External peer review / formalization	OPEN	The OpenAI repository itself states that unformalized results may still contain issues; Family 048 has no listed Lean formalization.

Policy 37 (Publication ceiling)**.** *Within Appendix NC, the mathematical conclusion is recorded as a* proof-level negative resolution *with status C2-high. This is stronger than the earlier conditional-negative status because the actual source proof has now been audited rather than only its abstract. It is not labeled formally certified or peer reviewed until an external referee or formal proof independently certifies the source construction.*

C.9 Consequences for Appendix NC

The Universal Shadow-Defect Gate proposed in NC.18 cannot hold for all isolated normal singularities. The audited free-tangent local surface has

$$D_{R/\mathbf{C}} = R\langle \mathrm{Der}_{\mathbf{C}}(R)\rangle,$$

so every Rees-liftable operator lies in the saturated derivation shadow and the universal defect must vanish. This does not affect NC.17: the strong escape theorem for the homogeneous-ICIS class remains a scoped positive result. The correct global picture is therefore bifurcated:

homogeneous ICIS: derivation-generated equality fails (NC.17),
general normal finite-type algebras: equality can hold at singular points (NC.20).

Declaration C.5 (NC.20 final internal verdict)**.** No fatal defect was found in the source-level audit of the September 2026 free-tangent surface construction. Combined with Becker–Rego and NC.19, it yields a singular characteristic-zero local algebra satisfying the Nakai generation equality. Appendix NC therefore records the unrestricted Nakai conjecture as negatively resolved at proof level, with external/formal certification still pending.

D Reflexive Escape–Collapse Boundary Theory

Source abstract

Appendix NC.21 isolates the exact algebraic boundary between the two regimes identified in NC.17–NC.20. Let k be a characteristic-zero field and let A be a finitely generated normal integral k-algebra. Write

$$D_{A/k} = \bigcup_{q\geq 0} D^q_{A/k}, \qquad \Delta_A = A\langle \mathrm{Der}_k(A)\rangle \subseteq D_{A/k}.$$

For each order q define the *order-filtered derivation-generated module*

$$M_q(A) := \Delta_A \cap D^q_{A/k}.$$

The principal theorem proves the exact identity

$$\boxed{D^q_{A/k} \cong M_q(A)^{**}} \qquad (q \geq 0).$$

Hence the discrepancy between all differential operators and those generated by first-order derivations is exactly the failure of $M_q(A)$ to be reflexive:

$$\boxed{\mathfrak{B}_q(A) := M_q(A)^{**}/M_q(A) \cong D^q_{A/k}/M_q(A).}$$

Consequently

$$D_{A/k} = \Delta_A \iff \mathfrak{B}_q(A) = 0 \text{ for every } q \iff M_q(A) \text{ is reflexive for every } q.$$

Because A is normal, each $M_q(A)$ is torsion-free; therefore collapse is equivalent to the Serre condition (S_2) for every $M_q(A)$. The defect is supported in codimension at least two. For an isolated normal local singularity $(A, \mathfrak{m})$ one further obtains

$$\mathfrak{B}_q(A) \cong H^1_{\mathfrak{m}}(M_q(A)),$$

so Escape versus Collapse becomes a local-cohomological depth dichotomy.

NC.21 also strengthens NC.19 from free tangent modules to projective tangent modules, records that reflexivity of $\mathrm{Der}_k(A)$ and codimension-one local freeness are automatic in the normal setting and therefore cannot distinguish the boundary, relates the intrinsic defect to the NC.18 Rees–Nakai defect, and introduces the first-escape order and the isolated-singularity boundary profile as new audit-compatible invariants. No literature-priority claim is made: the exact boundary formulation is presented as an internal AK theorem and is compared with the classical Becker–Rego mechanism, principal-parts theory, reflexive-module technology, and current Nakai literature.

Declaration D.1 (Post-resolution role)**.** NC.21 is a post-Nakai structural theorem package: it classifies the intrinsic reflexive defect separating Escape from Collapse. It is logically independent of claiming AK ownership of the negative Nakai resolution.

D.1 NC.21 scope, standing hypotheses, and audit ceiling

D.1.1 Standing hypotheses

Unless explicitly stated otherwise, throughout NC.21:

(H1) k is a field of characteristic zero;

(H2) A is a finitely generated normal integral k-algebra;

(H3) $K = \mathrm{Frac}(A)$;

(H4) $D^q_{A/k} = \mathrm{Diff}^q_{A/k}(A, A)$ denotes Grothendieck differential operators of order at most q;

(H5) $\Delta_A = A\langle \mathrm{Der}_k(A)\rangle$ denotes the k-subalgebra of $D_{A/k}$ generated by multiplication operators and first-order k-derivations.

Since k has characteristic zero it is perfect. Therefore every regular localization of a finitely generated k-algebra is smooth over k. The standard smooth-generation theorem for differential operators may then be used in codimension one [31, 18, 35].

D.1.2 Why the order filtration must be used

The literature distinguishes several "generated in order q" modules. In particular, the word-length module generated by products of at most q derivations need not equal $\Delta_A \cap D^q_{A/k}$ on a singular ring; long words can suffer order drop or cancellation. Recent Nakai work explicitly records examples where these modules differ [34].

Accordingly NC.21 uses the intrinsic order filtration

$$M_q(A) := \Delta_A \cap D^q_{A/k}. \tag{D.1}$$

This is the exact filtration relevant to the algebra equality $D_{A/k} = \Delta_A$.

Declaration D.2 (Claim ceiling)**.** All theorems through the exact reflexive-boundary theorem are internal algebraic statements assembled from standard imported interfaces that are cited explicitly. The publication status is C2-high pending independent line-by-line source binding and external referee review. No claim is made that the terminology "Escape– Collapse boundary" or the defect notation is prior-art novel.

D.2 Reflexivity of finite-order differential operators

D.2.1 Principal parts

Let

$$I_\Delta = \ker(A \otimes_k A \to A), \qquad P^q_{A/k} := (A \otimes_k A)/I_\Delta^{q+1}.$$

Since A is of finite type over k, $P^q_{A/k}$ is a finite A-module. Grothendieck's principal-parts representation gives

$$D^q_{A/k} \cong \mathrm{Hom}_A(P^q_{A/k}, A). \tag{D.2}$$

See EGA IV, §16 [31]; the same representation is used throughout modern work on differential signature and rings of differential operators [29, 35].

Lemma D.3 (Finite reflexivity of D^q)**.** *For every $q \geq 0$, $D^q_{A/k}$ is a finite reflexive A-module.*

Proof. Finiteness follows from (D.2). Since A is normal it is a reflexive module over itself. If M is finite and N is finite reflexive over a Noetherian domain, then $\mathrm{Hom}_A(M, N)$ is reflexive [36, Tag 0AV3]. Apply this with $M = P^q_{A/k}$ and $N = A$. □

Remark D.4. The reflexivity of finite-order differential-operator modules on normal rings is also used explicitly in the modern D-module literature [35]. NC.21 therefore does not introduce a new reflexivity assertion for D^q; the new step is to compare D^q to the exact order-filtered derivation algebra.

D.3 Codimension-one coincidence

D.3.1 Localization interfaces

Differential operators localize for algebras essentially of finite type. In principal-parts language, finite presentation gives

$$(D^q_{A/k})_{\mathfrak{p}} \cong D^q_{A_{\mathfrak{p}}/k}. \tag{D.3}$$

Likewise

$$\mathrm{Der}_k(A)_{\mathfrak{p}} \cong \mathrm{Der}_k(A_{\mathfrak{p}}), \qquad (\Delta_A)_{\mathfrak{p}} = \Delta_{A_{\mathfrak{p}}}.$$

These are the same localization interfaces already used in NC.18.

Lemma D.5 (Localization of the intrinsic order filtration)**.** *For every prime $\mathfrak{p} \subset A$ and every $q \geq 0$,*

$$M_q(A)_{\mathfrak{p}} = \Delta_{A_{\mathfrak{p}}} \cap D^q_{A_{\mathfrak{p}}/k}$$

inside $D_{A_{\mathfrak{p}}/k}$.

Proof. The forward inclusion is immediate. For the reverse inclusion, let x belong to the right side. Choose

$$x = a/s = b/t$$

with $a \in \Delta_A$, $b \in D^q_{A/k}$ and $s, t \notin \mathfrak{p}$. Equality in the localization gives some $u \notin \mathfrak{p}$ with $u(ta - sb) = 0$ in $D_{A/k}$. The A-module $D_{A/k}$ is torsion-free: if $0 \neq c \in A$ and $cP = 0$, then $cP(f) = 0$ for every $f \in A$, and the domain hypothesis forces $P(f) = 0$ for all f. Hence $ta = sb$. Now $ta \in \Delta_A$ and $sb \in D^q_{A/k}$, so $ta \in M_q(A)$, and $x = (ta)/(st)$ belongs to $M_q(A)_{\mathfrak{p}}$. □

D.3.2 Height-one regularity

Lemma D.6 (Height-one equality)**.** *If* $\mathrm{ht}(\mathfrak{p}) = 1$*, then*

$$M_q(A)_{\mathfrak{p}} = D^q_{A_{\mathfrak{p}}/k} \qquad (q \geq 0).$$

Proof. A normal Noetherian domain satisfies (R_1), so $A_{\mathfrak{p}}$ is a regular local ring. Since k is perfect and $A_{\mathfrak{p}}$ is essentially of finite type over k, it is smooth over k. Grothendieck's smooth calculus therefore gives

$$D_{A_{\mathfrak{p}}/k} = A_{\mathfrak{p}}\langle \mathrm{Der}_k(A_{\mathfrak{p}})\rangle.$$

Intersecting with order at most q and using Lemma D.5 proves the claim. □

D.4 The exact Escape–Collapse Boundary Theorem

Theorem D.7 (NC.21 Reflexive Boundary Theorem)**.** *For every $q \geq 0$ there is a canonical equality inside the generic K-vector space $D^q_{A/k} \otimes_A K$:*

$$\boxed{D^q_{A/k} = M_q(A)^{**}.} \tag{D.4}$$

Equivalently, $D^q_{A/k}$ is the reflexive hull, or codimension-one saturation, of the order-q part of the derivation-generated algebra.

Proof. The module $M_q(A)$ is finite because it is an A-submodule of the finite module $D^q_{A/k}$ and A is Noetherian. It is torsion-free because $D^q_{A/k}$ is torsion-free. For a finite module over a Noetherian normal domain, the reflexive hull is the intersection of the height-one localizations inside the generic fiber [37, Tag 0AVC]:

$$M_q(A)^{**} = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} M_q(A)_{\mathfrak{p}}.$$

By Lemma D.6, every term in this intersection is $D^q_{A/k,\mathfrak{p}}$. Since $D^q_{A/k}$ is reflexive by Lemma D.3, the same codimension-one intersection formula gives

$$D^q_{A/k} = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} D^q_{A/k,\mathfrak{p}}.$$

The two intersections are identical, proving (D.4). □

Definition D.8 (Reflexive Escape defect)**.** Define

$$\boxed{\mathfrak{B}_q(A) := M_q(A)^{**}/M_q(A).} \tag{D.5}$$

By Theorem D.7, this is canonically

$$\boxed{\mathfrak{B}_q(A) \cong D^q_{A/k}/M_q(A).} \tag{D.6}$$

Corollary D.9 (Exact Collapse criterion)**.** *The following are equivalent:*

(i) $D_{A/k} = \Delta_A$*;*

(ii) $D^q_{A/k} = M_q(A)$ *for every* $q \geq 0$*;*

(iii) $\mathfrak{B}_q(A) = 0$ *for every* $q \geq 0$*;*

(iv) $M_q(A)$ *is reflexive for every* $q \geq 0$*;*

(v) $M_q(A)$ *satisfies Serre's condition* (S_2) *for every* $q \geq 0$*.*

Proof. The equivalence of (i) and (ii) follows by intersecting the algebra equality with the order filtration. The equivalence of (ii)–(iv) is Theorem D.7 and Definition D.8. Each $M_q(A)$ is torsion-free, and over a Noetherian normal domain a finite module is reflexive if and only if it is torsion-free and (S_2) [38, Tag 0AVB]. Thus (iv) and (v) are equivalent. □

Corollary D.10 (Exact Escape criterion)**.** *The following are equivalent:*

$$D_{A/k} \neq \Delta_A,$$

$$\exists q \geq 0 : \ \mathfrak{B}_q(A) \neq 0,$$

$$\exists q \geq 0 : \ M_q(A) \textit{ fails } (S_2).$$

Moreover

$$\mathrm{codim}_A \operatorname{Supp} \mathfrak{B}_q(A) \geq 2 \qquad \textit{for every } q. \tag{D.7}$$

Proof. Only the support statement requires comment. The height-one localizations of $M_q(A)$ and $D^q_{A/k}$ coincide by Lemma D.6; hence the quotient (D.6) vanishes at all primes of height at most one. □

Remark D.11 (What NC.21 changes conceptually)**.** For normal finite-type domains, the Escape–Collapse boundary is not hidden in codimension one. Codimension one is always smooth and both algebras agree there. Every failure of derivation generation is a codimension-at-least-two extension failure. NC.21 identifies that failure exactly as failure of the order-filtered derivation algebra to be reflexive.

D.5 Local cohomology and the isolated-singularity form

Let $(A, \mathfrak{m})$ now be a normal local domain essentially of finite type over k, of dimension at least two, with an isolated singularity. Put $U = \mathrm{Spec}(A) \setminus \{\mathfrak{m}\}$.

Lemma D.12 (Punctured-spectrum reconstruction)**.** *For every* q*,*

$$M_q(A)^{**} \cong \Gamma(U, \widetilde{M_q(A)}|_U).$$

Proof. On U the ring is regular and hence smooth over k. Therefore $M_q(A)|_U = D^q_{A/k}|_U$, which is locally free. For a coherent module on a normal scheme, the reflexive extension across a codimension-at-least-two set is j_*j^* [39, Tag 0AY6]. Taking global sections gives the displayed isomorphism. □

Theorem D.13 (Local-cohomological boundary)**.** *For every $q \geq 0$,*

$$\boxed{\mathfrak{B}_q(A) \cong H^1_{\mathfrak{m}}(M_q(A)).} \tag{D.8}$$

Consequently

$$D_{A/k} = \Delta_A \iff H^1_{\mathfrak{m}}(M_q(A)) = 0 \quad \forall q.$$

Proof. Because $M_q(A)$ is torsion-free,

$$H^0_{\mathfrak{m}}(M_q(A)) = 0.$$

The standard local-cohomology exact sequence for the punctured spectrum is the following. See the Stacks Project, Tag 0DWR.[40]

$$0 \to H^0_{\mathfrak{m}}(M_q) \to M_q \to \Gamma(U, \widetilde{M}_q|_U) \to H^1_{\mathfrak{m}}(M_q) \to 0.$$

Use Lemma D.12 and Definition D.8. □

Corollary D.14 (Normal surface form)**.** *If* $\dim A = 2$*, then*

$$\mathfrak{B}_q(A) = 0 \iff \operatorname{depth}_A M_q(A) = 2.$$

Thus, on an isolated normal surface singularity, Escape occurs precisely when some order-filtered derivation-generated module loses maximal Cohen–Macaulay depth.

Proof. A finite torsion-free module over a two-dimensional normal local domain has depth at least one. By Theorem D.13, vanishing of $\mathfrak{B}_q$ is equivalent to $H^1_{\mathfrak{m}}(M_q) = 0$, hence to depth at least two; this is the maximal possible depth. □

D.6 Collapse region: from free to projective tangent modules

D.6.1 The Becker–Rego mechanism

Becker and Rego proved the classical implication

$$\text{Nakai} \Longrightarrow \text{Zariski–Lipman},$$

where Zariski–Lipman asks whether projectivity of the derivation module forces regularity [21, 22, 24]. Hochster explicitly records the free-basis version and its reflexive-extension proof [23].

NC.19 independently reconstructed the same mechanism for a normal affine ring with a free derivation module. The next statement upgrades the NC.19 wording to the natural projective form.

Theorem D.15 (Projective-Tangent Collapse)**.** *Assume* $\operatorname{Der}_k(A)$ *is a finite projective A-module of rank* $\dim A$*. Then*

$$\boxed{D_{A/k} = \Delta_A.} \tag{D.9}$$

Hence every singular normal finite-type algebra with projective tangent module lies in the Collapse region.

Proof. Projectivity is locally freeness. Choose a Zariski affine cover by principal opens $D(f_i)$ on which $\operatorname{Der}_k(A)_{f_i}$ is free. Localization of derivations and differential operators gives

$$\operatorname{Der}_k(A_{f_i}) \cong \operatorname{Der}_k(A)_{f_i}.$$

The NC.19 free-tangent theorem (equivalently the Becker–Rego local argument) therefore gives

$$D_{A_{f_i}/k} = A_{f_i}\langle \operatorname{Der}_k(A_{f_i})\rangle$$

on every member of the cover. Equality of submodules of $D^q_{A/k}$ can be checked after such a cover for each finite q, so $D^q = M_q$ for all q. Taking the union gives (D.9). □

Corollary D.16 (Family 048 belongs to Collapse)**.** *Assuming the September 23, 2026 OpenAI construction as re-audited in NC.20, there exists a singular normal affine complex surface A such that*

$$\mathrm{Der}_{\mathbf{C}}(A) \cong A^2, \qquad D_{A/\mathbf{C}} = A\langle \mathrm{Der}_{\mathbf{C}}(A)\rangle.$$

For this ring

$$\mathfrak{B}_q(A) = 0 \qquad \forall q.$$

Proof. The current OpenAI Math catalogue states that Family 048 constructs a singular normal affine complex surface with free rank-two tangent sheaf [26]. NC.20 separately re-audits the construction chain. Apply Theorem D.15. □

D.6.2 Two tempting weakenings that do not distinguish the boundary

Proposition D.17 (Reflexivity of the tangent module is automatic)**.** *For every normal finite-type domain A as above,*

$$\mathrm{Der}_k(A) = \mathrm{Hom}_A(\Omega^1_{A/k}, A)$$

is reflexive.

Proof. The module $\Omega^1_{A/k}$ is finite. Apply the same Hom-into-reflexive lemma used in Lemma D.3. □

Proposition D.18 (Codimension-one tangent freeness is automatic)**.** *For every height-one prime $\mathfrak{p}$, $\mathrm{Der}_k(A)_{\mathfrak{p}}$ is free of rank* $\dim A$ *over $A_{\mathfrak{p}}$ in the relative smooth rank appropriate to the localization. In particular, "reflexive tangent" and "locally free in codimension one" cannot serve as Escape–Collapse boundary conditions in the normal category.*

Proof. The localization $A_{\mathfrak{p}}$ is regular and smooth over the perfect base field. Its tangent module is therefore finite locally free. The localization identity for derivations gives the claim. □

Remark D.19 (First structural narrowing)**.** The previous two propositions eliminate two obvious candidates. The projective-tangent sufficient condition is genuinely global across codimension at least two; reflexivity alone contains no such information.

D.7 Relation to the NC.18 Rees–Nakai defect

NC.18 defined, for an isolated local singularity $(R, \mathfrak{m})$, an exact Rees quantization sector $\mathfrak{Q}_q(R)$, a saturated derivation shadow $\mathfrak{S}_q(R)$, and

$$\mathfrak{R}_q(R) = \mathfrak{Q}_q(R)/\mathfrak{S}_q(R).$$

NC.18 proved

$$\mathfrak{R}_q(R) \neq 0 \Longrightarrow D_{R/k} \neq \Delta_R.$$

The NC.21 boundary theorem turns this into an intrinsic target statement.

Proposition D.20 (Rees defect detects reflexive defect)**.** *If $\mathfrak{R}_q(R) \neq 0$, then*

$$\mathfrak{B}_q(R) \neq 0.$$

Consequently, if R is in the Collapse region, then

$$\mathfrak{R}_q(R) = 0 \qquad \forall q.$$

Proof. The NC.18 strong-escape theorem produces an operator of order at most q that is not contained in Δ_R. Hence it lies in $D^q_{R/k} \setminus M_q(R)$, so the quotient $D^q_{R/k}/M_q(R) \cong \mathfrak{B}_q(R)$ is nonzero. The second statement is the contrapositive. □

Remark D.21. Thus the Rees defect is a degeneration-based sufficient detector, whereas $\mathfrak{B}_q$ is the exact intrinsic defect. NC.18 remains useful because it may make nonvanishing computable on a tangent cone even when the intrinsic module M_q is difficult to access directly.

D.8 Known and AK-controlled regions

Class	Region	Basis / audit status
Regular finite-type k-algebras	Collapse	Grothendieck smooth generation; $D = \Delta$ [31]
Normal rings with projective tangent module	Collapse	Theorem D.15; Becker–Rego mechanism [21, 22]
OpenAI Family 048 singular normal affine complex surface	Collapse	Free tangent rank two; external witness re-audited in NC.20 [26]
Smooth-projectivization homogeneous ICIS of codimension ≥ 2 covered by NC.17	Escape	AK NC.17 C2-high scoped theorem
Singular isolated homogeneous hypersurfaces in the 2026 Yau–Zhu–Zuo class	Escape	External Nakai-positive result: singular members cannot collapse [11]
Weighted homogeneous fewnomial isolated singularities and hypersurface cusps covered by Xiao–Yau–Zhu–Zuo	Escape	External scoped Nakai-positive results [20]
Normal toric singularities	Escape	Nakai is known in the normal toric class; singular points therefore escape [32, 41]
Finite invariant subrings in the classical Ishibashi scope	Escape when singular	Nakai-positive invariant-ring results [32, 33]
Irreducible algebraic curves	Escape when singular	Mount–Villamayor dimension-one theorem; note normal curves are regular [19]

Audit note D.22 (No extrapolation)**.** The table records only proved or explicitly imported classes. It does not imply that projective tangent modules exhaust the Collapse region, nor that all complete intersections lie in Escape. In particular, NC.21 does not promote NC.17 beyond its stated source class.

D.9 Boundary invariants

Definition D.23 (First escape order)**.** Define

$$\beta(A) := \inf\{q \geq 0 : \mathfrak{B}_q(A) \neq 0\} \in \mathbf{N} \cup \{\infty\}. \tag{D.10}$$

Then

$$\beta(A) = \infty \iff A \text{ is in Collapse},$$

while finite $\beta(A)$ is the first order at which derivation generation loses reflexive saturation.

Definition D.24 (Escape support strata)**.** For every q put

$$E_q(A) := \operatorname{Supp} \mathfrak{B}_q(A) \subseteq \operatorname{Spec} A.$$

Then each $E_q(A)$ is closed and has codimension at least two. The total Escape locus is the set-theoretic union

$$E(A) := \bigcup_{q \geq 0} E_q(A).$$

A localization $A_{\mathfrak{p}}$ is Collapse precisely when $\mathfrak{p} \notin E(A)$.

Proposition D.25 (Localization of the boundary defect)**.** *For every prime* $\mathfrak{p}$*,*

$$\mathfrak{B}_q(A)_{\mathfrak{p}} \cong \mathfrak{B}_q(A_{\mathfrak{p}}).$$

Proof. Localization commutes with M_q by Lemma D.5, with D^q by principal parts, and with reflexive hull for finite modules under flat localization. Alternatively use the quotient description (D.6) directly. □

Definition D.26 (Isolated boundary profile)**.** If $(A, \mathfrak{m})$ is an isolated normal local singularity, then $\mathfrak{B}_q(A)$ is supported only at $\mathfrak{m}$ and therefore has finite length. Define

$$\mu_q^{\mathrm{EC}}(A) := \operatorname{length}_A \mathfrak{B}_q(A) = \operatorname{length}_A H^1_{\mathfrak{m}}(M_q(A)). \tag{D.11}$$

The sequence

$$\left(\mu_q^{\mathrm{EC}}(A)\right)_{q\geq 0}$$

is the *Escape–Collapse boundary profile*.

Remark D.27 (Interpretation)**.** The profile is identically zero on Family 048 and on every regular ring. On any NC.17 Escape example, at least one entry is positive. NC.17 provides asymptotic existence of escape operators in a critical window but does not yet compute the minimal $\beta(A)$ or the exact lengths μ_q^{EC}. Those are concrete next computations.

D.10 The actual unresolved boundary

Theorem D.7 gives an exact module-theoretic classification of Escape versus Collapse, but it is not yet a geometric classification of which singularities make all M_q reflexive. The main open questions can now be stated without ambiguity.

Question D.28 (Projective-Tangent Exhaustion)**.** For a normal finite-type domain in characteristic zero, does

$$D_{A/k} = \Delta_A$$

force $\mathrm{Der}_k(A)$ to be projective?

A positive answer would identify the entire Collapse region with the projective-tangent locus. In view of Family 048 this would not restore Nakai; it would instead say that singular Collapse points are exactly singular Zariski–Lipman counterexamples.

Question D.29 (Finite-order detection)**.** For a fixed dimension or embedding dimension, is there a universal finite Q such that

$$\mathfrak{B}_q(A) = 0 \quad (0 \leq q \leq Q) \Longrightarrow \mathfrak{B}_q(A) = 0 \quad \forall q?$$

A positive answer would replace an infinite algebra-generation condition by a finite audit. Singh-type order-two phenomena for hypersurfaces suggest that such bounds may exist in restricted classes [24, 11].

Question D.30 (Geometric criterion for (S_2) of M_q)**.** Which singularity invariants control the (S_2) property of

$$M_q(A) = \Delta_A \cap D^q_{A/k}?$$

Candidates include depth of principal parts, cotangent-complex amplitude, Jacobian/Fitting strata, logarithmic tangent geometry on resolutions or blowups, and the NC.18 Rees obstruction.

Question D.31 (Boundary-profile growth)**.** For isolated singularities in Escape, what are the asymptotics of $\mu_q^{\mathrm{EC}}(A)$? Does polynomial growth recover or refine known invariants such as multiplicity, differential signature, or symmetric signature [29]?

Declaration D.32 (NC.21 research target after this appendix)**.** The next substantive goal is no longer "prove or disprove Nakai." The exact intrinsic boundary is already identified by Theorem D.7. The next goal is to replace the reflexive-module criterion by geometric and computable criteria, beginning with Question D.28 and the surface formula (D.8).

D.11 Adversarial checks and failure modes

D.11.1 Why normality is essential to the exact theorem

Normality supplies all three structural inputs used by the proof:

1. (R_1), hence smooth codimension-one localizations in characteristic zero;
2. (S_2) and the codimension-one intersection description of reflexive hulls;
3. torsion-free/reflexive extension across codimension at least two.

Without normality, singular behavior may already occur in codimension one and D^q need not be recovered from height-one data. The one-dimensional pseudo-geometric counterexamples of Brumatti–Lequain–Levcovitz–Simis illustrate how badly Nakai-like statements can behave outside the finite-type normal framework [30].

D.11.2 Why the word filtration is not substituted

The equality

$$\mathrm{der}^q(A) = \Delta_A \cap D^q_{A/k}$$

is false in general; explicit singular examples are recorded in modern Nakai papers [34]. Replacing M_q by a naive word-length module would therefore change the theorem and can create false boundary defects.

D.11.3 Why $\mathrm{Der}(A)$ reflexive does not imply Collapse

Proposition D.17 shows this condition is automatic for every normal finite-type domain. NC.17 supplies normal singular rings in Escape, so reflexivity of the tangent module is far too weak.

D.11.4 Why codimension-one agreement alone is not Collapse

Codimension-one agreement is also automatic. Collapse requires the codimension-two extension module M_q to be already saturated. The quotient $\mathfrak{B}_q$ is exactly the missing extension data.

D.12 Prior-art position

The classical literature establishes the following surrounding facts. Nakai formulated higher derivation questions in 1970 [18]; Mount–Villamayor gave the first standard statement of the conjectural regularity implication and proved the curve case [19]; Rego and Becker related Nakai to Zariski–Lipman [21, 22]; Singh developed explicit hypersurface differential operators [24]; Ishibashi proved invariant-subring cases [32, 33]; and more recent work proves additional singularity classes [20, 34, 11]. Modern principal-parts work emphasizes reflexive hulls on normal rings and the role of codimension-one smooth loci [29, 35].

The present prior-art search did not locate a source phrasing the unrestricted normal-domain identity

$$D^q = (\Delta \cap D^q)^{**}$$

and the corresponding Escape–Collapse iff criterion in the exact NC.21 form. This absence is *not* a novelty proof. A dedicated MathSciNet/zbMATH and journal-level priority search remains mandatory before publication.

D.13 Claim register

ID	Claim	Status
NC21.1	$D^q_{A/k}$ finite reflexive for normal finite-type A	C1 imported / C2 assembly
NC21.2	M_q localizes as the intrinsic order filtration	C2 internal
NC21.3	M_q and D^q agree in height one	C1 smooth input / C2 assembly

ID	Claim	Status
NC21.4	$D^q = M_q^{**}$	C2-high internal
NC21.5	Collapse iff all M_q reflexive iff all M_q satisfy (S_2)	C2-high internal
NC21.6	Escape defect supported in codimension ≥ 2	C2 internal
NC21.7	Isolated local formula $\mathfrak{B}_q \cong H^1_{\mathfrak{m}}(M_q)$	C2-high internal
NC21.8	Projective tangent implies Collapse	C1 Becker–Rego / C2 NC assembly
NC21.9	Reflexivity of $\mathrm{Der}(A)$ automatic in normal setting	C1 standard
NC21.10	NC.18 nonzero Rees defect implies nonzero reflexive defect	C2 internal
NC21.11	First escape order / boundary profile	Definition, no classification claim
NC21.12	Projective-Tangent Exhaustion	OPEN
NC21.13	Finite-order detection	OPEN
NC21.14	Geometric characterization of boundary	OPEN

D.14 Final synthesis

NC.17 and NC.20 exhibited opposite singular behaviors:

$$\text{Escape: } D \neq \Delta, \qquad \text{Collapse: } D = \Delta.$$

NC.21 identifies the exact algebraic wall between them. For a normal finite-type characteristic-zero domain,

$$\boxed{D^q_{A/k} = \left(\Delta_A \cap D^q_{A/k}\right)^{**}}$$

for every q. Therefore the full ring of differential operators carries no additional codimension-one information beyond derivations. Escape is exactly the failure of the derivation-generated order module to extend reflexively across codimension at least two; Collapse is exactly the absence of that defect at every order.

For isolated normal local singularities this becomes

$$\boxed{D^q/M_q \cong H^1_{\mathfrak{m}}(M_q),}$$

turning the boundary into a concrete depth problem. Projective tangent modules lie entirely on the Collapse side, while normal homogeneous ICIS and several classical singularity families lie on the Escape side. The remaining problem is therefore not the original yes/no Nakai question but a classification of the codimension-two saturation mechanism.

Declaration D.33 (NC.21 final status)**.** NC.21 closes the *intrinsic algebraic* Escape–Collapse boundary for normal finite-type domains: the exact defect is the reflexive saturation quotient $\mathfrak{B}_q$. It does not yet close the *geometric classification* of that boundary. The strongest next target is to determine whether projectivity of the tangent module exhausts Collapse, and, if not, to characterize the additional Collapse mechanisms.

E Projective Exhaustion and Silent-Collapse Boundary

Source abstract

NC.22 is reconstructed as the missing bridge between the reflexive Escape–Collapse boundary of NC.21 and the torsion-free enveloping reduction of NC.23. Let k be a field of characteristic zero and let A be a normal integral k-algebra essentially of finite type. Put

$$D = D_{A/k}, \qquad \Delta = A\langle \mathrm{Der}_k(A)\rangle, \qquad M_q = \Delta \cap D^q.$$

NC.21 proves $D^q = M_q^{**}$ and therefore identifies

$$\mathfrak{B}_q = D^q/M_q$$

as the intrinsic Escape defect. NC.22 adds the word filtration W_q generated by words of at most q first-order derivations and separates the additional order-compression defect

$$\mathfrak{C}_q = M_q/W_q.$$

It also introduces the tangent projectivity defect

$$\mathfrak{P}(A) = \operatorname{coker}\big(\operatorname{Der}_k(A) \otimes_A \operatorname{Der}_k(A)^\vee \longrightarrow \operatorname{End}_A(\operatorname{Der}_k(A))\big).$$

The evaluation map criterion gives $\mathfrak{P}(A) = 0$ if and only if the tangent module is finite projective. Projective tangent annihilates both differential defects in every order:

$$\boxed{\mathfrak{P}(A) = 0 \Longrightarrow \mathfrak{B}_q(A) = \mathfrak{C}_q(A) = 0 \quad \forall q.}$$

Consequently the Escape locus is contained in the tangent nonprojectivity locus. The only possible mismatch is the Silent-Collapse locus

$$\mathfrak{S}_{\mathrm{sil}}(A) = \mathrm{NF}(\operatorname{Der}_k A) \setminus \bigcup_q \operatorname{Supp} \mathfrak{B}_q.$$

For every localization-stable class of normal finite-type algebras, the following are equivalent: Collapse implies projective tangent; the Silent-Collapse locus is always empty; and the Escape locus equals the tangent nonprojectivity locus. Thus the post-Nakai geometric classification problem is reduced exactly to the existence or nonexistence of the fourth quadrant

$$\operatorname{Der}_k(A) \text{ nonprojective}, \qquad D_{A/k} = A\langle \operatorname{Der}_k(A)\rangle.$$

No converse is asserted. Brown's theorem requires an additional differential-prime condition, and order-wise derivation generation cannot be identified with algebra generation when $\mathfrak{C}_q \neq 0$. NC.23 is therefore required to analyze the remaining torsion/saturation mechanism.

Declaration E.1 (Post-resolution role)**.** NC.22 separates order-compression, reflexive Escape, and tangent-projectivity defects. The converse $D = \Delta \Rightarrow \operatorname{Der}_k(A)$ projective is not proved and is not silently consumed.

E.1 Scope, standing hypotheses, and inherited boundary

E.1.1 Standing hypotheses

Unless a statement explicitly says otherwise:

(H1) k is a field of characteristic zero;

(H2) A is a normal integral k-algebra essentially of finite type;

(H3) $K = \operatorname{Frac}(A)$;

(H4) $D^q = D^q_{A/k}$ is the Grothendieck module of differential operators of order at most q, and $D = \bigcup_{q\geq 0} D^q$;

(H5) $L = \operatorname{Der}_k(A)$ and $\Delta = A\langle L\rangle \subseteq D$;

(H6) $M_q = \Delta \cap D^q$ is the intrinsic order filtration on the generated algebra.

Declaration E.2 (Inherited NC.21 theorem)**.** NC.22 uses the following NC.21 identity as an already proved internal input:

$$\boxed{D^q = M_q^{**} \qquad (q \geq 0).} \tag{E.1}$$

Hence

$$\mathfrak{B}_q(A) := D^q/M_q \simeq M_q^{**}/M_q \tag{E.2}$$

is supported in codimension at least two, and

$$D = \Delta \quad \Longleftrightarrow \quad \mathfrak{B}_q(A) = 0 \text{ for every } q. \tag{E.3}$$

Remark E.3 (Why NC.22 is a genuinely new layer)**.** The algebra equality $D = \Delta$ does not by itself identify how much word length is needed to represent an operator of a fixed actual order. A long word can drop to a smaller differential order on a singular algebra. Therefore the filtration generated by word length must be separated from $M_q = \Delta \cap D^q$.

E.2 Word filtration and the order-compression defect

Definition E.4 (Word filtration)**.** Let $W_q \subseteq \Delta$ be the A-submodule generated by composites

$$\delta_1\delta_2\cdots\delta_j, \qquad 0 \le j \le q, \quad \delta_i \in L,$$

where $j = 0$ means multiplication by an element of A. Then

$$W_0 = A, \qquad W_q \subseteq W_{q+1}, \qquad W_qW_r \subseteq W_{q+r}.$$

Every word of length at most q has differential order at most q, so

$$W_q \subseteq M_q. \tag{E.4}$$

Definition E.5 (Order-compression defect)**.** Define

$$\boxed{\mathfrak{C}_q(A) := M_q/W_q.} \tag{E.5}$$

This module measures exactly the difference between actual differential order and the naive word-length filtration inside the algebra generated by derivations.

Proposition E.6 (Two-defect exact sequence)**.** *For every $q \ge 0$ there is a canonical short exact sequence*

$$0 \longrightarrow \mathfrak{C}_q(A) \longrightarrow D^q/W_q \longrightarrow \mathfrak{B}_q(A) \longrightarrow 0. \tag{E.6}$$

Proof. The inclusions $W_q \subseteq M_q \subseteq D^q$ give the standard quotient exact sequence

$$0 \to M_q/W_q \to D^q/W_q \to D^q/M_q \to 0.$$

Substitute the definitions of $\mathfrak{C}_q$ and $\mathfrak{B}_q$. □

Corollary E.7 (Strong order-wise generation versus algebraic Collapse)**.** *The order-wise equality*

$$D^q = W_q \qquad \forall q$$

is equivalent to the simultaneous vanishing

$$\mathfrak{B}_q = \mathfrak{C}_q = 0 \qquad \forall q.$$

By contrast, algebraic Collapse $D = \Delta$ is equivalent only to $\mathfrak{B}_q = 0$ for every q and does not formally force $\mathfrak{C}_q = 0$.

Audit note E.8 (Brown guard-rail)**.** Brown's 1974 regularity criterion assumes both (i) absence of nonminimal differential prime ideals and (ii) order-wise generation by composites of at most the same order [42]. It therefore cannot be read as a proof that the weaker algebra equality $D = \Delta$ forces regularity or projective tangent. The distinction is precisely the role of $\mathfrak{C}_q$.

E.3 The tangent projectivity defect

Definition E.9 (Projectivity defect)**.** Let $L^\vee = \mathrm{Hom}_A(L, A)$ and let

$$\mathrm{ev}_L : L \otimes_A L^\vee \longrightarrow \mathrm{End}_A(L), \qquad x \otimes \varphi \longmapsto (y \mapsto \varphi(y)x).$$

Define

$$\boxed{\mathfrak{P}(A) := \mathrm{coker}(\mathrm{ev}_L).} \tag{E.7}$$

Theorem E.10 (Exact projectivity detector)**.** *For the finite A-module $L = \mathrm{Der}_k(A)$,*

$$\boxed{\mathfrak{P}(A) = 0 \quad \Longleftrightarrow \quad L \text{ is finite projective.}} \tag{E.8}$$

Moreover

$$\mathrm{Supp}\,\mathfrak{P}(A) = \mathrm{NF}(L). \tag{E.9}$$

Proof. If L is finite projective, the canonical evaluation map $L \otimes L^\vee \to \mathrm{End}(L)$ is an isomorphism, so $\mathfrak{P} = 0$. Conversely, if $\mathfrak{P} = 0$, the identity 1_L lies in the image. Hence there are finite families $x_i \in L$ and $\varphi_i \in L^\vee$ such that

$$y = \sum_i \varphi_i(y) x_i \qquad (y \in L).$$

This is the dual-basis criterion for finite projectivity. Localization commutes with the evaluation map and cokernel, so the same argument at every prime gives (E.9). □

Remark E.11. Normality already makes L reflexive in the standard finite-type setting used in NC.21. Reflexivity and codimension-one local freeness are therefore too weak to replace $\mathfrak{P}$: projectivity is the codimension-two/global condition that remains relevant.

E.4 Projective tangent annihilates both differential defects

Theorem E.12 (Projective-tangent defect annihilation)**.** *Under the standing hypotheses,*

$$\boxed{\mathfrak{P}(A) = 0 \quad \Longrightarrow \quad \mathfrak{B}_q(A) = \mathfrak{C}_q(A) = 0 \quad \text{for all } q \geq 0.} \tag{E.10}$$

Equivalently, finite projectivity of $\mathrm{Der}_k(A)$ *implies both*

$$D = \Delta \qquad \text{and} \qquad D^q = W_q \quad \forall q.$$

Proof. By Theorem E.10, L is finite projective. Localizing on $\mathrm{Spec}\, A$, it is free on an open cover. The Becker–Rego mechanism, equivalently the NC.19/NC.21 projective-tangent extension, gives

$$D_{A/k} = A\langle L \rangle = \Delta,$$

so $\mathfrak{B}_q = 0$ for all q by (E.3).

It remains to control word length. On an open set on which L is free, the standard PBW/principal-symbol argument identifies order-q operators generated by the frame with words of length at most q. Thus $(W_q)_\mathfrak{p} = (D^q)_\mathfrak{p}$ for every prime $\mathfrak{p}$. Since equality of finite modules is local, $W_q = D^q = M_q$. Hence $\mathfrak{C}_q = 0$ for every q. □

Remark E.13 (Classical direction only)**.** The direction in Theorem E.12 is the one consistent with the Becker–Rego result recalled by Singh: the Nakai conjecture would imply Zariski–Lipman because projective tangent supplies the relevant derivation-generation property [21, 22, 24]. None of these sources supplies the reverse implication "derivation generation implies projective tangent."

E.5 Escape, compression, and nonprojectivity loci

Definition E.14 (Three intrinsic loci)**.** Set

$$\mathcal{E}(A) := \bigcup_{q \geq 0} \mathrm{Supp}\, \mathfrak{B}_q(A), \tag{E.11}$$

$$C(A) := \bigcup_{q \geq 0} \mathrm{Supp}\, \mathfrak{C}_q(A), \tag{E.12}$$

$$\mathcal{N}(A) := \mathrm{NF}(\mathrm{Der}_k A) = \mathrm{Supp}\, \mathfrak{P}(A). \tag{E.13}$$

Theorem E.15 (Locus inclusions)**.** *For every A in the standing class,*

$$\boxed{\mathcal{E}(A) \subseteq \mathcal{N}(A), \qquad C(A) \subseteq \mathcal{N}(A).} \tag{E.14}$$

Proof. Let $\mathfrak{p} \notin \mathcal{N}(A)$. Then $L_\mathfrak{p}$ is projective over $A_\mathfrak{p}$. Differential operators, derivations, the word filtration, and the intrinsic order filtration commute with localization in the present essentially finite-type setting. Applying Theorem E.12 to $A_\mathfrak{p}$ gives $(\mathfrak{B}_q)_\mathfrak{p} = (\mathfrak{C}_q)_\mathfrak{p} = 0$ for all q. Hence $\mathfrak{p}$ lies in neither support union. □

Definition E.16 (Silent-Collapse locus)**.** The residual locus is

$$\boxed{\mathfrak{S}_{\mathrm{sil}}(A) := \mathcal{N}(A) \setminus \mathcal{E}(A).} \tag{E.15}$$

A point $\mathfrak{p} \in \mathfrak{S}_{\mathrm{sil}}(A)$ is exactly a point at which $\mathrm{Der}_k(A)_{\mathfrak{p}}$ is nonprojective while

$$D_{A_{\mathfrak{p}}/k} = A_{\mathfrak{p}}\langle \mathrm{Der}_k(A_{\mathfrak{p}})\rangle.$$

Remark E.17 (Compression can remain inside Silent-Collapse)**.** The definition removes the Escape defect $\mathfrak{B}$ but does not force $\mathfrak{C}$ to vanish. Thus a hypothetical Silent-Collapse point may realize low-order operators by longer derivation words whose excess length disappears only after singular relations are used. NC.23 identifies this phenomenon with saturation/torsion in the Lie–Rinehart enveloping algebra.

E.6 Projective Exhaustion and exact class-level equivalences

Conjecture E.1 (Projective-Exhaustion Conjecture)**.** *On the declared class,*

$$\boxed{D_{A/k} = A\langle \mathrm{Der}_k(A)\rangle \quad \Longrightarrow \quad \mathrm{Der}_k(A) \text{ projective}.} \tag{E.16}$$

Theorem E.18 (Silent-Collapse equivalence theorem)**.** *Let $\mathscr{C}$ be any class of normal integral essentially finite-type characteristic-zero k-algebras closed under localization. The following are equivalent.*

(i) For every $A \in \mathscr{C}$, Collapse implies projective tangent.

(ii) For every $A \in \mathscr{C}$, $\mathfrak{S}_{\mathrm{sil}}(A) = \varnothing$.

(iii) For every $A \in \mathscr{C}$, $\mathcal{E}(A) = \mathcal{N}(A)$.

(iv) For every $A \in \mathscr{C}$ and every prime $\mathfrak{p}$, nonprojectivity of $\mathrm{Der}_k(A)_{\mathfrak{p}}$ forces Escape after localization at $\mathfrak{p}$.

Proof. The inclusion $\mathcal{E}(A) \subseteq \mathcal{N}(A)$ is Theorem E.15. Therefore $\mathfrak{S}_{\mathrm{sil}}(A) = \varnothing$ is equivalent to $\mathcal{E}(A) = \mathcal{N}(A)$, proving (ii)⇔(iii). By definition, a point in $\mathfrak{S}_{\mathrm{sil}}(A)$ is precisely a localization at which tangent projectivity fails but Collapse holds. Since $\mathscr{C}$ is localization-stable, the nonexistence of such points is exactly (i), and equivalently (iv). □

Corollary E.19 (Exact status of the geometric classification problem)**.** *The intrinsic Escape–Collapse boundary of NC.21 is complete independently of Conjecture E.1. A purely tangent-geometric classification is complete if and only if Projective Exhaustion is proved. No other logical gap remains between the two locus descriptions.*

E.7 The isolated-singularity four-quadrant table

Suppose $(A, \mathfrak{m})$ is a normal isolated singularity in the standing class. The local possibilities are:

Tangent at $\mathfrak{m}$	D versus Δ	Status
projective/free	Collapse	exists; regular examples and, conditional on the audited external theorem, Family 048 singular examples
projective/free	Escape	impossible by Theorem E.12
nonprojective	Escape	exists; NC.17 and other positive Nakai classes supply examples
nonprojective	Collapse	**unresolved**; exactly Silent-Collapse / Projective Exhaustion

Declaration E.20 (Fourth-quadrant reduction)**.** The post-Nakai classification problem is therefore not "understand arbitrary singularities." It is the single existence question

$$\boxed{\mathrm{Der}_k(A) \text{ nonprojective} \quad \text{and} \quad D_{A/k} = A\langle \mathrm{Der}_k(A)\rangle.} \tag{E.17}$$

Every proof of Projective Exhaustion must exclude (E.17); every counterexample must construct it.

E.8 Compatibility with the negative resolution of unrestricted Nakai

Proposition E.21 (No conflict with NC.20)**.** *The unresolved fourth quadrant has no bearing on the NC.20 negative resolution of the unrestricted Nakai implication, provided the audited Family 048 external premise is accepted at its declared evidence level.*

Proof. Family 048 is used through the opposite tangent sector: its tangent module is free of rank two. Theorem E.12 then gives Collapse. Since the ring is singular, this is already a Nakai counterexample. Whether an additional counterexample can occur with nonprojective tangent does not alter that negative implication. □

E.9 Why PBW cannot be reversed at NC.22

Audit note E.22 (PBW and word/order separation)**.** Classical PBW arguments with projective L justify the sufficient direction in Theorem E.12. They do not justify the converse. In particular:

(a) algebraic Collapse controls $\mathfrak{B}_q$ but not formally $\mathfrak{C}_q$;

(b) long derivation words may acquire smaller actual order because singular relations create filtration torsion;

(c) a proof that $\operatorname{gr} D$ is generated by degree-one symbols under a projectivity hypothesis cannot be reversed without a theorem recovering projectivity from the filtered quotient;

(d) Brown's differential-prime hypothesis is an independent condition and may not be deleted.

NC.23 is therefore assigned the exact task of replacing a naive reverse-PBW argument by a torsion-free enveloping and saturation analysis.

E.10 NC.22 claim register and terminal handoff

Claim	Status	Basis
$W_q \subseteq M_q \subseteq D^q$	proved	definitions and order subadditivity
$0 \to \mathfrak{C}_q \to D^q/W_q \to \mathfrak{B}_q \to 0$	proved	quotient exact sequence
$\mathfrak{P} = 0 \iff \operatorname{Der}_k(A)$ projective	proved	dual-basis criterion
$\mathfrak{P} = 0 \Rightarrow \mathfrak{B}_q = \mathfrak{C}_q = 0$ for all q	proved	projective-tangent generation and local PBW
$\mathcal{E} \subseteq \mathcal{N}$ and $C \subseteq \mathcal{N}$	proved	localization of the previous theorem
Projective Exhaustion $\Leftrightarrow \mathfrak{S}_{\mathrm{sil}} = \varnothing \Leftrightarrow \mathcal{E} = \mathcal{N}$ on localization-stable classes	proved	locus definitions plus localization
Brown theorem proves Projective Exhaustion	rejected reading	Brown has an additional differential-prime condition and order-wise hypothesis
PBW alone proves Projective Exhaustion	rejected reading	reverse implication unavailable; word/order defect remains
Silent-Collapse is empty	**open at NC.22**	exact residual fourth quadrant
Unrestricted Nakai negative resolution	unaffected	NC.20 uses free/projective tangent sector

Declaration E.23 (NC.22 closure)**.** NC.22 is mathematically closed at the following exact strength:

intrinsic Escape locus $\subseteq$ tangent nonprojectivity locus, difference = Silent-Collapse.

It neither proves nor assumes that the difference is empty. The complete remaining question is exported to NC.23 in the equivalent Projective-Exhaustion / torsion-free enveloping form.

F Torsion-Free Enveloping Reduction and Final NC Closure

Source abstract

Appendix NC.23 continues the final-gate analysis of NC.21–NC.22. The aim is not to assume the Projective-Exhaustion Conjecture, but to identify exactly what remains after every formal and generic obstruction is removed.

Let k be a characteristic-zero field, let A be a normal integral k-algebra essentially of finite type, and put

$$L = \mathrm{Der}_k(A), \qquad D = D_{A/k}, \qquad \Delta = A\langle L\rangle.$$

Let $U = U_A(L)$ denote the Lie–Rinehart universal enveloping algebra and $\rho : U \to D$ its natural action. Lamkin's 2025 PBW theorem, specialized to the ordinary Lie–Rinehart case over the field k, removes the old projectivity hypothesis from the PBW map:

$$\mathrm{gr}_F U \simeq \mathrm{Sym}_A L.$$

The essential singular phenomenon is therefore not failure of the PBW map. It is torsion and saturation over A.

The first main theorem identifies the action kernel exactly. If $\tau_A(U)$ denotes the left A-torsion of U, then

$$\boxed{\ker \rho = \tau_A(U).}$$

In particular $\tau_A(U)$ is automatically a two-sided ideal and

$$\boxed{\Delta \simeq \overline{U}_A(L) := U_A(L)/\tau_A(U_A(L)).}$$

Thus Collapse has the exact enveloping formulation

$$\boxed{D = \Delta \iff \overline{U}_A(\mathrm{Der}_k A) \xrightarrow{\sim} D_{A/k}.}$$

The second main theorem identifies the mismatch between PBW word length and actual differential order. Let F_qU be the PBW filtration, $W_q = \rho(F_qU)$, and $M_q = \Delta \cap D^q$. Then

$$\boxed{M_q = \mathrm{Sat}_A(W_q;\Delta) = \{P \in \Delta : \exists 0 \neq a \in A,\ aP \in W_q\}.}$$

Combining this with NC.21 gives the canonical three-stage chain

$$\boxed{W_q \subseteq \mathrm{Sat}_A(W_q;\Delta) = M_q \subseteq M_q^{**} = D^q.}$$

Accordingly, the order-compression defect $\mathfrak{C}_q = M_q/W_q$ is precisely a saturation defect, whereas the Escape defect $\mathfrak{B}_q = D^q/M_q$ is precisely a reflexive defect.

This settles several false leads. PBW itself does not fail; a full D-module action on a reflexive module does not force local freeness; and normality/reflexivity alone does not eliminate the residual case. The only remaining geometric completion problem is the following exact implication:

$$\boxed{\overline{U}_A(\mathrm{Der}_k A) \simeq D_{A/k} \quad \Longrightarrow ? \quad \mathrm{Der}_k(A) \text{ projective.}}$$

Equivalently: can a normal finite-type singularity have nonprojective tangent module while its entire ring of differential operators is the torsion-free Lie–Rinehart enveloping algebra of that tangent module? This is precisely the Silent-Collapse problem.

The appendix also records exclusion mechanisms: singular D-simple rings are Escape; singular normal toric varieties are Escape; isolated homogeneous hypersurfaces are Escape by the 2026 Yau–Zhu–Zuo theorem; and the NC.17 homogeneous complete-intersection class of codimension at least two is Escape. Projective tangent remains a sufficient Collapse condition, and the audited Family 048 surface remains the canonical singular Collapse example.

No proof that Silent-Collapse is empty, and no Silent-Collapse example, is claimed here. Instead, NC.23 closes Appendix NC at its maximal proved boundary: the original Nakai target and the intrinsic Escape–Collapse classification are terminal, while the strictly stronger Projective-Exhaustion/TFEP implication is exported as Successor Problem SC.1 and is not treated as an unresolved NC proof obligation. This is scope closure, not a false proof of the converse. The publication ceiling is C2-high for the internal reduction theorems, with external literature inputs explicitly identified.

Declaration F.1 (Terminal scope)**.** NC.23 closes the NC.1–NC.23 theorem DAG only at its declared evidence strength. Silent-Collapse/TFEP is exported as Successor Problem SC.1. "Closed" does not mean that the converse $D = \Delta \Rightarrow \mathrm{Der}_k(A)$ projective has been proved.

F.1 Scope, notation, and inherited NC results

Throughout, k is a field of characteristic zero and A is a normal integral k-algebra essentially of finite type. Let

$$K := \mathrm{Frac}(A), \qquad L := \mathrm{Der}_k(A), \qquad D := D_{A/k} = \bigcup_{q\geq 0} D^q_{A/k}, \qquad \Delta := A\langle L\rangle \subseteq D.$$

The module L is finite because $\Omega^1_{A/k}$ is finitely presented and $L = \mathrm{Hom}_A(\Omega^1_{A/k}, A)$.

For

$$M_q := \Delta \cap D^q,$$

NC.21 proved

$$\boxed{D^q = M_q^{**}}, \qquad \mathfrak{B}_q := D^q/M_q \simeq M_q^{**}/M_q. \tag{F.1}$$

Hence

$$D = \Delta \iff \mathfrak{B}_q = 0 \quad \text{for every } q. \tag{F.2}$$

NC.22 introduced the PBW/word image W_q and

$$\mathfrak{C}_q := M_q/W_q,$$

but deliberately did not identify M_q intrinsically in terms of W_q. That missing identification is one of the main results below.

Declaration F.2 (Claim ceiling)**.** Theorems F.9, F.10, and F.13 are the new internal algebraic core of this appendix. They do not prove the Projective-Exhaustion Conjecture. The final implication "Collapse $\Rightarrow$ projective tangent" remains open in this framework.

F.2 PBW without projective tangent

Let $U = U_A(L)$ be the Lie–Rinehart enveloping algebra of the pair (A, L), with PBW filtration $F_{-1}U = 0$, $F_0U = A$, and F_qU generated on the left by words containing at most q elements of L.

Rinehart's original PBW theorem assumed that L is projective as an A-module [43]. Lamkin proved a stronger PBW theorem for Lie–Rinehart superalgebras over a characteristic-zero commutative base C: if A and L are free as C-supermodules, the PBW map is an isomorphism [44, Theorem 5.7]. In the ordinary purely even case and with $C = k$, both A and L are automatically free as k-vector spaces. Therefore:

Theorem F.3 (Field-base PBW theorem)**.** *There is a canonical graded A-algebra isomorphism*

$$\boxed{\mathrm{Sym}_A L \xrightarrow{\sim} \mathrm{gr}_F U_A(L).} \tag{F.3}$$

No projectivity of L over A is required.

Remark F.4 (What PBW does and does not say)**.** Equation (F.3) does not say that U is A-torsion-free. The symmetric powers $\mathrm{Sym}^q_A L$ themselves may carry A-torsion when L is nonprojective. Thus PBW moves the final obstruction from "failure of PBW" to the A-module geometry of the PBW pieces.

F.3 The generic Lie–Rinehart action

The derivation module is compatible with passage to the fraction field. Since $\Omega^1_{A/k}$ is finitely presented,

$$L \otimes_A K \simeq \mathrm{Hom}_K(\Omega^1_{A/k} \otimes_A K, K) \simeq \mathrm{Der}_k(K). \tag{F.4}$$

Write $L_K := \mathrm{Der}_k(K)$ and $U_K := U_K(L_K)$. The pair map $A \to K$, $L \to L_K$ induces a filtered algebra homomorphism

$$\lambda : U_A(L) \longrightarrow U_K(L_K). \tag{F.5}$$

Lemma F.5 (Kernel of genericization)**.** *Let*

$$\tau_A(U) := \{u \in U : \exists 0 \neq a \in A \text{ with } au = 0\}$$

be the left A-torsion subset. Then

$$\boxed{\ker \lambda = \tau_A(U).} \tag{F.6}$$

In particular $\tau_A(U)$ is a two-sided ideal.

Proof. If $au = 0$ with $0 \neq a \in A$, then a is invertible in K, so $\lambda(u) = 0$. Thus $\tau_A(U) \subseteq \ker \lambda$.

Conversely, let $u \in \ker \lambda$ and choose the least q with $u \in F_q U$. Under the PBW isomorphism, the degree-q symbol $\sigma_q(u)$ lies in $\mathrm{Sym}^q_A L$. Compatibility of PBW with the map to K shows that $\sigma_q(u)$ maps to zero in $\mathrm{Sym}^q_K(L \otimes_A K)$. The kernel of localization of an A-module at the fraction field is precisely its A-torsion. Hence there is $0 \neq a_q \in A$ with

$$a_q \sigma_q(u) = 0,$$

so $a_q u \in F_{q-1}U$. Repeating the argument at degrees $q-1, q-2, \ldots, 0$ produces a nonzero product $a \in A$ with $au = 0$. Thus $u \in \tau_A(U)$.

Since $\ker \lambda$ is a two-sided ideal, the last assertion follows. □

The next lemma is the generic faithfulness needed to compare PBW degree with actual differential order.

Lemma F.6 (Faithfulness and strictness over the fraction field)**.** *The natural action*

$$\rho_K : U_K(\mathrm{Der}_k K) \longrightarrow D_{K/k} \tag{F.7}$$

is injective. Moreover it is strict for the PBW and differential-order filtrations:

$$\rho_K(v) \in D^q_{K/k} \quad \Longrightarrow \quad v \in F_q U_K. \tag{F.8}$$

Proof. Choose a separating transcendence basis $t_1, \ldots, t_d$ for the finitely generated field extension K/k. Since $\mathrm{char}\, k = 0$, the extension $K/k(t_1, \ldots, t_d)$ is finite separable, and the coordinate derivations extend uniquely to a K-basis $\partial_1, \ldots, \partial_d$ of $\mathrm{Der}_k K$.

Rinehart PBW (or Theorem F.3) gives symbols ∂^α in U_K. Suppose an element of PBW degree m acts as a differential operator of order $< m$. Taking m successive commutators with multiplication by the t_i kills every lower-degree term and evaluates the top symmetric symbol. Varying the sequence of the t_i detects every coefficient of that degree-m symmetric tensor. Characteristic zero makes the relevant factorials nonzero. Hence the top symbol must vanish, contradicting minimality of m. This proves strictness; taking the zero operator proves injectivity. □

Remark F.7. The proof uses only the generic field and PBW. It does not assume that the tangent module is projective on $\mathrm{Spec}\, A$.

F.4 The action-kernel theorem

Let

$$\rho : U_A(L) \longrightarrow D_{A/k}$$

be the action induced by multiplication and derivations.

Lemma F.8 (Torsion-freeness of D over A)**.** *For every q, $D^q_{A/k}$, and hence $D_{A/k}$, is torsion-free as a left A-module.*

Proof. If $0 \neq a \in A$ and $aP = 0$ in D, then for every $x \in A$, $aP(x) = 0$. Since A is a domain, $P(x) = 0$ for all x, so $P = 0$. □

Theorem F.9 (Action-kernel theorem)**.** *For a normal integral essentially finite-type k-algebra in characteristic zero,*

$$\boxed{\ker \rho = \tau_A(U_A(L)).} \tag{F.9}$$

Consequently the A-torsion-free enveloping algebra

$$\overline{U}_A(L) := U_A(L)/\tau_A(U_A(L)) \tag{F.10}$$

acts faithfully on A, and the induced action identifies

$$\boxed{\overline{U}_A(L) \xrightarrow{\sim} \Delta.} \tag{F.11}$$

Proof. If $u \in \ker \rho$, then the induced operator on the fraction field is zero. Compatibility with (F.5) gives $\rho_K(\lambda(u)) = 0$. Lemma F.6 yields $\lambda(u) = 0$, so Lemma F.5 gives $u \in \tau_A(U)$.

Conversely, if $u \in \tau_A(U)$, choose $0 \neq a \in A$ with $au = 0$. Then $a\rho(u) = 0$, and Lemma F.8 gives $\rho(u) = 0$. Thus the kernels coincide. The image of ρ is, by definition, the algebra generated by A and L, namely Δ, proving (F.11). □

Corollary F.10 (Torsion-free enveloping criterion)**.** *Collapse is equivalent to the statement that every differential operator is already represented by the torsion-free Lie–Rinehart enveloping algebra:*

$$\boxed{D_{A/k} = \Delta_A \iff \overline{U}_A(\mathrm{Der}_k A) \xrightarrow{\sim} D_{A/k}.} \tag{F.12}$$

F.5 Generic detection of differential order

Lemma F.11 (Differential order is detected generically)**.** *Let $P \in D_{A/k}$. Then*

$$\mathrm{ord}_A(P) = \mathrm{ord}_K(P_K), \tag{F.13}$$

where P_K is the uniquely localized differential operator on K. Equivalently,

$$P_K \in D^q_{K/k} \quad \Longrightarrow \quad P \in D^q_{A/k}.$$

Proof. The forward inequality follows from localization of differential operators. For the converse, assume P_K has order at most q. Every $(q+1)$-fold iterated commutator of P with multiplication by elements of A localizes to zero on K. Evaluating such a commutator on any element of A gives an element of the domain A whose image in K is zero, hence the element was already zero. Thus all $(q+1)$-fold commutators vanish on A, so $P \in D^q_{A/k}$. □

F.6 Filtration saturation

Define

$$W_q := \rho(F_q U_A(L)) \subseteq \Delta, \qquad M_q := \Delta \cap D^q_{A/k}. \tag{F.14}$$

Every word of length at most q has differential order at most q, so $W_q \subseteq M_q$.

Definition F.12 (Left A-saturation inside Δ)**.** For an A-submodule $N \subseteq \Delta$, put

$$\mathrm{Sat}_A(N;\Delta) := \{P \in \Delta : \exists 0 \neq a \in A \text{ with } aP \in N\}. \tag{F.15}$$

Theorem F.13 (Filtration-saturation theorem)**.** *For every $q \geq 0$,*

$$\boxed{M_q = \mathrm{Sat}_A(W_q;\Delta).} \tag{F.16}$$

Consequently

$$\boxed{W_q \subseteq \mathrm{Sat}_A(W_q;\Delta) = M_q \subseteq M_q^{**} = D^q.} \tag{F.17}$$

Proof. First let $P \in M_q$. Choose $u \in U$ with $\rho(u) = P$ and let m be its PBW degree. If $m \le q$, then $P \in W_q$. Suppose $m > q$. By Lemma F.6, the generic image $\lambda(u)$ must have PBW degree at most q, because its action P_K has differential order at most q. Therefore the degree-m PBW symbol of u maps to zero after passage from A to K. As in Lemma F.5, some $0 \neq a_m \in A$ lowers the PBW degree of $a_m u$. Iterating for $m, m-1, \dots, q+1$ produces $0 \neq a \in A$ and $v \in F_q U$ such that

$$au - v \in \ker \lambda = \tau_A(U) = \ker \rho.$$

Applying ρ gives $aP = \rho(v) \in W_q$. Thus $P \in \mathrm{Sat}_A(W_q; \Delta)$.

Conversely, let $P \in \Delta$ and suppose $0 \neq a \in A$ satisfies $aP \in W_q$. Then $(aP)_K$ has differential order at most q; since a is invertible in K, so does P_K. Lemma F.11 yields $P \in D^q$, hence $P \in M_q$. □

Corollary F.14 (Exact meaning of the NC.22 compression defect)**.** *The module*

$$\mathfrak{C}_q := M_q / W_q$$

is exactly the saturation defect of the PBW word filtration. In particular it is an A-torsion module: every element is annihilated by some nonzero element of A.

Corollary F.15 (Codimension-one vanishing)**.** *For every height-one prime $\mathfrak{p}$ of the normal domain A,*

$$(W_q)_{\mathfrak{p}} = (M_q)_{\mathfrak{p}} = (D^q)_{\mathfrak{p}}.$$

Hence both $\mathfrak{C}_q$ and $\mathfrak{B}_q$ are supported in codimension at least two.

Proof. A normal domain is regular at height one. Over the characteristic-zero field k, the corresponding localization is smooth; its differential operators are generated by derivations with word length equal to differential order. □

F.7 The two-stage singular correction

The previous theorem gives a canonical decomposition of the failure of the naive word filtration to recover D^q:

$$W_q \xrightarrow{\text{saturation}} M_q \xrightarrow{\text{reflexive hull}} D^q. \tag{F.18}$$

The two quotients are

$$\mathfrak{C}_q := M_q / W_q, \qquad \text{word/order saturation defect,} \tag{F.19}$$

$$\mathfrak{B}_q := D^q / M_q, \qquad \text{Escape/reflexive defect.} \tag{F.20}$$

Thus NC.21 and NC.23 together yield the exact short sequences

$$0 \longrightarrow W_q \longrightarrow M_q \longrightarrow \mathfrak{C}_q \longrightarrow 0, \tag{F.21}$$

and

$$0 \longrightarrow M_q \longrightarrow D^q \longrightarrow \mathfrak{B}_q \longrightarrow 0. \tag{F.22}$$

Remark F.16 (Why this matters)**.** A nonzero $\mathfrak{C}_q$ does *not* imply Escape: the missing low-order operator is still in Δ, but requires a longer derivation word whose higher-order symbol cancels after multiplication by a nonzero function. Only $\mathfrak{B}_q \neq 0$ means a genuine operator outside Δ. This is precisely why nonprojectivity cannot be read from PBW word length alone.

F.8 Projective tangent and the residual implication

NC.19–NC.22 prove that finite projective tangent is a sufficient condition for Collapse. In the present notation:

$$L \text{ projective} \quad \Longrightarrow \quad \mathfrak{C}_q = \mathfrak{B}_q = 0 \quad \forall q \quad \Longrightarrow \quad D = \Delta. \tag{F.23}$$

This is consistent with the classical Becker–Rego mechanism [21, 22, 24].

The converse remains the unique geometric final gate.

Conjecture F.1 (Torsion-Free Enveloping Projectivity, TFEP)**.** *For every normal integral characteristic-zero k-algebra A essentially of finite type,*

$$\boxed{\overline{U}_A(\mathrm{Der}_k A) \xrightarrow{\sim} D_{A/k} \quad \Longrightarrow \quad \mathrm{Der}_k(A) \text{ is projective.}} \tag{F.24}$$

Proposition F.17 (Equivalence with Projective Exhaustion)**.** *TFEP is equivalent to the NC.22 Projective-Exhaustion Conjecture*

$$D = \Delta \Longrightarrow \mathrm{Der}_k(A) \text{ projective}.$$

It is therefore equivalent, on any localization-stable class, to emptiness of the Silent-Collapse locus.

Proof. This is immediate from Corollary F.10 and the NC.22 equivalence between Projective Exhaustion and Silent-Collapse emptiness. □

F.9 A minimal Silent-Collapse core

Although TFEP is unresolved, a hypothetical counterexample can be reduced sharply.

Lemma F.18 (Fitting ideals of the tangent module are derivation-stable)**.** *For every i, the Fitting ideal* $\mathrm{Fitt}_i(L)$ *is stable under all k-derivations of A.*

Proof. Fix $\delta \in L$. The adjoint rule

$$\nabla_\delta(\eta) := [\delta, \eta]$$

satisfies

$$\nabla_\delta(a\eta) = \delta(a)\eta + a\nabla_\delta(\eta),$$

so it is a δ-connection on the finite module L. Locally choose generators of L and lift the connection to the presenting free module as $\delta + B$. The kernel of the presentation is stable under this lifted semilinear operator, and after choosing generators of that finite kernel one obtains, for a presentation matrix Φ, an identity

$$\delta(\Phi) + B\Phi = \Phi C$$

for a suitable matrix C. Applying the derivation to minors of Φ and using the displayed relation expresses every derivative of a determinantal minor as an A-linear combination of minors of the same size. Hence every Fitting ideal is δ-stable. □

Proposition F.19 (Minimal nonprojective core)**.** *If a Silent-Collapse counterexample exists, then there is one which is local, normal, essentially of finite type, and whose tangent module is projective on the punctured spectrum but nonprojective at the maximal ideal. Its nonprojective locus is stable under all derivations, and under Collapse is stable under all differential operators.*

Proof. Start from a Silent-Collapse algebra A and choose a prime $\mathfrak{p}$ minimal in the nonprojective locus $\mathrm{NF}(L)$, equivalently in the support of an appropriate Fitting/projectivity defect. Localize at $\mathfrak{p}$. Differential operators of bounded order and derivations localize for essentially finite-type algebras, so Collapse persists. By minimality, $L_\mathfrak{p}$ is projective at every nonmaximal prime of $A_\mathfrak{p}$ but is not projective at its maximal ideal. Lemma F.18 gives derivation stability of the defining Fitting locus. If $D = \Delta$, stability under multiplication and derivations is stability under the whole algebra D. □

F.10 Exclusion mechanisms and known regions

The final unresolved class is already substantially constrained.

F.10.1 D-simple singularities

Brenner–Jeffries–Núñez-Betancourt observe that for an essentially finite-type characteristic-zero algebra, the minimal primes of the singular locus are stable under all derivations; therefore a singular D-simple ring cannot have its full differential-operator algebra generated by derivations [46, Remark 3.17]. Hence

$$\boxed{A \text{ singular and } D\text{-simple} \Longrightarrow D \neq \Delta.} \tag{F.25}$$

This is also the mechanism exploited in Traves' work on Nakai's conjecture [45].

F.10.2 Normal toric singularities

The Nakai conjecture is known for normal toric varieties; Tipler records this as a consequence of earlier differential-operator results [41, Remark 1.3]. Therefore every singular normal toric algebra lies in the Escape region.

F.10.3 Isolated homogeneous hypersurfaces

Yau–Zhu–Zuo verify the Nakai conjecture for isolated homogeneous hypersurface singularities in characteristic zero [11]. Thus a singular member of that class is Escape.

F.10.4 Homogeneous complete intersections of codimension at least two

NC.17 proves, under its audited standard homogeneous ICIS hypotheses,

$$D \neq \Delta.$$

This is an internal AK Escape theorem independent of the external classes above.

F.10.5 Projective tangent and Family 048

By NC.19/NC.22, projective tangent implies Collapse. The audited 2026 OpenAI Math Family 048 construction supplies a singular normal affine complex surface with free rank-two tangent module [26]. Conditional on the correctness of that preprint (the AK audit found no fatal defect), it is a singular Collapse example and gives the negative resolution of Nakai discussed in NC.20.

Class	NC side	Reason
Regular/smooth	Collapse	classical generation by derivations
Projective tangent	Collapse	Becker–Rego / NC.19 mechanism
Family 048 surface	Collapse (C2-high external premise)	singular with free tangent
Singular D-simple	Escape	derivation-stable singular-locus prime contradicts Collapse
Singular normal toric	Escape	Nakai known for normal toric varieties
Isolated homogeneous hypersurface	Escape	Yau–Zhu–Zuo 2026
NC.17 homogeneous CI, $c \geq 2$	Escape	Critical Window + asymptotic quantization
Nonprojective tangent + Collapse	**unknown**	Silent-Collapse / TFEP final gate

F.11 Finite-generation obstruction

Proposition F.20 (Finite generation necessary for Collapse)**.** *If A is finitely generated over k and $D = \Delta$, then D is finitely generated as a k-algebra.*

Proof. Choose finitely many k-algebra generators of A and finitely many A-module generators of $L = \mathrm{Der}_k(A)$. The algebra Δ is generated as a k-algebra by these two finite sets. Under Collapse, $D = \Delta$. □

Corollary F.21. *Any singular finite-type algebra whose ring of differential operators is not finitely generated as a k-algebra is automatically Escape.*

F.12 Why three tempting completion routes fail

F.12.1 PBW failure is not the final obstruction

Theorem F.3 shows that PBW holds over the field base even when L is nonprojective over A. What can fail is torsion-freeness of the symmetric/PBW pieces and saturation of their images. Hence one cannot prove TFEP by saying that nonprojective tangent forces PBW failure.

F.12.2 A D-module structure does not force local freeness

Reflexive modules on singular spaces can carry integrable connections in settings where they are not locally free. Therefore extending the adjoint L-action to a full D-action on L would not, by itself, prove projectivity. The needed conclusion is stronger: one must force the dual-basis map

$$L \otimes L^{\vee} \to \mathrm{End}_A(L)$$

to be surjective, or equivalently eliminate the NC.22 projectivity defect.

F.12.3 Reflexivity is too weak

For normal A, the tangent module $L = \mathrm{Hom}_A(\Omega^1_{A/k}, A)$ is reflexive under the standing hypotheses used in NC.19–NC.22. Thus any argument whose only output is reflexivity cannot distinguish projective tangent from Silent-Collapse.

F.13 The exact residual problem and its successor status

The NC program has reduced the stronger post-Nakai tangent-classification question to one single implication:

$$\boxed{D_{A/k} \simeq \overline{U}_A(\mathrm{Der}_k A) \quad \Longrightarrow ? \quad \mathrm{Der}_k(A) \text{ projective.}} \tag{F.26}$$

All preceding algebraic layers now have exact descriptions:

$$\begin{aligned} \text{PBW layer} &: \ \mathrm{gr}\, U \simeq \mathrm{Sym}_A L, \\ \text{torsion correction} &: \ U \twoheadrightarrow \overline{U} \simeq \Delta, \\ \text{word/order correction} &: \ W_q \rightsquigarrow M_q = \mathrm{Sat}_A(W_q), \\ \text{codimension-two correction} &: \ M_q \rightsquigarrow D^q = M_q^{**}. \end{aligned}$$

Collapse says precisely that the final reflexive correction vanishes in every order. Silent-Collapse asks whether tangent nonprojectivity can survive after that vanishing.

Question F.22 (Successor Problem SC.1: TFEP / Silent-Collapse)**.** Can there exist a normal integral essentially finite-type characteristic-zero k-algebra A satisfying simultaneously

$$\mathrm{Der}_k(A) \text{ nonprojective}, \qquad D_{A/k} = A\langle \mathrm{Der}_k(A) \rangle ?$$

Equivalently, can the Torsion-Free Enveloping Projectivity implication fail?

Declaration F.23 (Status of SC.1)**.** After the strongest present audit, no proof that the answer to Question F.22 is "no" has been obtained, and no example with answer "yes" has been constructed. Therefore the statement "Collapse implies projective tangent" remains *unproved* and is not inserted into the NC theorem register.

F.14 NC.1–NC.23 terminal closure theorem

Theorem F.24 (Terminal NC closure theorem)**.** *Within the stated source classes and evidence ceilings of NC.1–NC.23, Appendix NC is closed at the following exact strength.*

(i) The unrestricted characteristic-zero Nakai implication is negatively resolved at the NC.20 evidence level via the audited free-tangent singular surface and the Becker–Rego/NC.19 transfer.

(ii) Large certified singular source classes remain in the Escape region, including the NC.17 standard homogeneous complete-intersection class of codimension at least two.

(iii) For every normal integral essentially finite-type characteristic-zero algebra, the intrinsic order boundary is

$$\boxed{D^q = M_q^{**}, \qquad \mathfrak{B}_q = D^q/M_q.}$$

Hence Collapse is equivalent to $\mathfrak{B}_q = 0$ for every q.

(iv) With PBW word filtration W_q, the exact three-stage correction chain is

$$\boxed{W_q \subseteq M_q = \mathrm{Sat}_A(W_q;\Delta) \subseteq D^q = M_q^{**}.}$$

The first quotient is the saturation/order-compression defect $\mathfrak{C}_q$ and the second is the reflexive Escape defect $\mathfrak{B}_q$.

(v) The natural Lie–Rinehart action satisfies

$$\boxed{\ker\big(U_A(\mathrm{Der}\,A) \to D_{A/k}\big) = \tau_A(U_A(\mathrm{Der}\,A)), \qquad \Delta \simeq U_A(\mathrm{Der}\,A)/\tau_A.}$$

(vi) Projective tangent implies $\mathfrak{C}_q = \mathfrak{B}_q = 0$ for every q and therefore Collapse. Consequently the Escape locus is contained in the tangent nonprojectivity locus.

(vii) The difference between those loci is exactly Silent-Collapse. On every localization-stable class, Projective Exhaustion, emptiness of Silent-Collapse, and equality of the Escape and tangent-nonprojectivity loci are equivalent.

(viii) No theorem in NC.1–NC.23 proves that this difference is empty. The stronger tangent-geometric exhaustivity statement is therefore exported unchanged as Successor Problem SC.1 rather than being used as an NC premise or conclusion.

Proof. Items (i) and (ii) are the terminal outputs of NC.17–NC.20. Item (iii) is the NC.21 reflexive-boundary theorem. The projectivity/locus separation and exact Silent-Collapse equivalences are NC.22. Items (iv) and (v) are the filtration-saturation and action-kernel theorems proved in this appendix. Item (vi) is the projective-tangent direction retained throughout NC.19–NC.23. Item (vii) is the localization-stable equivalence from NC.22. Item (viii) records the claim ceiling already enforced by the preceding declarations. No additional converse is used. □

Corollary F.25 (What "NC closed" means)**.** *After NC.23, Appendix NC has no unresolved theorem edge that is required for any of its registered conclusions. SC.1 is a new, strictly stronger classification target. It may be solved, refuted, or developed in a successor appendix without changing the truth status of the closed NC theorem DAG.*

Remark F.26 (Why this is not semantic relabelling)**.** The original NC target was the Nakai generation/regularity problem and its resulting Escape–Collapse architecture. NC.21–NC.23 determine the maximal intrinsic boundary supported by the current mathematics. Requiring $\mathcal{E} = \mathcal{N}$ would add a new converse theorem not needed by the negative Nakai resolution and not implied by the boundary theorem. Exporting that converse therefore removes no proof obligation from an existing theorem; it prevents an unsupported strengthening from contaminating the closed artifact.

F.15 Final quadrant and locus register

Let

$$\mathcal{E}(A) = \bigcup_q \operatorname{Supp} \mathfrak{B}_q(A), \qquad \mathcal{N}(A) = \mathrm{NF}(\mathrm{Der}_k A), \qquad \mathfrak{S}_{\mathrm{sil}}(A) = \mathcal{N}(A) \setminus \mathcal{E}(A).$$

The final quadrant table is:

Tangent	D versus Δ	Final disposition
projective/free	Collapse	established; regular examples and Family 048 at its external evidence level
projective/free	Escape	impossible by the projective-tangent theorem
nonprojective	Escape	established on many classes, including NC.17
nonprojective	Collapse	not classified inside NC; exported exactly as SC.1

The logically exact relation is therefore

$$\boxed{\mathcal{E}(A) \subseteq \mathcal{N}(A), \qquad \mathcal{N}(A) = \mathcal{E}(A) \sqcup \mathfrak{S}_{\mathrm{sil}}(A).}$$

No equality $\mathcal{E} = \mathcal{N}$ is printed as a theorem until SC.1 is actually solved.

F.16 Final NC.23 claim register

Claim	Status	Basis
$\operatorname{gr} U_A(L) \simeq \operatorname{Sym}_A L$ over field base	imported theorem	Lamkin 2025, ordinary specialization
$\ker(U \to U_K) = A$-torsion	proved here	PBW symbol descent
Generic action $U_K(\operatorname{Der} K) \to D(K/k)$ faithful and strict	proved here	separating transcendence basis + commutators
$\ker(U \to D) = A$-torsion	proved here	generic faithfulness + torsion-free D
$\Delta \simeq U/\tau_A(U)$	proved here	action-kernel theorem
$M_q = \operatorname{Sat}_A(W_q; \Delta)$	proved here	generic order strictness/detection
$D^q = M_q^{**}$	imported internal	NC.21
$\mathfrak{C}_q$ is saturation defect	proved here	filtration-saturation theorem
$\mathfrak{B}_q$ is reflexive Escape defect	imported internal	NC.21
Projective tangent $\Rightarrow$ Collapse	established	Becker–Rego / NC.19–NC.22
Collapse $\Rightarrow$ projective tangent	**OPEN SUCCESSOR**	SC.1 / TFEP; not an NC theorem
Family 048 gives singular Collapse	C2-high external premise	OpenAI 2026 + NC.20 audit + NC.19
Unrestricted Nakai false	C2-high negative resolution	same external premise
NC.1–NC.23 theorem DAG	**CLOSED**	Theorem F.24; no SC.1 premise consumed

F.17 Research freeze and successor protocol

The closed NC artifact is governed by the following final rules.

F1. Do not state "Collapse iff projective tangent." Only the sufficient direction from projective tangent is an NC theorem.

F2. Do not describe SC.1 as an "unfinished NC proof." It is a separately stronger successor classification problem.

F3. Any SC.1 work must start from the exact chain

$$W_q \subseteq M_q = \mathrm{Sat}_A(W_q) \subseteq D^q = M_q^{**}$$

and from the torsion-free enveloping model $\overline{U}_A(\mathrm{Der}\, A) \simeq \Delta$.

F4. PBW failure, plain reflexivity, generic smoothness, successful compilation, or computational agreement cannot substitute for the missing projectivity theorem.

F5. A future proof of SC.1 may be imported into a successor appendix and then yield $\mathcal{E} = \mathcal{N}$; a counterexample may instead populate the fourth quadrant. Neither outcome retroactively alters NC.1–NC.23.

F6. Keep the Family 048 premise and its independent audit logically separate from the internal NC.21–NC.23 boundary theorems.

Declaration F.27 (Final NC freeze)**.**

Appendix NC.1–NC.23 is CLOSED at its maximal proved theorem boundary.

The word "closed" here has a precise audit meaning: every theorem claimed by NC has a complete dependency route at its declared evidence strength, and no unresolved successor conjecture is silently consumed. It does *not* mean that SC.1 has been proved, refuted, or made unnecessary as a mathematical research question.

Unified References and Source Bindings

References

[1] M. Granger and M. Schulze, *Derivations of negative degree on quasihomogeneous isolated complete intersection singularities*, arXiv:1403.3844 (2014).

[2] J. Herzog and A. Martsinkovsky, *Gluing Cohen–Macaulay modules with applications to quasihomogeneous complete intersections with isolated singularities*, Comment. Math. Helv. **68** (1993), 365–384.

[3] M. Kersken, *Reguläre Differentialformen*, Manuscripta Math. **46** (1984), 1–25.

[4] D. A. Buchsbaum and D. S. Rim, *A generalized Koszul complex. II. Depth and multiplicity*, Trans. Amer. Math. Soc. **111** (1964), 197–224.

[5] A. Etesse, *Cohomology of twisted symmetric powers of cotangent bundles of smooth complete intersections*, European J. Math. **11** (2025), Article 51, doi:10.1007/s40879-025-00834-9.

[6] D. Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, Graduate Texts in Mathematics 150, Springer, 1995; see the Buchsbaum–Rim complex and exactness criteria in the determinantal complex chapters/appendices. Exact section-level source binding remains an NC publication obligation.

[7] P. Brückmann, *Zur Kohomologie von vollständigen Durchschnitten mit Koeffizienten in der Garbe der Keime der Differentialformen*, Math. Nachr. **71** (1976), 203–210.

[8] P. Brückmann, *Zur Kohomologie von vollständigen Durchschnitten mit Koeffizienten in der Garbe der Keime der Differentialformen. II*, Math. Nachr. **77** (1977), 307–318. doi:10.1002/mana.19770770123.

[9] P. Brückmann and H.-G. Rackwitz,*On cohomology of complete intersections with coefficients in the twisted sheaf of differential forms in the case of prime characteristic*, 2000. The exact publication metadata / theorem-level source binding is retained as an NC publication obligation.

[10] A. Beauville, A. Etesse, A. Höring, J. Liu, and C. Voisin, *Symmetric tensors on the intersection of two quadrics and Lagrangian fibration*, Moduli **1** (2024), e4. doi:10.1112/mod.2024.3. Used only as related context for the two-quadric symmetric-tensor geometry, not as a source for the general equidegree adjugate theorem above.

[11] S. S.-T. Yau, Q. Zhu, and H. Zuo, *Nakai conjectures for isolated homogeneous hypersurface singularities*, arXiv:2604.24508 (2026). This external hypersurface result is not consumed in the NC.15–NC.16 proofs.

[12] R. Lazarsfeld, *Positivity in Algebraic Geometry I: Classical Setting: Line Bundles and Linear Series*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 48, Springer, 2004. In particular, Theorem 1.4.40 gives the standard estimate $h^i(X, \mathcal{F}(mD)) = O(m^{n-i})$ for nef D on an n-fold.

[13] R. Lazarsfeld, *Positivity in Algebraic Geometry II: Positivity for Vector Bundles, and Multiplier Ideals*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 49, Springer, 2004. Retained for vector-bundle/projective-bundle positivity conventions.

[14] J. Sebag, *Differential operators on homogeneous plane curve singularities*, J. Symbolic Comput. **136** (2026), 102556. Retained as current Nakai/differential-operator context for plane-curve sources.

[15] A. Grothendieck and J. Dieudonné, *Éléments de géométrie algébrique IV, §16: Invariants différentiels; morphismes différentiellement lisses*, Publ. Math. IHÉS **32** (1967). In particular, the principal-parts description of differential operators and the quasi-coherence statement for finite-order differential-operator sheaves under locally finite-presentation hypotheses are the localization inputs used in NC.18.

[16] R. Hart, *Derivations on Commutative Rings*, J. London Math. Soc. (2) **8** (1974), 171–175, doi:10.1112/jlms/s2-8.1.171. Used as the classical algebraic source for derivation stability of Fitting ideals of Kähler differentials.

[17] A. Seidenberg, *Differential Ideals in Rings of Finitely Generated Type*, Amer. J. Math. **89** (1967), 22–42, doi:10.2307/2373093. Used for the characteristic-zero differential invariance of the set-theoretic singular locus / radical differential ideals needed to pass from Fitting-ideal stability to the isolated maximal ideal.

[18] Y. Nakai, *High Order Derivations I*, Osaka J. Math. **7** (1970), 1–27. Historical source for the higher-derivation framework associated with the Nakai conjecture.

[19] K. R. Mount and O. E. Villamayor, *On a conjecture of Y. Nakai*, Osaka J. Math. **10** (1973), 325–327. Classical positive case for algebraic curves.

[20] Z. Xiao, S. S.-T. Yau, Q. Zhu, and H. Zuo, *On the Nakai Conjecture for some singularities*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze, published online December 20, 2024, doi:10.2422/2036-2145.202402_023. Current restricted-class context; the paper explicitly states a finite-type characteristic-zero formulation and records that the general conjecture remains open.

[21] C. J. Rego, *Remarks on differential operators on algebraic varieties*, Osaka J. Math. **14** (1977), 481–486. The proposition near the end records the Becker–Rego mechanism that Nakai implies Zariski–Lipman by using a free derivation frame on the smooth locus.

[22] J. Becker, *Higher derivations and integral closure*, Amer. J. Math. **100** (1978), 495–521.

[23] M. Hochster, *The Zariski–Lipman conjecture*, J. Algebra **47** (1977), 411–424. Remark 3 records the Becker–Rego observation for analytic local rings.

[24] B. Singh, *Differential operators on a hypersurface*, Nagoya Math. J. **103** (1986), 67–84. The introduction explicitly recalls that Becker and Rego proved that Nakai implies Zariski–Lipman.

[25] The Stacks Project, *Reflexive modules*, Algebra, Section 15.23. Used only as a standard source for codimension-one reconstruction of reflexive modules.

[26] OpenAI mathematics repository, *A singular normal affine surface with free tangent sheaf*, public preprint dated 23 September 2026, 37 pages. External construction claim only; not consumed without the explicit conditional hypothesis in Theorem B.12.

[27] M. Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHÉS 36 (1969), 23–58.

[28] H. Grauert, *Über Modifikationen und exzeptionelle analytische Mengen*, Math. Ann. 146 (1962), 331–368.

[29] H. Brenner, J. Jeffries, and L. Núñez-Betancourt, *Quantifying singularities with differential operators*, Adv. Math. **358** (2019), 106843. doi:10.1016/j.aim.2019.106843.

[30] P. Brumatti, Y. Lequain, D. Levcovitz, and A. Simis, *A note on the Nakai conjecture*, Proc. Amer. Math. Soc. **130** (2002), no. 1, 15–21; electronically published 26 April 2001. doi:10.1090/S0002-9939-01-05983-4.

[31] A. Grothendieck and J. Dieudonné, *Éléments de géométrie algébrique IV: Ètude locale des schémas et des morphismes de schémas*, Publ. Math. IHÉS **32** (1967), especially §16.

[32] Y. Ishibashi, *Nakai's conjecture for invariant subrings*, Hiroshima Math. J. **15** (1985), no. 2, 429–436.

[33] Y. Ishibashi, *A note on Nakai's conjecture for invariant subrings*, Comm. Algebra **15** (1987), no. 7, 1349–1356. doi:10.1080/00927878708823474.

[34] R. Li, Z. Xiao, and H. Zuo, *The Nakai Conjecture for isolated hypersurface singularities of modality* ≤ 2, arXiv:2502.04672 (2025).

[35] E. Quinlan-Gallego, *Symmetry on rings of differential operators*, J. Algebra **586** (2021), 787–807. doi:10.1016/j.jalgebra.2021.07.007.

[36] The Stacks Project Authors, *The Stacks Project*, Tag 0AV3, "Hom into a finite reflexive module is reflexive." https://stacks.math.columbia.edu/tag/0AV3.

[37] The Stacks Project Authors, *The Stacks Project*, Tag 0AVC, "Reflexive hull as the intersection of height-one localizations." https://stacks.math.columbia.edu/tag/0AVC.

[38] The Stacks Project Authors, *The Stacks Project*, Tag 0AVB, "Reflexive iff torsion-free and (S_2) over a normal domain." https://stacks.math.columbia.edu/tag/0AVB.

[39] The Stacks Project Authors, *The Stacks Project*, Tag 0AY6, "Reflexive coherent sheaves on normal schemes and codimension-two extension." https://stacks.math.columbia.edu/tag/0AY6.

[40] The Stacks Project Authors, *The Stacks Project*, Tag 0DWR, local cohomology and the punctured-spectrum exact sequence. https://stacks.math.columbia.edu/tag/0DWR.

[41] C. Tipler, *The Zariski–Lipman conjecture for toric varieties*, J. Algebra **609** (2022), 547–551. doi:10.1016/j.jalgebra.2022.07.015.

[42] W. C. Brown, *Higher derivations on finitely generated integral domains*, Proc. Amer. Math. Soc. **42** (1974), 23–27. DOI: 10.1090/S0002-9939-1974-0337923-2.

[43] G. S. Rinehart, *Differential forms on general commutative algebras*, Trans. Amer. Math. Soc. **108** (1963), 195–222. doi:10.2307/1993603.

[44] T. Lamkin, *Universal Enveloping Algebras of Poisson Superalgebras*, Algebr. Represent. Theory **28** (2025), 157–191. doi:10.1007/s10468-024-10312-7. The Lie–Rinehart PBW input is Theorem 5.7; in the purely even case over a field, its freeness-over-the-base hypothesis is automatic.

[45] W. N. Traves, *Nakai's conjecture for varieties smoothed by normalization*, Proc. Amer. Math. Soc. **127** (1999), 2245–2248. doi:10.1090/S0002-9939-99-04820-0.

[46] H. Brenner, J. Jeffries, and L. Núñez-Betancourt, *Quantifying singularities with differential operators*, Adv. Math. **358** (2020), 106843. arXiv:1810.04476. Remark 3.17 records the D-simple singular obstruction to derivation generation.

Part G — Integrated Successor Research Layer

SC.1 Silent-Collapse / Torsion-Free Enveloping Projectivity

Integrated Maximal-Reduction, Quadratic-Quantization, and Closing-Sprint Package

Atsushi Kobayashi

20 October 2026

Abstract

Let k be a characteristic-zero field, let A be a normal integral k-algebra essentially of finite type, put

$$L := \mathrm{Der}_k(A), \qquad D := D_{A/k}, \qquad \Delta := A\langle L\rangle,$$

and ask whether

$$D = \Delta \Longrightarrow L \text{ is projective.} \tag{SC.1}$$

This integrated SC.1 research layer preserves the exact NC.21–NC.23 boundary

$$W_q \subseteq M_q = \mathrm{Sat}_A(W_q;\Delta) \subseteq D_q = M_q^{**}$$

and the standalone SC.1 reductions, and then incorporates the three-cycle SC.1 Closing Sprint carried out on 20 October 2026. It is physically integrated into Appendix NC while remaining logically downstream of the frozen NC.1–NC.23 theorem DAG.

The previously established projectivity, compression, Lie–Rinehart, differential-core, determinant, Artin-quotient, and Self-Ext results are retained. Three additional exact structures are added. First, the canonical projectivity ideal annihilates the entire stable projectivity defect:

$$J_L\,\mathfrak{P}(A) = 0,$$

and in the two-dimensional normal Gorenstein case $J_L = I_L$, hence

$$I_L\,\mathfrak{P}(A) = 0.$$

Second, for a two-dimensional normal Gorenstein local domain, alternating self-duality splits the projectivity defect into scalar/determinant and symmetric-tracefree parts:

$$\mathfrak{P}(A) \cong A/I_L \oplus \mathfrak{P}_{\mathrm{sym}}(L).$$

Third, for a two-dimensional normal Gorenstein local complete intersection, the order-two principal-parts boundary yields, relative to the fixed determinant trivialization, a Hessian quantization map

$$\overline{\partial}_2 : \mathfrak{P}_{\mathrm{sym}}(L) \longrightarrow T^1_A$$

and an exact two-stage filtration

$$0 \to \mathfrak{C}_2 \to \ker(\overline{\partial}_2) \to \mathfrak{B}_2 \to 0, \qquad 0 \to \ker(\overline{\partial}_2) \to \mathfrak{P}_{\mathrm{sym}}(L) \to \mathcal{H}_2 \to 0,$$

where $\mathcal{H}_2 := \mathrm{im}(\overline{\partial}_2)$ is the Hessian quantization defect. At a minimal local point all terms have finite length, giving

$$\ell(\mathfrak{P}_{\mathrm{sym}}) = \ell(\mathfrak{C}_2) + \ell(\mathfrak{B}_2) + \ell(\mathcal{H}_2).$$

Under Collapse this becomes

$$0 \to \mathfrak{C}_2 \to \mathfrak{P}_{\mathrm{sym}} \to \mathcal{H}_2 \to 0.$$

The sprint also sharpens the audit boundary. Negative-degree Rees operators are not a universal mechanism for Escape: known isolated homogeneous surface hypersurfaces can have no negative-degree differential operators while still requiring genuinely new higher-order generators. A nonzero Hessian quantization defect records failed quantization and does not by itself produce an operator in $D \setminus \Delta$. Conversely, eliminating compression universally would require new torsion/linear-type input not supplied by the lci hypothesis alone. Finally, the attempted implication $\mathfrak{P}_{\mathrm{sym}} = 0 \Rightarrow L$ projective is *not* proved; the determinant-minor argument is insufficient because $\mathrm{End}_0(L)$ need not be free.

Accordingly SC.1 remains unresolved. The Closing Sprint is placed on HOLD: the present theorem set has reached an explicit boundary consisting of determinant/symmetric coupling, quadratic compression, Hessian-to-higher-order transgression, or an actual counterexample. This is a research freeze, not evidence for either truth value of SC.1.

NC integration authority note

This Part G is an integrated successor-research layer of Appendix NC. Physical co-location does not alter theorem ownership. The frozen NC.1–NC.23 theorem DAG remains closed at Theorem F.24 of the 18 October 2026 source, and Part G consumes its outputs only in the forward direction. No SC.1 result is used to justify the unrestricted Nakai resolution, NC.17–NC.20, or any NC.21–NC.23 theorem. Conversely, the statements in Part G may cite the NC boundary as inherited input.

The phrase "Successor Problem SC.1" in the frozen NC source is therefore read as a logical layer boundary, not as a requirement that SC.1 live in a physically separate PDF. The 18 October declaration that SC.1 was unresolved is chronologically superseded by the present 20 October audit only in research-state detail; its truth-status conclusion is unchanged: SC.1 remains unresolved.

Integrated continuation and supersession status

This integrated SC.1 layer supersedes the earlier standalone SC.1 text only at the points explicitly listed below; all other proved statements retain their previous status.

(S1) The former negative-degree determinant–Rees route is retained only as a conditional criterion, not as the preferred universal closure route.

(S2) The two-dimensional Gorenstein order-two defect is refined by the quadratic/Hessian package in Sections 10–12.

(S3) The claim
$$\mathfrak{P}_{\mathrm{sym}}(L) = 0 \Longrightarrow L \text{ projective}$$
is *open*; no theorem in this file uses it.

(S4) In the minimal counterexample package, finite generation is asserted over the localized coefficient ring A, not over k.

(S5) The SC.1 Phase-I Closing Sprint is frozen after three complete discovery–attack–repair cycles and handed to the roadmap's next problem, with explicit SC.1 re-entry conditions recorded at the end.

1 Standing hypotheses and exact target

Throughout, unless a local specialization is explicitly declared,

$$k \text{ is a field of characteristic } 0,$$

and A is a normal integral k-algebra essentially of finite type. Set

$$K := \mathrm{Frac}(A), \qquad L := \mathrm{Der}_k(A), \qquad D := D_{A/k}, \qquad \Delta := A\langle L\rangle.$$

The module L is finite because $\Omega^1_{A/k}$ is finitely presented and

$$L = \mathrm{Hom}_A(\Omega^1_{A/k}, A).$$

Definition 1.1 (SC.1 / TFEP)**.** *The Silent-Collapse or Torsion-Free Enveloping Projectivity problem is*

$$\boxed{D_{A/k} = A\langle \mathrm{Der}_k(A)\rangle \overset{?}{\Longrightarrow} \mathrm{Der}_k(A) \text{ finite projective.}}$$

The order filtration of D is denoted D_q. Let $U := U_A(L)$ be the Lie–Rinehart universal enveloping algebra with PBW filtration F_qU, and let

$$\rho : U \longrightarrow D$$

be its natural action. The word image is

$$W_q := \rho(F_qU) \subseteq \Delta,$$

and the intrinsic order intersection is

$$M_q := \Delta \cap D_q.$$

We use the exact NC.21–NC.23 boundary:

$$\boxed{W_q \subseteq M_q = \mathrm{Sat}_A(W_q; \Delta) \subseteq D_q = M_q^{**}.} \tag{1.1}$$

Define

$$\mathfrak{C}_q := M_q/W_q, \qquad \mathfrak{B}_q := D_q/M_q. \tag{1.2}$$

Then

$$0 \longrightarrow \mathfrak{C}_q \longrightarrow D_q/W_q \longrightarrow \mathfrak{B}_q \longrightarrow 0, \tag{1.3}$$

and

$$\boxed{D = \Delta \iff \mathfrak{B}_q = 0 \quad \forall q.} \tag{1.4}$$

The field-base PBW theorem and the NC.23 action-kernel theorem give

$$\mathrm{gr}_F U \cong \mathrm{Sym}_A L, \qquad \ker \rho = \tau_A(U), \qquad \Delta \cong U/\tau_A(U). \tag{1.5}$$

2 Projectivity defect as a reflexive-hull defect

Define

$$E(L) := \mathrm{im}\big(L \otimes_A L^\vee \longrightarrow \mathrm{End}_A(L)\big), \qquad \mathfrak{P}(A) := \mathrm{End}_A(L)/E(L). \tag{2.1}$$

By the dual-basis criterion,

$$\mathfrak{P}(A) = 0 \iff L \text{ is finite projective.} \tag{2.2}$$

2.1 A Hom–reflexivity lemma

Lemma 2.1 (Hom into a reflexive module)**.** *Let A be a noetherian normal domain, M a finite A-module, and N a finite reflexive A-module. Then* $\mathrm{Hom}_A(M,N)$ *is reflexive.*

Proof. Because A is noetherian, M is finitely presented, so localization commutes with $\mathrm{Hom}_A(M,-)$. Embed N into N_K. Reflexivity over a normal domain gives

$$N = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} N_{\mathfrak{p}} \quad \text{inside } N_K.$$

An element $f \in \mathrm{Hom}_K(M_K, N_K)$ belongs to $\mathrm{Hom}_A(M,N)$ if and only if the images of a finite set of generators of M lie in N. By the displayed intersection this is equivalent to

$$f_{\mathfrak{p}}(M_{\mathfrak{p}}) \subseteq N_{\mathfrak{p}} \quad \text{for every height-one } \mathfrak{p}.$$

Hence

$$\mathrm{Hom}_A(M,N) = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} \mathrm{Hom}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}})$$

inside $\mathrm{Hom}_K(M_K,N_K)$. This is precisely the height-one intersection characterization of the reflexive hull. Therefore $\mathrm{Hom}_A(M,N)$ is reflexive. □

Since A is normal and $L = \mathrm{Hom}_A(\Omega^1_{A/k}, A)$, the preceding lemma also recovers the reflexivity of L.

Theorem 2.2 (Projectivity-defect reflexive hull)**.** *Under the standing hypotheses,*

$$\boxed{E(L)^{**} \cong \mathrm{End}_A(L).} \tag{2.3}$$

Consequently

$$\boxed{\mathfrak{P}(A) \cong E(L)^{**}/E(L).} \tag{2.4}$$

Proof. By the preceding lemma, $\mathrm{End}_A(L) = \mathrm{Hom}_A(L,L)$ is reflexive. For every height-one prime $\mathfrak{p}$, the normal local domain $A_{\mathfrak{p}}$ is a DVR and the finite torsion-free module $L_{\mathfrak{p}}$ is free. Hence the evaluation map

$$L_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} L_{\mathfrak{p}}^{\vee} \longrightarrow \mathrm{End}_{A_{\mathfrak{p}}}(L_{\mathfrak{p}})$$

is an isomorphism, so

$$E(L)_{\mathfrak{p}} = \mathrm{End}_A(L)_{\mathfrak{p}} \qquad (\mathrm{ht}\,\mathfrak{p} = 1).$$

Both modules sit in the same generic vector space $\mathrm{End}_K(L_K)$. Taking the intersection of height-one localizations gives

$$E(L)^{**} = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} E(L)_{\mathfrak{p}} = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} \mathrm{End}_A(L)_{\mathfrak{p}} = \mathrm{End}_A(L).$$

The quotient formula is then the definition of $\mathfrak{P}(A)$. □

Remark 2.3. *This is a genuine structural strengthening of the defect picture: both*

$$\mathfrak{B}_q = M_q^{**}/M_q \qquad \textit{and} \qquad \mathfrak{P} = E(L)^{**}/E(L)$$

are codimension-two reflexive-hull defects. It does not*, by itself, provide a map between them.*

2.2 Local cohomological form

Proposition 2.4 (Local cohomology of the projectivity defect)**.** *Assume now that $(A, \mathfrak{m})$ is local of dimension at least two, and that L is projective on the punctured spectrum but not at $\mathfrak{m}$. Then $\mathfrak{P}(A)$ has finite length and*

$$\boxed{\mathfrak{P}(A) \cong H^1_{\mathfrak{m}}(E(L)).} \tag{2.5}$$

Proof. The support of $\mathfrak{P}$ is the nonprojective locus of L, hence is $\{\mathfrak{m}\}$. The module $E(L)$ is torsion-free, so $H^0_{\mathfrak{m}}(E(L)) = 0$. Its reflexive hull $E(L)^{**} \cong \mathrm{End}_A(L)$ has depth at least two, hence

$$H^0_{\mathfrak{m}}(E(L)^{**}) = H^1_{\mathfrak{m}}(E(L)^{**}) = 0.$$

Apply local cohomology to

$$0 \longrightarrow E(L) \longrightarrow E(L)^{**} \longrightarrow \mathfrak{P} \longrightarrow 0.$$

Since $\mathfrak{P}$ has finite length, $H^0_{\mathfrak{m}}(\mathfrak{P}) = \mathfrak{P}$, and the connecting map gives the claimed isomorphism. □

3 Compression is associated-graded torsion

Lemma 3.1 (Order is unchanged by a nonzero left coefficient)**.** *Let $0 \neq a \in A$ and $P \in D$. Then*

$$\mathrm{ord}(aP) = \mathrm{ord}(P).$$

Proof. For every $b \in A$,

$$[aP, b] = a[P, b],$$

because a and b commute. Iterating commutators shows that every $(q+1)$-fold commutator of aP is a times the corresponding commutator of P. Since A is a domain, multiplication by a is injective on the A-valued outputs. Thus aP has order at most q if and only if P has order at most q. □

Theorem 3.2 (Compression–associated-graded torsion identity)**.** *For every $m \geq 1$,*

$$\boxed{\tau_A(\mathrm{gr}^W_m \Delta) \cong \frac{W_m \cap M_{m-1}}{W_{m-1}}.} \tag{3.1}$$

Proof. Write $\mathrm{gr}^W_m \Delta = W_m / W_{m-1}$. Suppose $x + W_{m-1}$ is A-torsion. Then for some $0 \neq a \in A$,

$$ax \in W_{m-1} \subseteq D_{m-1}.$$

By the preceding lemma, $x \in D_{m-1}$. Since $x \in W_m \subseteq \Delta$, this gives

$$x \in W_m \cap M_{m-1}.$$

Conversely, if $x \in W_m \cap M_{m-1}$, then

$$x \in M_{m-1} = \mathrm{Sat}_A(W_{m-1}; \Delta),$$

so some $0 \neq a \in A$ satisfies $ax \in W_{m-1}$. Thus the class of x in $\mathrm{gr}^W_m \Delta$ is torsion. □

Corollary 3.3 (Compression vanishing criterion)**.**

$$\boxed{\mathfrak{C}_q = 0 \ \forall q \iff \mathrm{gr}^W \Delta \text{ is } A\text{-torsion-free.}} \tag{3.2}$$

Proof. If $\mathfrak{C}_{m-1} \neq 0$, choose $x \in M_{m-1} \setminus W_{m-1}$ and then choose the minimal $r \geq m$ with $x \in W_r$. The class of x in $\mathrm{gr}_r^W \Delta$ is nonzero and is torsion by Theorem 3.1. Conversely, nonzero torsion in $\mathrm{gr}_m^W \Delta$ is represented by an element of $M_{m-1} \setminus W_{m-1}$, hence gives $\mathfrak{C}_{m-1} \neq 0$. □

3.1 PBW torsion survivor

Let

$$T := \tau_A(U) = \ker(U \to D).$$

For the induced filtration on T, write

$$\mathrm{in}_m(T) \subseteq \mathrm{gr}_m^F U \cong \mathrm{Sym}_A^m L$$

for the degree-m initial module.

Theorem 3.4 (PBW compression quotient)**.** *For every m,*

$$\boxed{\tau_A(\mathrm{gr}_m^W \Delta) \cong \frac{\tau_A(\mathrm{Sym}_A^m L)}{\mathrm{in}_m(T)}.} \tag{3.3}$$

Hence

$$\boxed{\frac{\tau_A(\mathrm{Sym}_A^m L)}{\mathrm{in}_m \tau_A(U)} \cong \frac{W_m \cap M_{m-1}}{W_{m-1}}.} \tag{3.4}$$

Proof. Because $W_\bullet$ is the quotient filtration induced by $U \to U/T \cong \Delta$,

$$\mathrm{gr}^W \Delta \cong \mathrm{gr}^F U / \mathrm{in}(T) \cong \mathrm{Sym}_A L / \mathrm{in}(T).$$

If $t \in T \cap F_m U$, then some $0 \neq a \in A$ kills t. Passing to the principal symbol shows that a kills $\mathrm{in}_m(t)$. Thus

$$\mathrm{in}_m(T) \subseteq \tau_A(\mathrm{Sym}_A^m L).$$

For an A-module M and a submodule $N \subseteq \tau_A(M)$ over a domain,

$$\tau_A(M/N) = \tau_A(M)/N.$$

Indeed, if $a(m+N) = 0$, then $am \in N$ and some nonzero b kills N elementwise on the cyclic element am; hence $ba\,m = 0$, so $m \in \tau_A(M)$. The reverse inclusion is immediate. Applying this with $M = \mathrm{Sym}_A^m L$ and $N = \mathrm{in}_m(T)$ gives (3.3), and Theorem 3.1 gives (3.4). □

Definition 3.5 (PBW torsion survivor)**.** *The module*

$$S_m^{\mathrm{PBW}}(L) := \frac{\tau_A(\mathrm{Sym}_A^m L)}{\mathrm{in}_m \tau_A(U_A(L))} \tag{3.5}$$

is the degree-m PBW torsion survivor.

Corollary 3.6.

$$S_m^{\mathrm{PBW}}(L) \neq 0 \iff \frac{W_m \cap M_{m-1}}{W_{m-1}} \neq 0.$$

In particular, the family of PBW torsion survivors detects compression*, not projectivity defect directly.*

3.2 A single word cannot cause compression

Proposition 3.7 (Single-word order strictness)**.** *Let $\delta_1, \dots, \delta_m \in L$ be nonzero. Then*

$$\delta_1 \cdots \delta_m$$

has exact differential order m.

Proof. After tensoring with K, each δ_i is a nonzero k-derivation of K. The degree-m principal symbol of the composition is the product

$$\sigma_m(\delta_1 \cdots \delta_m) = \delta_1 \cdots \delta_m \in \operatorname{Sym}^m_K \operatorname{Der}_k(K).$$

The symmetric algebra of the K-vector space $\operatorname{Der}_k(K)$ is a domain, so this product is nonzero. Therefore the operator has order m over K, and hence over A. □

Remark 3.8. *Compression must therefore come from cancellation among a nontrivial A-linear combination of words, not from spontaneous order loss of one nonzero word.*

4 A stable Lie–Rinehart action on the projectivity defect

For $\delta \in L$, define

$$\nabla^L_\delta(x) := [\delta, x] \qquad (x \in L). \tag{4.1}$$

This is a δ-connection:

$$\nabla^L_\delta(ax) = \delta(a)x + a\nabla^L_\delta(x).$$

For $T \in \operatorname{End}_A(L)$ define

$$\nabla^{\mathrm{End}}_\delta(T) := \nabla^L_\delta \circ T - T \circ \nabla^L_\delta. \tag{4.2}$$

Lemma 4.1 (Stability of finite-rank endomorphisms)**.** *$E(L)$ is stable under $\nabla^{\mathrm{End}}_\delta$ for every $\delta \in L$.*

Proof. For a rank-one endomorphism

$$R_{x,\phi}(y) = \phi(y)x \qquad (x \in L,\ \phi \in L^\vee),$$

a direct calculation gives

$$\nabla^{\mathrm{End}}_\delta(R_{x,\phi}) = R_{[\delta,x],\phi} + R_{x,\delta\cdot\phi},$$

where

$$(\delta \cdot \phi)(y) := \delta(\phi(y)) - \phi([\delta, y]).$$

The latter is again A-linear in y, so both terms lie in $E(L)$. □

Theorem 4.2 (Stable Lie–Rinehart action)**.** *The quotient*

$$\mathfrak{P}(A) = \operatorname{End}_A(L)/E(L)$$

carries a canonical flat Lie–Rinehart L-connection. Explicitly,

$$\nabla^{\mathfrak{P}}_\delta([T]) := [\nabla^{\mathrm{End}}_\delta(T)] \tag{4.3}$$

is well defined, satisfies

$$\nabla^{\mathfrak{P}}_\delta(a[T]) = \delta(a)[T] + a\nabla^{\mathfrak{P}}_\delta([T]),$$

and is A-linear in δ:

$$\nabla^{\mathfrak{P}}_{a\delta} = a\nabla^{\mathfrak{P}}_\delta. \tag{4.4}$$

Its curvature is zero.

Proof. Only A-linearity in δ requires care. For $x \in L$,

$$[a\delta, x] = a[\delta, x] - x(a)\delta.$$

Hence

$$\nabla^L_{a\delta} - a\nabla^L_\delta = -\big(x \mapsto x(a)\delta\big),$$

and the correction is a rank-one endomorphism, hence lies in $E(L)$. Therefore the induced operator on $\mathfrak{P}$ is A-linear in δ. Flatness follows from the Jacobi identity:

$$[\nabla^{\mathrm{End}}_\delta, \nabla^{\mathrm{End}}_\eta] = \nabla^{\mathrm{End}}_{[\delta,\eta]}.$$

Thus $\mathfrak{P}$ is a Lie–Rinehart module. □

Let

$$e := [\mathrm{id}_L] \in \mathfrak{P}(A). \tag{4.5}$$

Corollary 4.3 (Horizontal identity class)**.**

$$\boxed{\nabla^{\mathfrak{P}}_\delta(e) = 0 \qquad (\delta \in L).} \tag{4.6}$$

Proof. $\nabla^{\mathrm{End}}_\delta(\mathrm{id}_L) = 0$. □

4.1 The canonical projectivity ideal

Definition 4.4. *Define*

$$\boxed{J_L := \mathrm{Ann}_A(e).} \tag{4.7}$$

Then

$$Ae \cong A/J_L. \tag{4.8}$$

Proposition 4.5 (Geometric meaning of J_L)**.**

$$\boxed{V(J_L) = \mathrm{NF}(L).} \tag{4.9}$$

Moreover J_L is stable under every derivation in L.

Proof. After localization at $\mathfrak{p}$,

$$e_{\mathfrak{p}} = 0$$

if and only if $\mathrm{id}_{L_{\mathfrak{p}}}$ lies in the image of $L_{\mathfrak{p}} \otimes L^\vee_{\mathfrak{p}} \to \mathrm{End}(L_{\mathfrak{p}})$. By the dual-basis criterion, this is equivalent to $L_{\mathfrak{p}}$ being projective. Thus the support of Ae is exactly $\mathrm{NF}(L)$, proving (4.9).

If $a \in J_L$, then $ae = 0$. Applying the connection and using horizontality of e gives

$$0 = \nabla_\delta(ae) = \delta(a)e.$$

Hence $\delta(a) \in J_L$. □

4.2 The identity quotient really descends through the torsion-free enveloping quotient

The preceding connection makes $\mathfrak{P}$, hence Ae, a $U_A(L)$-module.

Theorem 4.6 (Identity-quotient descent)**.** *The map*

$$A \longrightarrow Ae, \qquad a \longmapsto ae \tag{4.10}$$

is $U_A(L)$-linear when A carries its natural differential action. Consequently the torsion kernel

$$T = \tau_A(U_A(L)) = \ker(U_A(L) \to D)$$

acts trivially on Ae. Therefore

$$\boxed{A/J_L \cong Ae \quad \textit{is canonically a } \Delta\textit{-module.}} \tag{4.11}$$

If $D = \Delta$, it is canonically a D-module.

Proof. For $a \in A$ and $\delta \in L$,

$$\delta \cdot (ae) = \delta(a)e + a(\delta \cdot e) = \delta(a)e.$$

This is exactly the image of the natural action of δ on $a \in A$. The same is immediate for multiplication by elements of A. Since A and L generate $U_A(L)$, the map is U-linear. Every element of T acts as zero on A because $T = \ker(U \to D)$. Hence it acts as zero on the quotient Ae. □

Corollary 4.7 (Full D-stability under Collapse)**.** *If $D = \Delta$, then*

$$\boxed{P(J_L) \subseteq J_L \qquad (P \in D).} \tag{4.12}$$

Theorem 4.8 (Stable-annihilator theorem)**.** *The canonical projectivity ideal annihilates the whole projectivity defect:*

$$\boxed{J_L\, \mathfrak{P}(A) = 0.} \tag{4.13}$$

Equivalently, $\mathfrak{P}(A)$ is canonically an A/J_L-module.

Proof. Let $a \in J_L$. By definition,

$$a\,[\mathrm{id}_L] = 0 \quad \text{in } \mathfrak{P}(A),$$

so $a\,\mathrm{id}_L \in E(L)$. Every element of $E(L)$ is a finite sum of rank-one maps and therefore factors through a finite free A-module. For any $T \in \mathrm{End}_A(L)$,

$$aT = T \circ (a\,\mathrm{id}_L),$$

which again factors through a finite free module. Hence $a[T] = 0$ in $\mathfrak{P}(A)$. Since T was arbitrary, $a\mathfrak{P}(A) = 0$. □

Remark 4.9. *Theorem 4.8 is stronger than the cyclic statement $J_L e = 0$: it shows that the same canonical ideal controls every stable endomorphism class, not only the identity class.*

5 A differential-core obstruction

Assume $(A, \mathfrak{m})$ is local. Define the full differential core of $\mathfrak{m}$ by

$$\operatorname{core}_D(\mathfrak{m}) := \{a \in A : \ P(a) \in \mathfrak{m} \text{ for every } P \in D\}. \tag{5.1}$$

It is the largest D-stable ideal contained in $\mathfrak{m}$.

Theorem 5.1 (Differential-core obstruction to Silent-Collapse)**.** *Suppose $(A, \mathfrak{m})$ is local and L is projective on the punctured spectrum. If*

$$D = \Delta \quad \textit{and} \quad L \textit{ is nonprojective at } \mathfrak{m},$$

then

$$\boxed{0 \neq J_L \subseteq \operatorname{core}_D(\mathfrak{m}) \subseteq \mathfrak{m}.} \tag{5.2}$$

Consequently,

$$\boxed{\operatorname{core}_D(\mathfrak{m}) = 0 \Longrightarrow \big(D = \Delta \Longrightarrow L \textit{ is projective at } \mathfrak{m}\big).} \tag{5.3}$$

Thus a local minimal Silent-Collapse counterexample necessarily has nonzero differential core.

Proof. Under the displayed nonprojectivity hypothesis, $V(J_L) = \{\mathfrak{m}\}$, so J_L is proper $\mathfrak{m}$-primary. Because the generic tangent module is free, a nonprojective prime cannot be minimal; in the present normal setting it has codimension at least two. Hence A is not a field, and the proper $\mathfrak{m}$-primary ideal J_L is nonzero. By Corollary 4.7, Collapse makes J_L D-stable. As $J_L \subseteq \mathfrak{m}$, the defining maximality of $\operatorname{core}_D(\mathfrak{m})$ yields (5.2). The implication (5.3) is its contrapositive under the punctured-spectrum hypothesis. □

Remark 5.2. *This recovers the familiar D-simple obstruction as a special case, but the hypothesis* $\operatorname{core}_D(\mathfrak{m}) = 0$ *is strictly formulated at the local maximal ideal and is the exact property used in the proof.*

6 Two-dimensional Gorenstein specialization

Assume now that $(A, \mathfrak{m})$ is a two-dimensional normal Gorenstein local domain. Then L is reflexive of rank two. On the regular locus,

$$\det L \cong \omega_A^{-1},$$

and reflexive extension gives

$$(\bigwedge^2 L)^{**} \cong \omega_A^{-1} \cong A. \tag{6.1}$$

Fix a trivialization

$$\omega : \ (\bigwedge^2 L)^{**} \xrightarrow{\sim} A.$$

Define the determinant image ideal

$$I_L := \operatorname{im}\Big(\bigwedge^2 L \longrightarrow (\bigwedge^2 L)^{**} \xrightarrow{\omega} A\Big). \tag{6.2}$$

Lemma 6.1 (Alternating self-duality)**.** *The pairing*

$$L \times L \longrightarrow A, \qquad (x, y) \longmapsto \omega(x \wedge y)$$

induces an isomorphism

$$\boxed{L \xrightarrow{\sim} L^{\vee}.} \tag{6.3}$$

Proof. On the punctured spectrum L is locally free of rank two and the determinant pairing is perfect. Both L and $L^\vee$ are reflexive, so an isomorphism in codimension one extends uniquely over the normal domain. □

Theorem 6.2 (Determinant identification of the projectivity ideal)**.** *Under the two-dimensional Gorenstein hypotheses,*

$$\boxed{J_L = I_L.} \tag{6.4}$$

Proof. Use the self-duality (6.3) to identify

$$L \otimes L^\vee \cong L \otimes L.$$

Let

$$\Theta : L \otimes L \longrightarrow \mathrm{End}_A(L)$$

be the resulting evaluation map. Over K, choose a basis e_1, e_2 with $\omega(e_1 \wedge e_2) = 1$. A direct calculation gives

$$\Theta(e_1 \otimes e_2 - e_2 \otimes e_1) = -\mathrm{id}_{L_K}. \tag{6.5}$$

More generally, the alternating tensor corresponding to $u \in \bigwedge^2 L$ maps to $-\omega(u)\mathrm{id}_L$ inside the generic endomorphism space.

Thus every $a \in I_L$ satisfies $a\,\mathrm{id}_L \in E(L)$, so $I_L \subseteq J_L$.

Conversely suppose $a \in J_L$. Then $a\,\mathrm{id}_L$ lies in the image of Θ. Over K,

$$L_K \otimes L_K = \mathrm{Sym}^2_K L_K \oplus \bigwedge^2 L_K,$$

and under Θ the scalar identity lies entirely in the alternating summand by (6.5). Therefore the symmetric component of any tensor representing $a\,\mathrm{id}$ maps to zero over K. Since $\mathrm{End}_A(L)$ is torsion-free over the domain A, that symmetric image is already zero over A. The alternating component is represented by an element $u \in \bigwedge^2 L$ with $\omega(u) = \pm a$. Hence $a \in I_L$. Thus $J_L = I_L$. □

Corollary 6.3 (Concrete surface counterexample condition)**.** *If a two-dimensional normal Gorenstein local Silent-Collapse counterexample exists, then*

$$I_L = J_L$$

is a proper $\mathfrak{m}$-primary ideal and is stable under every $P \in D$.

Remark 6.4. *This converts the abstract identity class $[\mathrm{id}_L]$ into a concrete determinant defect. It does not yet produce an operator outside Δ. The missing step is a transverse-lifting theorem from the determinant defect to a genuine differential operator.*

7 The decisive logical audit

This section records the point at which the previous research narrative must be corrected.

7.1 Leakage is equivalent to SC.1

Proposition 7.1 (Leakage equivalence)**.** *The statement*

$$\boxed{\mathfrak{P}(A) \neq 0 \Longrightarrow \exists q \geq 0 : \mathfrak{B}_q(A) \neq 0} \tag{7.1}$$

for every A in the standing class is logically equivalent to SC.1.

Proof. By (2.2),

$$\mathfrak{P}(A) = 0 \iff L \text{ is projective.}$$

By (1.4),

$$\mathfrak{B}_q(A) = 0 \ \forall q \iff D = \Delta.$$

Therefore the contrapositive of (7.1) is exactly

$$D = \Delta \Longrightarrow L \text{ projective.}$$

□

Audit Conclusion 7.2. *Calling (7.1) the "last Leakage lemma" is harmless as terminology, but it must not be described as a strictly weaker reduction of SC.1. It* is *SC.1 in defect notation.*

7.2 RDS is also equivalent, not a reduction

Since $D_q = M_q^{**}$, one has

$$D = \Delta \iff M_q = M_q^{**} \quad \forall q. \tag{7.2}$$

By Theorem 2.2,

$$L \text{ projective} \iff E(L) = E(L)^{**}. \tag{7.3}$$

Proposition 7.3 (RDS equivalence)**.** *The universal implication*

$$\boxed{M_q \text{ reflexive for all } q \Longrightarrow E(L) \text{ reflexive}} \tag{7.4}$$

is equivalent to SC.1.

7.3 Compression does not yet exhaust the hidden defect

Define the Escape, compression, and nonprojectivity loci

$$\mathcal{E}(A) := \bigcup_q \operatorname{Supp} \mathfrak{B}_q, \qquad \mathcal{C}(A) := \bigcup_q \operatorname{Supp} \mathfrak{C}_q, \qquad \mathcal{N}(A) := \operatorname{Supp} \mathfrak{P}. \tag{7.5}$$

The known direction is

$$\mathcal{E}(A) \subseteq \mathcal{N}(A), \qquad \mathcal{C}(A) \subseteq \mathcal{N}(A). \tag{7.6}$$

No proved theorem gives the reverse inclusion into either union.

Definition 7.4 (Two logical Silent-Collapse branches)**.** *At a local point with $\mathfrak{P} \neq 0$ and $\mathfrak{B}_q = 0$ for every q, distinguish:*

$$\begin{aligned} &\textit{SC-C:} \quad \mathfrak{C}_q \neq 0 \textit{ for some } q, \\ &\textit{SC-U:} \quad \mathfrak{C}_q = 0 \textit{ for every } q. \end{aligned}$$

SC-C is the compression-hidden *branch. SC-U is the* ultra-silent *branch.*

Proposition 7.5 (PBW survivors detect exactly the SC-C channel)**.** *For a Silent-Collapse candidate,*

$$\exists m:\ S_m^{\mathrm{PBW}}(L) \neq 0 \iff \exists q:\ \mathfrak{C}_q \neq 0. \tag{7.7}$$

Thus PBW torsion survivors detect precisely the compression-hidden branch, not the full Silent-Collapse problem.

Proof. Combine Theorem 3.3 with Corollary 3.2. □

Audit Conclusion 7.6 (Critical correction)**.** *There is currently no proved implication*

$$\mathfrak{P} \neq 0 \Longrightarrow \exists q : \mathfrak{C}_q \neq 0. \tag{7.8}$$

Therefore a hypothetical counterexample is not *presently forced to carry a nonzero PBW torsion survivor. The ultra-silent branch*

$$\mathfrak{P} \neq 0, \qquad \mathfrak{B}_q = \mathfrak{C}_q = 0 \quad \forall q \tag{7.9}$$

has not been constructed, but it has also not been excluded by the current theorem set.

Any proof strategy based solely on PBW compression must first prove (7.8), or independently rule out SC-U.

8 Exact conditional escape criteria

8.1 Artin quotient lifting criterion

Assume for this subsection that $(A, \mathfrak{m})$ is local with residue field $A/\mathfrak{m} = k$ and that J_L is proper $\mathfrak{m}$-primary. For $r \geq 2$, put

$$B_r := A/J_L^r, \qquad \mathfrak{n}_r := \mathfrak{m}/J_L^r. \tag{8.1}$$

Lemma 8.1 (Differential operators on an Artin local k-algebra)**.** *Let B be a finite-dimensional local k-algebra with residue field k. Then*

$$\boxed{D_{B/k} = \mathrm{End}_k(B).} \tag{8.2}$$

Proof. Let $I = \ker(B \otimes_k B \to B)$ be the diagonal ideal. Since B is Artin local with residue field k, the radical of $B \otimes_k B$ is nilpotent and contains I. Hence $I^N = 0$ for $N \gg 0$. For $q \geq N - 1$, the module of principal parts is

$$P^q_{B/k} = (B \otimes_k B)/I^{q+1} = B \otimes_k B.$$

Therefore

$$D^q_{B/k} = \mathrm{Hom}_B(B \otimes_k B, B) \cong \mathrm{Hom}_k(B, B).$$

□

Lemma 8.2 (Radicals of differential ideals in characteristic zero)**.** *Let R be a characteristic-zero k-algebra and let $I \subseteq R$ be stable under every k-derivation under consideration. Then $\sqrt{I}$ is stable under the same derivations.*

Proof. This is the standard characteristic-zero radical-differential-ideal theorem; see Seidenberg [9]. It is invoked here only for $I = J_L$ and $\sqrt{J_L} = \mathfrak{m}$. □

Theorem 8.3 (Artin quotient lifting obstruction)**.** *Assume $D = \Delta$. Then each J_L^r is D-stable. For every $r \geq 2$ the natural reduction map*

$$\mathrm{red}_r : D_{A/k} \longrightarrow D_{B_r/k} \tag{8.3}$$

is not *surjective.*

Proof. The ideal J_L is derivation-stable, hence every power J_L^r is derivation-stable. Multiplication by A also preserves J_L^r. Thus Δ preserves J_L^r, and under $D = \Delta$ so does D.

By Lemma 8.2, the equality $\sqrt{J_L} = \mathfrak{m}$ implies that $\mathfrak{m}$ is derivation-stable; hence every induced derivation on B_r preserves $\mathfrak{n}_r$. Therefore every operator in the image of $\Delta = D$ preserves $\mathfrak{n}_r$.

For $r \geq 2$, $\mathfrak{n}_r \neq 0$. By Lemma 8.1,

$$D_{B_r/k} = \mathrm{End}_k(B_r),$$

and the latter contains a k-linear endomorphism carrying a nonzero element of $\mathfrak{n}_r$ to 1. Such an endomorphism does not preserve $\mathfrak{n}_r$ and hence cannot lie in the image of red_r. □

Remark 8.4. *This theorem is a necessary condition for Silent-Collapse, not a contradiction. The missing theorem would be a special lifting result for the projectivity-ideal thickenings A/J_L^r. No such general surjectivity theorem is presently available.*

8.2 Projectivity-Rees transverse escape criterion

Let

$$\mathcal{R}_{J_L}(A)$$

be the extended J_L-adic Rees algebra and

$$G_J := \mathrm{gr}_{J_L} A$$

its associated graded ring (equivalently, the $u = 0$ fiber of the chosen extended Rees degeneration).

Proposition 8.5 (Transverse escape criterion)**.** *Suppose an order-q operator*

$$\overline{P} \in D^q_{G_J/k}$$

has strictly negative J_L-degree and admits an actual Rees quantization to an operator

$$P \in D^q_{A/k}.$$

Then

$$P \notin \Delta.$$

Consequently $D \neq \Delta$.

Proof. Every derivation preserves J_L, hence preserves each power J_L^n. Therefore every word in derivations, and hence every element of Δ, has nonnegative J_L-adic degree on the associated graded ring. A lift whose associated-graded $u = 0$ action has negative degree cannot belong to Δ. □

Remark 8.6. *Thus a complete Rees proof of SC.1 would require a theorem of the form*

$$J_L \neq A \Longrightarrow \exists \text{ negative-degree Rees-liftable operator on } \mathrm{gr}_{J_L} A.$$

This is precisely the unproved transversality step. For arbitrary graded singularities it is too strong without further hypotheses.

9 The two-dimensional Gorenstein Self-Ext route

Under the hypotheses of Section 6, the stable endomorphism module is

$$\underline{\mathrm{End}}_A(L) := \mathrm{End}_A(L)/\{\text{maps factoring through finite free modules}\}. \tag{9.1}$$

By definition of $E(L)$,

$$\boxed{\mathfrak{P}(A) \cong \underline{\mathrm{End}}_A(L).} \tag{9.2}$$

For a complete or Henselian two-dimensional isolated Gorenstein local ring with Hom-finite stable MCM category, standard stable-category/Auslander–Reiten duality (see, e.g., Buchweitz [17]) gives, up to the conventional canonical-module twist,

$$D_k \,\underline{\mathrm{Hom}}_A(L, L) \cong \mathrm{Ext}^1_A(L, L), \tag{9.3}$$

and the twist is trivial in the Gorenstein surface case. Thus nonprojectivity predicts a nonzero self-extension.

Audit Conclusion 9.1 (Exact status of the Self-Ext route)**.** *Even granting the standard duality (9.3), no canonical theorem has been obtained that sends*

$$0 \neq \mathrm{Ext}^1_A(L, L)$$

to

$$0 \neq \mathfrak{B}_q = D_q/M_q$$

for some q. The missing map is not a matter of notation: $\mathrm{Ext}^1_A(L, L)$ is a deformation/stable-category obstruction, whereas $\mathfrak{B}_q$ is a defect in the order-filtered ring of scalar differential operators. A source-faithful transversality or quantization theorem is still required.

10 Quadratic splitting in the two-dimensional Gorenstein case

Continue under the hypotheses of Section 6. Since $\mathrm{char}\, k = 0$, the scalar 2 is invertible in A and the trace map splits:

$$\mathrm{End}_A(L) = A\,\mathrm{id}_L \oplus \mathrm{End}_{0,A}(L), \tag{10.1}$$

where

$$\mathrm{End}_{0,A}(L) := \ker\big(\mathrm{tr} : \mathrm{End}_A(L) \to A\big).$$

Using the alternating self-duality $L \simeq L^\vee$, decompose

$$L \otimes_A L = \mathrm{Sym}^2_A L \oplus \bigwedge^2 L. \tag{10.2}$$

Definition 10.1 (Symmetric tracefree image)**.** *Let*

$$\beta : \mathrm{Sym}^2_A L \longrightarrow \mathrm{End}_{0,A}(L) \tag{10.3}$$

be the symmetric component of the evaluation map under the alternating self-duality, and set

$$S_2(L) := \mathrm{im}(\beta), \qquad \mathfrak{P}_{\mathrm{sym}}(L) := \mathrm{End}_{0,A}(L)/S_2(L). \tag{10.4}$$

Concretely, after fixing the determinant pairing ω, one may write, up to the harmless unit normalization chosen in the self-duality,

$$\beta(x \odot y)(z) = \omega(x \wedge z)y + \omega(y \wedge z)x. \tag{10.5}$$

This endomorphism has trace zero.

Theorem 10.2 (Scalar–symmetric splitting of the projectivity defect)**.** *Under the two-dimensional normal Gorenstein hypotheses,*

$$\boxed{E(L) = I_L \,\mathrm{id}_L \oplus S_2(L).} \tag{10.6}$$

Consequently

$$\boxed{\mathfrak{P}(A) \cong A/I_L \oplus \mathfrak{P}_{\mathrm{sym}}(L).} \tag{10.7}$$

Moreover, by Theorem 4.8 and $J_L = I_L$,

$$\boxed{I_L\,\mathfrak{P}(A) = 0, \qquad I_L\,\mathfrak{P}_{\mathrm{sym}}(L) = 0.} \tag{10.8}$$

Proof. Over the fraction field K, choose e_1, e_2 with $\omega(e_1 \wedge e_2) = 1$. A direct 2×2 calculation shows that the alternating summand $\bigwedge^2 L_K$ maps to scalar endomorphisms, while the symmetric summand $\mathrm{Sym}^2_K L_K$ maps to trace-zero endomorphisms. The image of $\bigwedge^2 L$ in the scalar line is exactly $I_L \,\mathrm{id}_L$ by the definition of I_L and Theorem 6.2. The image of the symmetric summand is $S_2(L)$ by definition. Since the scalar and trace-zero summands intersect trivially, (10.6) follows. Quotienting the split target (10.1) by (10.6) gives (10.7), and (10.8) follows from Theorem 4.8. □

Audit Conclusion 10.3 (No determinant-to-symmetric coupling theorem)**.** *The decomposition (10.7) does* not *prove*

$$I_L \neq A \Longrightarrow \mathfrak{P}_{\mathrm{sym}}(L) \neq 0. \tag{10.9}$$

An attempted determinant-minor argument was rejected during the Closing Sprint: the target $\mathrm{End}_{0,A}(L)$ *is a rank-three reflexive module and need not be free, so surjectivity onto it does not force its actual determinant image to be the unit ideal. Therefore*

$$\mathfrak{P}_{\mathrm{sym}}(L) = 0 \Longrightarrow L \textit{ projective} \tag{10.10}$$

remains open in the present framework and is not used below.

11 Order-two principal parts and Hessian quantization

Assume now, in addition, that $(A, \mathfrak{m})$ is a two-dimensional normal Gorenstein local complete intersection essentially of finite type over k. Let

$$\mathfrak{d} := \ker(A \otimes_k A \to A)$$

be the diagonal ideal and

$$P^q_{A/k} := (A \otimes_k A)/\mathfrak{d}^{q+1}$$

the module of principal parts of order q; see also the standard principal-parts description of differential operators in [15].

The principal-parts exact sequence

$$0 \longrightarrow \mathfrak{d}^2/\mathfrak{d}^3 \longrightarrow P^2_{A/k} \longrightarrow P^1_{A/k} \longrightarrow 0 \tag{11.1}$$

gives, after applying $\mathrm{Hom}_A(-, A)$,

$$0 \longrightarrow D^1_{A/k} \longrightarrow D^2_{A/k} \longrightarrow \mathrm{Hom}_A(\mathfrak{d}^2/\mathfrak{d}^3, A) \xrightarrow{\partial_2} \mathrm{Ext}^1_A(P^1_{A/k}, A). \tag{11.2}$$

11.1 Removal of the diagonal quadratic defect

Multiplication in the diagonal ideal gives a natural surjection

$$\operatorname{Sym}^2_A \Omega_{A/k} \twoheadrightarrow \mathfrak{d}^2/\mathfrak{d}^3. \tag{11.3}$$

On the punctured spectrum A is regular, hence smooth over the characteristic-zero base field, so (11.3) is an isomorphism there. Its kernel is therefore supported at $\mathfrak{m}$ and has finite length. Since A is two-dimensional Cohen–Macaulay,

$$\operatorname{Hom}_A(K, A) = \operatorname{Ext}^1_A(K, A) = 0$$

for every finite-length module K. Dualizing (11.3) therefore gives an isomorphism

$$\boxed{\operatorname{Hom}_A(\mathfrak{d}^2/\mathfrak{d}^3, A) \cong H_2(A) := \operatorname{Hom}_A(\operatorname{Sym}^2_A \Omega_{A/k}, A).} \tag{11.4}$$

Thus the separate diagonal quadratic defect vanishes in this source class.

Lemma 11.1 (Quadratic symbol target)**.** *With the fixed determinant trivialization from Section 6, there is a reflexive identification*

$$\boxed{H_2(A) \cong \operatorname{End}_{0,A}(L).} \tag{11.5}$$

Proof. The module $H_2(A)$ is reflexive by the Hom–reflexivity lemma. The module $\operatorname{End}_{0,A}(L)$ is a direct summand of the reflexive module $\operatorname{End}_A(L)$ and is therefore reflexive. On the punctured regular locus, $L \simeq \Omega^{\vee}_{A/k}$ is locally free of rank two with trivial determinant; there

$$H_2(A) \simeq \operatorname{Sym}^2_A L \simeq \operatorname{End}_{0,A}(L).$$

Reflexive extension over the normal surface gives (11.5). □

Because $P^1_{A/k} \cong A \oplus \Omega_{A/k}$ as a left A-module,

$$\operatorname{Ext}^1_A(P^1_{A/k}, A) \cong \operatorname{Ext}^1_A(\Omega_{A/k}, A).$$

Set

$$T^1_A := \operatorname{Ext}^1_A(\Omega_{A/k}, A). \tag{11.6}$$

Combining (11.2)–(11.6) yields the canonical order-two boundary

$$\boxed{0 \longrightarrow D^1 \longrightarrow D^2 \longrightarrow H_2(A) \xrightarrow{\partial_2} T^1_A.} \tag{11.7}$$

11.2 Hessian form of the boundary for an lci presentation

Choose a smooth ambient presentation

$$A = S/(f_1, \dots, f_c), \tag{11.8}$$

where $f_1, \dots, f_c$ is a regular sequence and, locally, S has coordinates $x_1, \dots, x_{c+2}$. The conormal resolution identifies

$$T^1_A \cong \operatorname{coker}\Big(A^{c+2} \xrightarrow{J_f} A^c \Big), \tag{11.9}$$

where $J_f = (\partial f_\alpha / \partial x_i)$.

Proposition 11.2 (Hessian boundary formula)**.** *Let $B \in H_2(A)$ be represented in the ambient coordinates by a symmetric matrix (b_{ij}) satisfying the cotangent relations. Then $\partial_2(B)$ is represented in the cokernel (11.9) by*

$$\boxed{\left(\sum_{i,j} b_{ij}\,\frac{\partial^2 f_\alpha}{\partial x_i \partial x_j}\right)_{\alpha=1}^{c},} \tag{11.10}$$

up to the invertible factor determined by whether mixed terms in the principal symbol are normalized with $1/2$. In particular, the kernel of ∂_2 is exactly the sector of quadratic symbols that quantize to actual order-two differential operators on A.

Proof. Lift the coefficients of B to S and choose an ambient second-order operator with principal part represented by B. Because B already defines a bilinear form on $\Omega_{A/k}$, the principal symbol of the commutator with each defining equation vanishes: the order-one part of $[P_B, f_\alpha]$ is the contraction of B with df_α, which is zero in $\Omega_{A/k}$. The remaining order-zero obstruction to preserving f_α modulo $(f_1, \dots, f_c)$ is therefore the second derivative term (11.10), modulo the image of the Jacobian map. Adding a first-order correction changes precisely this order-zero vector by an element of the Jacobian image. The connecting morphism in (11.7) records this class. Hence its vanishing is equivalent to the existence of the required first-order correction and therefore to quantizability. □

11.3 Exact order-two defect decomposition

Under (11.5), the image $S_2(L)$ from Section 10 is the principal-symbol image of products of first-order derivations. Since such products are actual differential operators,

$$S_2(L) \subseteq \ker(\partial_2). \tag{11.11}$$

Thus ∂_2 descends to

$$\overline{\partial}_2 : \mathfrak{P}_{\mathrm{sym}}(L) \longrightarrow T^1_A. \tag{11.12}$$

Define

$$\mathcal{H}_2(A, L) := \mathrm{im}(\overline{\partial}_2) \tag{11.13}$$

and call it the *Hessian quantization defect.*

Let

$$Q_2 := \ker(\partial_2) \subseteq H_2(A). \tag{11.14}$$

Since $D^1 = W_1$, the order-two symbol sequence gives

$$\frac{D^2}{W_2} \cong \frac{Q_2}{S_2(L)}. \tag{11.15}$$

Combining (11.15) with the inherited NC exact sequence

$$0 \to \mathfrak{C}_2 \to D^2/W_2 \to \mathfrak{B}_2 \to 0$$

yields the following.

Theorem 11.3 (Quadratic defect filtration)**.** *For a two-dimensional normal Gorenstein local complete intersection,*

$$\boxed{0 \longrightarrow \mathfrak{C}_2 \longrightarrow \ker(\overline{\partial}_2) \longrightarrow \mathfrak{B}_2 \longrightarrow 0,} \tag{11.16}$$

and

$$\boxed{0 \longrightarrow \ker(\overline{\partial}_2) \longrightarrow \mathfrak{P}_{\mathrm{sym}}(L) \longrightarrow \mathcal{H}_2(A, L) \longrightarrow 0.} \tag{11.17}$$

At a minimal isolated nonprojective point all modules displayed are finite length, and therefore

$$\boxed{\ell_A(\mathfrak{P}_{\mathrm{sym}}) = \ell_A(\mathfrak{C}_2) + \ell_A(\mathfrak{B}_2) + \ell_A(\mathcal{H}_2).} \tag{11.18}$$

If $D = \Delta$, then $\mathfrak{B}_2 = 0$ and hence

$$\boxed{0 \longrightarrow \mathfrak{C}_2 \longrightarrow \mathfrak{P}_{\mathrm{sym}}(L) \longrightarrow \mathcal{H}_2(A, L) \longrightarrow 0.} \tag{11.19}$$

Proof. Equation (11.16) is the NC order-two exact sequence after the identification (11.15). Equation (11.17) is the definition of $\mathcal{H}_2$ together with

$$\ker(\overline{\partial}_2) = Q_2/S_2(L).$$

The length formula follows by additivity. Under Collapse, $\mathfrak{B}_2 = 0$ by the exact Escape–Collapse criterion, so (11.16) identifies $\ker(\overline{\partial}_2)$ with $\mathfrak{C}_2$, yielding (11.19). □

Audit Conclusion 11.4 (Direction of the Hessian defect)**.** *A nonzero class in $\mathcal{H}_2$ is a class of a quadratic symbol that* fails *to quantize to an order-two differential operator. Therefore*

$$\mathcal{H}_2 \neq 0$$

does not, by itself, imply the existence of an operator in $D \setminus \Delta$. Any route converting $\mathcal{H}_2 \neq 0$ into Escape requires an additional higher-order transgression theorem. No such theorem is proved here.

12 Closing-Sprint route audit

12.1 Negative-degree Rees is not a universal Escape mechanism

The transverse criterion of Section 8 remains correct as a conditional implication: a Rees-liftable operator of strictly negative J_L-degree lies outside Δ. What is rejected is its use as a universal supply theorem. In the classical graded surface-hypersurface control

$$R = k[x, y, z]/(f), \qquad f \text{ homogeneous of degree at least } 3$$

with an isolated singularity, Vigué's results, as recalled and extended in [16], show simultaneously that there are no differential operators of negative graded degree and that genuinely new higher-order generators occur in every order. Thus strong Escape need not be witnessed by negative degree. The implication

$$\text{Escape} \Longrightarrow \text{negative-degree Escape}$$

is false even in a central calibration family, so negative-degree detection is demoted from universal architecture to one optional sufficient mechanism.

12.2 Compression route

The inherited PBW theorem gives

$$\mathfrak{C}_q = 0 \ \forall q \iff \mathrm{gr}^W \Delta \text{ is } A\text{-torsion-free.} \tag{12.1}$$

A strong sufficient condition is torsion-freeness of the relevant symmetric-power algebra before the PBW action-kernel quotient. Linear-type conditions on the tangent module would provide one route to such torsion control, but the lci hypothesis alone supplies no theorem in this package forcing that condition. Hence universal elimination of the Q-C/compression channel remains open.

12.3 Closing-Sprint status

The three-cycle sprint produced new exact structure, but did not prove SC.1 and did not construct a counterexample. The remaining high-value bridges are now:

(R1) **determinant-to-symmetric coupling:** decide whether $I_L \neq A$ forces $\mathfrak{P}_{\mathrm{sym}} \neq 0$ for tangent modules in the declared source class;
(R2) **compression elimination:** prove a source-faithful torsion/linear-type theorem forcing $\mathfrak{C}_2 = 0$ (or all $\mathfrak{C}_q = 0$);
(R3) **Hessian transgression:** turn $\mathcal{H}_2 \neq 0$ into a genuine higher-order Escape class by an actual theorem;
(R4) **counterexample construction:** realize all minimal Silent-Collapse conditions simultaneously.

No current theorem closes any one of these universally. Accordingly the SC.1 Phase-I sprint is placed on HOLD after Section 15; this is a research-management status, not a mathematical truth-value statement.

13 Maximal necessary conditions for a counterexample

Theorem 13.1 (Minimal Silent-Collapse counterexample package)**.** *If SC.1 is false, then after localizing at a prime minimal in* $\mathrm{NF}(L)$ *there exists a local normal domain* $(A, \mathfrak{m})$ *satisfying all of the following:*

(i) *A is essentially of finite type over a characteristic-zero field;*
(ii) *$L = \mathrm{Der}_k(A)$ is reflexive and is projective on* $\operatorname{Spec} A \setminus \{\mathfrak{m}\}$*;*
(iii) *L is nonprojective at $\mathfrak{m}$;*
(iv) *$D = \Delta$;*
(v) *$\mathfrak{B}_q = 0$ for all q, equivalently every M_q is reflexive;*
(vi) *J_L is a nonzero proper $\mathfrak{m}$-primary ideal;*
(vii) *J_L is stable under every operator in D;*
(viii) $\mathrm{core}_D(\mathfrak{m}) \neq 0$*;*
(ix) *under Collapse, $D = \Delta$ is finitely generated as an A-algebra;*
(x) *if, in addition, the localized residue field satisfies $A/\mathfrak{m} = k$, then every reduction map*

$$D_{A/k} \to D_{(A/J_L^r)/k} \qquad (r \geq 2)$$

fails to be surjective;
(xi) *either the compression-hidden branch SC-C occurs, in which case a PBW torsion survivor is nonzero, or the ultra-silent branch SC-U occurs, in which case*

$$\mathrm{gr}^W \Delta$$

is A-torsion-free.

Proof. Items (i)–(v) are the NC minimal-core and Escape–Collapse reductions. Items (vi)–(viii) are Sections 4–5. For (ix), choose finite A-module generators of L. They generate $\Delta = A\langle L\rangle$ as an A-algebra, hence under Collapse they generate D as an A-algebra. No claim is made that the localized essentially-finite-type ring A or D is finitely generated as a k-algebra. Item (x) is the conditional residue-field refinement supplied by Theorem 8.2; it is not asserted when $A/\mathfrak{m} \neq k$. Item (xi) is the exhaustive dichotomy of Section 7 together with the PBW quotient theorem. □

Corollary 13.2 (Two-dimensional Gorenstein counterexample package)**.** *In addition to the preceding conditions, a two-dimensional normal Gorenstein local counterexample must satisfy*

$$J_L = \mathrm{im}\Big(\bigwedge^2 L \to (\bigwedge^2 L)^{**} \cong A\Big),$$

and this determinant ideal must be proper, $\mathfrak{m}$-primary, and D-stable.

14 Routes that are closed, routes that remain open

Route	Status	Exact reason
PBW failure from nonprojectivity	Rejected	Field-base PBW holds without A-projectivity.
Plain reflexivity of L	Rejected	L is already reflexive over normal A.
Full D-module structure on $\mathfrak{P}$	Not automatic	U-torsion need not annihilate all of $\mathfrak{P}$.
Identity cyclic quotient Ae	Proved	$Ae \cong A/J_L$ descends through $U/\tau_A(U) \cong \Delta$.
Stable annihilator $J_L\mathfrak{P} = 0$	Proved	Factorization of $a\,\mathrm{id}_L$ through a finite free module propagates to every endomorphism class.
PBW compression survivor	Proved as detector	Detects $\mathfrak{C}_\bullet$, not $\mathfrak{P}$ directly.
$\mathfrak{P} \neq 0 \Rightarrow$ PBW survivor	Open	Ultra-silent branch is not excluded.
Brown direct route	Rejected as universal solver	Differential-prime hypothesis is independent and fails on the minimal projectivity locus.
Artin quotient route	Conditional	Silent-Collapse forces quotient-lifting maps to fail; no general surjectivity theorem is available.
Negative-degree J_L-Rees route	Conditional / demoted	The transverse criterion is valid, but known graded hypersurface Escape need not occur in negative degree.
2D Gorenstein determinant identification	Proved	$J_L = I_L$.
Scalar–symmetric projectivity split	Proved	$\mathfrak{P} \cong A/I_L \oplus \mathfrak{P}_{\mathrm{sym}}$.
$I_L \neq A \Rightarrow \mathfrak{P}_{\mathrm{sym}} \neq 0$	Open	The rejected determinant-minor argument does not control the determinant image of nonfree $\mathrm{End}_0(L)$.
2D Gorenstein lci quadratic filtration	Proved	$\mathfrak{P}_{\mathrm{sym}}$ is filtered by $\mathfrak{C}_2$, $\mathfrak{B}_2$, and the Hessian quantization defect $\mathcal{H}_2$.
$\mathcal{H}_2 \neq 0 \Rightarrow$ Escape	Open / wrong direction directly	$\mathcal{H}_2$ records failed quantization; a higher-order transgression theorem is required.
Universal compression elimination	Open	Requires new torsion/linear-type input not supplied by the lci hypothesis alone.
Self-Ext transversality	Open	No proved map $\mathrm{Ext}^1_A(L,L) \to \mathfrak{B}_q$.
Leakage theorem	Open / equivalent target	It is exactly SC.1 in contrapositive defect form.

15 Final theorem and Closing-Sprint verdict

Theorem 15.1 (Integrated maximal internal closure for SC.1 as of 20 October 2026)**.** *From the standing hypotheses, the NC.21–NC.23 boundary, the original standalone reductions, and*

the Closing-Sprint arguments proved in this integrated edition, one obtains the following exact package:

(a) *projectivity defect and Escape defect are reflexive-hull defects;*
(b) *compression is exactly associated-graded A-torsion and is measured by PBW torsion survivors;*
(c) *the projectivity defect carries a canonical stable Lie–Rinehart action;*
(d) *the horizontal identity class produces the canonical projectivity ideal J_L, which under Collapse is D-stable;*
(e) *the stronger annihilation relation $J_L\mathfrak{P} = 0$ holds without assuming Collapse;*
(f) *a local minimal Silent-Collapse candidate forces a nonzero differential core;*
(g) *in dimension two and the normal Gorenstein case, $J_L = I_L$ and*

$$\mathfrak{P} \cong A/I_L \oplus \mathfrak{P}_{\mathrm{sym}};$$

(h) *for two-dimensional normal Gorenstein local complete intersections, order two admits the exact filtration*

$$0 \to \mathfrak{C}_2 \to \ker(\overline{\partial}_2) \to \mathfrak{B}_2 \to 0, \qquad 0 \to \ker(\overline{\partial}_2) \to \mathfrak{P}_{\mathrm{sym}} \to \mathcal{H}_2 \to 0;$$

(i) *under Collapse the preceding filtration reduces to*

$$0 \to \mathfrak{C}_2 \to \mathfrak{P}_{\mathrm{sym}} \to \mathcal{H}_2 \to 0;$$

(j) *a nonzero Hessian quantization defect does not itself yield Escape, and the lci hypothesis alone does not eliminate compression;*
(k) *the implication $I_L \neq A \Rightarrow \mathfrak{P}_{\mathrm{sym}} \neq 0$ remains open;*
(l) *the Leakage and Reflexive Defect Synchronization statements remain equivalent to SC.1, not weaker reductions.*

No theorem in this file excludes every Silent-Collapse branch, and no algebra satisfying the corrected full counterexample package is constructed. Therefore

SC.1 remains UNRESOLVED in the present internal framework.

Audit Conclusion 15.2 (Closing-Sprint HOLD decision)**.** *The three-cycle SC.1 Phase-I sprint satisfies the roadmap HOLD condition. It produced reusable exact structure, but the remaining progress would require at least one genuinely new bridge of a type not presently available:*

(1) *determinant-to-symmetric coupling for tangent modules;*
(2) *source-faithful elimination of quadratic or all-order compression;*
(3) *Hessian/nonquantization-to-higher-order-Escape transgression;*
(4) *stable-to-differential quantization;*
(5) *or an explicit Silent-Collapse counterexample satisfying the corrected package.*

Accordingly SC.1 is placed on HOLD*. This status records a research boundary only. It neither proves SC.1 nor supplies evidence that SC.1 is false.*

Remark 15.3 (Re-entry condition)**.** *SC.1 should be reopened when a later AK problem supplies a new multi-projection, torsion-control, transgression, or quantization mechanism that attacks one of the five HOLD obligations above. The roadmap therefore moves next to Terao's conjecture to develop Poset-Indexed Multi-Projection, with the explicit intention of later returning that machinery to the NC/SC.1 boundary.*

16 Non-claims

This integrated note does *not* claim:

(i) that every Silent-Collapse counterexample has nonzero compression;
(ii) that PBW torsion survivors are unavoidable under tangent nonprojectivity;
(iii) that $D(A) \to D(A/J)$ is generally surjective;
(iv) that every proper determinant/projectivity ideal produces a negative-degree differential operator on its associated graded ring;
(v) that negative degree is necessary for Escape;
(vi) that $\mathfrak{P}_{\mathrm{sym}} = 0$ forces L to be projective;
(vii) that $I_L \neq A$ forces $\mathfrak{P}_{\mathrm{sym}} \neq 0$;
(viii) that $\mathcal{H}_2 \neq 0$ alone produces an operator outside Δ;
(ix) that the lci hypothesis forces L or its symmetric algebra to be of linear type;
(x) that Auslander–Reiten duality alone produces scalar differential operators outside Δ;
(xi) that order-by-order non-generation is identical with algebraic Escape;
(xii) that the absence of a known counterexample proves SC.1;
(xiii) that the Closing-Sprint HOLD decision is a mathematical truth-value statement.

17 Research freeze and handoff

The integrated SC.1 research layer is frozen at the boundary proved above. The former instruction to choose immediately among SC-U exclusion, determinant–Rees transversality, or counterexample construction is superseded by the Closing-Sprint audit.

The exact frozen re-entry ledger is

SC1R1 :	$I_L \neq A \implies \mathfrak{P}_{\mathrm{sym}} \neq 0$?
SC1R2 :	$\mathfrak{C}_2 = 0$ (or $\mathfrak{C}_q = 0\ \forall q$) from a new tangent-module theorem?
SC1R3 :	$\mathcal{H}_2 \neq 0 \implies \exists q \geq 3 : \mathfrak{B}_q \neq 0$?
SC1R4 :	$\underline{\mathrm{End}}_A(L)$ or $\mathrm{Ext}^1_A(L, L) \longrightarrow \mathfrak{B}_q$ by a proved quantization map?
SC1R5 :	construct a normal characteristic-zero Silent-Collapse counterexample.

Until one of these inputs is supplied, repeating PBW, plain-reflexivity, Brown, negative-degree-only Rees, or unquantized Hessian arguments is prohibited as a claimed SC.1 closure route.

The roadmap therefore advances to the next problem, Terao's conjecture, for the explicit purpose of constructing Poset-Indexed Multi-Projection. Any later return to SC.1 must state which frozen re-entry item is being attacked and which genuinely new mechanism has become available.

References

[1] A. Kobayashi, *AK High-Dimensional Projection Structural Theory, Appendix NC — Complete Re-audited Edition: NC.1–NC.23: Nakai Resolution Ownership, Escape Theory, Rees Transfer, Collapse Mechanisms, and the Escape–Collapse Boundary*, AK-HPDST Project, 18 October 2026.

[2] G. S. Rinehart, *Differential forms on general commutative algebras*, Trans. Amer. Math. Soc. **108** (1963), 195–222.

[3] T. Lamkin, *Universal Enveloping Algebras of Poisson Superalgebras*, Algebr. Represent. Theory **28** (2025), 157–191. The Lie–Rinehart PBW input is Theorem 5.7.

[4] W. C. Brown, *Higher derivations on finitely generated integral domains*, Proc. Amer. Math. Soc. **42** (1974), 23–27.

[5] C. J. Rego, *Remarks on differential operators on algebraic varieties*, Osaka J. Math. **14** (1977), 481–486.

[6] J. Becker, *Higher derivations and integral closure*, Amer. J. Math. **100** (1978), 495–521.

[7] M. Hochster, *The Zariski–Lipman conjecture*, J. Algebra **47** (1977), 411–424.

[8] R. Hart, *Derivations on Commutative Rings*, J. London Math. Soc. (2) **8** (1974), 171–175.

[9] A. Seidenberg, *Differential Ideals in Rings of Finitely Generated Type*, Amer. J. Math. **89** (1967), 22–42.

[10] H. Brenner, J. Jeffries, and L. Núñez-Betancourt, *Quantifying singularities with differential operators*, Adv. Math. **358** (2020), 106843.

[11] The Stacks Project Authors, *The Stacks Project*, Tag 0AV3: Hom into a finite reflexive module is reflexive.

[12] The Stacks Project Authors, *The Stacks Project*, Tags 0AVB, 0AVC: reflexive modules over normal domains and height-one reconstruction.

[13] The Stacks Project Authors, *The Stacks Project*, Tag 0AY6: codimension-two extension of reflexive coherent sheaves.

[14] The Stacks Project Authors, *The Stacks Project*, Tag 0DWR: local cohomology and the punctured-spectrum exact sequence.

[15] The Stacks Project Authors, *The Stacks Project*, Tag 09CH, "Finite order differential operators," including modules of principal parts.

[16] R. N. Diethorn, J. Jeffries, C. Miller, N. Packauskas, J. Pollitz, H. Rahmati, and S. Vassiliadou, *Resolutions of differential operators of low order for an isolated hypersurface singularity*, Michigan Math. J. **76** (2026), no. 2, 261–337, doi:10.1307/mmj/20236386.

[17] R.-O. Buchweitz, *Maximal Cohen–Macaulay Modules and Tate Cohomology*, Mathematical Surveys and Monographs, vol. 262, American Mathematical Society, 2021.

[18] S. S.-T. Yau, Q. Zhu, and H. Zuo, *Nakai conjectures for isolated homogeneous hypersurface singularities*, arXiv:2604.24508 (2026).